



Multi-affordance Dynamics on Music Streaming Platforms. A Case Study on Deezer

Dougal Shakespeare

► To cite this version:

Dougal Shakespeare. Multi-affordance Dynamics on Music Streaming Platforms. A Case Study on Deezer. Computer Science [cs]. École des hautes Études en Sciences sociales (EHESS), 2025. English. ⟨NNT : ⟩. ⟨tel-05461977⟩

HAL Id: tel-05461977

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HAL Authorization

École doctorale de l'EHESS

Centre d'analyse et de mathématique sociales (EHESS)

Doctorat

Discipline : Mathématiques et informatique

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Multi-affordance Dynamics on Music Streaming Platforms

A Case Study on Deezer

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Acknowledgements

I would like to thank all those who have supported me throughout the PhD journey in one way or another. In no particular order, thanks to Camille Roth for being an amazing and insightful supervisor. You left me enough space to explore my own ideas thoroughly whilst also offering me the guidance I needed to put me back on track when I went too deep down the dreaded but sometimes insightful *rabbit hole*. Thanks to the whole Computational Social Science team at the Centre Marc Bloch which sadly is no more. Thanks to Telmo for putting up with the endless server requests after the lab was no more. Thanks to Jérémie for our Japan and Berlin collaboration and teaching me the magic of qualitative research. Thanks to Lena for the helpful conversations we had about the PhD and also, for listening to me when I was stressed! Thanks to everyone who had to put up with me rambling about the PhD in particular, Myriam who had to put up with the vast majority of it. Thanks again to Myriam. Not only did the PhD bring me on an interesting academic journey but also, it allowed me to meet one of the most important people of my life. Thanks to Lorenzo for giving me the idea and motivation to do the PhD. If it wasn't for working with you and your supervision during my time at the Pompeu Fabra University's Music Technology Group in Barcelona, I doubt I would be on the path I am now, whatever that may be. Thanks to everyone in RECORDS including particularly, Manuel for putting up with our endless data requests; I promise this is now over. Thanks to Thomas for offering some valuable insights and support in the thesis committee. Thank you to Marion for giving us the opportunity to host the Networks and Music Session at Sunbelt. Thank you all for occasionally (I joke) resorting to English. Thanks to CNRS for the French lessons. Thank you to Bilge for the industry experience and allowing me to further dip my toes into the world of UX research. Even if things didn't end as we both would have wanted it was still a pleasure to work with you and I wish you all the best! Thank you to the rapporteuses for taking the time to read over this thesis and what's more, offer your insights and experience. Thanks to my Mum for always being there to support and encourage me. And finally, thanks to my Dad, who

sadly isn't here to see me defend. I know it doesn't compete with your thesis on Tardi, but I hope you would have been happy to see what I have done over the last few years and that I am bringing this to a close. On to the next chapter.

Abstract

Consumers of modern music have multiple ways to discover and engage with music online. Platforms like Deezer and Spotify offer a broad range of affordances such as algorithmic recommendations, personal playlists, search functions, and editorial playlists, that enable users to interact with music in highly personalised and moreover, varied ways. Despite the potential for these platforms to transform music consumption practices, existing research exploring how exactly these platforms are adopted and how their users consume music in a guided or autonomous manner, and furthermore, in a diverse or concentrated manner, remains poorly understood to this day.

At the same time, the multi-affordance nature of these platforms also offers novel opportunities to examine the influence of algorithmic devices on users' listening diversity. By studying what users do *organically* (i.e., in the relative absence of platform recommendations), a novel strand of the state of the art has emerged, aiming to exploit data on interaction with multi-affordance platforms to assess the impact of algorithmic guidance. This approach offers a novel framework to assess algorithmic confinement by accounting for organic user behaviour, which has indeed escaped prior attempts to conceptualise and measure the effects known as “filter bubbles”. Through the large-scale analysis of users' selection and algorithms' exposure logs on the French music streaming platform Deezer, this thesis aims to provide an in-depth analysis of the use of multi-affordance platforms in practice and the role of algorithmic recommendation systems in promoting diverse or rather confined listening practices. The contribution of this thesis lies at the intersection of Human-Computer Interaction, Information Retrieval and the Sociology of Consumption, offering a new and unique perspective on the interplay between streaming platform design, algorithmic systems and cultural consumption.

Keywords Music Information Retrieval, Human Computer Interaction, Filter Bubbles

Résumé

Les consommateurs de musique moderne disposent de multiples moyens pour découvrir et apprécier la musique en ligne. Des plateformes comme Deezer et Spotify offrent un large éventail de possibilités - recommandations algorithmiques, listes de lecture personnelles, fonctions de recherche et listes de lecture éditoriales - qui permettent aux utilisateurs d'interagir avec la musique de manière hautement personnalisée et, de surcroît, potentiellement variée. Malgré le potentiel de ces plateformes à transformer les pratiques de consommation musicale, la manière exacte dont ces plateformes sont adoptées et dont leurs utilisateurs consomment de la musique de manière guidée ou autonome et, en outre, de manière diversifiée ou concentrée, reste à ce jour mal comprise. Dans le même temps, la nature multi-affordance de ces plateformes offre également de nouvelles possibilités d'examiner l'influence des dispositifs algorithmiques sur la diversité d'écoute des utilisateurs. En étudiant ce que les utilisateurs font « organiquement » (c'est-à-dire en l'absence relative de recommandation de la part de la plateforme), un nouveau volet de l'état de l'art a vu le jour, visant à exploiter les données d'interaction avec les plateformes multi-affordances afin d'évaluer l'impact du guidage algorithmique. Cette approche offre un nouveau cadre pour évaluer le confinement algorithmique en tenant compte du comportement organique de l'utilisateur, qui a en effet échappé aux tentatives antérieures de conceptualisation et de mesure des effets dits de « bulles de filtre ». Grâce à une analyse à grande échelle des historiques d'écoute des utilisateurs et notamment des modes d'accès au contenu (algorithmique ou non) sur la plateforme française de streaming musical Deezer, cette thèse vise à fournir une analyse approfondie de l'utilisation des plateformes multi-affordances en pratique et du rôle des dispositifs de recommandation algorithmique dans la promotion de pratiques d'écoute diversifiées ou plutôt confinées. La contribution de cette thèse se situe à l'intersection des études des interactions homme-machine, de la recherche d'information et de la sociologie de la consommation, offrant une perspective nouvelle et inédite sur l'interaction entre la conception des plateformes de streaming, les dispositifs algorithmiques et la consommation culturelle.

Mots-clés Music Information Retrieval, Human Computer Interaction, Filter Bubbles

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Introduction

"Each subsequent [music] format has less physical presence while allowing for more storage and greater possibilities for user programming"
(McCourt, 2005, p. 249)

Background

With music streaming platforms now accounting for over two thirds of global music revenues (International Federation of the Phonographic Industry (IFPI), 2025), their influence on music retrieval and consumption is not to be understated. Music streaming platforms provide unprecedented access to ever-expanding catalogues of music: Spotify alone hosts more than 100 million tracks (Spotify Newsroom, 2025), while the French-based service Deezer offers over 120 million (Deezer, 2025). More music than could be listened to in an entire life time. As such, today's music consumer is no longer constrained by *scarcity* as was once the case in a market dominated by physical sales. Instead, in absence of the right kind of platform design, they may even be limited by *choice overload* (Romero Meza and D'Urso, 2024).

To help users navigate these vast online cultural repositories, it has become commonplace for platforms to integrate various so-called programmed *affordances* (Davis, 2020) into their design. As with traditional radio listening, the music they surface is typically pre-programmed, curated into thematically coherent lists, and presented in a ready-to-consume format. At the same time, programmed affordances differ from traditional radio in one crucial respect: the music they deliver is often highly personalised, shaped by algorithmic systems drawing on users' prior consumption patterns. Typical examples of these kinds of affordances include algorithmic recommendation devices like Spotify's "Discover Weekly" or Deezer's "Flow" and human/hybrid editorial playlists variously assigned

labels such as "Top 50 Global", "RapCaviar" and "Rock Essentials".¹ These kinds of programmed affordances are in a sense, new forms of cultural intermediaries which have both evolved and taken shape alongside the widespread adoption of music streaming platforms. Despite a growing number of works that lean towards the idea that these affordances, in particular those of the algorithmic variety, may facilitate passive (Hesmondhalgh, 2022; Prey, 2019) and moreover, confined consumption practices (Pariser, 2011), the impact they can have upon music consumption *in practice rather than principle* remains to date, under documented.

At the same time, existing literature exploring how in practice users make use of modern day music streaming platforms indicates that for most users, programmed affordances do not play a marked role in facilitating music retrieval and consumption. In practice, music streaming platforms do not demand that users adopt programmed affordances. These platforms are relatively unique in that they also commonly offer several possibilities for *organic* music engagements (i.e., without programmed guidance) and furthermore, this is one of the most widely adopted modes of access on-platform (Villermet et al., 2021; Beuscart et al., 2019). For instance, organic personal playlists allow users to save and revisit their favourite music whilst classical features like the search bar allow users to perform organic query driven retrievals of music much akin to a Google search. In principle, a user may go the entirety of their platform lifespan without leveraging programmed affordances.

In short, music streaming platforms have become *multi-affordance* environments, offering diverse ways of discovering and listening to music; they are, in many respects, the “Swiss army knife” of digital platforms. This diversity in design, crucially *afforded* but not imposed, opens up, as we shall demonstrate through the contribution of this thesis, several novel methodological opportunities to examine and re-evaluating existing anxieties around algorithmic guidance and music streaming platforms more generally. From this perspective, and by drawing on the unique combination of consumption and rejection logs derived from interactions with organic, algorithmic and editorial affordances on the Deezer platform, this thesis both underscores the organic agency of users in creating confined listening experience as well as the potentially diversifying role which programmed affordances can in several instances, play. Taken together, the scientific contributions of this thesis cascade to construct a significant challenge to existing techno-determinist narratives that have long shaped discourse in fields such as Human Computer Interaction

¹Even the work of editorials is frequently supported by algorithmic tools (Eriksson, 2020), enabling them to stay current with new releases and emerging trends.

and Media Studies.

Motivation & Research Goals

Advancing the emerging field of multi-affordance streaming platform studies (Anderson et al., 2020; Villermet et al., 2021; Epps-Darling et al., 2020; Mok et al., 2022) through the analysis of anonymised retrospective, multi-affordance usage logs from the French music streaming platform, Deezer, the contribution of this thesis is two-fold. Firstly, it seeks to understand how in practice, multi-affordance music streaming platforms are adopted by end-users and what kinds of consumption and information retrieval practices they facilitate. This provides a valuable contribution to the Sociology of Consumption and Media Studies. Secondly, this thesis aims to illustrate how the development of multi-affordance platforms and the analysis of multi-channel, granular data derived from users' on-platform interactions can provide new insights into the longstanding filter bubble debate (Michiels et al., 2022; Borgesius et al., 2016; Pariser, 2011) by shedding light on how users behave both when adopting and in relative absence of algorithmic devices.

To this avail, we centre this thesis around the following research questions.

- **(RQ1)** How do users adopt music streaming platforms and more precisely, how does this vary under the different affordances they embody?
- **(RQ2)** To what extent can multi-affordance user-platform interaction data provide novel insights into the role of music streaming platforms' personalized algorithmic recommendation devices in fostering either confined or diverse consumption practices?
- **(RQ3)** How do choices in scale and information representation fundamentally influence the conclusions drawn from such analyses?
- **(RQ4)** To what degree are users' selections of algorithmic recommendations also responsible for inducing confined consumption practices?

Addressing each of these research questions provides a nuanced understanding of the interplay between algorithmic design and cultural consumption, forming the foundation of this thesis.

Scientific Contributions

The research within this thesis was undertaken as part of the ANR-funded project, "RECORDS" ² which, alongside one of its main partners Deezer, facilitated confidential access to Deezer listening logs which form the basis for much of the analysis conducted. In the framework of a joint Centre d'analyse et de mathématique sociales (CAMS), Centre national de la recherche scientifique (CNRS) and Centre Marc Bloch (CMB) affiliation, research was physically undertaken in a large part at the CMB's Computational Social Science team ³ and hosted and supervised by Camille Roth. Thesis funding was provided through the team via a mix of ANR-funding stemming from the "RECORDS" grant (ANR-2019-CE38-0013) and thanks to other grants secured by the team.

The contribution of this thesis is based upon four academic articles. Since the thesis is paper-based, sections are mainly built upon these publications.

The first article engages with RQ1 and was published in the International Society of Music Information Retrieval Conference in 2021. It provides a preliminary exploration of how different affordances attract varied engagement dynamics and furthermore, their inter-dependence.

Shakespeare, Dougal, and Camille Roth. "*Tracing Affordance and Item Adoption on Music Streaming Platforms.*" Proceedings of the 22nd International Society for Music Information Retrieval Conference. 2021. (Shakespeare and Roth, 2021)

The second article, published in CompleNet in 2024, then goes on to present a more theoretical analysis of the interpretability of high dimensional item embedding spaces which are deployed later to construct dense concentration measurements. This chapter, although not directly relating to our main research questions, acts as a necessary pre-requisite to conduct much of these analyses that will follow by establishing that the main tools used to capture diversity are grounded and interpretable.

Shakespeare, Dougal, and Camille Roth. "*Interpreting Node Embedding Distances Through n -Order Proximity Neighbourhoods.*" Complex Networks XV. CompleNet-Live 2024. Springer Proceedings in Complexity. Springer, 2024. (Shakespeare and Roth, 2024) ⁴

²<https://records.huma-num.fr/>

³<https://cmb.huma-num.fr/>

⁴Reproduced with permission from Springer Nature.

The third article, published in Nature’s Scientific Reports in 2025, addresses RQ2 and RQ3 and focuses on conducting a thorough analysis of how users’ adopt music streaming platform affordances to achieve varied levels of concentration and furthermore, how this varies depending on the type of information representation and scale deployed.

Shakespeare, Dougal, Victor Chareyron and Camille Roth. "*Reframing the Filter Bubble Through Diverse Scale Effects in Online Music Consumption.*" Scientific reports, 2025. (Shakespeare et al., 2025)

The final article of this thesis follows a mixed method approach and engages mainly with RQ4. This research makes use of rich, large-scale survey data from the RECORDS project (Renisio et al., 2024) to shed light on how users typically vary in their openness - captured via on-platform behaviours and off-platform attitudes - to novelty in algorithmically generated music recommendations. This research presented in this thesis is currently in pre-print format. It is planned to be submitted to the International World Wide Web Conference 2026.

Shakespeare, Dougal, Camille Roth. "*Tracing Desired, Exposed and Adopted Algorithmic Novelty on Music Streaming Platforms*", Pre-print, 2025.

Part I

Related Work and Background

What Do Music Streaming Platforms Afford?

"Stunning amounts of music are created and produced in any year, and the vast majority of it is not destined for attentive engagements."

(Kassabian, 2013, p. 1)

As a first step towards contextualising the contributions of this thesis, we centre our attention on understanding what kind of interactions music streaming platforms *afford* and furthermore, which kinds are more and less common. We commence with a more theoretical literature review of *affordances*, a terminology and framework which we shall use throughout this thesis to refer to the different opportunities for listening that music streaming platforms offer but do not demand from a user. To this end, we outline affordance theory's trajectory through different academic disciplines, leading up to its applications to technology and the study of platforms. Building on this foundation, we then progress to describe literature on two distinct forms of interaction that music streaming platforms are understood to afford: different kinds of information retrieval and different kinds of attention. To this end, we focus on understanding these interaction types through a *passive-active* dichotomy which is both well established in existing social science and media studies literature on music consumer behaviour and also, broad enough to encapsulate more quantitative studies which shed light on how users make use of platforms in practice rather than principle. Together, these two perspectives offer a distinct albeit complementary framing which sheds light on how music streaming platforms are increasingly embedded in users' *everydayness* and the degree to which users are willing to relinquish *organic* control in favour of external guidance, be that human or algorithmic.

The Framework of Affordance

The framework associated with *affordances* whilst currently well established in Media Studies and Human Computer Interaction, was first introduced by the cognitive and ecological psychologist J.J. Gibson in his work "*The Senses Considered as Perceptual Systems*" (Gibson, 1966) and more thoroughly developed in his later endeavours (Gibson, 2014b). Whilst the verb *afford* existed prior to Gibson's formulation, "affordance" was created to fill what Gibson saw as a semantic void. More precisely and in Gibson's own words:

"The affordances of the environment are what it offers the animal, what it provides or furnishes, either for good or ill. The verb to afford is found in the dictionary, but the noun affordance is not. I have made it up. I mean by it something that refers to both the environment and the animal in a way that no existing term does. It implies the complementarity of the animal and the environment." (Gibson, 2014a, p. 119)

Put simply, Gibson describes an affordance to be a relational construct which captures the action possibilities which an environment (e.g. a platform) offers to an agent (e.g. a platform user). According to Gibson, these are said to exist independently of agents' perceptions (i.e., the fact that no agent perceived them does not negate their existence). By means of a more concrete classical example, a door can *afford* the action of pulling or pushing, which can be mediated by its design (e.g. the design of the handle). Likewise, music streaming platforms *afford* different types of guidance and listening (e.g. those that stem from algorithmic devices, searching, editorial playlists etc.) which likewise, can be influenced by how a platform is designed. Affordance theory captures that "*technologies produce fields of action (including unexpected actions), but that not all actions are possible*" (Jordan, 2012, p. 302). This framing positions Gibson's affordance theory between technological determinism and social constructivism - a balanced, intermediate stance that is often cited as one of the theory's key strengths (Hutchby, 2001; Scarlett and Zeilinger, 2019).

Following the development of affordances by Gibson, the theory underwent somewhat of semantic drift as it became more widely studied by scholars, first within the field of Design studies (Norman, 1988, 1999; Davis, 2020) and then later, Human Computer Interaction (Gaver, 1991) and Media Studies (Scarlett and Zeilinger, 2019). Regarding

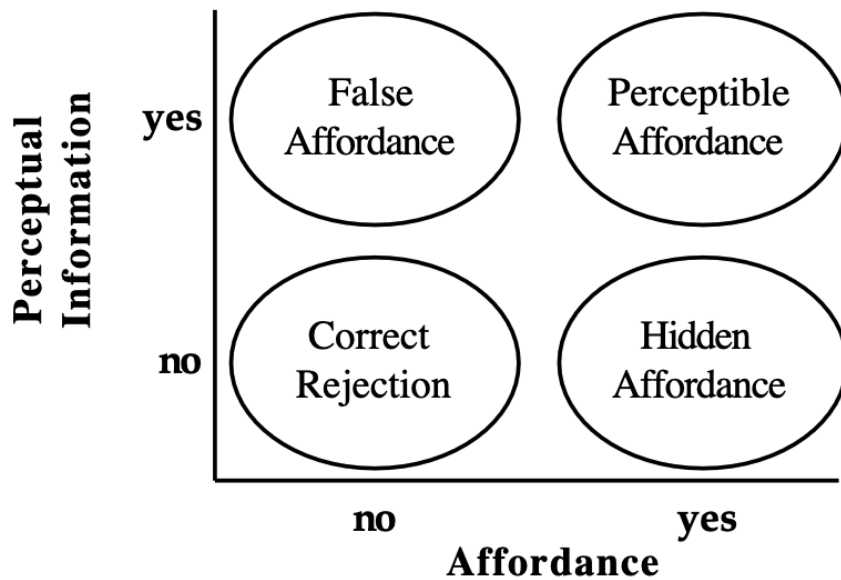


Figure 1: Distinction between affordances which are perceived and actionable as defined in the work of Gaver (1991)

the former, perhaps the most influential development of affordance theory stems from the work of Norman (Norman, 1988, 1999). In contrast to Gibson's arguably more fuzzy theoretical framing, Norman takes a more practical, design-focused perspective, defining an affordance to be "*the perceived and actual properties of the thing, primarily those fundamental properties that determine just how the thing could possibly be used*" (Norman, 1988, p. 9). More precisely:

"Affordances provide strong clues to the operations of things. Plates are for pushing. Knobs are for turning. ... When affordances are taken advantage of, the user knows what to do just by looking: no picture, label or instruction is required." (Norman, 1988, p. 9)

Norman in fact, contends that affordances can be developed *by design*, leading to a greater focus on affordances which are perceived. This marks a significant departure from Gibson's original formulation which positions affordances as independent of agents' perceptions.

Since Norman, numerous researchers have taken up the task of reformulating the affordance framework, particularly with the goal of studying technologies and more recently, online platforms. Of note, Gaver (1991) develop the idea of "*Technology Affordances*" (Gaver, 1991) which places a particular focus on affordances in the context of user interfaces. As depicted in Figure 4, Gaver draws a clear distinction between not only affordances which are and are not perceived - the key distinction between Norman and Gibson - but also, between those which were wrongly or correctly perceived, adding further nuance

to the affordance framework. Similarly, Hutchby (2001) provides a thorough development of affordances in relation to technologies through a more sociological lens. More recently and directly related to the investigation of the effects of algorithmic guidance, Panigutti et al. (2025) outline how algorithmic auditing studies geared at understanding the effects of an interface’s design are, in fact, commonly structured around analysing the outcomes of technological affordances.

In the context of platform studies, utilising the framework of affordance allows for not only the material properties of platforms but also, users’ agency to be acknowledged - a duality which had commonly evaded contemporary attempts to describe human-machine interactions on platforms (Davies, 2018; Michiels et al., 2022; Pariser, 2011). In this thesis, we adopt the term affordance precisely because it captures the diverse ways users engage with platforms while foregrounding the role of user agency in shaping these interactions and downstream behaviours. Specifically, we use affordance to describe the various means by which platforms enable users to stream music on-platform. As depicted in Figure 5, there exist three main affordances which can be adopted to facilitate music listening and discovery on music streaming platforms like Deezer: *organic* (*O*), *algorithmic* (*A*) and *editorial* (*E*) (Villermet et al., 2021; Beuscart et al., 2019; Louail and Barthelemy, 2017). These affordances are typically prominently positioned on platforms’ home screens. As such, we contend that users generally *perceive* these mechanisms as available, even if they do not actively adopt them. In this sense, our usage of affordance aligns with both Gibson’s and Norman’s formulations since the set of perceived affordances approximates the set of available affordances, effectively dissolving the ontological distinction between perceived and actual affordances in the context of this thesis.

Music Streaming and Passive Information Retrieval

Modern music streaming platforms host an unprecedented abundance of music, with catalogues so vast that no listener could consume more than a fraction within a single lifetime. Faced with this overwhelming abundance, these platforms are designed to guide users toward music they are likely to enjoy, thereby alleviating *choice overload* (Romero Meza and D’Urso, 2024). To achieve this, platforms commonly provide various forms of so-called “*programmed*” affordances (Anderson et al., 2020), such as algorithmic recommender systems and human-curated editorial playlists, which help users discover new music aligned with their tastes. At the same time, platforms do not demand that users adopt programmed guidance; users are also able to listen to music *organically* (i.e., in relative

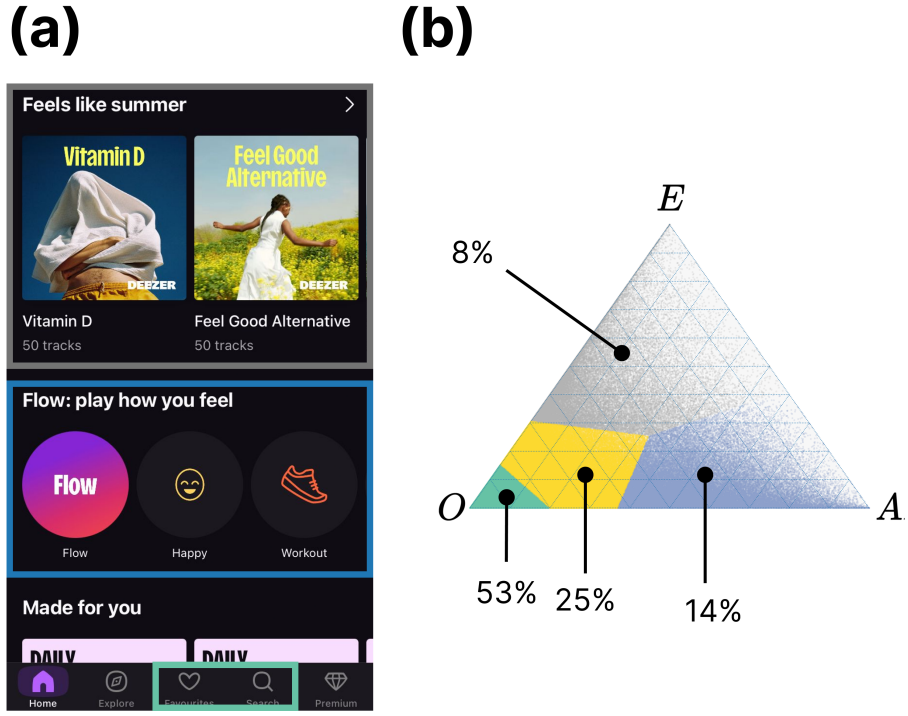


Figure 2: **Affordances on the Deezer Platform.** Figure (a) shows the current Deezer home screen which offers several different *organic* (O , green), *algorithmic* (A , blue) and *editorial* (E , grey) affordances to stream music. Figure (b) shows the distribution of 500K random Deezer users in ternary space based on their average proportion of streams from O , A and E affordances. Boundaries are drawn using k-means ($k = 4$).

absence of programmed guidance) through organic affordances such as the search-bar, their own personal playlists or, trawling manually through various artists' discographies. Put simply, music streaming platforms have expanded not just vertically (i.e., with depth, specialising in one feature), but also, horizontally (i.e., with breadth, diversifying across many features) such that now, there is no one means of reaching content on-platform.

Despite the fact that programmed affordances simplify navigation through platforms' vast catalogues, they also, by construction, make users less in control and thus, more passive in information retrieval processes. For instance, when using an algorithmic device such as Deezer's Flow, users are not directly in control of retrieval; tracks are selected automatically and users are not capable of directly selecting next-track recommendations from a pre-curated list. In contrast, when adopting organic personal playlists to consume music, users are both in control of selecting not only at a more general semantic level (e.g. which of their genre or mood themed playlists to consume) and track level but also, in the construction of the playlists itself. Similarly, editorial playlists are somewhere in between - users are not in control of what tracks are in the playlists but they do get to choose what to consume at a more general semantic level (e.g. to consume "90's Boom-Bap", "80s Throwback") and track level. In short, modern music streaming platforms afford a

wide variety of ways to retrieve and listen to music, each of which offload the information retrieval process to varying degrees. Nonetheless, it is not necessarily *demanded* that users make use of the platform in any one way. Studying users' relative adoption of more programmed and more organic affordances thus offers one novel, fruitful means of analysing how in control modern-day music consumers prefer to be when listening to recorded music online.

While several scholars have expressed concerns that music streaming platforms encourage consumers to passively outsource "*the creation, maintenance, and storage of their music collections*" (Morris, 2015, p. 168), existing empirical research regarding how users actually behave on-platform suggests that such concerns may be overstated (Beuscart et al., 2019; Villermet et al., 2021). For instance, on Deezer, so-called organic streams - stemming from users' interactions with affordances like their own personal playlists and the search bar - have been shown to constitute the majority of activity both at the platform scale (Beuscart et al., 2019) and the average user scale (Villermet et al., 2021). Furthermore, similar effects have also been observed on the Spotify platform. In an internal report produced by Spotify in 2018, only 31% of all streams were said to result from programmed affordances such as algorithmic recommendation devices and editorial playlists (United States Securities and Exchange Commission, 2018) - presumably the rest were organic. Taken together, existing literature suggests that across several of the most widely adopted music streaming platforms, users often exhibit a preference for active, in-control information retrieval.

Why might music streaming users prefer to stay in-control of information retrieval and thus, frequently adopt organic affordances? What kind of behaviour does being in-control of these processes allow for? One potential explanation for these findings supported by existing literature may lie in understanding the level of exploitative consumption, which may be denoted as "discrete concentration" since it consists in the repeated consumption of items, which organic affordances support. Although we shall not yet go deep into this literature since we provide a more thorough review shortly, it remains relevant to already provide a high-level overview in order to begin to tackle these kinds of questions. Beuscart et al. (2019) show that a significant fraction of a Deezer user's organic streams typically comes from interactions with their own personal playlists and thus by construction, these streams presumably reflect interactions with music already known to the user. Similarly, Hosey et al. (2019) and Li et al. (2019) show that users often initiate searches on music streaming platforms with a specific track in mind, suggesting that the content accessed has frequently been consumed previously perhaps even on-platform. Villermet et al. (2021)

show that organic streams tend to be more redundant than algorithmic or editorial ones adding further support for this hypothesis. Similarly, Beuscart et al. (2019) find that tracks discovered through organic affordances are more likely to be re-exploited than those discovered through algorithmic affordances on Deezer. Taken together, this body of literature suggests that organic in-control music information retrieval may play a distinct role in fostering redundant, exploitative forms of engagement. Such reliance on these kinds of affordances may signal a broader orientation toward exploitation in music streaming practices, echoing usage patterns associated with legacy media like iPods and MP3 players.

In summary, while publicly available research on users' preferences for maintaining active control over information retrieval on music streaming platforms remains limited, the existing literature consistently points in the same direction. When given the choice, users tend to engage with predominately organic affordances, suggesting users desire to remain actively involved in their listening practices (Villermet et al., 2021; Beuscart et al., 2019; United States Securities and Exchange Commission, 2018). This challenges existing techno-deterministic narratives that portray users as passive recipients of online algorithmic guidance (Davies, 2018) and underscores that, at least on music streaming platforms, users can and do exercise agency over whether to adopt these devices. This once again reinforces our choice of using "affordance theory" to describe the facets for retrieval and listening offered by a platform.

While this literature offers a strong baseline to build upon, it could benefit from a deeper understanding of the inter-dependencies between affordances. For example, if a user's personal playlist largely consists of tracks discovered through algorithmic recommendations, it becomes more complex to assess whether the use of such affordances reflects a preference for active or passive information retrieval. This thesis aims to address this gap by examining how items move between affordances, helping to clarify what organic engagement truly entails. In doing so, it seeks to contribute to a more nuanced understanding of user agency within platform-mediated music consumption.

Music Streaming and Passive Listening

Another closely connected question related to how end-users appropriate and engage with music streaming platforms concerns perception rather than behaviour. When users adopt streaming platforms like Spotify or Deezer to play music, to what extent are they using them to actively listen to music? Is music commonly the centre of attention, positioned

in the foreground or rather, to what extent does it tend to wash over listeners and perhaps even serve as a complementary stimulus for another activity? To what extent do platforms enable and even foster these practices? To address these questions, we begin by revisiting the various ways in which diverse patterns of music listening have been conceptualised and categorised, even prior to the transition of recorded music from physical to digital formats and, then subsequently, to platforms. This allows us to have a more holistic perspective on contemporary concerns that music streaming platforms may foster passive listening (Prey, 2019).

In the Social Sciences, music consumers' listening practices have long since been meticulously characterised through an *passive* vs *active* dichotomy (Hesmondhalgh, 2022). Much of this literature can be traced back to the seminal German philosopher and musicologist Theodor Adorno, who made a substantial effort in describing consumers by the attentiveness of their listening and understand the forces which produce different listening practices. This is most clearly exemplified in his 1945 publication "*A Social Critique of Radio Music*" (Adorno, 1945) in which consumers are classified under an active-passive dichotomy as a means of theorising about the effects of increased standardisation in music production on listening. According to Adorno, as societies transitioned from feudal to industrial systems, music likewise, shifted from serving a social function to being a mass-produced, commodified good. Put simply, cultural production is said to be subject to the same standardisation and efficiency logics as material production which is thought to result, over time, in a rise in passive *commodity listening* designed "*to dispense as far as possible any effort on the part of the recipient*" (Adorno, 1945, p. 231). Technological developments which facilitate music's diffusion en masse such as, at the time of writing, the Radio are thought to reinforce this process. Even though these devices offer consumers the illusion of choice, in practice according to Adorno, the curations they offer are often predictable, formulaic and standardised, thus promoting passive, inattentive, background listening. This reasoning is highly reminiscent of contemporary critiques which contend that algorithmic recommendation devices and platforms function in much the same way (Seaver, 2019; Prey, 2019).

As Hesmondhalgh (2022) note, a reading of "*A Social Critique of Radio Music*" offers several highly valuable insights even today, which can be useful for contextualising contemporary literature geared at understanding the relation between music streaming platforms, their affordances and users' listening attentiveness. Firstly, the text demonstrates that concerns about technological advancements designed to distribute music en masse and more precisely, fears that these technologies may encourage passive listening,

are not necessarily reserved for music streaming platforms. Secondly, it also suggests that if contemporary concerns are largely influenced by an Adornonian thinking, they may be unfounded. Although the text offers some insightful descriptions regarding the increased commodification of music, more causal claims regarding the effect this can have on listener attention remain less robust (Sterne, 2003; Hesmondhalgh, 2022). For instance, the genres thought to demand attentive listening have been shown to be strongly related to skill, modernity and class (Bourdieu, 1984), raising questions regarding the extent to which Adorno’s causal reading of the world may be strongly biased by his own social position. Thus as scholars, we should be particularly cautious of literature claiming that music streaming platforms *cause* passive listening. Such claims may simply echo longstanding anxieties that were arguably not well-founded in the first place (Hesmondhalgh, 2022).

Building on this caveat, contemporary research on passive listening in the context of music streaming platforms typically avoids broad causal claims about the effects of production standardisation on listener attention. Instead, recent scholarship has increasingly focused on understanding how music is integrated into the *everyday* (DeNora, 2000), and the ways in which music streaming platforms enable and support this mode of parallel, sometimes background, engagement. This literature, spanning broadly across Sociology (DeNora, 2000), Media Studies (Pedersen, 2020; Prey, 2019) and Marketing (Fuentes et al., 2019) connects closely to the study of active / passive listening since by proxy, it focuses on scenarios where music listening is not necessarily the sole focus. Fuentes et al. (2019) study 15 Swedish music consumers’ activity-based listening practices, which they coin “*Soundtracking*”. Through participant interviews, the authors identify activity-based listening as the primary function of music streaming platforms.⁵ Similar results have also been found both in the early days of streaming platforms (Schedl et al., 2015) and even prior to their widespread adoption (Kamalzadeh et al., 2012; North et al., 2004).

Examining the design of music streaming platforms’ algorithmic recommendation devices also provides evidence to support the notion that music streaming platforms are commonly used for activity-based listening. Symbolic of the so-called “*contextual turn*” in recommender system design over the past decade (Pagano et al., 2016), Deezer R&D team has placed substantial effort on developing user embeddings to capture listening contexts and activities, which may potentially be deployed to enhance future recommender systems on-platform (Marey et al., 2024). Similarly, since 2021, Deezer’s “Flow” algorithmic recommendation device is capable of utilising track mood embeddings (Deezer,

⁵This practice may be facilitated by the *in-pocket*, portable nature of the devices commonly used to access music streaming platforms (now smartphones) - on Deezer, two thirds of a user’s streams come from mobile listening (Beuscart et al., 2019).

2023), which in turn, allow users to tailor recommendations to align with not only specific activities and contexts (e.g. high-energy tracks for running) but also their own specific psychological states. These developments echo the same conclusions reached by Pedersen (2020) who claim that music streaming platforms recommend music in ways that ‘*amplify and encourage user practices of ubiquitous listening*’. Together, this literature underscore the prevalence of activity-based music listening and the role which music streaming platforms play in facilitating this practice.

In recent years, several scholars have also started to more concretely draw connections between music streaming platforms capacity for on-the-go, activity-based listening and inattentive listening (Kamalzadeh et al., 2012; Pedersen, 2020; Prey, 2019) otherwise coined “*lean-back listening*” (Pelly, 2018). In particular, Prey (2019) discuss the connection between users outsourcing music retrieval to streaming platforms and passive listening. Prey describes how the design of music streaming platforms has developed to be increasingly curation-based, frequently relying on playlist-based affordances, which are put at the forefront of platforms’ visual brand and design. This shift is said to result in music exposure (and at times production) becoming increasingly oriented around moods and activities. As music is designed to mould to the consumer, rather than the other way around, listening is implied to become less attentive and thus more passive. While such arguments may appear intuitive - particularly given cognitive theories of attention bottlenecks (Tombu et al., 2011) - empirical evidence supporting these claims remains limited. Currently, the vast majority of existing studies either rely on small sample interviews with music consumers or literature reviews which often cite other non-empirical works. Whilst activity-based listening has the potential to transition music listening to a supporting activity which may induce more inattentive listening so might an activity support a listening experience. At the same time, the theory that music streaming users’ listening attentiveness is more often passive is also at tension with existing empirical literature on users’ skipping behaviours. In a public data driven report produced by Spotify in 2014, approximately 50% of all tracks played were skipped (Whitman, 2014) implying that users are very much *behind the controls* of streaming platforms and that active evaluations are being made.

In summary, inspecting qualitative literature and the design of music streaming platforms provides substantial evidence that music streaming platforms are commonly used in parallel with other activities. The devices commonly adopted to host music streaming platforms (smartphones) inevitably facilitate this practice. Nonetheless, the extent to which this translates to passive listening remains in question despite some scholars

theorising in this direction (Prey, 2019). It may be possible, as Hesmondhalgh (2022) would contend, that these are simply reproductions of *old anxieties*. Testing such claims empirically, although beyond the scope of this thesis, remains a significant challenge for the Music Information Retrieval community in the years to come.

The Filter Bubble Hypothesis

"Metaphors can be genuinely useful in approaching a complex set of issues, but only if they are well-chosen, well explained, and kept from short-circuiting more in-depth analysis through their apparent intuitive simplicity."

(Bruns, 2019, p. 8)

In the following section, we focus on outlining the filter-bubble hypothesis as outlined by Pariser (2011). We then move on to discuss more recent critical readings of the theory before proposing our own definition of *algorithmic confinement* which is utilised in the subsequent sections within this literature review.

Hypothesis Outline

Critical research on so-called "new media" - such as in their time, radio, film, television and now, music streaming - is deeply influenced by the sociopolitical climate in which they are produced. In periods of relative stability, for instance the post-war era where the "*consumer society*" flourished (Baudrillard, 1998), research tends towards less critical appraisals whilst in times of instability research opts for a more critical lens, often seeking to uncover strong media effects from the outset (Silverstone, 2005). This perspective provides a valuable backdrop to contextualise the seemingly endless stream of academics (Brinkmann et al., 2023), policymakers (The New York Times, 2017), and mainstream media outlets (Levy, 2023) that have in recent years, claimed albeit often with very different intents, that social media and more recently, streaming platforms' algorithmic devices constrain rather than foster diverse communication and media consumption practices.

At first glance, this outcry may seem novel, but its origins can be traced all the way back to similar concerns voiced during the early days of the Internet. In 1995, Negroponte (2015) cautioned that the web would inevitably produce "*a kind of newspaper printed in an edition of one*" and similarly, Sunstein (2006) argued that the web isolates users in

"*information cocoons*". These concept helped to inform the tone and contents of journalist and internet activist Eli Pariser's widely cited 2011 publication, "*The Filter Bubble: What the Internet Is Hiding from You*" (Pariser, 2011) which remains to date, one of the most widely cited theories concerning the role of online algorithmic guidance in fostering confined consumption practices. Released shortly after Google incorporated algorithmic personalisation into their search engine for the first time (Areeb et al., 2023), Pariser's thesis states that online personalised algorithmic devices function to expose users to diversity lacking information thereby limiting the extent to which they are exposed to diverse content such as for instance, news articles from the opposite side of the political spectrum or music genres outside of one's current preference or taste. This notion is what Pariser now famously calls the *filter bubble*. More precisely, according to its principles, the flow of information between users and algorithms operates as follows in an online environment. Algorithmic devices filter information to generate personalised recommendations for users. Users then engage in a secondary filtering process, choosing which recommendations to select, consume and in some cases reject. This interaction is logged by the platform and processed as so-called *implicit* feedback for the algorithm (Oard et al., 1998). The algorithm is then subsequently retrained using this feedback, and at a later iteration, delivers further, more personalised recommendations to users and so on and so forth. As depicted in Figure 6, this establishes a continuous feedback loop between users and algorithms that according to Pariser, can result in algorithmic devices exposing less and less diverse recommendations to users over time.⁶ These kinds of feedback loops are now almost universally found on platforms (Brinkmann et al., 2023) and as such, the study of filter bubble like dynamics is not reserved for one platform type in particular but rather, acts a general framework to understand human-machine interactions and the effect this can have on consumption diversity.

In the case of music streaming platforms, filter-bubble-like effects can have major repercussions for a wide range of stakeholders, including platforms, artists, and, perhaps most importantly, users. From a platform perspective, literature has demonstrated that diverse on-platform consumption practices tend to be associated with higher engagement (Anderson et al., 2020) and furthermore, that algorithmic recommendation diversity plays a substantial role in ensuring that users are satisfied with recommendations (Nguyen et al., 2014). As such, ensuring an appropriate balance of diversity in algorithmic recommendations may, from a platform's perspective, seem like an effective strategy to drive engagement and reduce churn. Such provides a motivating example of why filter bub-

⁶This is said to have a destabilising effect upon the democratic process although how exactly remains largely up to the imagination.

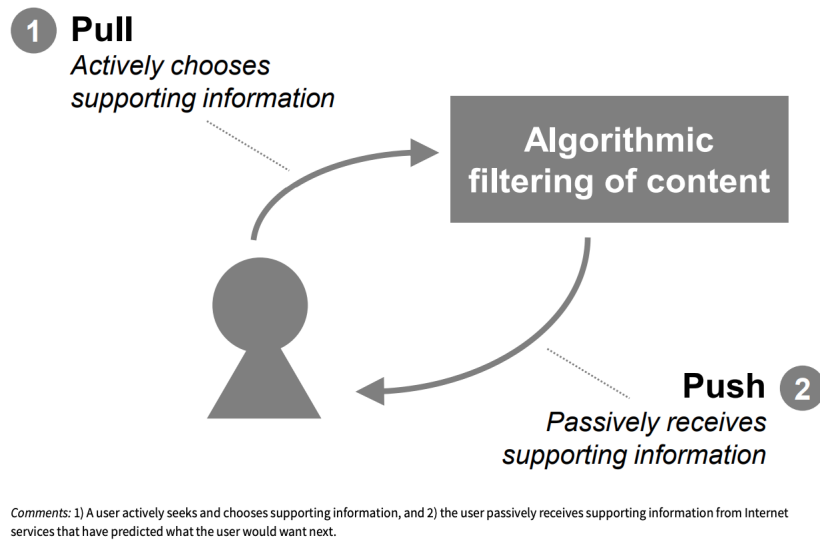


Figure 3: Theorised feedback loop within the Filter Bubble hypothesis. Image from the works of Dahlgren (2021).

ble dynamics might be a cause for concern for these actors. Similarly, from an artist's perspective, filter-bubble effects may also not be desirable. More confined, personalised recommendations can reduce the chances for artists to reach diverse audiences through the algorithm, potentially forcing them to rely on a single market segment and increasing the risk of failure if preferences shift within that group. Finally, from a user perspective, lower-diversity recommendations may not only lead to lower satisfaction (Nguyen et al., 2014), but could also disrupt existing class structures, which several scholars argue are closely tied to diverse, omnivorous tastes (Peterson, 1997). In short, the effect of filter-bubbles on music streaming platforms should not be understated and, just as has been demonstrated to be the case when studying online political confinement (Robertson et al., 2023), may have both foreseen and unforeseen repercussions for a wide range of actors.

Critical Readings

Whilst filter bubbles might sound like a cause for concern for those connected to the music streaming echo-system, to what extent are they as a framework, scientifically sound? Despite Pariser's thesis being eagerly adopted and regurgitated by mainstream media outlets (Levy, 2023; The New York Times, 2017) that is not to say it was met with the same positive reception within academia. With some scholars even going so far as to describe filter bubbles as "*metaphors that failed us*" (Bruns, 2019), researchers promptly took up the task of critically inspecting the foundations of Pariser's thesis on theoretical grounds even prior to those that are empirical (Bruns, 2019; Michiels et al., 2022; Dahlgren, 2021; Davies, 2018). Most notably, Dahlgren (2021) and then later Michiels et al. (2022) both

conduct poignant readings and thereafter, critiques of Pariser’s original text.

At the more critical end of the spectrum, Dahlgren (2021) call for the complete abandonment of the terminology due to their belief that it carries an excessive amount of unfounded negative connotations. Furthermore, according to Dahlgren it remains unclear if the confinement dynamics which Pariser envisioned would emerge even in the absence of algorithmic devices. As such, Dahlgren argues that the very concept of a filter bubble is one that stems from "*technological determinism*" (Davies, 2018) - if an algorithm is involved in a dynamic, must it be that the algorithm is necessarily causing this? Dahlgren also critiques Pariser’s formulation of the mechanisms which create a filter bubble and furthermore, the theorised effect which this can have upon a society. For instance, Pariser contends that filter bubbles emerge from substantial personalisation at both the user and platform side thus, creating a feedback loop in which content becomes increasingly homogeneous as algorithmic engagement increases over time. Nevertheless, on the user side, literature suggests that personalization processes may not be so straightforward. While several empirical studies support the role of personalization in media consumption, aligning with the theory of *selective exposure* - a psychological framework suggesting that individuals gravitate toward media reinforcing their prior beliefs (Katz et al., 2017)⁷ - this does not imply that consumers entirely avoid engaging with diverse content (Garrett, 2009). Thus, Pariser’s formulation of user personalisation within this feed back loop is likely too strict and, ultimately, neglects the role of human agency within so-called *organic* selection processes (Roth, 2019).

Despite Dahlgren making a strong case for the abolition of the filter bubble as a *signifier* (Barthes, 1957), other scholars are more reformist in their critique. In particular, Michiels et al. (2022) recognises that the terminology still holds some utility which can be resurrected. Building upon the notion that a filter bubble is a dynamic rather than static process, the authors propose a refined definition of the filter bubble which goes as follows:

"A technological filter bubble is a decrease in the diversity of a user’s recommendations over time, in any dimension of diversity, resulting from the choices made by different recommendation stakeholders." (Michiels et al., 2022, p. 2)

Succinctly, this definition is built around 4 dimensions which are said to assist the theory’s

⁷Historically, most works empirically exploring the theory are motivated by empirically testing the psychological theory of cognitive dissonance (Morvan and O’Connor, 2017) which asserts that when individuals are confronted with contrasting ideas, certain mental defence mechanisms are activated to produce harmony between new ideas and pre-existing beliefs, which results in cognitive equilibrium. More recently empirical works have also been motivated by testing social identity theory (Knobloch-Western et al., 2020) which hypothesises that consumers prefer media that contains messages that present their in-group in a favourable light. (Tajfel et al., 1979)

operationalisation. The first dimension is *diversity*. A filter bubble is said to be a decrease in "*any dimension of diversity*" (Michiels et al., 2022) be that source diversity (i.e., a news source), topic diversity (e.g. semantics of a news article), viewpoint diversity (e.g. sentiments towards the topic) or in more general computational terms, diversity built upon different information representations (Mok et al., 2022). Their second dimension, *recommendation*, makes explicit that a filter bubble is a decrease in diversity of a user's recommendations rather than consumptions. Their third dimension, *time*, emphasises that the output of a filter bubble ultimately stems from that of a longitudinal process. Finally, they introduce their final dimension, *recommendation stakeholders*; the authors advocate for a multi-stake holder perspective for instance, by considering users as well as providers to further shed light on how a filter bubble came to be.

Whilst it remains difficult to dispute that Michiels et al's conception of the filter bubble is more grounded than Pariser's, it also comes with its own unique set of limitations. Firstly, claiming that filter bubbles should be evaluated solely in terms of algorithmic exposures significantly limits the capacity for most researchers to engage in empirical work on this topic. In practice, access to this form of empirical data - algorithmic exposure signals - is, although leveraged within this thesis, relatively hard to attain for researchers working outside of private partnerships with platforms. Secondly, focusing on algorithmic exposures alone may fail to account for the role of user driven personalisation in causing confinement. Consider by means of example, a lifelong music streaming user who as they age, follow the seemingly common pattern of developing increasingly confined listening tastes and preferences (Harrison and Ryan, 2010; Glevarec et al., 2020) which would have happened also in a counterfactual world where streaming platforms never existed. Even if a recommender system operated in a way that was not inherently diversity constraining (e.g. by simply mirroring a user's organic activity), studying solely exposure signals would not necessarily indicate this.

In light of these limitations - both regarding Pariser's original formulation (Pariser, 2011; Dahlgren, 2021) and later extrapolations (Michiels et al., 2022) - we choose to largely restrain from using the terminology "filter bubble" within this literature review and to some degree thesis. Instead, we opt more commonly for the term "*algorithmic confinement*" which compares algorithmic impact relative a (possibly counterfactual) baseline where algorithms are partially or completely absent. More precisely, we define algorithmic confinement as follows.

Algorithmic confinement: *Algorithmic confinement is the setting in which the*

media algorithmically exposed to or selected by users is more concentrated than selections/exposures which involve a partial or complete absence of algorithms.

We not only contend but also, demonstrate that this formulation is general enough to capture literature geared at assessing not only the effect of algorithmic devices but also platforms. Furthermore, this definition is capable of capturing literature which relies on both temporal and aggregate methods and also state of the art multi-affordance literature.

Testing Algorithmic Confinement

"Making sense then of the work and effects of algorithms needs to be sensitive to their contextual, contingent unfolding across situation, time and space. What this means in practice is that single or limited engagements with algorithms cannot be simply extrapolated to all cases."

(Kitchin, 2019, p. 16)

We now turn our attention to more quantitatively minded literature geared at testing algorithmic confinement dynamics. To what extent does the literature suggest that *algorithmic confinement* is common across platforms and, in particular, on music streaming platforms such as Spotify and Deezer, the focus of this work? Perhaps surprisingly, the answer to this question is unclear. Even though some research finds evidence in support of this dynamic (Anderson et al., 2020; Bandy and Diakopoulos, 2020) other literature suggests it is uncommon - several works demonstrate that following algorithmic guidance actually fosters diverse media consumption practices (Robertson et al., 2023; Bakshy et al., 2015; Cardenal et al., 2019; Möller et al., 2018; Epps-Darling et al., 2020; Villermet et al., 2021; Zhou et al., 2010; Fortunato et al., 2006; Aiello and Barbieri, 2017; Nguyen et al., 2014; Datta et al., 2018; Haim et al., 2018; Beuscart et al., 2019). These seemingly contradictory results may stem at least in part from different approaches regarding *how* concentration scores are computed and compared between algorithmic and less algorithmic settings. In the literature, this has principally been done via *aggregate* or *temporal* comparisons. To respect this difference, we segment this section of our review into two parts based on *how* concentration scores are computed and compared. For a concise review of this literature we refer the reader to table 2.

Author	Platform / RS type	Analysis	Data Type	Algorithms / Platforms Limit Diversity?
Lathia et al. (2010)	Video RS	Temporal	Simulation	$y + n$ (depends on activity and scale)
Nguyen et al. (2014)	Video RS	Temporal	Empirical	n
Su et al. (2016)	Social Media	Aggregate	Experimental	y (with + personalisation)
Datta et al. (2018)	Music Streaming	Aggregate	Empirical	n
Geschke et al. (2019)	Social Media	Temporal	Simulation	y
Aridor et al. (2020)	n/a	Temporal	Simulation	n
Roth et al. (2020)	Video Streaming	Temporal	Simulation	y
Holtz et al. (2020)	Music Streaming	Aggregate	Experimental	y (with + personalisation)
Brown et al. (2022)	Video Streaming	Temporal	Experimental	y
Huszár et al. (2022)	Social Media	Aggregate	Empirical	y (with + personalisation)
Godinot and Tarissan (2023)	Music Streaming	Temporal	Simulation	n
Chen et al. (2024)	Music Streaming	Aggregate	Experimental	no effect (with + diverse RS)
Sutkennik et al. (2024)	Video Streaming	Temporal	Empirical	$y + n$ (depends on scale)
Coavoux and Aussant (2024)	Streaming (music, video)	Aggregate	Empirical	n

Multi-affordance Literature

Sharma et al. (2015)	E-commerce	Aggregate	Empirical	n
Aiello and Barbieri (2017)	Blogging	Aggregate	Empirical	n
Beuscart et al. (2019)	Music Streaming	Aggregate	Empirical	n (less top-head concentration)
Anderson et al. (2020)	Music Streaming	Aggregate / Temporal	Empirical	y
Epps-Darling et al. (2020)	Music Streaming	Aggregate	Empirical	n
Villemet et al. (2021)	Music Streaming	Aggregate / Temporal	Empirical	n
Mok et al. (2022)	Music Streaming	Aggregate / Temporal	Empirical	n/a

Table 1: Summary of the past decade’s most relevant and impactful research concerned with capturing the effect of platforms’ algorithmic devices on users’ engagement diversity. Literature is detailed in chronological order and segmented based upon if the study was focused on the multi-affordance aspect of a platform.

Temporal Comparisons

One method of appraising algorithmic confinement that has been widely applied in existing literature has been to track the diversity of algorithms’ exposures or users’ selections over time or over users’ *platform lifecycles* (Mok et al., 2022). This research addresses algorithmic confinement since it compares the diversity of algorithmic selections/exposures to those in earlier time periods where, by construction, users had received less algorithmic guidance. Brown et al. (2022) explore how YouTube’s recommender system drives users toward increasingly confined ideological spaces with increased adoption. To investigate this, their study recruits participants who are active YouTube users before the trial begins and assigns them randomly to both a seed video and a set of rules dictating how to adopt recommendations (e.g. follow the second recommendation on-screen), thereby controlling for users’ current activity which may influence the diversity of recommendations they receive. Tracking recommendation exposures across a chain of interactions, the authors find that regardless of the initial seed video or user behaviour, YouTube’s recommender system gradually directs users towards more and more confined (moderately conservative) videos. Also on the topic of video streaming, Sukiennik et al. (2024) study how personalised algorithmic recommendations on a major Chinese video streaming platform tend to vary in concentration over time. Their work reveals that adopted recommendations tend on average, to become more topic-confined over time however, to a great extent, this depends on the granularity of topic classification deployed - at a more granular levels of topic classification increasing concentration is less apparent.

In the absence of user-level/usage empirical data, it is also typical for research of this nature to rely on simulating users’ and sometimes algorithms’ item selection behaviours (Lathia et al., 2010; Roth et al., 2020; Godinot and Tarissan, 2023). This line of research (while often based on oversimplified assumptions about user selection behaviours) is highly valuable since it helps to disentangle the effect of user and algorithmic filtering in fostering algorithmic confinement and furthermore, pinpoint which user behaviours lead to different diversity outcomes. Lathia et al. (2010) study the relation between collaborative filtering-based recommender systems becoming more confined over time and users’ on-platform behaviours using a simulation-based framework. Measuring concentration from a discrete, repetition-based perspective, their work finds that algorithmic confinement is strongly conditional on how users interact with platforms. Lower within-session activity (captured via the number of items rated) is associated with receiving more concentrated, repetitive recommendations whilst at an overall aggregate scale, lower activity (captured

via the total number of rated items) is associated with receiving more diversity. Similarly, on the Youtube platform, Roth et al. (2020) investigate how, at the short-term one-recommendation-to-the-next scale, following recommendations leads to topic confinement. Querying Youtube’s video recommendations to generate a recommendation networks and then simulating user consumptions on these networks, their findings suggest that in this setting, following Youtube’s recommendations reinforces topic confinement especially for items tagged as music and for items that are more niche. Nevertheless, research suggests that more long-term confinement dynamics may be less trivial. Generating collaborative filtering-based recommendations from retrospective music listening logs and measuring diversity in a network theoretic manner built upon the relation between users, items and item categories, Godinot and Tarissan (2023) shows how theoretically, music streaming users’ consumption diversity over time, increases sub-linearly with the number of recommendations adopted. Nonetheless, once again, this dynamic is found to vary depending on how users interact with the platform in the simulation.

Whilst several of the works outlined point towards users’ interactions with algorithmic recommendations becoming more concentrated over time (Roth et al., 2020; Lathia et al., 2010; Brown et al., 2022; Sukiennik et al., 2024), to what extent might users’ consumptions have grown even more concentrated had they not followed the algorithm? In the 2014 landmark study, Nguyen et al. (2014) conduct "*the first study looking at recommended content diversity, user behaviour over the time, and the effect of taking recommendations on consumed content diversity*" (Nguyen et al., 2014). Although somewhat dated in terms of the algorithmic devices and platforms studied ⁸ their work nevertheless, provides a solid intuition of how early recommender systems might have influenced users’ media consumption diversity over time. Focusing on jointly analysing users’ of the MovieLens platforms’ consumption diversity and recommendation adoption histories in the period of 2008-2010 their work finds that independent of whether users adopted the platform’s movie recommendations or not, recommendations became increasingly confined over time presumably, as the platform’s recommender system fine-tuned to users’ preferences. At the same time, users’ consumptions were also found to become increasingly confined however crucially, this was lessened for users who *adopted* algorithmic recommendations. The authors interpret these findings by suggesting that algorithmic recommendation devices may in fact, contrary to traditional filter bubble narratives (Pariser, 2011), have a dampening effect upon users’ narrowing consumption practices which may nonetheless, emerge *organically* even in the absence of algorithmic recommendation devices.

⁸Rudimentary collaborative filtering recommender systems and the MovieLens platform where media consumptions are performed off-platform.

More recently, the notion that users’ online media consumptions tend to become increasingly confined over time even without algorithmic devices has been supported by several empirical (Mok et al., 2022) and simulation-based studies (Aridor et al., 2020; Geschke et al., 2019). From a more theoretical perspective, Aridor et al. (2020) explore the mechanisms which drive the narrowing of diversity in both users’ consumptions and algorithmic recommendations. By developing a mathematical model of user behaviour that incorporates risk aversion as well as idiosyncratic and common-value components of item evaluations, their work supports the findings of Nguyen et al. (2014), which suggest that algorithmic recommendations help to limit the narrowing that would otherwise have occurred *organically*. When simulating a scenario in which users blindly follow recommendation,⁹ the adoption of recommendations leads to a clear narrowing dynamic. However, when user selection processes are modelled in a manner that more closely resembles users’ decision making processes in practice, the results are consistent with Nguyen et al. (2014). Similarly, Geschke et al. (2019) use an agent-based model to simulate the decision-making processes that influence users’ consumptions of recommended content on social media platforms. Driven by cognitive mechanisms like confirmation bias their model demonstrates that confinement dynamics emerge even in the absence of additional social or technological filters however the presence of these additional filters further amplifies concentration in users’ engagements. Similarly, Mok et al. (2022) demonstrate that on the Spotify platform, users’ organic streams become increasingly exploitative from the moment they register.

In summary, existing literature appears to suggest that in several instances, users’ interactions with algorithmic guidance can become more confined with increased adoption (Roth et al., 2020; Lathia et al., 2010; Brown et al., 2022; Sukiennik et al., 2024; Mok et al., 2022). Nonetheless, this may not always be caused by algorithmic guidance. Several works have demonstrated that not only do users’ consumptions tend to become increasingly confined over time even in the absence of algorithmic devices (Geschke et al., 2019) but also in several instances, the adoption of algorithmic guidance may slow this process thus, helping keep users’ consumptions diverse for longer (Nguyen et al., 2014; Aridor et al., 2020).

⁹Surprisingly, this is frequently assumed across many algorithmic auditing studies (Aridor et al., 2020)

Aggregate Comparisons

Another closely related strand of literature focuses on appraising algorithmic confinement via the study of aggregate, rather than temporal dynamics. Literature of this form is typically geared at analysing the effect of adopting more personalised algorithmic devices in place of existing less personalised ones (Holtz et al., 2020; Huszár et al., 2022; Su et al., 2016; Chen et al., 2024) or, more generally, adopting platforms in place of no platform at all (Coavoux and Aussant, 2024; Datta et al., 2018).

Regarding the former, across several platforms, literature suggests that increasing personalisation within algorithmic recommendation devices may likely cause increased concentration in users' consumptions. The methodologies utilised to reach these conclusions tend to rely on either randomised field experiments (Holtz et al., 2020; Huszár et al., 2022; Chen et al., 2024) or natural experiments that establish random or as-if random assignment to treatment and control conditions (Su et al., 2016). Huszár et al. (2022) study how the introduction of a more personalised news feed algorithm on Twitter may cause users' news media consumption to become more confined in terms of partisan homogeneity. Focusing on 5% of all Twitter users and committing a randomized control group to a reverse chronological feed and others to a more personalised, algorithmic version their work finds that the introduction of the latter more personalised device caused users' consumptions¹⁰ to become more skewed towards wing news sources. Similarly, Su et al. (2016) show that more personalised recommendation algorithms on social media platforms can also result in algorithmic confinement not just at the individual but also collective level. Adopting a semi-randomised natural experiment approach and utilising observational Twitter data for causal analysis, their work studies the effect of introducing more personalised friend recommendations on users' following behaviours. Exploring variations in the follower network structure before and after the introduction of the algorithmic "Who to Follow" feature during mid-2010, the authors find that the introduction of a more personalised algorithm likely caused users' following behaviours to become more skewed towards popular users thus encouraging a "rich gets richer dynamic".

In the music streaming domain, the work of Holtz et al. (2020) also provides convincing evidence that increasing personalisation can cause users' consumptions to become more confined. Studying the case of podcast recommendation on Spotify the authors conduct a large-scale randomized field experiment across 900,000 users. Their work finds that increasing recommender system personalisation causes both users' podcast consump-

¹⁰Captured via "lingering" on the media for a fraction of a second.

tion diversity to decrease and furthermore, become more distinct from other users - the archetypical filter bubble paradigm that Pariser (2011) envisioned. Nevertheless, these effects are not found to be long-lasting. Upon reverting the treatment groups' podcast recommendations back to the less personalised default algorithm, consumption diversity quickly restores to prior levels suggesting that users' consumption practices display a certain level of plasticity and adaptivity to platform developments. Closely connected, Chen et al. (2024) also study how modifications to music streaming platform's algorithmic recommendation devices change the diversity of users' consumptions. Performing a large-scale field experiment across 10 million users of the popular Chinese music streaming platform NetEase Cloud Music, the authors study the extent to which modifying a recommender system to be more geared towards diversity rather than simply *accuracy* - see (Kaminskas and Bridge, 2016) for a comprehensive review of these novel "beyond accuracy" recommender systems - can likewise, cause users' music consumptions to diversify. Exposing a randomly selected cohort of users unknowingly to the diversified device the authors find no evidence to suggest that on the whole, recommender system diversity fosters diversity in users' consumptions however it might, contrary to what prior research claims (Anderson et al., 2020; Holtz et al., 2020), reduce engagement.

Whilst literature appears to suggest that on both social media (Huszár et al., 2022; Su et al., 2016) and music streaming platforms (Holtz et al., 2020; Chen et al., 2024) increasing personalisation within algorithmic devices may have a negative effect on users' consumption diversity¹¹, to what extent is this also the case if we focus more broadly, on the effect of platform adoption? For instance, to what extent does the literature suggest that a consumer's music consumption practices are more or less diverse in the counterfactual world in which they never adopted Spotify or say, Deezer? In the recent works of Coavoux and Aussant (2024) this setting is thoroughly explored. Analysing French music consumers and making use of the 2018 "*Pratiques Culturelles*" (Cultural Practices) survey their work attempts to identify the effect of platform adoption on diversity whilst controlling for reverse causation effects (e.g. a causal effect of the diversity of cultural tastes on the likelihood of using streaming). Across three domains - music, movies, and TV series - the adoption of streaming platforms is found to increase consumers' consumption diversity albeit to a lesser extent for music. Similarly, Datta et al. (2018) demonstrate, using a panel of music consumers' listening histories across various digital music platforms, that the adoption of streaming platforms is associated with significant increases in both the quantity and diversity of music consumed during the first few months post-adoption.

¹¹Perhaps as recommendations get more accurate and loose serendipity.

While these effects gradually diminish, even after six months, adopters were found to continue to play substantially more music with a greater diversity than before.

In summary, existing literature suggests that whilst introducing more personalised algorithmic devices to platforms may cause users' online media and social interactions to narrow and become more confined (Holtz et al., 2020; Su et al., 2016; Huszár et al., 2022), for the time being, adopting platforms which contain these devices may still lead to richer, more diverse media interactions (Datta et al., 2018; Coavoux and Aussant, 2024).

Testing Algorithmic Confinement Using Organic Comparisons

"[Research on filter bubbles] has moved from considering algorithmic curation as decisive force shaping online engagement towards acknowledging the influence of users' own choices on social media ranking algorithms and search engines."

(Brinkmann et al., 2023, p. 16)

In the following chapter, we outline the novel, state-of-the-art branch of algorithmic confinement literature which focuses on applying *organic* benchmarks to assess the relative effects of algorithmic guidance on consumption diversity. This literature is especially relevant to this thesis for several reasons. Firstly, the platforms studied are more often than not, music streaming platforms, making the insights derived particularly applicable to the study of Deezer. Secondly, throughout the majority of the scientific contributions in this thesis, we follow a similar methodological framework to much of this literature, comparing algorithmic to organic behaviour in terms of streams, selections, or sometimes, counter-factual organic scenarios. As such, the methods and conclusions derived within this literature are also highly translatable to this thesis.

Drawing Organic Comparisons From Multi-Affordance Signals

Since Deezer's inception as "Blogmusik" in 2007 (Deezer Jobs, 2025), the platform has come a long way in terms of the affordances which it offers users to explore, exploit and curate music. Users are now able to access music much like on Spotify, through a wide array of organic, algorithmic and editorial affordances which has in turn, brought about novel opportunities for researchers to study algorithmic confinement via rich multi-affordance signals generated through users interactions with these devices (Anderson et al., 2020; Epps-Darling et al., 2020; Villermet et al., 2021; Mok et al., 2022; Beuscart et al., 2019;

Aiello and Barbieri, 2017; Sharma et al., 2015; Chang and Ugander, 2022). On the whole, this research (mostly centred around music streaming platforms) offers several pathways to study algorithmic confinement on multi-affordance platforms.

For instance, to conduct *aggregate appraisals*, users' behaviour under organic affordances (O) can be compared to behaviour under other more or less algorithmic affordances in two ways. (1) *Descriptively*, thereby acknowledging that users' O behaviours are conditional on their adoption of other affordances (Anderson et al., 2020; Villermet et al., 2021; Epps-Darling et al., 2020). (2) *Counterfactually*, which requires applying "matching" methods to approximate from O the counterfactual O without the existence of the algorithm (i.e., $O \setminus A$) (Aiello and Barbieri, 2017; Sharma et al., 2015; Chang and Ugander, 2022). Regarding the former more descriptive strand, this type of research tends to focus on either conducting concentration measurements across affordances whether that be within users' consumptions (i.e., comparing a user's O diversity to their A diversity) (Anderson et al., 2020) or between users (i.e., how do users who adopt O compare on average to those who adopt A) (Villermet et al., 2021). In rare cases, research has also compared users' activity across platforms which offer varied levels of social and/or algorithmic assistance (Nikolov et al., 2015) although we shall not further extrapolate upon this research since it does not concern multi-affordance platforms per se.¹² Similarly, *temporal appraisals* can be conducted by studying how, after controlling for necessary confounders, a user's consumption diversity varies alongside their relative proportions of affordance adoption (e.g. p_A , p_O , etc.) (Villermet et al., 2021; Anderson et al., 2020).

Existing multi-affordance literature presents a mixed consensus regarding the extent to which streaming platforms and the algorithmic devices they contain foster "*algorithmic confinement*". Largely, this appears to be dependent on the scale at which diversity measurements are conducted and information representations constructed¹³.

Anderson et al. (2020) study how 100 million users of the Spotify platform's listening diversity differs across so-called "programmed" (e.g. algorithmic or editorial) and organic affordances. Using a high-dimensional embedding space to capture track similarity - broadly a proxy for semantic similarity - and thus, concentration their work delves into conducting both aggregate and temporal appraisals of algorithmic confinement. Regarding the former, the authors show that the music users' consumption through "pro-

¹²Nikolov et al. (2015) illustrates how, in the early days of the web (2006-2010), sites that prioritised social filtering (social media with chronological news feeds) rather than organic information retrieval (search engines) differed in the levels of diversity they encouraged. Using anonymised click logs from over 100,000 American university students, the authors find that social media-facilitated information retrieval exhibited consistently higher concentration compared to that facilitated by search engines. This finding, captured using a discrete entropy-based framework, reveals a pattern of repetition and focus that may emerge from the introduction of *human social filters* even before the introduction of more personalised algorithmic devices.

¹³For instance, dense high-dimensional or discrete item or category based.

grammed" affordances is, on average, less diverse (i.e. less spread in high dimensional item embedding space) than the music they consume through organic affordances. Similarly, on the topic of temporal appraisals, the authors demonstrate that reducing algorithmic adoption is associated with increasing consumption diversity. In short, users who adopt algorithms less frequently in their platform usage tend to be more diverse. Whilst the authors speak more in terms of correlation rather than causation, these findings provide a first intuition regarding the extent to which programmed guidance fosters confinement relative to other ways in which users listen to music on-platform.

On the other hand, literature has also demonstrated that when users' listening concentration is measured using discrete measures (which more closely capture redundancy), algorithmic confinement is less strong (Villermet et al., 2021; Mok et al., 2022). Epps-Darling et al. (2020) demonstrate that Spotify streams are more male-skewed when users adopt organic rather than algorithmic affordances. Similarly, at a more aggregate scale, Villermet et al. (2021) explore how varying levels of affordance usage on Deezer lead to different degrees of discrete concentration. Categorising platform affordances, much like within this thesis, into three types - organic, algorithmic, and editorial - their study finds that organic affordances are actually the least diverse when diversity is captured in a discrete, redundancy-based manner. Conducting a temporal appraisal of algorithmic confinement by comparing relative affordance adoption frequencies to listening redundancies within these affordances adds further nuance to this finding. In contrast to the work of Godinot and Tarissan (2023) who find that using a simulation-based framework, users' overall diversity increases sub-linearly with recommendation adoption, the authors find that in practice, algorithmic confinement increases alongside adoption.¹⁴ Interestingly, Villermet et al. also demonstrate that this dynamic is not exclusive to the algorithmic affordance. For instance, a user who adopts algorithmic affordances more heavily may exhibit greater discrete concentration within their algorithmic consumptions compared to other users but fundamentally, the same dynamic can also be observed for organic affordances and furthermore, to an even greater extent. These findings suggest that under a discrete formulation of concentration, increased usage of organic devices tends to be associated with more concentration than increased use of algorithmic devices. In summary, the work of Villermet et al. hints much like Nguyen et al. (2014) at the organic origins of online media confinement dynamics and also at a potential discrepancy between diversity

¹⁴One explanation for this apparent contradiction might be that Villermet et al. consider concentration within an affordance rather than overall item concentration. Thus, it might well be that algorithmic listening tends to become more saturating with adoption; however, the effect on a user's overall concentration may be a dampening one if there were to be minimal overlap between organic and algorithmic affordances. To date, this topic remains, for the most part, not well explored and again, touches upon the current lack of understanding regarding the inter-dependence between affordances in literature.

measurements that rely on different information representation methods (e.g. dense vs discrete).

In a similar vein, Mok et al. (2022) conclude there is *"little connection between the two perspectives [i.e., diversity built on dense vs discrete information representations]"*. Their study identifies that Spotify users who are the most explorative and thus, achieve discrete diversity are not the ones who are the most diverse in a dense geometric sense. In addition, the authors highlight several findings that are similar to that of Villermet et al. (2021) and Datta et al. (2018) - *narrowing* of consumptions is not a phenomena that is unique to algorithmic affordances but rather, from the point after users register on the platform, algorithmic and to an even greater extent, organic listening is found to grow increasingly concentrated as users settle into the platform. Whilst this might be a mechanical effect at the beginning of users' platform lifecycles (i.e., since all content is novel), understanding why this dynamic is sustained over time remains non-trivial.

Regarding more causally oriented aggregate appraisals which leverage O signals to approximate $O \setminus A$, the degree to which algorithmic confinement effects exist remains just as unclear and dependent on scale and information representation. Aiello and Barbieri (2017) investigate how algorithmic recommendations influence the diversity of users' following networks on Flickr and Tumblr. To approximate causal effects, they employ a matching approach that pairs users with nearly identical local connectivity up to their first k neighbours. These matched pairs are then divided into two subgroups: one where the connection to the $(k+1)$ neighbour occurred via algorithmic recommendation and the other organically. Assessing diversity of the resulting networks using topic entropy, the authors find that algorithmic recommendations may actually cause increased short-term diversity. On the other hand, Sharma et al. (2015) find that at a more long-term level, algorithmic recommendations may have little effect on user behaviour on the Amazon platform. Predicting $O \setminus A$ from O (self-driven query) and A ("similar items") the authors estimate that as many as 75% of algorithmically accessed items would have been viewed organically even in the absence of algorithmic guidance.

Together, this literature demonstrates how music streaming platforms afford differently depending on how users make use of them (i.e., adopt affordances), the kind of diversity measured and the scale at which measurements are derived. At the same time, this literature challenges the overly simplistic view of algorithmic devices as inherently diversity-constraining forces. By beginning to understand what users do and, furthermore, might have done in relative or complete absence of algorithmic devices, this literature of-

fers a more nuanced appraisal of algorithmic confinement. This can, as we shall go on to demonstrate, help to reframe existing filter-bubble narratives.

Accounting for Organic Selections

So far we have seen how there is evidence to suggest that whilst the adoption of streaming platforms can boost users' consumption diversity (Coavoux and Aussant, 2024; Datta et al., 2018), this effect appears to be mediated by the extent to which on-platform algorithmic devices are designed to be personalised (Holtz et al., 2020; Huszár et al., 2022; Su et al., 2016), how users utilise platforms (Villermet et al., 2021; Anderson et al., 2020; Nguyen et al., 2014; Aridor et al., 2020) and the type of diversity measurements being conducted (Mok et al., 2022) (e.g. discrete (Villermet et al., 2021) or dense (Anderson et al., 2020)). Nonetheless, the vast majority of this literature is based on the analysis of consumption signals. As such, the role of users' selections in each of these dynamics, in particular algorithmic confinement, remains mostly unclear. While these topics remain relatively unexplored in music streaming literature, Political Science has actively sought to understand these kinds of questions by comparing differences in the concentration of online media which users *select* in comparison to that which they are *exposed* to algorithmically (Robertson et al., 2023; Bakshy et al., 2015; Mummolo, 2016). Although primarily focused on social media platforms and political information, this body of work provides valuable inspiration and a strong foundation for this thesis.

Typically, this research is driven by the theory of *selective exposure*, a core concept in communication and media studies which describes the propensity for individuals to consume media that reinforces their prior beliefs (Katz et al., 2017) thereby limiting their exposure to cross-cutting items and furthermore, potentially increasing their risk of being confined in (mostly partisan) echo chambers (Iyengar and Hahn, 2009; Sunstein, 2001). These studies explore the degree to which algorithmic devices and organic selections of these algorithmic devices align more or less with users' prior preferences (e.g. political leanings) (Bakshy et al., 2015; Robertson et al., 2023). Typically, prior item preferences are assessed by means other than consumption, such as by analysing the distribution of political ideologies over followed pages on a social media platform. Alignment is thus operationalised by the extent to which users' consumption and exposure logs in the algorithmic affordance remain more or less confined within users' prior preferences (Bakshy et al., 2015).

Existing empirical research from this field suggests that, perhaps surprisingly, on both search engines (Robertson et al., 2023), social media platforms (Bakshy et al., 2015; Aiello and Barbieri, 2017) and digital news outlets (Haim et al., 2018), concentration may tend to emerge mainly from users’ self-imposed *organic* rather than *algorithmic* personalisation. In the works of Bakshy et al. (2015), users’ selections of news articles on Facebook are examined as a sequential chain of information filtering. Firstly, by one’s social network (i.e., their friends whose posts appear in their newsfeed), secondly, by algorithmic devices (i.e., the newsfeed) which filter how this information is exposed to users and then finally, in terms of users’ own selections of these items. The authors find that users are exposed to 15% less cross-cutting content in newsfeeds due to algorithmic ranking (relative to that shared by their social network). Nevertheless, crucially, this pails in magnitude in comparison to what is actually selected by users - users consume 70% less cross-cutting content due to their own selections.

The notion that users’ selections can be even more confined than algorithmic exposures is also corroborated in the more recent work of Robertson et al. (2023). In this work, the authors show that when searching on Google users select both more identity-congruent and unreliable news sources than they are exposed to within their search results. This provides support for the selective exposure hypothesis and furthermore, brings into question existing research efforts which rely only on selection logs to appraise algorithmic confinement, such as, for instance, is typical in music streaming literature. At the same time, recent research has also shown that item topics may play a more crucial role in deciding what users do and don’t consume, hinting at a plethora of inter-dependent selection processes which can reinforce or rather, undermine one another (Mummolo, 2016).

On the whole, these findings offer strong evidence that, in the context of online consumption of algorithmically recommended, politically charged media, users *organically* limit the diversity of information they consume through their selection and rejection behaviours. However, the extent to which these insights apply to the music streaming domain remains, to the best of our knowledge, largely unexplored and hence, a research area which this thesis contributes towards.

Contextualising Organic Confinement

"Music is prophecy. Its styles and economic organisation are ahead of the rest of society because it explores, much faster than material reality can, the entire range of possibilities in a given code"

(Attali, 1985, p. 11).

On theoretical (Dahlgren, 2021; Bruns, 2019) and empirical grounds (Nguyen et al., 2014; Aridor et al., 2020; Coavoux and Aussant, 2024; Datta et al., 2018; Villermet et al., 2021), literature has provided substantial evidence to support the notion that confined music consumption practices may emerge *organically* even in the absence of algorithmic devices. Furthermore, literature has also demonstrated that users' interactions with algorithmic devices (i.e., their selections and rejections) may also limit the diversity of their media consumptions online (Bakshy et al., 2015; Robertson et al., 2023). To begin to understand what drives these processes, we turn our attention away from more computational endeavours and towards the rich body of qualitative literature that, for the most part, stems from Psychology and the Sociology of Consumption. In the following section, we focus on understanding why concentration within consumers music consumptions might emerge organically from two perspectives - dense and discrete. This distinction reflects the existing empirical platform-centric literature that demonstrates how measuring consumption diversity under both perspectives can lead to remarkably different conclusions (Mok et al., 2022; Anderson et al., 2020; Villermet et al., 2021).

The Discrete Perspective

Variety may well be "*the spice of life*" (Sheldon et al., 2013), but according to a rich body of literature from Psychology, familiarity also appears to be a substantial driver of music enjoyment and consumption even in the absence of algorithmic influence. To begin to understand why users' music consumptions might organically confine in the discrete

sense and thus lack novelty, we centre our attention on understanding how affect for and consumption of a song varies alongside the number of times it is exposed. In line with literature within this field, we outline two different types of exposures. Those that are externally determined and those that are self-determined.

Considering first how consumers affect for music varies alongside repeated exposures that are externally determined, a well established body of research from Psychology has theorised that song preference increases *organically* with exposures due to increased familiarity - what is denoted as the "*mere-exposure effect*" (Zajonc, 1968; Chmiel and Schubert, 2017). Put simply, the mere exposure effect is a general theory of preference formation through familiarity that applies to a variety of stimuli spanning across visual, social and auditory domains. Even when a song is not chosen to be consumed, research has demonstrated that consumers' affect for this song initially increases with the number of exposures (Chmiel and Schubert, 2017; Peretz et al., 1998).¹⁵ Nevertheless, this effect has its limits; research also suggests that music goes "stale" for consumers. After a certain threshold of exposures, research has found affect to decrease possibly as a result of satiation/boredom overpowering habituation drivers (Berlyne, 1970). Phrased differently, affect gain from one exposure to the next decreases with the number of exposures until the point at which after a certain threshold where affect peaks, further exposure may only leads to affect reduction (Chmiel and Schubert, 2017)¹⁶ This results in the well-documented inverted-U curve whereby affect varies alongside stimulus exposure.

The extent to which these dynamics may be reproducible in the context of self-driven exposure, more prevalent on streaming platform environments, remains less clear. As O'Brien (2021) pertinently highlight, "*few (if any) studies in the mere-exposure literature examine people's free choices to repeatedly experience the stimulus in question*". Research that focuses on self-driven exposures tends to be on the whole, more quantitative and rely on retrospective usage logs from platforms. As such, within this form of literature, it is more typical to focus on outlining how consumers' consumption behaviours (rather than affect) vary alongside song exposures. Sguerra et al. (2022) verify that on the Deezer platform, inverted-U dynamics emerge even if stream rates and self determined exposures are considered in place of affect and externally driven exposures respectively.¹⁷ The authors study how "fans" (users who have > 40 exposures to a given track) probability of

¹⁵Chmiel and Schubert (2017) conducted a comprehensive meta-analysis of studies examining the relationship between the inverted-U curve and music preference, encompassing research spanning an entire century. Their findings reveal that 87.72% of the literature supports the validity of the inverted-U model.

¹⁶This process is closely related to "*hedonic adaptation*" processes (Frederick and Loewenstein, 1999).

¹⁷Given that on the Deezer platform, the majority of activity is organic (Villermet et al., 2021), it is likely that a marked proportion of these streams stemmed from either users-driven selections whether that be in terms of curation or song retrieval.

streaming a track varies from the 1st to 40th exposure. The authors find that inverted-U curves also commonly emerge in this setting. One way in which these findings could be interpreted is in relation to the previously detailed lab insights (Chmiel and Schubert, 2017) - users may be more likely to re-engage (up until the apex) due to an affect increase which occurred at the prior exposure. Moreover, at a more granular level, Whitman (2014) demonstrate that on Spotify the probability of streaming a track as a function of stream time also follows an inverted-U shape. Nevertheless, the extent to which this association may result from similar psychological processes (i.e., boredom, craving familiarity) or rather, distinct publics remains unclear. Similarly, in recent years, there have also been several notable efforts to statistically model the dynamics of users' online music consumptions (Anderson et al., 2014; Benson et al., 2016). Unlike the previously detailed works, these efforts generally seek to understand and thus model repeat consumptions as a function of elapsed time between past same item consumptions (Anderson et al., 2014; Benson et al., 2016) or through the intrinsic properties of an item itself (e.g. quality captured via an item's popularity (Anderson et al., 2014)).

The concept of self-driven repeated media consumptions has also been thoroughly explored in Consumer Psychology literature. This literature is on the whole, more theoretical than empirical and focuses on theorising about the Psychological processes which drive repeated consumption. Put simply, these works suggest that repetitive consumption may be a learning process that exposes novelty, be that at the side of the *consumed* or *consumer* (O'Brien, 2021). Regarding the former, repeated consumption practices are suggested to function through stimulus novelty, for instance, via novel discovery (e.g. appreciating a song's baseline for the first time) or rediscovery within the song. At the same time, repeated consumption is also said to have the potential to produce novel understandings of the self on the side of the consumer. For instance, repeated media consumption has been shown to be associated with increased feelings of mastery, fandom and identity affirmation (O'Brien, 2019), which may be pleasurable and thus, drive consumers to repeat consumptions.

In summary, literature from Psychology suggests that music streaming users' tendencies to repeat consumption and remain discretely confined may not necessarily be a platform-specific nor an algorithmically driven phenomenon at that. Instead, there is solid evidence to suggest that this behaviour may stem from consumers' underlying cognitive and psychological processes (O'Brien, 2019, 2021; Zajonc, 1968; Chmiel and Schubert, 2017), which in turn, may increase affect and drive users to repeat engagements in the future. In contrast to other media like TV and films, this is a particularly distinct property

of what music affords (Sguerra et al., 2022). This literature, although perhaps trivial to avid music consumers, provides a solid theoretical basis for contextualising why algorithmic devices may not always be the direct cause of the discrete concentration observed in users’ on-platform consumption.

The Dense Perspective

While the field of music psychology offers valuable insights into why music consumers tend to repeatedly engage with the same songs, sociological perspectives are especially useful for understanding how different social groups may distribute their attention across music with varying degrees of semantic similarity (e.g., spanning multiple genres). Given that semantic similarity is often operationalised through distances in item embedding spaces, such sociological literature provides a useful lens for interpreting why users may *organically* exhibit different spreads in dense vector space (Anderson et al., 2020).

Historically, the study of consumption in the social sciences was often secondary to the analysis of production, or when addressed, framed primarily through critiques of the “consumer society” (Slater, 2005). However, in recent years, consumption has emerged as a central focus, particularly as media consumption is increasingly recognised as a key site for the articulation of elite distinction, that is, the use of cultural preferences, including musical taste, as a means of signalling and maintaining one’s social position. Early emulation-based theories of distinction, such as those proposed by Veblen (1917) and later Simmel (1957), suggested that elites maintain their social position by continually evolving their tastes in response to imitation from lower social groups, thereby preserving the exclusivity and symbolic power of their preferences. More recently, misrecognition-based theories, most notably developed by Pierre Bourdieu (Bourdieu, 1984), have argued that distinction is less about reactive shifts in taste and more about the ability of dominant social groups to institutionalise their preferences as legitimate through influence over cultural production, education, media, and other structures of symbolic power (Friedman and Reeves, 2020).

Amongst other things, the outcomes of these processes are differences in cultural taste and consumption practices between those who occupy different social positions (e.g. different classes). In more computational terms, there remains to this day, active debate amongst misrecognition scholars as to whether the mechanisms of distinction result in all classes exhibiting concentrated, low entropy genre tastes or rather, the elite exhibiting

high entropy, balanced genre tastes whilst those of lower economic capital maintain their low entropy, skewed tastes. Regarding the former school of thought, under the Bourdieusian framing, each class is said to have narrow exclusive tastes. For instance, at the time of writing Bourdieu's distinction (1970s), the French upper class were more likely to express taste for so-called high-brow genres such as Classical music whereas those from lower classes were more strongly associated with expressing taste for so-called low-brow genres such as pop music, rock and roll and French chanson.

Since Bourdieu a seemingly endless stream of Sociologists from France and, particularly since the 1990s', the anglophone world, have sought to assert the reproducibility of these findings and build upon the homology thesis. Perhaps the most notable and furthermore, drastic challenge ¹⁸ to Bourdieu's homology thesis stems from the "*omnivore*" thesis, first noted by DiMaggio (1987) and later extrapolated in the work of Peterson (Peterson, 1997, 2005). More than 30 years later and studying class distinction within America rather than France, Peterson's thesis proclaims that a new outcome of class distinction has emerged which relies not on expressing *univore* high-brow taste but rather, broad *eclectic* taste which spans across both low and high-brow culture. Once again, translating these findings to more concentration centric terminology more apt for this thesis, Peterson's thesis pronounces that higher class yields more "omnivorous", spread taste ¹⁹ whilst lower class is associated with, similar to what Bourdieu suggests, "univore" low entropy tastes.

In more recent years, there have been several attempts to focus a Bourdiesian lens on music streaming platforms (Webster, 2020, 2023; Airoidi, 2021; Coavoux and Aussant, 2024). Whilst scarce, and indeed more research is needed, existing literature suggests that when adopted, the relative effect which music streaming platforms can have upon the class structured omnivore-univore divide is either de-amplifying (Webster, 2023) or neutral (Coavoux and Aussant, 2024). Webster (2023) claims that the adoption of music streaming platforms can undermine processes of distinction by taking over the labour of music curation and compressing the time needed to appreciate music for its own sake. Similarly, Coavoux and Aussant (2024) show that the effect of platform adoption on music consumption diversity is independent of one's social position - adoption marginally increases diversity either way, and thus, presumably if all were to adopt streaming platforms, this would have a negligible effect on the pre-existing omnivore-univore divide. Nevertheless, while in theory, the effect of streaming platforms on consumption diversity

¹⁸Within the anglophone scholarly world there exists some disputable as to the degree to which Peterson's thesis challenges Bourdieu's homology theory (Erickson, 1996; Chan and Goldthorpe, 2017), or rather, aligns with his overall framework despite the specificities changing (Holt, 1997; Lizardo and Skiles, 2015).

¹⁹Perhaps as a result of boundaries of cultural preferences becoming more fluid, perhaps as a result of reduced reliance on distaste to delineate social identity.

is independent of economic capital, the adoption of these platforms clearly is not. Just as in the early 2000s, higher social positioning was associated with greater adoption of listening devices that gave users more flexibility (López-Sintas et al., 2008) likewise, does the same hold for the adoption of music streaming platforms (Coavoux and Aussant, 2024). For an advert-free, usable experience users are required to pay a monthly subscription fee thus creating a substantial barrier to entry for those with less economic capital. As such, paywalls may in fact amplify class distinctions since these diversifying opportunities are only available to a select few (DiMaggio, 2004; Coavoux and Aussant, 2024). In a more theoretical direction, Airol di (2021) make a notable effort of translating a Bourdieusian thinking to a socio-technical perspective, thereby treating algorithmic devices as social agents that actively take part in social life. We refer the reader to this text if they wish to gain a more thorough understanding of how state-of-the-art Sociology of Consumption is currently approaching the study of music streaming platforms and algorithmic recommendation devices.

In summary, what may initially appear to be a purely mathematical choice between measuring music consumption diversity using discrete or continuous geometric item representations is, as shown throughout this section, deeply connected to broader disciplinary perspectives. The discrete approach tends to align with psychological perspectives, offering insight into the meanings and origins of diversity at the individual level. In contrast, the continuous, dense geometric approach aligns more closely with sociological literature aimed at understanding how diversity varies across larger social groupings. Together, these perspectives provide essential context for the decision to employ both approaches in this thesis and help clarify why organic concentration may still emerge within each framework.

Part II

Scientific Contributions

Tracing Affordance and Item Adoptions on Music Streaming Platforms

Foreword

This research was published at the International Society of Music Information Retrieval conference in 2021, the beginning of my PhD journey. Building upon prior research from the RECORDS project (Villermet et al., 2021; Beuscart et al., 2019) it explores the relatively under-examined phenomenon of item transfers across affordances - specifically those that originate from programmed discoveries (e.g., via the *A* or *E* affordance) and later reappear in organic consumptions - and further develops upon the user taxonomies proposed by Villermet et al. (2021).

The motivation for studying item transfers across affordances came from an interest in understanding just how organic platform interactions truly were on Deezer. Given that at the time of publication, organic behaviour were increasingly being used to benchmark algorithmic concentration against (Anderson et al., 2020), the broader goal of this research was to allow for a better understanding of if such approaches were sensible or at the least, may require a different framing. Together, this made this research a natural starting point for my PhD and served as an opportunity to test foundational assumptions before pursuing them further.

Ultimately, this research demonstrates that for the vast majority of Deezer users, organically streamed songs rarely originate from discoveries made via programmed affordances such as *A* recommendation devices or *E* playlists. Therefore, using either organically streamed tracks or overall organic listening as a benchmark for assessing algorithmic in-

fluence appears to be a reasonable and defensible approach. These findings also further support the view that music streaming users generally prefer to remain *in control* when engaging with music streaming platforms (Beuscart et al., 2019). Crucially, organic activity continues to account for the majority of platform activity, even after taking into account the original source affordance adopted for discovery.

TRACING AFFORDANCE AND ITEM ADOPTION ON MUSIC STREAMING PLATFORMS

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ABSTRACT

Popular music streaming platforms offer users a diverse network of content exploration through a triad of affordances: *organic*, *algorithmic* and *editorial* access modes. Whilst offering great potential for discovery, such platform developments also pose the modern user with daily adoption decisions on two fronts: platform affordance adoption and the adoption of recommendations therein. Following a carefully constrained set of Deezer users over a 2-year observation period, our work explores factors driving user behaviour in the broad sense, by differentiating users on the basis of their temporal daily usage, adoption of the main platform affordances, and the ways in which they react to them, especially in terms of recommendation adoption. Diverging from a perspective common in studies on the effects of recommendation, we assume and confirm that users exhibit very diverse behaviours in using and adopting the platform affordances. The resulting complex and quite heterogeneous picture demonstrates that there is no blanket answer for adoption practices of both recommendation features and recommendations.

1. INTRODUCTION

Today, the modern music streaming platform architecture is a far cry from the digital repository which it once was. Increased diversification at a platform level has resulted in a now common triad of platform affordances, *organic* (*O*), *algorithmic* (*A*) and *editorial* (*E*), which allow users to explore a platform's ever expanding musical catalogue through novel paths. *A* affordances refer to the platform's plethora of Recommender System (RS) architectures (e.g., the popular Flow playlist on Deezer) whilst *E* affordances correspond to mostly human curated playlists (such as recommended playlists variously called "10s electronic", "Rock & Chill", etc.). We characterise all remaining modes of access under *O* including for instance the search bar, user-constructed playlists, and more broadly, modes of content access which do not utilise any degree

of recommendation. No longer is it necessary for the avid music listener to spend hours on end organically trawling through digital repositories to find their 'niche' but rather, they are free to draw upon *A* and *E* affordances to further complement or even replace their exploration in this space. Nonetheless, while it is true that affordance diversification yields increased potentials for exploration this comes at the price of a greater emphasis on the user's to utilise the platform - questions of what affordance to utilise and what items therein to adopt quickly become frequent decisions the modern user must face in their daily platform usage.

In our work we centre our attention on this notion of adoption on two fronts: *affordance adoption* (O, A, E) and *item adoptions* therein (i.e. adoption into one's organic catalogue: $A \rightarrow O \parallel E \rightarrow O$) through the introduction of the novel metric of adoption. Tracing user listening practices on the popular music streaming platform, Deezer our work sheds light on the varied and often heterogeneous nature of adoption and behavioural differences amongst users. The contribution of this work sits within the growing body of state of the art literature which appraises the interconnected effects of human behaviour and algorithmic influence through an organic comparison – a contribution of high relevance to music streaming providers and user's alike to comprehend the bi-directional affect of recommendation upon user preference and downstream platform behaviour.

2. LITERATURE REVIEW

2.1 Appraising Algorithmic Influence via an Organic Comparison

Whilst historically the primary role of a music Recommender System (mRS) on streaming platforms was to facilitate the efficient personalised exploration of a platform's often vast musical catalogue thereby minimising the risk of *choice overload*, a substantial body of multi-disciplinary literature [1–4] points towards the same conclusion that user exploration nonetheless may remain confined to a minute fraction of homogeneous musical content – a phenomenon famously denoted as '*filter bubble*' [5]. Similarly, simulation based RS literature has additionally explored the tendency for feedback loops to emerge in a plethora of domains [6–9]. Nonetheless, while such practices are clearly outlined in literature, a less trivial second order question still remains ambiguous:



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To what degree is user platform behaviour primarily a product of algorithmic influence or rather, an autonomous organic process imposed by the user themselves?

In light of such questions, a novel branch of the state of art has sought to measure algorithmic influence by drawing parallels to a user’s organic platform behaviour that is, without algorithmic influence. Roth conceptualises this debate through what he coins the ‘*ROM-COM dichotomy*’ [10] – the tendency for filtering algorithms to either Read Our Minds (ROM) acting as cognitive aids to facilitate organic exploration or rather, Change Our Minds (COM) algorithmically distorting a prior organic preference.

Literature appraising algorithmic influence through an organic reference has been applied to a range of multi-disciplinary contexts. In the works of Bakshy et al. [11], the potential for Facebook’s NewsFeed algorithm to expose users to cross-partisan content is appraised by drawing parallels to user organic explicit preference and subsequent item consumption. Their work finds filtering algorithms to have minimal effect in reducing cross-partisan information in comparison to the organic selection processes imposed by the user suggesting a user’s diversity limitation may be principally due to human pre- and post-selection. In the music domain, Epps-Darling et al. [12] study the role of Spotify’s algorithmic influence on gender representation through an organic reference point. Their findings show user’s organic preference towards male artists to be marginally stronger than that recommended by Spotify’s mRS, again suggesting the human organic bias to be stronger than that generated algorithmically. Anderson et al. [13] also analyse diversity with respect to organic vs programmed (algorithmic/editorial) listening events on Spotify. Through the generation of a usage-based embedding space, they find algorithmically-driven exploration in this space to be less diverse than organic, providing evidence for a COM effect.

Overall, current literature paints a picture of divergent music streaming platform practices dependent upon the user’s degree of organic-algorithmic usage and actualised with respect to context. Thus, we commence our work with the prior hypothesis that platform use behaviour is largely varied - there is no *average user* for which a blanket answer to algorithmic influence may be applied.

2.2 Item Adoption in Recommender System

Whilst item adoption in terms of consumption confined to a given affordance has been covered extensively and critically both in terms of user studies [14, 15] and user modelling [16–18], literature concerning item adoption in relation to the dynamic transfer of items across affordance (e.g. from an external affordance into ones organic catalog) remains to this date, sparse. Nonetheless, at a higher level, the adoption of music streaming platforms as such, independent of the affordances they offer, has been notably

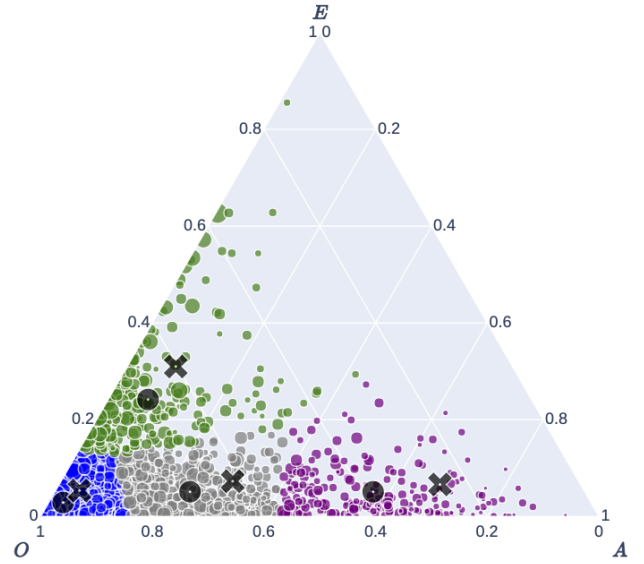


Figure 1. Ternary plot of affordance adoption classes $\emptyset+$ (blue), o (grey), e (green), a (purple), disks represent centroid positions. Crosses represent corrected centroid positions after taking into account the pre-adoption origin of plays (see Sec. 5).

studied in the field of Cultural Studies. In the works of Datta et al. [19], streaming adoption is shown to lead to substantial increases in both the quantity and diversity of music consumed by a user. Similarly, Rushan et al. [20] also study factors of the platform interface which in itself, determine a consumer’s decision to adopt music streaming platforms as a result of increased platform familiarisation. Still, adoption related to transitions across platform affordances remains largely unexplored in the literature - a gap this work aims to address

2.3 Temporal Dimensions of mRS Usage

Time of day information has been evidenced to be an important signal in disentangling platform behaviour [21] and thus, has in recent years become a signal commonly utilised in context dependent RS literature [22–24] to provide increased personalisation and accuracy. What is more, user studies have also revealed that both the time of day and week can play a substantial role in mediating user platform experience and downstream projected user behaviour [25, 26]. Utilising such rich temporal signals encapsulated within listening logs, our work seeks to explore the degree to which adoption on both fronts may differ across temporal daily usages.

3. METHODOLOGY

3.1 Listening Events Data Set

We work with about 2 years of listening histories from about 13k Deezer users who registered within a 1 month period in September 2017. We further focus on users who remained active over the entire observation period: formally, we impose a maximum inter-event time threshold

of 10 days thus ensuring that users rely on the platform for a regular source of music. This eventually yields 2701 users.

We discard listening events <30s as these are deemed as so-called ‘skips’. We further merge unique song identifiers which share identical audio embeddings in a pre-build latent space supplied by Deezer (see [27] for use case / generation details). This prevents double counting of identical songs which may have been mis-labeled as distinct. For each listening event we characterise the mode of access used to retrieve the content of which on Deezer there exists a triad of affordances: *organic* (O), *algorithmic* (A) and *editorial* (E).

3.2 Defining user classes

Affordance Adoption. We capture user adoption of platform affordances through the proportion of content accessed via each of the platform’s three main affordances after aggregating listening histories. For a given user, their listening history is represented by the temporally-ordered list of listening events over our observation period, formally defined as:

$$P = ((s_i, t_i, f_i))_{i \leq n}$$

where s_i is the song ID, t_i the timestamp, and $f_i \in \{A, E, O\}$ the affordance used for the i -th listening event. We accordingly denote the sublist of P restricted to a given affordance F as $P_F = ((s_i, t_i, f_i))_{i \leq n \wedge f_i = F}$. Considering the proportion of plays accessed *organically*, *algorithmically* and *editorially* respectively, the affordance profile of a given user is defined by the triplet $(|P_O|/|P|, |P_A|/|P|, |P_E|/|P|)$ which sums to 1 and can be represented as a barycentric coordinate in ternary space (see Figure 1). Performing a k-means clustering ($k = 4$) across affordance profiles yields 4 distinct classes of users which we label as follows: very organic ‘o+’ (1786 / 65.98%), organic ‘o’ (429 / 15.85%), algorithmic ‘a’ (224 / 8.27%), organic/editorial ‘æ’ (268 / 9.90%). We note that we rather deal with bins, areas or classes than with well-separated clusters per se. Thus, from herein we refer to affordance adoption clusters as classes. Already, a highly varied picture of affordance adoption on the platform emerges along with a shared preliminary benchmark: for all classes, users on average display some degree of O adoption whilst the same cannot be said for A and E . Indeed, at first sight it appears Deezer users do not typically adopt affordances on all fronts but rather use the platform predominately as an organic catalogue much the way one would search through a traditional song library, albeit a much larger one here. However, as we shall later detail, a user’s tendency to consume mostly O content may be misleading: if a significant proportion of a user’s O catalogue is a product of A or E adoption the very definition of what it means to adopt O affordances is brought into question.

Concentration Measures. Beyond sheer user activity over the entire observation period denoted by play counts $|P|$, we consider a notion of *redundancy* [26] quantified as

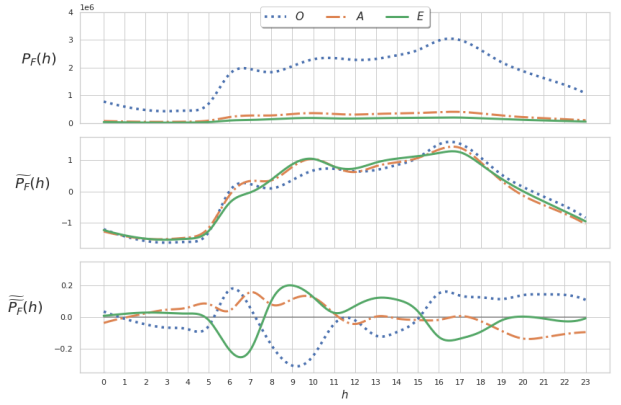


Figure 2. Normalised platform affordance time-of-day activity. Aggregate levels are shown (top) followed by z-normalised activity (middle) and residual de-trended activity levels (bottom).

a measure of how much a given user saturates their listening catalogue (i.e., plays the same songs repeatedly), formally defined as $R = 1 - |S|/|P|$ where $S = \{s | (s, t, f) \in P\}$ is the set of unique songs in P .

We additionally characterise the diversity of a user’s exploration using a pre-built 32 dimensional latent space $\mathcal{E}$ constructed from low-level audio features via metric learning [27]. The audio embeddings were primarily used by Deezer for the task of artist disambiguation (i.e., where artists had the same name but were stylistically unique). Thus, the construction of $\mathcal{E}$ maximises acoustically dissimilar artists while acoustically similar artists remain close in this space. For each user, we compute their average pairwise *cosine distances* between audio embeddings $\mathcal{E}_s$ for each $s \in S$ and ultimately, report average values.

Temporal Time of Day Analysis. Platform usage also varies substantially over the elapsed day [26]. We consider platform activity with respect to *time-of-day* at an aggregate level across all affordances (Fig. 2, top row). We observe that it predominantly consists of O followed by A and E at much lower magnitudes, consistent with what has been previously detailed. Without loss of generality, we denote $\mathcal{P}$ as the platform-level history for all users (i.e., all streams from the whole dataset). To analyse temporal trends independent of magnitude, we consider hourly play counts for each affordance $\mathcal{P}_F(h) = \left| ((s_i, t_i, f_i))_{i \leq n \wedge (t_i \text{'s hour} = h) \wedge (f_i = F)} \right|$, to which we subsequently apply a z-normalisation relative to daily play count averages defined as:

$$\widetilde{\mathcal{P}}_F(h) = \frac{\mathcal{P}_F(h) - \langle \mathcal{P}_F(h) \rangle_h}{\sigma_{\mathcal{P}_F(h)}}$$

These normalised activity levels (Fig. 2, second row) reveal three peaks of gradually increasing magnitude, respectively in the early morning, morning, and afternoon (16-17:00), from which a gradual decay in activity is experienced. To capture temporal adoption variations of affordances with respect to one another, we finally apply so-

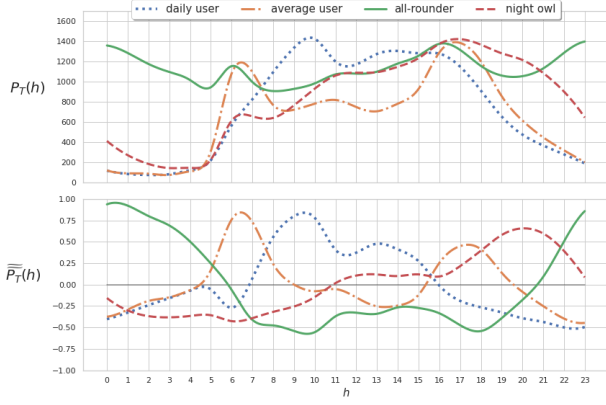


Figure 3. Normalised time-of-day activity levels for each time-of-day class. Aggregate levels are shown (top) followed by residual de-trended activity levels (bottom).

called ‘*detrending*’ as per [28] by comparing the above z-normalisations $\tilde{P}_F(h)$ relative to other affordances, formally defined as:

$$\tilde{\tilde{P}}_F(h) = \tilde{P}_F(h) - \langle \tilde{P}_F(h) \rangle_F$$

Affordance adoption at a platform level clearly varies across hours of the day (Fig. 2, bottom row). For one, in the early morning and from the afternoon hours and on into the evening, we observe the tendency to favour O to be more prominent, respective to normalized A and E affordance activity in the same time blocks (i.e. independent of magnitude). Otherwise either A or E seem to be favoured (again, in relative trend) and essentially appear to exhibit bi-modal relative adoption peaks across the day, albeit at different moments (rather in the morning for A and in the early morning and early afternoon for E).

We complement the temporal platform-level analysis by examining daily patterns at the user level, as users will commonly have varied activity levels on music streaming platforms [29]. To this end we consider user-level hourly activity aggregated over all affordances $P(h) = \left| \left((s_i, t_i, f_i) \right)_{i \leq n \wedge (t_i \text{'s hour} = h)} \right|$. As in [26], we subsequently cluster users using k-means ($k = 4$ again) applied on $P(h)$ as a 24-dimensional unit vector. We observe 4 distinct user-centric behavioural dynamics which we label as follows: the ‘*average*’ user (726 / 26.82%), the ‘*night owl*’ (928 / 34.28%), the ‘*all rounder*’ (245 / 9.05%) and the ‘*daily*’ user (808 / 29.85%). We represent the variations of $P(h)$ on Fig. 3 by applying the same type of normalization as used above for platform-level quantities $\mathcal{P}(h)$, except that we consider temporal clusters T instead of affordances F i.e., $\mathcal{P}_T(h)$ in place of $\mathcal{P}_F(h)$.

3.3 Item Adoption

To measure the tendency for users to adopt algorithmic or editorial songs into their organic catalogue we define the novel measure of *adoption* α . At a high level, an item can be said to be adopted the first time it has been played organically by a user given that the song was first recommended through some affordance F and not played organically as

a prior. Since we are interested in item adoption and thus unique songs, we now focus on song sets rather than lists of plays i.e., S . We outline two possible mutually exclusive song sets, denoted ϕ and ρ , to differentiate songs which could have been adopted yet were not, from the ones which were actually adopted:

– **Adoption feasible**, ϕ , denoting the set of songs which were played through F but not via O and thus, had the potential to be adopted yet were not:

$$\phi_F = \left\{ s \in S \mid \exists (s, t, F) \in P, \nexists (s, t', O) \in P \right\}$$

– **Adoption realised**, ρ , denoting the set of songs which were played via F as a prior and were consumed through O at least once subsequently:

$$\rho_F = \left\{ s \in S \mid \exists (s, t, F) \in P, t < \min_{(s, t', O) \in P} t' \right\}$$

Furthermore, if a user is exposed to more recommendations before ultimately making the decision to adopt, this may indicate a weaker influence of the platform’s affordance or that adoption is less likely to be a direct product of it (for instance the user might be more likely to have heard the song from an external source such as the radio). To capture this intuition, we introduce $r_F(s)$, the number of recommendations of song s which appeared through F before organic adoption:

$$r_F(s) = \left| \left\{ (s, t, F) \in P \mid t < \min_{(s, t', O) \in P} t' \right\} \right|$$

to which we apply a polynomial scaling function which decays to give more weight to lower numbers of recommendations - a similar practice to how listening counts are often scaled logarithmically in mRS literature [30, 31].

We assess the relative impact of item adoption at two levels of abstraction. Foremost, with respect to the number of items which both could have been and were adopted by the user through F i.e., $|\phi_F| + |\rho_F|$. Formally let this be defined by:

$$\alpha_F = \frac{\sum_{s \in \rho} r_F(s)^{-\lambda}}{|\phi_F| + |\rho_F|}$$

where $\lambda \in (0, 1]$ is a hyperparameter which affects the degree of polynomial decay with respect to algorithmic impact. In our experiments we set the value of $\lambda = 0.5$. We note our choice of λ is cautious and should in future work be more refined with statistical and qualitative user studies exploring the role of repeated affordance recommendation prior to adoption.

Secondly, we normalise adoption with respect to the number of unique items consumed via O , thereby capturing the relative impact of algorithmic adoption in a user’s overall organic listening catalogue. Formally,

$$\alpha'_F = |\rho_F| / |\{s \in S \mid \exists (s, t, O) \in P\}|$$

We note that this value is bounded by the number of organic streams in a user’s listening history but nonetheless, we deem this to be a useful measure to capture the influence of item adoption in bringing into question the very meaning of what is deemed organic.

	o+					o				
	ar	au	no	du	all, (znorm)	ar	au	no	du	all, (znorm)
$ P $	*26.62K	15.59K	*20.87K	15.94K	18.58K, (0.21)	26.09K	15.63K	19.68K	16.88K	18.01K, (0.18)
R	*0.90	0.85	0.86	*0.83	0.85, (0.02)	*0.82	0.77	0.77	0.77	0.77, (0.02)
α_A	0.28	0.25	0.28	0.25	0.25, (0.06)	0.15	0.14	0.15	0.14	0.14, (0.07)
α_E	0.26	0.27	0.24	*0.21	0.25, (0.10)	0.18	0.19	0.16	0.15	0.17, (0.08)
α'_A	0.01	0.02	0.02	0.02	0.01, (0.05)	0.06	0.06	0.06	0.05	0.06, (0.03)
α'_E	0.02	0.02	0.02	0.02	0.02, (0.08)	0.03	0.02	0.02	0.03	0.02, (0.04)
$\mathcal{E}$ dist.	0.28	0.29	0.29	0.29	0.28, (0.02)	0.28	0.29	0.30	0.29	0.29, (0.01)

	a					æ				
	ar	au	no	du	all, (znorm)	ar	au	no	du	all, (znorm)
$ P $	41.20K	15.06K	16.18K	19.75K	19.01K, (0.40)	23.80K	15.40K	19.68K	23.09K	20.14K, (0.15)
R	*0.85	0.77	0.74	0.77	0.77, (0.04)	0.77	0.74	0.77	0.76	0.76, (0.01)
α_A	0.10	0.09	0.09	0.09	0.09, (0.04)	0.17	0.16	0.20	0.15	0.17, (0.08)
α_E	0.20	0.14	0.15	0.13	0.14, (0.13)	0.14	0.14	0.16	0.13	0.14, (0.08)
α'_A	0.08	0.08	0.08	0.08	0.08, (0.02)	0.03	0.02	0.02	0.02	0.02 , (0.14)
α'_E	0.03	0.03	0.02	0.03	0.03, (0.04)	0.05	0.05	0.05	0.05	0.05 , (0.04)
$\mathcal{E}$ dist.	0.29	0.30	0.29	0.30	0.30, (0.01)	0.28	0.29	0.28	0.29	0.29, (0.01)

Table 1. Experimental results across two static affordance and temporal *time of day* user classes. Values in bold represent the top value, while marked with * are results where the difference is statistically significant (two tailed t-test, $\alpha = 0.05/n$ after Bonferroni correction).

4. RESULTS

Temporal Affordance Adoption Variations. We first examine the distribution of affordance classes across time-of-day classes. As shown in Figure 4 we observe two fundamental preliminary findings: (1) daily users are more heavily composed of both *a* and α users respective to other time-of-day classes. (2) *o+* users are more likely to reside within the all rounder and, to a lesser extent, night owl class. Framed differently, users who adopt almost solely *O* are more likely to favour platform activity in the evening hours of the day whilst users who more heavily *A/E*-adopt are more likely to favour activity in the day time hours. Once again, our findings reiterate what was observed from our temporal platform evaluation – the use of recommendation affordances corresponds to different categories of temporal use as well as, we contend, different types of users.

Characterising platform behaviour. To further disentangle the respective use cases we now characterise behavioural dynamics for each time-of-day and affordance class. Focusing first on affordances, we attain results that go against the grain of a diversity-constraining narrative (see Table 1). Users who *A*-adopt more frequently are found to have more diverse exploration in $\mathcal{E}$ whilst maintaining relatively low redundancy levels as measured by R . On the contrary, *o+* users are found to saturate their listening catalogue reflected in the high R levels attained. From these findings, it appears possible that the consumption of *A* content may have a diversify effect upon a user’s P at both a behavioural and deeper content-based level.

Regarding item adoption within affordance classes, both *a* and α users exhibit low levels of α_A and α_E , respectively. This may indicate that higher adoption of *A* and *E* affordances corresponds to a more passive, less engaged mode of usage. In this scenario—reminiscent of tradi-

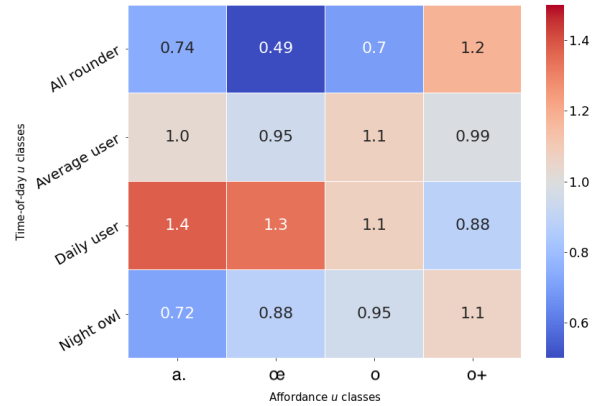


Figure 4. Affordance vs. time-of-day distributions. Values are normalised such that above or below 1 indicate respectively similar, over- or under- representation of affordance classes relative to those found globally.

tional radio listening—the user appears to take a backseat, adopting a "let it play" approach to interaction. Nonetheless, when *a* and α users do take the decision to adopt this makes a substantial impact to their *O* catalogue and thus, the dispersion of users in the *A, E, O* ternary space as we shall later detail. Drawing parallels to a more pure organic behaviour through the *o+* user, we observe polar opposite dynamics in comparison to the *a* and α users. Whilst *o+* users *A/E*-adopt sparsely, their ultimate downstream platform use is less recommendation-skeptic reflected in the much higher levels of item adoption rates for *A* and *E* affordances attained ($\alpha_A = 0.25$, $\alpha_E = 0.25$) but with minimal impact to the constitution of their overall *O* catalogue. From the detailed findings it is clear that our preliminary assumption that users display varied behavioural dynamics holds true: users diverge dramatically in both their affor-

dance adoption and adoption of items therein and perhaps most importantly, with varying degrees of impact to their overall O catalogue.

We next consider the extent to which a given affordance class varies behaviourally alongside temporal preference for platform usage. For instance, does an $o+$ user who has preference for daily usage behave the same as one who has preference for nightly activity? For a given affordance class and behavioural measure, we test if each temporal class differs significantly from all other temporal classes in the same affordance group. We perform a two-tailed Welch’s unequal-variance paired t-test [32]. Table 1 marks a value as significant if $p > \alpha/3$ (i.e. after adjusting for errors via Bonferroni correction to control for Type I errors) for all remaining temporal counterparts. For instance, considering $(o+, |P|)$, we check if we observe a greater marked or significant difference between all rounders and all other time-of-day classes. Results are shown in Table 1. Largely, we observe time-of-day classes to have no significant effect upon distorting behavioural dynamics for each affordance class. However, one outlier remains, the *all rounder*. Remarkably, this class appears to display distinct behavioural dynamics for almost all affordance classes bar α , divulging in high levels of platform usage but at the cost of saturating listening catalogue reflected in the high average $|P|$ and R levels (respectively $27.32K$ and 0.88). Such findings suggest time-of-day preferences can have a significant effect in mediating surface level activity amongst affordance classes but with no clear downstream propagation to a user’s deeper musical preference in terms of varied audio content consumed and preference for item adoption.

Disentangling heterogeneous platform behaviour. To disentangle the influence of time-of-day preference and affordance adoption on users’ item adoption and reactions to recommendations we next perform a factorial ANOVA, shifting each behavioural attribute to be the dependent variable whose variance we seek to explain. We primarily fit our data to an OLS model $Y = \beta_0 + \beta_1 F + \beta_2 T + \beta_3 FT + \varepsilon$ (where F and T represent affordance and time-of-day labels respectively) before subsequently applying a factorial ANOVA. Due to space constraints, the full ANOVA results table is not included however we now detail the most relevant results to this work. As hypothesised, we observe the only effect size (η_p^2) for which time-of-day classes may have a both moderate and significant effect is with regard to a user’s activity $|P|$. On the contrary, affordance classes offer a moderate-to-high explanatory factor for the variance of the remaining behavioural attributes, foremost adoption. Perhaps most interestingly, the effect of affordance classes on α_A is particularly strong (0.11) implying that a user’s decision to adopt items into their organic catalogue may, as hypothesised, be principally a product of adopting recommendation affordances.

For completeness we ultimately examine the effect of sequential time-of-day and affordance adoption influence on the notion of what is meant by an organic stream. Contrary to our preliminary belief that organic access acted as

a benchmark for Deezer platform exploration we observe users to actually be more algorithmic and editorial than first thought, albeit indirectly. Reassigning organic streams to now belong to the affordance in which it was initially adopted from we recompute centroids for each affordance adoption class. In cases where a stream was both adopted via A and E we deem that the item was adopted via the affordance which had the most streams prior to adoption. As shown in Fig. 1, we observe all affordance adoption classes centroids to experience a marked shift towards A and E poles – even more so for users who are already closer to these poles i.e., particularly for a and α users.

5. CONCLUDING REMARKS

In a time where the modern music streaming platform encapsulates a myriad of modes to access content, users may and do personalise their platform use in highly varied ways. By acknowledging, assuming and confirming the diversity of user platform behaviour, our work traces the interconnected yet surprisingly sequential factors which drive affordance and item adoption. Our results paint a highly complex picture of user platform behaviour whereby time-of-day preference mediates low-level platform behaviour (activity levels) and affordance adoption, while affordance adoption preference mediates the ultimate higher-level decision to adopt content into one’s O catalogue, a factor which is indeed more reflective of musical taste.

Coming full circle, the heterogeneity of item adoption and its subsequent impact on one’s organic catalogue brings into question the nature of what constitutes an organic stream - after taking into consideration the role of adoption, users are indeed found to be markedly less organic than was initially thought. This, in turn, may redefine what affordance adoption really is. Although beyond the scope of this work, we also suggest that a fruitful direction for future work would be to appraise item adoption relative to affordance adoption, first and foremost by differentiating the impact of repeated exposure along user behavioural classes or by estimating the bidirectional transfer of items between A and E affordances. What is more, we also recommend further investigating the role of temporal preference at different scales such as weekly, monthly or longitudinal. Overall, this work highlights an often-overlooked nuance in how recommender systems influence the diversity of users’ media consumption: the ways in which users engage with streaming platforms may significantly shape how recommendation mechanisms are adopted, interpreted, and experienced. In practice, this type of work and approach could be utilised at the platform level to further the development of context-dependent RS, providing musical recommendations which are far more suited to the high variety of user’s driving use cases.

6. REFERENCES

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Interpreting Node Embedding Distances Through n-Order Proximity Neighbourhoods

Foreword

Prior research within and connected to the RECORDS project has primarily relied on discrete measures of concentration to assess users' listening diversity under various platform affordances (Louail and Barthelemy, 2017; Beuscart et al., 2019; Villermet et al., 2021). Whilst fruitful in its own right, these approaches fail to account for the varying degrees of similarity between items and as such, there exists a semantic gap between how listening diversity is perceived and how it is quantitatively assessed (Porcaro et al., 2022). To address this, the contribution of this thesis focuses on applying a dense, high-dimensional item embedding space to capture similarity between artists / tracks and then later downstream, users' listening concentrations under different affordances. This allows for a more granular, geometric measure of item similarity which can as we shall shortly demonstrate, offer a distinct perspective on the role of algorithmic devices in fostering listening concentration.

In this chapter, presented at the Complex Networks Conference in 2024 and published through Springer in the conference proceedings ²⁰, we focus on identifying the most suitable model to generate an item embedding space to be used to these ends. The construction of embedding spaces is not unique to the study of music listening; a wide range of techniques has emerged from fields such as complex networks and natural language processing (NLP). Many of these methods can be readily adapted to music-related data, provided some level of abstraction is applied between music objects (e.g. artists or tracks)

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and analogous objects in other domains. For example, treating a playlist as a document and tracks as terms allows for the straightforward application of NLP-based embedding models to listening event data. Similarly, the application of node embedding models is also trivial given that usage based artist/song co-occurrence matrices are treated as the adjacency matrix of a network. The latter perspective informed our exploration of various node embedding techniques for constructing an artist-based embedding space. In this study, we assess the suitability of different embeddings by examining their interpretability within the original co-occurrence network. This focus was intentional: the network itself encodes rich structural information, ranging from first-order (direct) connections to higher-order (multi-hop) relationships, which we sought to preserve and meaningfully reflect in the learned embedding space.

Our findings revealed that the classical singular value decomposition (SVD) outperforms more sophisticated methods such as node2vec and DeepWalk in preserving both local and global network structure. Despite being less tailored to network-specific tasks, SVD demonstrated superior capacity to retain first-, second-, and higher-order neighbourhoods. For our purposes, SVD offers the best of both worlds: a method that is both interpretable and effective in capturing complex structural relationships within the network.

Interpreting Node Embedding Distances Through n -Order Proximity Neighbourhoods



Dougal Shakespeare and Camille Roth

Abstract In the field of node representation learning the task of interpreting latent dimensions has become a prominent, well-studied research topic. The contribution of this work focuses on appraising the interpretability of another rarely-exploited feature of node embeddings increasingly utilised in recommendation and consumption diversity studies: inter-node embedded distances. Introducing a new method to measure how understandable the distances between nodes are, our work assesses how well the proximity weights derived from a network before embedding relate to the node closeness measurements after embedding. Testing several classical node embedding models, our findings reach a conclusion familiar to practitioners albeit rarely cited in literature—the matrix factorisation model SVD is the most interpretable through 1, 2 and even higher-order proximities.

Keywords Interpretability · Node embedding · Paradigmatic

1 Introduction

In the field of node representation learning, proximity-based embedding models have frequently been build upon or share many parallels with NLP architectures such as the skip-gram model [16] leveraged by deepwalk [21] and node2vec [9]. Whilst the representations learnt by such models have proven to be well-suited for a variety of downstream tasks [5], the extent to which these representations may be at the same time, *interpretable* and not just “black boxes” has increasingly become a cause for concern [5, 6, 13, 22]. To date, research efforts in this direction have largely, much like in NLP, focused on latent dimension interpretation which although fruitful in its own right, neglects the interpretation of *inter-node distances* per se, the focus of this

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work. With a growing number of consumption diversity (see the *GS* score [2, 28]) and recommendation [3, 4, 17, 18] studies now making use of inter-node distances out-of-the-box, being able to interpret their meaning in a simple, comprehensible manner is a more relevant task than ever. The present work explores one simple strategy for interpreting inter-node distances: interpretation through a network’s n -hop proximities. The *goodness* of a simple interpretation is partially dependent on the existence of a sufficient one-to-one mapping between nodes’ proximities in a graph G and closenesses in an embedding space e . If such a condition is fulfilled, the relation between both variables is monotonic and thus, given a closeness score between two nodes it is possible to *interpret* an associated proximity in G without further decoding (and thus, it is simple) given knowledge of the underlying function which relates both variables. Likewise, vice versa from proximity to closeness. Building upon this understanding, our work introduces and applies a novel framework to evaluate the interpretability of e by assessing the degree to which there exists a distinct mapping between 1, 2 and higher-order proximities in G and closeness in e . Applying four classical node embedding models, deepwalk [21], node2vec [9], SDNE [29] and Truncated Singular Value Decomposition (SVD) to two new music streaming-based artist co-occurrence networks, our work provides a novel account of node embedding models’ interpretability which we hope will further inform researchers’ model selection processes particularly in domains where interpretation is paramount.

2 Related Work

2.1 n -Proximity Semantics

To interpret inter-node distances in vector space we turn our attention to the field of structural linguistics which has inspired the development of many NLP and thus, node proximity-based embedding models. Of particular relevance to this work is the concept of *syntagmatic* and *paradigmatic* relations [12] which in more graph theoretic terms, closely corresponds to first- and second-order proximities (i.e., hops) [8]. In linguistics, syntagmatic relations refer to a co-occurrence between two words in a sentence (e.g., ‘eat’, ‘food’) whilst paradigmatic relations refer broadly to substitution. More precisely, two words are said to be strongly paradigmatically connected if they frequently occur in similar contexts, for instance the synonyms “big” and “large”. Inspired by the ‘distributional semantic hypothesis’ [12], classical NLP models such as skip-gram [16] typically try to operationalise paradigmatic relations resulting in their reflection in embedded geometry [25]. Building upon this understanding, recent works in the fields of explainable manifold learning [10] as well as graph embedding evaluation [11] have sought to interpret embedded distances by their respective correspondences to paradigmatic weights by applying measures such as cosine similarity.

Besides having a strong grounding in structural linguistics, paradigmatic weights also closely relate to the notion of *structural similarity* first introduced in the field of social network analysis [14], a relaxed notion of the stricter concept of ‘structural equivalence’ applied to segment nodes into sets of equivalent ‘roles’ [11, 23]. Closely connected to this work, Jin et al. [11] study correspondences between the similarity of nodes’ roles (of which one is structural) and inter-node nearest neighbour distances in embedded space. Our work differs in that we focus exclusively on proximity strengths rather than the more general notion of roles and such, we appraise not just second-order paradigmatic proximities (denoted by Jin et al. as structural roles) but also higher-order proximities [26].

2.2 Node Embedding Interpretability

Research concerning the *post-hoc* interpretation of node representations can broadly be conceptualised under two taxonomies: (1) interpretation of single dimensions and (2) interpretation of collections of dimensions (i.e., structural representations). Regarding the former, Gogoglou et al. [6] study how individual dimensions of embedded space tend to correspond to both a network’s internal structure and node labels assigned externally. Similarly, Piaggese et al. [22] develop and apply novel measures that capture the global interpretability of embedding vectors based on the marginal contribution of embedding dimensions in predicting a network’s structure. Regarding the latter, Dalmia et al. [5] explore how several node embedding models encode elementary node properties such as Page Rank, node degree, closeness centrality and clustering coefficients through a prediction task. In a similar direction, Liu et al. [13] leverage the entire structural representation of node embedding models by formulating interpretation as a problem of taxonomy induction from a node’s vector representations. More closely connected to this work, Park et al. [20] introduce a novel framework to interpret node representations through other target nodes in the treated network. Whilst similar in many respects, our research differs in that we focus on interpreting inter-node distances rather than dimensions.

3 Datasets

We experiment with embedding two undirected positive point wise mutual information (PPMI) networks of artist co-occurrences constructed from a random sample of anonymised user consumption and curation logs supplied in confidence by the Deezer music streaming platform. We focus on co-occurrence networks since typically, these are the form of networks embedded in recommendation and consumption diversity studies where inter-node distance measurements are widely applied. For instance, in the context of nearest neighbour-based recommendation, suggestions of items simi-

lar to a given target item are generated by querying its neighbourhood in latent vector space.

DeezerPL: Building upon outstanding efforts treating human-generated playlists as a natural language [2, 15, 19] we focus on a cohort of approximately 1 million anonymised private Deezer playlists randomly sampled from the platform at the end of 2022. Each playlist is represented by an ordered list of artist ids (393,624 unique at the time of extraction), containing an average number of songs of $\langle |\mathbb{P}| \rangle = 58.32$, $\sigma_{|\mathbb{P}|} = 139.32$. Upon further examination of this dataset we observe a marked number of extremely long outlier playlists which have a length of up to approximately 10K songs. Despite their length, artist redundancy [27] is amongst the highest for these playlists. We theorise that these may be generated by parties other than users and as such, filter playlists where $|\mathbb{P}| > 2\sigma_{|\mathbb{P}|}$ to remove these outliers from our analysis set. We then focus on the subset of playlists that have at least 10 unique artists and thus, contain rich co-occurrence information. After applying both filterings 245,556 playlists and 285,028 artists remain. Due to computational constraints we then employ a random sampling of 5K playlists from our filtered dataset which yields 11,637 unique artists across all playlists. Finally, since our analysis is not concerned with self-directed co-occurrences we reduce each playlist to only contain one occurrence of each of the artists featured in the playlist and construct an artist-artist co-occurrence matrix populated with within-session co-occurrence counts. As is typical in an NLP pipeline we remove unrepeated co-occurrences as these are deemed to be hapaxes and cast our matrix to a Positive Point Wise Mutual Information (PPMI) matrix to account for connectivity due to sheer popularity effects. More formally:

$$S_{i,j} = \max \left(\log_2 \frac{p(i,j)}{p(i)p(j)}, 0 \right) \quad (1)$$

where $p(i,j)$ is the proportion of co-occurrences which contain both nodes i and j , $p(i)$ denotes the proportion of co-occurrences that involve artist node i and thus, $p(i)p(j)$ denotes the probability of co-occurrence if the within-session occurrence of i and j is assumed to be independent. This forms our final first-order so-called syntagmatic network S .

DeezerSess: Alongside our playlist-based network we also construct a session-based artist co-occurrence network from 7859 randomly sampled anonymised Deezer user session logs in the month of January 2018. New sessions of continuous platform engagement are defined when a platform drop off longer than 20min occurs and as per [2], streams lasting less than 30s are filtered as ‘skips’. Before any further filtering our dataset consists of 396,100 sessions and 29,012 unique artists. We proceed by filtering artists whose name (utilised for sense checking the semantics of each embedded space) could not be assigned and then subsequently, filter so-called *fandom* sessions which only contain one artist. Phrased differently, applying both of these filtering strategies means that we only analyse sessions that contain ≥ 2 artists with assigned names. Removing users who no longer have any sessions after applying this filtering reduces our user set size to 708. We then impose a further

Table 1 Network properties after imposing the various filtering criteria. Q and Q_γ denote genre modularity scores before and after low-degree filtering respectively (DeezerSess: $\gamma = 14$, DeezerPL: $\gamma = 15$)

	n -order	$ V $	$ E $	Q	Q_γ	$\langle W_1 \rangle$	$\langle W_2 \rangle$	$\langle W_3 \rangle$	$\langle W_4 \rangle$
DeezerPL	S	5950	306,984	0.40	0.41	1.43	3.60	5.90	8.86
	P	/	6,564,746	/	0.21	0.01	0.07	0.16	0.36
	H	/	10,642,266	/	0.18	0.01	0.13	0.41	0.84
DeezerSess	S	2681	101,989	0.24	0.22	1.32	3.30	5.33	8.00
	P	/	2,168,791	/	0.09	0.01	0.05	0.13	0.35
	H	/	1,423,672	/	0.08	0.04	0.16	0.45	0.80

filtering constraint that users must have >50 sessions (equating to a lower bound of on-average 1.61 sessions per day) to focus on users who actively use the platform as a source of music consumption. This reduces our user set to 532 users who collectively consume 9225 unique artists across 32, 526 sessions. For each user we then compute an artist-artist co-occurrence matrix of size 9225×9225 which we normalise by the number of sessions they engage in (i.e., the upper bound for a co-occurrence) to account for user activity. We then compute the element-wise summation over all user matrices and cast the summed matrix to a PPMI matrix forming our final network, S .

Low-Degree Node Filtering: As a last processing step we perform a network-level filtering of low-degree nodes which have a degree $< \gamma$ where $\gamma \in \mathbb{N}$ thus, elevating the mechanical effect of low-degree nodes which lack signal to compute node attraction scores (see Sect. 5.2). For each network we assign γ to be the maximum value possible whilst still ensuring $\leq 50\%$ of nodes are filtered. As depicted in Table 1, this filtering strategy does not appear to affect the overall genre-homogeneous community structure typical of music co-occurrence networks [1] reflected by the minimal variance in genre modularity scores before and after filtering. After performing this filtering strategy the DeezerPL network consists of 4 connected components, the largest of which spans 5880 nodes and has an average shortest path of 2.35. By contrast the DeezerSess network only consists of one connected component and has an average shortest path of 2.69.

4 Node Embedding Models

We test four classical node embedding models: (1) SVD, (2) DeepWalk, (3) node2vec, (4) SDNE, using the GEM python package [7, 8]. Each model (aside from SVD which does not require iterative learning) is trained to output dense 128 dimensional real vectors for 100 epochs.

SVD—As a paradigmatic baseline we apply SVD to S , an industry standard employed by music streaming platform to construct user and item embeddings [1]. More formally, SVD is a factorisation technique which seeks to approximate a matrix $A \in \mathbb{R}^{b \times c}$ as the product of three matrices, $B \in \mathbb{R}^{b \times m}$, $\Sigma \in \text{diag}(\mathbb{R}_+^m)$ and $C \in \mathbb{R}^{c \times m}$. In practice, the resulting factorisation corresponds to finding an optimal solution which minimises the $\|A - \hat{A}\|_F^2$ loss where $\hat{A} = B\Sigma C^*$, B and C are orthonormal by columns and $\hat{A}$ is a low rank approximation of A . In our analysis we make use of the $B\Sigma$ component as our latent item embedding.

DeepWalk (dw) [21]—The seminal node embedding model leverages the skip-gram [16] architecture from NLP by treating vertex paths traversed by random walks as sentences and the vertices therein as terms to be embedded. We focus on tuning 2 hyperparameters—the walk length (l) and context size (c)—by fixing one of the hyperparameters to be assigned to its default value ($l = 50, c = 10$) and adjusting the other parameter within the following discrete parameter space: $l \in \{20, 50, 140\}$ and $c \in \{2, 6, 10, 14\}$. For both the DeezerPL and DeezerSess networks we select a final hyperparameter configuration ($c = 10, l = 20$) which yields the minimum average rank whereby models are ranked by I score (see Sect. 6) for syntagmatic, paradigmatic and higher proximities.

node2vec (n2v) [9]—Similar to dw, n2v performs random walks on a network which later act as input for the skip-gram model. It however differs in its introduction of two additional parameters p and q which bias the random walking process. The former constrains two-hop redundancy whilst the latter biases random walks to be broad or deep. Assigning $p, q = 1$ to be both parameters default values we perform a hyperparameter tuning in the range $p, q \in \{1/4, 1, 4\}$ in a manor which is equivalent to that described for dw. Applying the same hyperparameter selection criteria detailed for n2v we find that for the DeezerSess network ($p = 0.25, q = 1$) yields the best performance whereas a hyperparameter tuning more biased towards a breadth-first exploration ($p = 1, q = 4$) is more optimal for the DeezerPL network.

Structural Deep Network Embedding (SDNE) [29]—In addition to skip-gram models we also apply SDNE due to its high levels of flexibility regarding syntagmatic and paradigmatic preservation controlled by the α hyperparameter. The model is comprised of both an unsupervised deep autoencoder which constructs an embedding from which a vertex neighbourhood can be reconstructed and a supervised part which applies a loss when first-order syntagmatic vertices' latent representations are dissimilar. To explore the effect of introducing syntagmatic-paradigmatic bias variations we construct two versions of the SDNE model: SDNE_P ($\alpha = 0.01$) and SDNE_S ($\alpha = 0.5$).

5 Proximity and Attraction

5.1 n -Proximities

We first centre our attention on the extent to which a node's neighbourhood typically remains geometrically close or distant in embedding space across different proximities—what we denote as *attraction* (or *repulsion*) dynamics due to the movement of nodes during embedding training. To this end, we focus on three n -order (or n -hops) inter-node proximities, in short n -proximities: first-order syntagmatic ($n = 1$), second-order paradigmatic ($n = 2$) and higher-order (with $n = 4$) proximities. To evaluate the weight of such proximities we construct three weighted un-directed proximity networks which are represented by their adjacency matrices $S, P, H \in \mathbb{R}^{|V|} \times \mathbb{R}^{|V|}$ based on a node set V (see Table 1). As previously described, edge weights in our first-order proximity network S simply correspond to PPMI scores between nodes. To capture second-order paradigmatic proximities we form a structural paradigmatic weighted adjacency matrix $\hat{P}$ by querying node-node distances using cosine similarity such that:

$$\hat{P}_{ij} = \begin{cases} \frac{S_i \cdot S_j}{\|S_i\| \|S_j\|} & \text{if } i \neq j \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

whereby self-loops are removed. We similarly build a higher-order proximity network $\hat{H}$ by computing paradigmatic relations of paradigmatic relations, whereby 2 nodes are connected when they exist in a 4-hop relation in S . More formally:

$$\hat{H}_{ij} = \begin{cases} \frac{P_i \cdot P_j}{\|P_i\| \|P_j\|} & \text{if } i \neq j \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

We then compute the final versions P and H of the paradigmatic and higher order adjacency matrices by imposing the additional constraint that a relation can only exist in the first order it appears and not beyond (i.e., in absentia [24]). More formally, $P_{ij} = \hat{P}_{ij}$ if $S_{ij} = 0$, and 0 otherwise; likewise, $H_{ij} = \hat{H}_{ij}$ if $P_{ij} = 0$, and 0 otherwise. For each of our 3 proximity networks we then segment non-zero edges (i.e., node tuples) by their associated edge weights into four sets of increasing proximity—(W_1, W_2, W_3, W_4)—by applying k-means with $k = 4$. Average weights in each proximity class are depicted in Table 1. In addition, we compute a further control class W_0 which contains all node tuples which do not feature as edges in S, P and H (i.e., $S_{i,j} = P_{i,j} = H_{i,j} = 0$).

5.2 Neighbourhood Attraction Score

Given a node embedding model to be evaluated, we first appraise the extent to which typically, a node's n -proximity neighbourhoods (i.e., neighbourhoods in S , P , and H) are attracted towards or repulsed from that node in embedded space. To this end, we define a series of cumulative observation window sizes common to the whole dataset. Denoting D as the set of all pairwise inter-node cosine distances ($1 - \cos$) derived from our embedding space, we define a series of $|V| + 1$ growing window sizes $(w_i)_{i \in \{0, \dots, |V|\}}$:

$$w_i = \min(D) + \frac{i(\max(D) - \min(D))}{|V|} \quad (4)$$

thus configuring a uniform window expansion rhythm featuring strides of constant size $\frac{\max(D) - \min(D)}{|V|}$. We consider each node in V as a *target node* (t) from which we shall compute attractions towards, also examining the extent to which node tuples in each weighted proximity set ($W_0, \dots, W_4$) typically exhibit distinct *attraction* dynamics. For each $t \in V$ we denote $N_j = \{v | (t, v) \in W_j\}$ as its W_j -filtered neighbourhood. Together N_j are non-overlapping sets spanning over $V \setminus \{t\}$. The pipeline that follows is detailed for the N_1 neighbourhood however equivalent logic applies for all other neighbourhoods N_j . We first assert that N_1 contains a sufficient number of nodes for plotting and curve fitting that will follow by imposing the filter that $|N_1| > 0.005 \cdot |V|$. Given that this condition holds we then compute a cumulative *hit* vector of size $|V|$ whereby each element corresponds to the proportion of the neighbourhood captured in window w_i :

$$h_i = \frac{|\{v \in N_1 : d(t, v) < w_i\}|}{|N_1|} \quad (5)$$

In other words, the h vector describes the growing amount of nodes of the N_1 neighbourhood which remain within a growing radius around t . For curve fitting purposes we then associate this h vector with an x vector whereby $x_i = -6 + \frac{12(i-1)}{|V|}$ and fit a sigmoid function on x to h using non-linear least squares:

$$f(x) = \frac{1}{1 + e^{-g(x-s)}} \quad (6)$$

whereby g and s are the parameters to be fitted that respectively affect growth and shift of the sigmoid. Using f we then compute node t 's neighbourhood attraction score w.r.t. nodes in N_1 simply as:

$$\delta_1 = \int_{-6}^6 f(x) \quad (7)$$

Greater values indicate stronger local attraction dynamics to the target node t (i.e., a bigger area under the cumulative hit curve), which denotes a better preservation of closeness in embedded space. Inversely, smaller values reflect stronger local repulsion dynamics away from t (i.e., a smaller area under the curve). To elicit comparisons of δ_1 between different models which may display unique item density distributions in vector space we normalise δ_1 by the neighbourhood attraction score ($\tilde{\delta}$) of a null model which reflects the average h over all nodes irrespective of the proximity class:

$$\dot{\delta}_1 = \log_2(\delta_1/\tilde{\delta})$$

and likewise for all other N_j and thus, δ_j and $\dot{\delta}_j$.

5.3 Results

We first centre our attention on attraction differences between neighbourhoods which exist in distinct n -proximities (i.e., between S , P and H). As depicted in Figs. 1 and 2, lower-order proximities (i.e., S) tend to experience on-average stronger attraction dynamics. This effect is particularly pronounced for SVD reflected in the high δ scores obtained even for the weakest neighbourhood (N_1). Regarding model differences, all models aside from dw and n2v display greater attractions to 1-, 2- and 4-proximities than other proximities reflected by the control set W_0 . SVD consistently exhibits the highest attraction scores across all n -proximities tested even in the case of neighbourhoods in higher proximity relations despite implicitly optimising structural 2-order relations. Indeed, it appears that SVD is capable of capturing both low and high-order relations in its geometry and, furthermore, distinguishing between different H neighbourhoods as we shall later demonstrate. Regarding the model's capacity to consistently locally position 1-order proximities we hypothesise that this effect may be due, rather, to the typical structure of co-occurrence networks which exhibit a strong correlation between S and P proximities (average S , P correlation over all nodes is $r = 0.643$ for DeezerSess and $r = 0.628$ for DeezerPL). Hence, optimising P relations can implicitly result in S optimisation by proxy. Nonetheless, the SDNE_S model which, by design, tries to closely embed nodes in strong S relations tends in fact, to display weaker first-order attraction dynamics than its second-order paradigmatic counterpart (SDNE_P). Interestingly it appears that, within the context of co-occurrence network embedding, trying explicitly to operationalise first-order relations may be both redundant and furthermore, counter-productive.

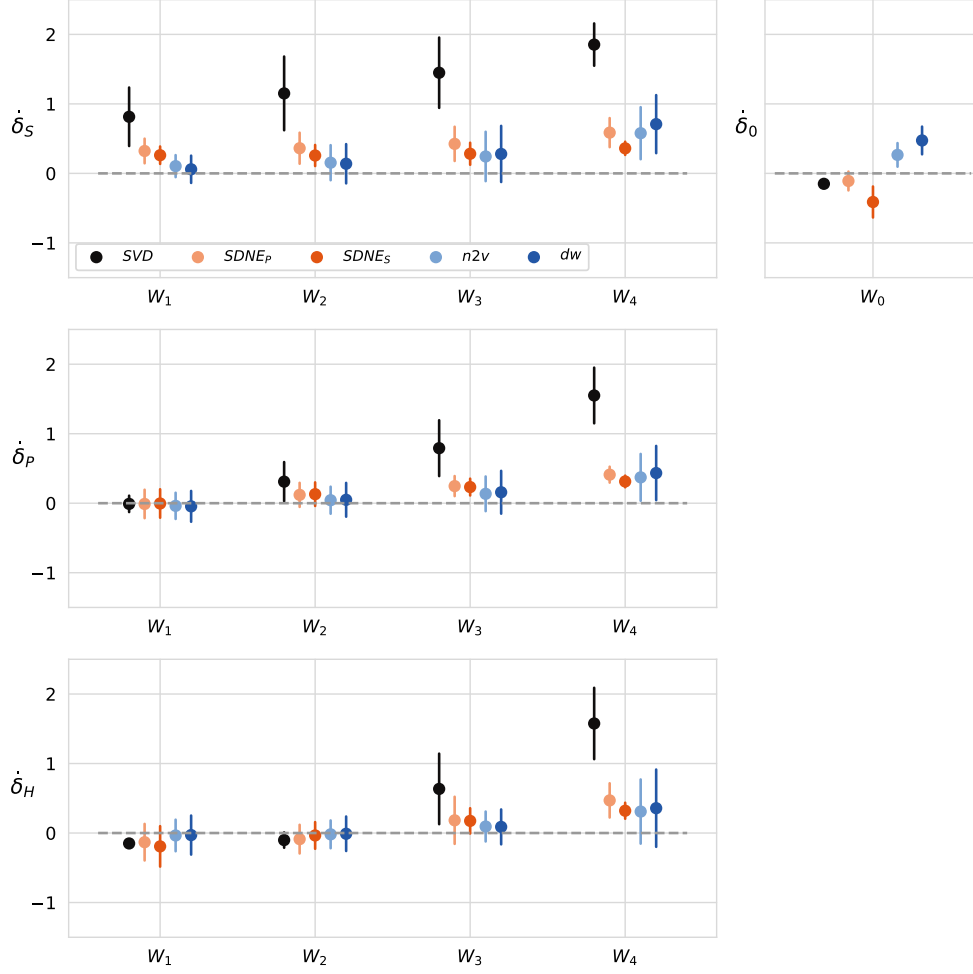


Fig. 1 Attraction scores for the DeezerSess network before z-normalisation for first-order ($\dot{\delta}_S$), second-order ($\dot{\delta}_P$), higher-order ($\dot{\delta}_H$) and other ($\dot{\delta}_0$) proximities which acts as a control set

6 Appraising n -Proximity Interpretability

The interpretability of an embedding model in a simple manner depends in part on the existence of a sufficient one-to-one mapping between proximity weights (from an n -order adjacency matrix) and embedding distances. For instance, a strong (network) proximity should not correspond to the same (embedding) distance as a weak proximity. To quantify this, we propose to compute the dissimilarity between attraction score distributions associated with each proximity class (W_j), the main focus of this work. In order to interpret a given inter-node distance with confidence, proximity class mappings should each be associated with embedding distance mappings that are as distinct as possible from one another. For all $t \in V$ we segment valid $\hat{\delta}$ into 5 sets of attraction scores— $\Delta_0, \Delta_1, \Delta_2, \Delta_3, \Delta_4$ which correspond to attraction scores for N_0, N_1, N_2, N_3, N_4 neighbourhoods respectively. To compare between models whose attraction scores span different ranges we perform a form of z-normalisation that allows us to place each model's attraction scores on a comparable scale whilst

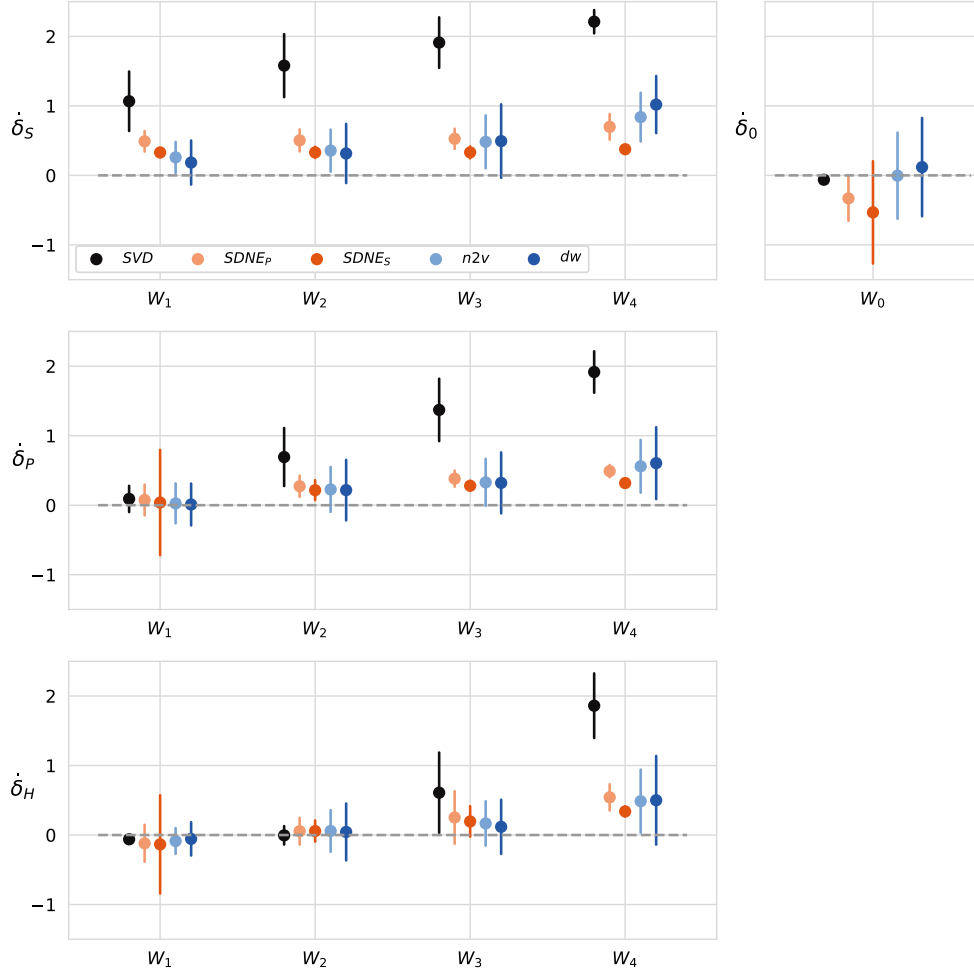


Fig. 2 Attraction scores for the DeezerPL network before z-normalisation for first-order ($\dot{\delta}_S$), second-order ($\dot{\delta}_P$), higher-order ($\dot{\delta}_H$) and other ($\dot{\delta}_0$) proximities which acts as a control set

still capturing attraction variations between each W_j as well as absolute differences between trends inter-nodes.

$$\tilde{\Delta}_i = \{(\dot{\delta} - \langle \mu_{\Delta} \rangle) / \sigma_{\mu_{\Delta}} : \dot{\delta} \in \Delta_i\} \quad (8)$$

where $\mu_{\Delta} = (\langle \Delta_0 \rangle, \langle \Delta_1 \rangle, \langle \Delta_2 \rangle, \langle \Delta_3 \rangle, \langle \Delta_4 \rangle)$.

Binning each $\tilde{\Delta}_i$ into a discrete histogram representation with a bin size of $\sigma_{\mu_{\Delta}}/4$ in the range $[-10\sigma_{\mu_{\Delta}}, \dots, 10\sigma_{\mu_{\Delta}}]$, we construct the following count normalised histograms $P = (P_0, P_1, P_2, P_3, P_4)$. We then compute our final interpretability score, I as:

$$I = \frac{\sum_{i=1}^{|P|-1} \sum_{j=i+1}^{|P|} JS(P_i, P_j)}{\binom{|P|}{2}} \quad (9)$$

where JS reflects the symmetric Jensen–Shannon score noting m the pointwise mean of p and q , and KL the Kullback–Leibler divergence,

Table 2 Interpretability results for both networks across S , P and H proximities (standard deviations in parentheses). Bold values mark maximums per proximity and * indicate that all pairwise comparisons between neighbourhood histograms for a model were significant using a two-sample Kolmogorov–Smirnov test ($\alpha = 0.05$ and Bonferroni correction applied)

	DeezerSess					DeezerPL				
	$SV D$	$SDNE_P$	$SDNE_S$	$n2v$	dw	$SV D$	$SDNE_P$	$SDNE_S$	$n2v$	dw
I_S	*0.69 (± 0.16)	*0.57 (± 0.25)	*0.53 (± 0.28)	*0.57 (± 0.17)	*0.60 (± 0.18)	*0.75 (± 0.12)	*0.56 (± 0.30)	0.46 (± 0.33)	*0.54 (± 0.17)	*0.51 (± 0.19)
I_P	*0.80 (± 0.05)	*0.69 (± 0.13)	*0.72 (± 0.13)	*0.60 (± 0.16)	*0.58 (± 0.18)	*0.77 (± 0.09)	*0.74 (± 0.11)	*0.70 (± 0.17)	*0.51 (± 0.13)	*0.46 (± 0.12)
I_H	*0.73 (± 0.19)	*0.57 (± 0.22)	*0.71 (± 0.13)	*0.55 (± 0.20)	*0.54 (± 0.23)	0.60 (± 0.33)	*0.67 (± 0.15)	*0.70 (± 0.13)	*0.53 (± 0.13)	*0.43 (± 0.13)

$$JS(p, q) = \sqrt{\frac{1}{2}(KL(p\|m) + KL(q\|m))}$$

To summarise, we compute z-normalised attraction score sets $\tilde{\Delta}_j$ for each neighbourhood N_j , which we bin into 80 one-standard-deviation regularly-spaced bins of size one-quarter-of standard deviation. This yields five histograms which we compare across pairs of neighbourhoods N_j . High values of I correspond to more distinct pairs P_i, P_j and thus, under this conceptualisation a higher interpretability.

Results. Regarding interpretability differences between n -proximities we observe that as depicted in Table 2, aside from dw and $n2v$ who display their highest I scores largely within 1-order proximities, the node embedding models we test have the highest interpretability scores within 2-order P proximities. Even weaker neighbourhood classes (i.e., N_1, N_2) are found to display distinct attraction dynamics with respect to 2-order proximities. Regarding model differences, just as we saw previously that the dw and $n2v$ models were outliers with respect to neighbourhood attraction dynamics likewise, are they distinct with regard to interpretability. Aside from S proximities, both models are found to yield the lowest I_P and I_H scores. By contrast, $SV D$ is the most interpretable model across almost all n -proximities with a slight exception for the DeezerPL dataset whereby the $SDNE_S$ model surprisingly, attains the highest I_H score hinting at a non-trivial, complex relationship between the optimisation of first- and higher-order proximities. Overall, our findings suggest that despite being less state-of-the-art, inter-node distances are consistently the most interpretable via n -proximities for the $SV D$ model. We suggests that thus, this model may be well-fitted for application domains where distance interpretation is both paramount and requires minimal computational efforts for instance, when generating on-line interpretable nearest neighbour recommendations from a given seed item. At the same time, we remark that $SDNE_S$ remains a fruitful alternative if both the interpretability and attraction of S proximities are rather counter-intuitively, not a priority.

7 Discussion

The contribution of this work introduces and applies a novel pipeline to study node embedding interpretability at the rarely exploited level of inter-node distances. Drawing influence from structural linguistics which inspired the development of several NLP and thus, node embedding models our work appraises the extent to which (embedded) closeness measurements derived from several classical node embedding models' representation spaces tend to correspond to distinct (network) proximities. In turn, our findings lead us to a conclusion often cited in industry [1] but yet surprisingly understated in academia: SVD is both the most interpretable model and furthermore, capable of capturing higher order relations within its geometry. In applications such as generating personalised recommendation or user diversity assessments where inter-node distances are applied out-of-the box and thus, required to be reflective of some semantic meaning in their own right, we suggest that the usage of SVD would be advisable. At the same time, our work also provides a valuable insight for researchers engaging in co-occurrence network embedding, a common practice in recommender system studies: optimising first-order proximities may be both redundant due to implicit correlations to second-order proximities and furthermore, sub-optimal to increase first-order attractions as was observed to be the case for the SDNE_S model. Whilst beyond the scope of this work, we suggest a fruitful direction for future works may be to expand upon the cohort of models and networks tested in this study to add further rigour to these findings. In addition, another direction may be found in exploring the effect of other distance measures (in the case of this work cosine distance) upon n -proximity interpretation and furthermore, the interplay this has with the embedding model deployed.

Disclosure of Interests. There were no conflicts of interest relating to the production of this work.

Acknowledgements This paper has been partially realized in the framework of the “RECORDS” grant (ANR-2019-CE38-0013) funded by the ANR (French National Agency of Research).

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Reframing the Filter Bubble Through Diverse Scale Effects in Online Music Consumption

Foreword

Published as an open-access article in *Nature's Scientific Reports* in 2025, this research presents a novel analysis of how temporal scale and information representation (dense or discrete) shape algorithmic confinement conclusions. In turn, we demonstrate that over time, the adoption of algorithmic devices tends to result in less discrete concentration (i.e., more novelty), yet concurrently, more dense concentration (i.e., less spread in dense vector space). This brings nuance to and enriches the filter bubble narratives.

The previous two chapters in this thesis strongly guide this research's methodological direction. Firstly, based on the understanding that organic streams are largely independent of algorithmic ones, (Shakespeare and Roth, 2021), we chose to apply *organic* benchmarks to appraise algorithmic confinement. Secondly, drawing on insights from the experiments conducted in Shakespeare and Roth (2024), we chose to apply SVD to generate dense item representations. This ensures that the dense measures of algorithmic confinement which we derive remain interpretable.

Following a series of discussions with the lead of Deezer R&D, we were also able to release both the code needed to reproduce the experiments from this paper as well as an anonymised subset of the dataset used. The dataset is released as open-access thereby facilitating reproducibility and also encouraging future research in this area. These kinds of datasets are highly valuable for the field of Music Information Retrieval since multi-affordance traces are usually scarcely available outside of partnerships between industry

and academia.



OPEN Reframing the filter bubble through diverse scale effects in online music consumption

Dougal Shakespeare^{1,2}✉, Victor Chareyron¹ & Camille Roth^{1,2}

The impact of platforms' algorithmic curation devices on users' consumption diversity has long since been the subject of debate in popular culture and academia alike. In recent years, methods to concretely appraise this debate have taken a comparative turn, with emerging research strands seeking to assess the influence of algorithmic recommendations relative to users' *organic* behaviour (i.e. those with a relative lack of algorithmic influence). Nevertheless, as the contribution of this work clearly shows, contradictory findings that algorithmic devices may both foster "bubbly" confinement dynamics and diverse exploration appear to stem, at least in part, from tensions surrounding information representation and scale. Taking into account various scales, affordances, and diversity measures, the contribution of this work conducts an empirical analysis of approximately 50,000 Deezer users' artist consumption histories across different affordances offering various degrees of algorithmic assistance. Transitioning through three scales of increasing aggregation—intra-sessions, inter-sessions and inter-affordances—our work underscores how shifts in perspective, particularly in information representation, can yield markedly different conclusions concerning the role of algorithmic devices in fostering diversity. However, when considered collectively, these perspectives coalesce to provide a nuanced update to the seminal filter bubble narrative: algorithmic curation devices may introduce more novelty than what users' achieve organically, yet concurrently, this novelty is more semantically confined. Consequently, our research significantly contributes towards an enriched understanding of the intricate dynamics surrounding on-platform algorithmic devices.

More and more information access is mediated by digital platforms and the various affordances they embody. Now ubiquitous personalised *algorithmic* curation devices (A) play a key role in this process, affecting what, how and when users are exposed to information on-platform. Even though some studies appear to confirm the seminal hypothesis that these devices may have a constraining effect upon diversity, isolating users in so-called filter bubbles¹ of overly personalised, homogeneous content^{2,3}, an array of prior works appear to suggest that they rather have the potential to increase consumption diversity^{4–16}. Notwithstanding, empirically testing the validity of such claims frequently requires defining a suitable baseline reflecting users' *organic* (O) consumptions, that is, those performed with a relative lack of algorithmic assistance. The type of usage traces utilised to this end depends largely on the type of diversity appraisal being conducted. Broadly, this can be classified under two strands of literature based on whether they appraise concentration dynamics in terms of *A consumption* vs. *A exposure*^{4,5,12,13}, or in terms of *A consumption* vs. *another lesser A affordance*^{2,8–11,16–18}.

The former is most commonly conducted within the field of Political Science and is driven by the theory of selective exposure, a core concept in communication and media studies which describes the propensity for individuals to consume media that aligns with their prior beliefs¹⁹ thereby limiting exposure to cross-cutting content. Empirical works in this direction typically explore how alignment differs within the information sets algorithmic devices expose users to, and then, downstream, within the information sets that users' organically select^{4,5}. Under this framing, alignment is thus operationalized by the extent to which users' consumption and exposure logs in the algorithmic affordance remain more or less confined within users' prior preference bubbles⁵. Whilst by design, algorithmic devices tend to generate recommendations that align with and thus, are *within* users' prior preference bubbles recent empirical results suggest that perhaps surprisingly, users' own organic selections of algorithmic recommendations can further constrain their consumption diversity^{4,5,12,15} sometimes, to an even greater extent^{4,5}.

The second strand of literature seeks to appraise algorithmic confinement by comparing users' consumption diversity across *different* affordances which each offer variable levels of algorithmic assistance. Following Davis formulation²⁰, an affordance describes *how* an object enables and constrains; in the context of online platforms,

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this generally corresponds to one way of accessing content or information. For instance, on music streaming platforms which are also the focus of many studies within this strand this includes organic personal playlist, algorithmic recommendation devices and human editorial curations to name a few^{2,8–10,16,17}. While most existing works focus on a variety of affordances within one platform, some perform cross-platform comparisons whereby each platform is also deemed an affordance in itself^{14,18}. Put simply, the diversity of the content consumed within an affordance would be compared with that of another affordance mostly empirical however sometimes hypothesised¹¹. Typically, this involves comparing users' consumptions within *A* and *O* affordances. It can also more broadly be conceptualised in terms of comparing consumptions within an algorithmic affordance to a lesser algorithmic one. Like several of the previously detailed *A* exposure works, existing research generally concludes that users' algorithmic consumptions are consistently either less^{9,16} or no less concentrated⁸ than their organic ones.

Whilst both branches of research appear to imply that confinement dynamics may be amplified by self-imposed organic personalisation a handful of research adds further nuance to this narrative regarding how concentration dynamics vary for specific content²¹, user categories⁹ and, perhaps most importantly, when relying on diversity measures which leverage more geometric feature representations^{2,13,17}. The latter hints at an important dichotomy regarding how information and, therefore, diversity, is being constructed: that is, with *discrete* or *dense* representations. Indeed, existing research tends to rely on discrete information representations built upon either item categories (e.g., left- vs. right-wing, music genres, etc.) or simply items themselves. Under this framing diversity is typically appraised through *novelty*, be that in terms of users' exploration vs. exploitation dynamics (i.e., exploring new items / exploiting known ones) or repetition-based measures whose methods rely on set-theoretic or entropy based approaches. In contrast, recent works² have also sought to study diversity with more dense, continuous item representations, essentially capturing users' diversity through the *spatial disparity* of the items they consume. Under this framing, research suggests that algorithmic devices may in fact have a constraining effect upon users' diversity — Anderson et al.² show that users' *A* consumptions are consistently less diverse than their *O* ones on the Spotify platform. These approaches highlight a potential tension which resides between diversity measures which rely on *discrete* or *dense* information representations and, thus, diversity measures which capture *novelty* or *spatial* disparity. This is in line with the recent findings of Mok et al.¹⁷ which suggest that at the user level, there is little connection between the two perspectives.

Nevertheless, Mok et al. do not disaggregate affordances when drawing these conclusions and furthermore, an attempt to compare their results to the affordance-centric perspective runs into the issue of scale variance — Mok et al. work at the short-term (per week) and long-term (4-year scale) whilst Anderson et al. work at the aggregate 1-month scale². This hints at another factor which may have a drastic effect upon the extent to which research may or may not support the “filter bubble” hypothesis: the temporal or user aggregation scale applied. Existing research typically groups effects for entire platforms^{10,11,14}, user groups^{13,22} or whole user consumption histories^{9,15} and thus, works at a broad aggregate scale. Whilst some studies distinguish short- vs. longer-term effects^{17,18}, little further refinement is commonly performed to segment consumption logs into temporally consistent units. In particular, despite the increasing interest in so-called ‘session-based recommender systems’ in the literature dedicated to the conception of algorithms²³, empirical works leveraging specifically the level of sessions congruent with the bursty behaviour of users remain quite rare.

The contribution of this work seeks to empirically appraise the extent to which diversity methods built upon discrete or dense information representations capture heterogeneous perspectives of users' consumption diversity across *organic* and lesser *organic* affordances and furthermore, explore how this varies with aggregation scale. To this avail, we perform an empirical multi-affordance analysis of the listening histories of approximately 50K users of the French based music streaming platform, Deezer. Leveraging short term *sessions* of continuous platform engagement we study diversity at 3 successive scales of increasing aggregation: (1) intra-sessions, (2) inter-sessions (the closest to existing aggregate analysis^{2,8,9,16}) and (3) inter-affordances. Consequently, our work shows how confinement conclusions can drastically differ dependent on the scale and information representation chosen. However, when parallel perspectives are considered en masse they can in many cases coalesce to form a more nuanced narrative. This allows us to provide a substantial update of the classical filter bubble narrative which we hope will contribute towards a more enriched understanding of the intricate dynamics surrounding on-platform confinement: algorithmic curation devices on music streaming platforms may introduce more novelty than what users achieve organically, yet concurrently, this novelty is more spatially confined.

The deezer platform

Our study falls into the strand of literature which seeks to compare users' consumption diversity between various on-platform affordances with different levels of algorithmic / organic assistance. To this end, we focus on the French-based music streaming platform Deezer which offers a fruitful environment due to the richness of on-platform affordances offered.

Dataset scope and affordance classification

Prior to performing any concentration computations, our initial pre-filtered dataset consists of 1 month (March 2023) of listening histories from a random sample of 48, 243 anonymised Deezer users who streamed at least once in that month and pay for a subscription to the platform. Although not analysed directly, this user set is used to compute our user- and affordance-centric datasets which form the basis of our analysis. This is further clarified in the *Affordances and User Centric Datasets* section below. We pre-process our dataset by filtering listening events shorter than 30 seconds as these are deemed to be so-called ‘skips’ rather than streams². After applying this data processing pipeline we work with 23, 944, 195 listening events covering 168, 685 unique artists. We then label each stream in accordance with the affordance adopted to instantiate it. More precisely, the Deezer platform keeps highly granular logs regarding where on platform a song was streamed, thus allowing us

to classify affordances in a more fine-grained manner than simply *organic* / “*programmed*” as outlined in prior works². In particular, we divide organic affordances into two finer grained classes which are semantically and, as we shall shortly demonstrate, behaviourally distinct: playlists (*P*) vs. query-based (*Q*). Indeed, *P* affordances refer to playlists which have either been *curated* or *saved* organically by the user whilst *Q* affordances refer to all other modes of organic access which most commonly result from a user’s direct queries to a platform’s database (e.g. search result pages^{24,25}) or browsing a platform’s informational landscape with minimal algorithmic assistance (e.g. on an artist’s profile page). Similarly, we are able to break down the “programmed” affordance further into algorithmic (*A*) and editorial (*E*) affordances based on the extent to which they rely on algorithmic or, rather, organic human curations from editors. *A* refers to the platform’s plethora of Recommender Systems (e.g., the popular never-ending mix called ‘Flow’ on Deezer) whilst *E* affordances correspond to playlists curated by Deezer Editorial staff including playlists variously called “10s electronic”, “Rock & Chill”, etc. All things considered this leaves us with 4 affordances which can be used to classify the origins of users’ streams, each of which offers a unique perspective regarding organic-algorithmic assistance: *organic playlists* (*P*), *organic query* (*Q*), *editorial curation* (*E*) and *algorithmic curation* (*A*).

The session scale

Deezer, like many other platforms, offers much flexibility regarding how users’ streams can be aggregated. Due to the bursty nature of users’ platform access streams typically cluster into so-called *sessions* of continuous platform engagement thus, providing a temporally cohesive scale to work at. In our work we define a new session for a user when the inter-listening event time between any two consecutive streams exceeds 20 minutes – this captures 91% of inter-stream times and results in 2, 209, 591 sessions. To understand how users’ affordance adoption preferences correspond to the session scale, we compute for each user the proportion of their sessions which are monads, dyads, triads or tetrads i.e., drawing on 1, 2, 3, or 4 distinct affordances (in any order). Surprisingly, the majority of users’ sessions only make use of one affordance. On average 73.9% of a user’s sessions are monadic and furthermore, more monadic users are more active: computing the percent of monadic sessions at the platform level we even observe an increased prevalence of this session type (83.3%). This simplifies our analysis, allowing us to focus exclusively on monadic sessions which, although not all, still make up the majority of users’ platform interactions whilst preserving users’ relative affordance preferences (see [Supplementary Information](#)).

Focusing from hereon out on analysing users’ diversity using solely monadic sessions, we shall compute and compare diversity between affordances at 3 scales of increasing aggregation. Firstly, at the session scale by conducting an intra-session diversity analysis. Secondly at the inter-session scale (the closest to existing aggregate analysis^{2,8,9,16}) treating sessions as cohesive units of measurement in their own right and then thirdly, we move to the affordance scale to consider inter-affordance similarity. These three stages of comparison form the groundwork of our analysis and provide various perspectives to study *diversity at scale*.

Affordance and user centric datasets

We subset our monadic dataset by applying two filtering strategies. The first constructs, for each affordance, the subset of users and their corresponding sessions who show marked levels of engagement (≥ 3 monadic sessions) with the *P*, *Q*, *A* or *E* affordance. The second filtering strategy constructs a subset of users and their associated sessions who are multi-affordance that is, they display marked engagement (≥ 3 monadic sessions) with *each* of the 4 platform affordances. Ultimately, the former proves useful for an *affordance-centric* analysis whilst the latter facilitates a *user-centric* analysis which can be used to sense-check the conclusions derived at the affordance scale, thus reducing the risk of ecological fallacy. For further details we refer the reader to the “[Methods](#)” section.

Although not detailed here due to space constraints we remark that we were also able to reproduce each of the experiments on two novel affordance-centric and user-centric datasets derived from a random 50K user dataset for the month of March 2022. Affordance differences detailed are largely consistent across each of the datasets tested as described in the [Supplementary Information](#) section.

Artist representation and diversity measurement approach

To explore the tensions between users’ consumption diversity dynamics both in terms of *novelty* and *spatial disparity*¹⁷ we construct both discrete and dense item representations upon which respectively novelty and spatial disparity-based diversity measurements can be performed. The former relies on set-theoretic approaches whilst the latter leverages a high-fidelity item embedding space. In both cases we work with artists rather than songs as items and thus, both spatial and novelty based diversity measures are computed either on artists representations in the former case or in the latter, directly on artists themselves.

Spatial disparity measures

To capture users’ consumption diversity we construct an artist-based embedding space by applying SVD to a popularity normalised artist-artist within-session co-occurrence matrix *M*. We choose to apply SVD over other more “state-of-the-art” embedding models since it is both more interpretable and reproducible²⁶. Furthermore, it is worth highlighting that SVD has also been shown to be closely connected to the Skip-Gram Negative Sampling (SGNS) variant of the word2vec model²⁷. As such, applying this method further connects our work to existing literature applying SGNS models to conduct consumption diversity studies on multi-affordance platforms^{2,28}. We provide in Fig. 1 a visual two-dimensional projection of our SVD embedding space to give a sense of the artist space we are working with and its likely spikiness.

We compute the disparity a given session by computing an activity-weighted “Generalist-Specialist” (GS) score^{2,28}. At a high level, this measure captures the expected cosine similarity between a session’s centroid in vector space and the vector representation of a randomly drawn stream from within that session. Higher GS score indicate that a session is more concentrated in vector space and therefore, more *specialist* whereas lower

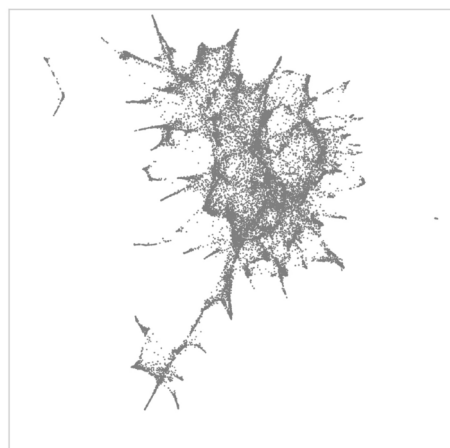


Fig. 1. PaCMAP²⁹ projection of the artist embedding space.

affordance F	n_F	$\langle GS(S) \rangle$	$\langle \gamma \rangle$	$\langle GS(S_F) \rangle$
A	12,411	0.829 (± 0.073)	*0.098 (± 0.159)	0.870 (± 0.078)
E	10,923	0.874 (± 0.081)	*0.130 (± 0.222)	0.852 (± 0.105)
P	28,896	*0.809 (± 0.083)	0.244 (± 0.255)	0.860 (± 0.079)
Q	36,004	0.919 (± 0.065)	0.134 (± 0.210)	*0.793 (± 0.106)

Table 1. Intra-session ($GS(S)$) and inter-session (γ and $GS(S_F)$) concentrations for affordance-centric sub-datasets. Standard deviations are mentioned inside parentheses and averages are reported over all users who displayed marked engagement with the affordance. Bold values mark minimums (i.e. highest diversity) per affordance and * denotes that the diversity of this affordance is significantly greater than all other affordance distributions across several pairwise one-tailed Kolmogorov-Smirnov tests ($\alpha = 0.001$ and Bonferroni correction applied). n_F denotes the number of users within each affordance group.

GS score indicate that a session is more spread in vector space and thus, more *generalist* and diverse. Since artists in our space tend to be clustered by genres, higher GS score associated with a session can likewise be interpreted as the sessions being more genre homogeneous and thus, semantically consistent whilst lower GS scores indicate that the session is generally, more genre diverse. For a more formal analysis of this relationship we refer readers to the [Supplementary Information](#) section. Similarly, it holds that a more spread, spatially diverse session with a lower GS score could be interpreted as also being more semantically diverse. This is due to the fact that so-called distributional semantic models such as SVD are known to capture structural paradigmatic semantics³⁰ within their geometry. As such, a session with a low GS score is likely to consist of artists which on average tend to have weak paradigmatic connections between them.

Novelty measures

The relation between spatial disparity and novelty is non-trivial: typically a session can be spatially disparate but made of recurrent elements only and, conversely, a spatially compact session could comprise exclusively novel elements. Put simply, spatial disparity is highly dependent on how exactly users navigate a vector space but does not indicate necessarily whether they go to novel or already-visited places and thus, whether they repeat consumptions or not. To address this issue, we configure a novelty-based appraisal of diversity in set theoretic terms. More precisely, we capture users' novelty (1) through the repetition of items within a session (when studying diversity intra-sessions) and (2) through the overlap between consumed item sets (when studying diversity inter-sessions and inter-affordances). The former relies on a repetition-based formulation and is captured with a simple length-adjusted Redundancy measure, whilst the latter is more set-theoretic and is measured with the overlap measure (γ) which makes use of the Bipartite Configuration Model (BiCM)³¹ that we detail further below and in the “[Methods](#)” section.

Intra-session diversity Spatial disparity analysis

We first centre our attention on analysing the diversity of non-fandom sessions (i.e. sessions containing more than > 1 artist) in SVD space using both our affordance-centric and user-centric datasets. Starting with the former and analysing $\langle GS(S) \rangle$ scores per affordance we observe marked variations in consumption diversity dependent on the affordance adopted. We present these results in [Table 1](#) whereby an affordance having a significantly greater diversity than all other affordances is appraised by applying several pairwise one-tailed Kolmogorov-

	Spatial disparity	Novelty
Intra-session	$Q \text{ — } E \text{ — } A - P$	$Q \text{ — } P - E \text{ — } A$
Inter-session	$A - P - E \text{ — } Q$	$P \text{ — } Q \text{ — } A$

Table 2. Summary of the findings for affordance hierarchy from the least to most diverse, from left to right (whereby line lengths are simply proportional to unidimensional distances between reported diversities in each case) for affordance-centric results, depending on intra- vs. inter-session scale, and on spatial disparity vs. novelty.

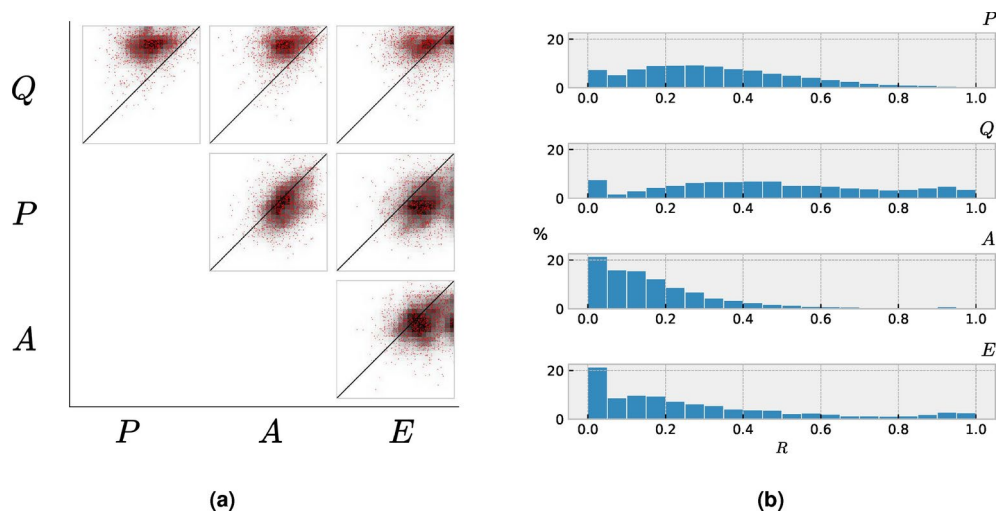


Fig. 2. Intra-session Diversity Analysis. Left (a) details distribution of users of the user-centric multi-affordance filtered dataset with respect to their GS scores for non-identical affordance tuples. Densities above (respectively, below) the line of equality indicate greater artist concentration i.e., lower diversity within sessions which utilise the affordance indicated in the grid rows (respectively, columns). Right (b) shows length adjusted intra-session Redundancy distributions computed for each affordance-centric user set corresponding to each of the 4 platform affordances.

Smirnov tests ($\alpha = 0.001$). To account for these multiple comparisons we correct α via Bonferroni correction. We chose to apply the Kolmogorov-Smirnov test since its non-parametric nature enables its application to non-normal distributions, which are particularly prevalent in our novelty-based inter-session analysis. As displayed in Table 1 and visually in Table 2, quite strikingly, there is a clear difference between the diversity of organic P and organic Q sessions, despite them previously being grouped under the more general *organic* umbrella term². In fact, they even display polar opposite diversity dynamics at this scale. More precisely, organic Q and, then, E sessions appear to be the least diverse whilst algorithmic A sessions and even more so, organic P sessions tend to be the most diverse, according to their GS scores. Put simply, at this level algorithmic sessions do not seem to be more bubbly than organic affordances. We observe comparable results with the user-centric perspective: for each multi-affordance user we compute $\langle \text{GS}(S) \rangle$ scores per each of the 4 affordances they engage with and then for each pairwise affordance combination we project users as points in GS space as depicted in Fig. 2a. Upon inspecting this figure it becomes clear that once again, users' Q sessions are consistently their least diverse whilst users' organic P sessions are, on the contrary, consistently their most diverse. In short, the diversity hierarchy observed during our previous affordance-centric analysis is found once again when considering differences between all affordances (see [Supplementary Information](#)).

Novelty analysis

To measure intra-session diversity using discrete item representations, therefore capturing novelty, we first compute the proportion of previously removed random sessions which reside within each session type thus, allowing us to appraise the extent to which different affordances tend to foster at least *some* artist novelty. Without loss of generality let S denote a user's session which is, more formally, a list of streamed artist ids. Similarly, let S_F denote the collection of monadic sessions which the user instantiated with affordance $F \in \{A, E, P, Q\}$. Thus, the proportion of random sessions within a given affordance is captured as:

$$\phi_F = \frac{|\{S \in S_F : |\text{supp } S| = 1\}|}{|S_F|} \quad (1)$$

where $\text{supp } S$ denotes the support of the session set S (i.e., the set of unique artist IDs reached in S). Focusing first, on our affordance-centric subsets we observe that, just as organic Q sessions were found to be the least diverse from a dense, spatial perspective, likewise are they the least diverse from a discrete, novelty-based perspective: they contain by far the most fandom sessions, with an average of 46.9% over users. To put things in perspective, on average 26.13%, 18.40% and 13.54% of a user's E , P and A sessions are fandom respectively (over the sets of users who engage with each affordance). There is little variance ($\pm 5\%$) in these results when using the user-centric dataset. In short, organic Q sessions are, in the most literal sense, the least novel whereas A sessions are, again in this literal sense, the most novel.

To explore the extent of this artist novelty we next apply a simple measure of redundancy, or repetition which captures a user's tendency to explore vs exploit artists within a session^{9,32}. Restricting our analysis again to only consider non-fandom sessions, we compute a session's redundancy as:

$$R(S) = 1 - \frac{|\text{supp } S|}{|S|} \quad (2)$$

When a user decides to exploit previously-consumed artists rather than explore new ones within a session their R score will be higher. To account for the fact that the upper bound of R is constrained by the length of a session (i.e. $|S|$) we adjust R to be relative to other sessions with comparable lengths. This is simply performed by computing the percentile rank of a session's R score relative to all other sessions in the same log length bin. After applying this adjustment we then compute per affordance average redundancy scores (R_F).

Considering first the affordance-centric perspective, as depicted in Fig. 2b, just as we found previously that both in terms of spatial disparity and fandom based novelty A sessions tend to be one of the most diverse, likewise do they tend to foster the highest artist exploration dynamics ($\langle R_A \rangle = 0.189$). At the intra-session scale, this provides strong evidence on all grounds against the claim that algorithms foster more *bubbly* artist dynamics than their organic counterparts. Similarly, even after removing sessions of the fandom kind, Q sessions, previously found to be the least diverse in terms of spatial disparity and fandom based novelty, likewise, exhibit the strongest exploitation dynamics ($\langle R_Q \rangle = 0.475$). If any affordance were to be deemed bubbly at the intra-session scale it would be the organic Q affordance.

Regarding other affordances we find that users' personal playlist (P) sessions are surprisingly redundant ($\langle R_P \rangle = 0.335$) despite having relatively low fandom rates and, moreover, being the most spatially diverse – although these session types typically foster at least some level of artist novelty they also encourage strong artist exploitation dynamics. We find that the inverse dynamic holds for E sessions, as users tend to use them in a more exploitative manner ($\langle R_E \rangle = 0.286$). They can nevertheless, on occasion, be highly artist-focused, reflected in their high fandom levels: we may simply hypothesise that this may be due to editorial playlists dedicated to a specific artist. Similarly, from the multi-affordance user-centric perspective we observe comparable differences between affordances: $\langle R_Q \rangle = 0.439$, $\langle R_P \rangle = 0.292$, $\langle R_A \rangle = 0.202$ and $\langle R_E \rangle = 0.288$.

In summary these findings suggest that, at the intra-session scale, there is some discrepancy between diversity measures which rely on discrete vs. dense item representations, particularly for the P affordance. Nevertheless, when considered in parallel, these perspectives offer a nuanced means of characterising and differentiating user behaviour under various levels of algorithmic, human and self-driven organic assistance. We furthermore observe a weak linear correlation between users' spatial disparity and novelty ($\langle r \rangle = 0.261$ over all affordances), confirming that the relationship users have with diversity and exploration is highly varied¹⁷ even when users' consumptions are disaggregated by affordances.

Inter-session diversity

Having shown that algorithmic devices may foster artist diversity rather than concentration at the intra-session scale, we now move on to explore how users' achieve inter-session diversity that is, how their sessions vary from one session to the next and how this differs under various levels of algorithmic / organic assistance. To this end, we firstly measure diversity from a spatial perspective, applying once again the GS score and then subsequently, from a discrete set-theoretic perspective by constructing an entropy-based monopartite projection of users' session-artist bipartite networks. At this stage of our analysis, fandom sessions are now included.

Spatial disparity analysis

We first consider affordance-centric subsets and compute, for each affordance F , weighted session centroids μ_S for all $S \in S_F$ over our 1-month observation window. We subsequently compute GS scores over all these centroids that we denote as $GS(S_F)$ since it is computed over the collection of sessions, S_F . We finally consider averages over users of each affordance-centric dataset. As depicted in Table 1, this scale adds further nuance to the previous intra-session analysis. Despite A sessions previously being found to be the most diverse intra-session, it appears that over time, these sessions tend to become concentrated, just like the P and E kind but to a lesser degree. On the other hand, at this scale organic Q sessions are clear outliers with respect to how users spread their attention around different regions of a platform's informational landscape: these sessions are the most spatially diverse over time despite the fact that they lack diversity at the intra-session scale, as we previously demonstrated. Likewise we observe remarkably similar dynamics when conducting a user-centric analysis (see [Supplementary Information](#)).

All in all these findings underline the importance of aggregation scale when conducting algorithmic diversity appraisals. As we have seen for the *A* affordance, this can have a drastic effect upon the extent to which confinement dynamics are found.

Novelty analysis

We next focus on capturing the overlap of users’ sessions within each platform affordance. To this end we start by modelling a user’s session-artist interactions as an undirected bipartite network whose edges reflect featurings of artists within sessions. These networks enclose co-listening patterns exhibited across sessions and are constructed per affordance: for each user in each of the 4 affordance-centric subsets, one bipartite network will be constructed for each user. In the user-centric phase of our analysis, 4 bipartite networks will be constructed per user since they engage with all 4 platform affordances.

To capture item concentration we generate monopartite projections, on the session side, of these bipartite networks, whereby two sessions are connected when they are deemed to be significantly similar in terms of the artists they share. To differentiate between session similarities which arise solely from users’ artist favoritism and session lengths even in the absence of any inter-dependence between sessions we apply the Bipartite Configuration Model (BiCM)³¹ as a null model. This model randomly re-configures edges between artists and sessions whilst at the same time preserving degree distributions on both the session and artist side, thus preserving a user’s relative artist preference and session length. This eventually allows us to connect sessions which are *significantly* similar and compute an index γ , a set-theoretic notion of inter-session similarity which trivially captures the proportion of sessions that are connected. Further technicalities are clarified in the “Methods” section.

As presented in Table 1, when conducting an affordance-centric analysis we observe that organic *P* sessions exhibit the greatest artist overlap and thus, greater tendency to exploit the same artist sets repeatedly over time. In contrast, *Q* and *E* sessions exhibit similarly moderate overlap whilst for the *A* session this effect is even more exaggerated. Likewise this effect was also observed when conducting a user-centric analysis. We hypothesise this difference may be due to the increased stability of a user’s personal playlists in comparison to algorithmic devices which, by contrast, might churn out novel item lists more frequently, rather than simply appending to existing curated artist sets (e.g., adding a song to one’s own playlist).

Combined with our prior dense, spatial disparity analysis this leads us to an updated filter-bubble narrative which once again, underscores the tension between diversity measures built upon discrete and dense item representations¹⁷ – over time music streaming platforms’ algorithmic devices may promote novelty even more so than what users attain organically but nonetheless, this novelty is more semantically confined.

Affordance similarity

Having understood that conclusions surrounding bubbly confinement are highly dependent on the scale at which diversity measurements are applied and furthermore, how diversity is framed we now move on to study inter-affordance similarity. In order to draw relative comparisons between all affordance combinations we concentrate solely on conducting a user-centric analysis (i.e. users who engage with all 4 affordances) when studying diversity in terms of novelty and spatial disparity.

Spatial disparity analysis

For each user, we compute an overarching per-affordance centroid from each of their session centroids (μ_S) constructed during our inter-session analysis. At a high level this captures a vector representation of a user’s average session within each of the 4 platform affordances. To capture similarity between affordances, we calculate the cosine similarity between each centroids which we then z-normalise per user to capture relative rather than absolute inter-affordance similarities.

As shown in Table 3 which reports average normed inter-affordance similarities we find that the artist regions users exploit with the *A*, *P* and *Q* affordances are on-average the closest in vector space whilst, quite strikingly, those exploited with the *E* affordance appear to consistently reach more distant regions of vector space which most likely reflect semantically distinct artists.

Measure type	Spatial Disparity			Novelty		
	<i>P</i>	<i>A</i>	<i>E</i>	<i>P</i>	<i>A</i>	<i>E</i>
<i>Q</i>	0.357	0.389	-0.434	0.421	0.243	-0.012
<i>P</i>		0.499	-0.490		0.011	-0.280
<i>A</i>			-0.320			-0.233

Table 3. Average z-normed inter-affordance similarities for each type of measures: spatial disparity (cosine similarity) vs. novelty (set-theoretic $\langle \gamma_{..} \rangle$). The subgraphs below summarize the findings by connecting each affordance to the one it is most similar to, with an arc of comparable length.

Succinctly, editorial human curated affordances appears to play a valuable role in helping users consume disparate artists styles more so than for affordances whose curations are either implicitly algorithmically personalised (A) or explicitly organically personalised by the user themselves (P , Q).

Novelty analysis

Once again, we utilise the entropy-based BiCM model to construct a session network upon which we can apply the γ measure to capture item similarity, yet not just within but now between affordances. To this end, we compute a monopartite session projection on the bipartite network that reflects all of a given user's sessions, irrespective of affordance. For each pair of affordances, we then filter to only consider nodes and edges which can be associated with the two given affordances. This yields a further session-session bipartite network whereby sessions on each side of the bipartite network all emerge from the same affordance. From this network we then compute γ measures on each side which, in turn, reflects how item sets are mirrored between affordances. We then compute average values over both γ computations. We eventually z-normalise all γ scores to consider relative differences between affordances and present average z-norms per affordance pair in Table 3, where we also provide closest neighbour subgraphs which summarise affordance relative proximities.

From this perspective we now observe that there tends to be the biggest artist overlap between users' engagements within organic affordances (P , Q). Combined with the fact that the artists users consume with organic affordances tend may be relatively dissimilar from a spatial perspective –more so than say, (P , A) or (A , Q)– this suggests that these affordances may play a dual role: both reinforcing user tastes through repetition, as well as bursting users out of their own semantic bubble, presumably through organic queries (Q) which tend to be the most diverse at the inter-session scale. For users' E sessions we observe a more consistent narrative when applying both spatial and novelty-based diversity appraisals — just as before they are consistently the most novel in comparison to other affordances. In essence, we find that at the inter-affordance scale, diversity measures built upon dense and discrete item representations, which capture spatial disparity and novelty respectively, provide a more homogeneous description of affordances' relative diversity differences, in comparison to what we observed at the previous lower aggregation scales.

Discussion

Studying music streaming users' on-platform consumption diversities across several unique affordances which each offer variable levels of algorithmic and human assistance, the contribution of this work provides strong empirical evidence that both scale and information representations may have a marked impact upon the conclusions reached when conducting organic appraisals of algorithmic confinement. Progressing from the short-term intra-session to aggregate inter-session scale our work demonstrates that diversity differences between affordances can quickly change particularly, in the case of algorithmic affordances. Whilst music streaming platforms' algorithmic devices appear to foster diversity at the short-term, bursty session level these sessions nevertheless, over time fail to provide a substantial level of spatial disparity and thus, do not particularly appear to assist users in exploring novel regions, genres, or artist styles any more than what they attain organically.

Notwithstanding, considering diversity measures built upon discrete (capturing novelty) and dense (capturing spatial disparity) item representations in parallel allowed us to add further nuance to this conclusion: both at the short term session scale and over time, algorithmic curation devices may promote more novelty than what users reach organically but nevertheless, this novelty is more semantically confined. Thus, we theorise that users' adoption of algorithmic devices can in some sense, break them out of their bubbles by exposing them to novelty whilst at the same time, reinforcing the boundaries of this bubble by ensuring that content is both spatially homogeneous from one session to the next and a close mirror of what they consume organically. Based on these findings we suggest that in order to reach more nuanced conclusions regarding the role of algorithmic devices in fostering bubbly confinement dynamics it may well moving forward, be fundamental to exploit several methods which leverage different aggregation scales and information representations.

Whilst we contend that this message is valuable for a broad variety of researchers studying the role of algorithmic devices on-platform it is also important to acknowledge that there are several limitations of this work. For one, we focus exclusively on the case of the Deezer music streaming platform. Whilst indeed, many of the affordances be that organic, algorithmic or editorial may be generalisable to other platform types, the inner workings of such devices and, furthermore, the interfaces they offer, may of course vary from platform to platform. Whilst so far most research performing multi-affordance organic appraisals of algorithmic confinement has tended to focus on this platform type we nevertheless suggest that a prominent direction for future work may also be found in testing the relation between different diversity measures (dense or spatial) on other platform types (e.g. social media platforms). In addition, we suggest that although outside the scope of this work future research should also focus on bridging the gap between A exposure and multi-affordance literature thereby acknowledging that an algorithmic consumption is at least in part also mediated by an organic human selection process.

Methods

SVD based feature space

We construct a co-occurrence matrix M based on intra-monadic session co-occurrences whereby rows define the artist structure to be embedded. We perform an over-sampling of affordances which are in minority classes, thus balancing the number of session per affordance. We proceed to populate M with within-session co-occurrence counts. We then cast M to a positive point wise mutual information matrix to account for co-consumption dynamics which may emerge from popularity effects alone. Finally, we apply Truncated Singular Value Decomposition (SVD) to the matrix to generate a low-dimensional ($k = 128$) embedding space. More

formally, SVD is a factorisation technique which seeks to approximate a matrix $M \in \mathbb{R}^{n \times m}$ as the product of three matrices, $U \in \mathbb{R}^{n \times k}$, $\Sigma \in \text{diag}(\mathbb{R}_+^k)$ and $V \in \mathbb{R}^{m \times k}$. In our analysis we make use of the $U\Sigma$ component as our latent item embedding which from hereon out we shall denote as L , an $n \times k$ matrix whose rows (i.e. $l_1, l_2, \dots, l_n$) correspond to the latent representations of artists. For simplicity let artist ids exist in the range $[1 \dots n]$ and let an artist with id i have the latent vector representation l_i .

GS score

We estimate spatial disparity through an activity-weighted so-called “Generalist-Specialist”, or GS, score^{2,28}. Let S represent the list of artist ids associated with a given user’s session. Let us first define the weighted centroid of a session in vector space as:

$$\mu_S = \frac{\sum_{j \in S} w_j l_j}{|S|} \quad (3)$$

where l_j denotes the latent representation of artist j and w_j denotes the number of plays associated with them within S (note that $\sum_j w_j = |S|$). To capture intra-session spatial disparity we compute the GS score of S as:

$$GS(S) = \frac{1}{|S|} \sum_{j \in S} w_j \frac{l_j \cdot \mu_S}{\|l_j\| \cdot \|\mu_S\|} \quad (4)$$

Similarly, we compute spatial disparity at the inter-session scale as:

$$GS(S_F) = \frac{1}{|S_F|} \sum_{S \in S_F} \frac{\mu_S \cdot \mu_F}{\|\mu_S\| \cdot \|\mu_F\|} \quad (5)$$

whereby for a given user and their monadic consumptions, S_F reflects the family of sessions associated with affordance $F \in \{A, E, P, Q\}$ and μ_F denotes the average location of F -session centroids in vector space.

For each subset (affordance-centric or user-centric) we only report GS scores for sessions that contain $< 20\%$ niche artists whereby an artist is defined as niche if they have been listened to by ≤ 10 users. In the case where a session contains more than one niche artist but still the session is composed of $< 20\%$ niche artists we simply drop the artists classified as niche from the set. This has a minimal effect on user sessions due to the long-tailed nature of popularity on-platform: on average across each affordance-centric user set 5.76% of a user’s average session is composed of “niche” artists. This ensures that we filter artists whose structure in M , and thus also in vector space, is poorly represented, thereby avoiding deriving spatial concentration conclusions which may have more to do with popularity than underlying semantics.

As detailed in the [Supplementary Information](#), we attain equivalent relative disparity differences between affordances when measuring within-session disparity with both normalised genre entropy and the GS score therefore, suggesting that our results would likely be the same if a genre-based item representation was applied throughout.

Bipartite configuration model (BiCM)

We use BiCM³¹ as a baseline to compute item diversity in the inter-session and inter-affordance cases i.e., where the significance of overlaps between consumption sets needs to be computed. BiCM interprets links as independent random variables and, as such, defines a grandcanonical probability measure (i.e. over all possible network configurations which preserve node degrees on both sides). To illustrate on inter-session diversity, and following the methodology defined in³³, we conceptualise session similarity using so-called V -motifs which reflect the number of shared artists between any two sessions and make use of the BiCM model to define significance of V -motifs w.r.t. to the expectation under the BiCM null model ($\alpha = 0.05$). (Solver convergence rates are very high – for each of the affordance-centric sets we observe that the solvers tested were on average able to converge on 94.46% of users tested. In cases of repeated non-convergence these users are dropped from the analysis set). For further details of methodological technicalities we refer readers to the work of^{31,33}.

We may now measure session homogeneity, in item terms, using each of the n projected session networks $\mathcal{S}_u$ of a user u engaging in n affordances. We construct an adjusted overlap measure γ which trivially captures the proportion of sessions $S_i \in \mathcal{S}_u$ that are connected, formally defined as:

$$\gamma = |\{S_i \in \mathcal{S}_u \text{ s.t. } \deg(S_i) > 0\}| / |\mathcal{S}_u| \quad (6)$$

An illustration of this pipeline is shown in Fig. 3. We proceed similarly to compute inter-affordance similarity, except that we configure affordance-based pseudo-sessions gathering all listening events associated with a given affordance. We then compute inter-pseudo-session similarity in the same way as before on pairs of affordance-based pseudo-sessions, which actually reflects a measure on the similarity between each pair of considered affordances.

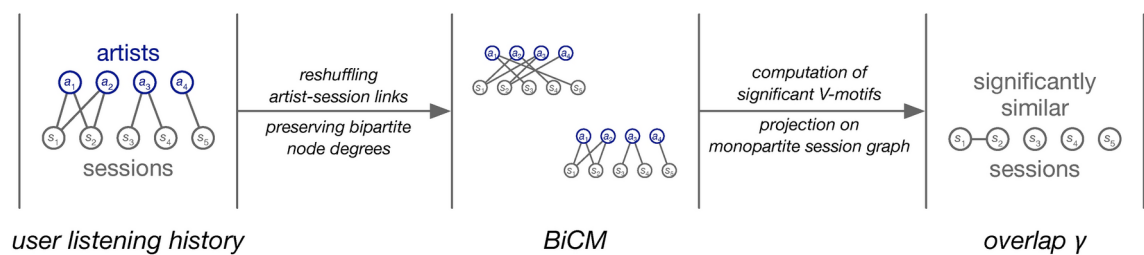


Fig. 3. Schematic illustration of the BiCM session overlap computation pipeline, from an empirical bipartite session-artist network to the computation of significant V-motifs or monopartite projection links i.e., significant connections between sessions in terms of their artist overlap.

Affordance	Sub-affordance
P	playlist_page
Q	album_page, playlist_radio, shuffled_collection, artist_top, profile_top, suggest_track, search_page_track, recently_played_playlist, profile_playlists, suggest_playlist, dynamic_page_album, dynamic_page_playlist, search_page, history_page, profile_artists, shuffled_offline, profile_albums, recently_played_user_radio, recently_played_album, cached_content, suggest_album, siri_album, artist_randomdiscography, siri_track, search_playlist, siri_artist_top, widget_album, artist_discography_subpage, siri_playlist, recently_played_user, recently_played_radio, widget_artist, suggest_radio, artist_discography,siri_radio, siri_artist_radio, player_default_playlist, search_album, artist_highlight_album, siri_shortcuts_shuffled_collection, album_discography, dynamic_page_track, search_show_episode, artist_similar_playlist, suggest_artist, menu_panel_notification_album, personalsong_page, search_artist, profile_radios, suggest_livestream, suggest_show_episode, artist_highlight_playlist, artist_related_playlists_subpage, recently_played_artist
	feed_user_radio, dynamic_page_user_radio, profile_user_radio, widget_flow, dynamic_page_smartracklist, dynamic_page_artist_radio, flow_tab_smartracklist, track_mix_contextual, siri_flow, dynamic_page_radio, artist_smartradio, widget_smartracklist, smartracklist_page, siri_shortcuts_user_radio, flow_tab_flow, radio_page, feed_playlist, dynamic_page_livestream, feed_track, track_mix_30s, smartradio_page
	dynamic_page_playlist, playlist_page, playlist_radio
A	feed_user_radio, dynamic_page_user_radio, profile_user_radio, widget_flow, dynamic_page_smartracklist, dynamic_page_artist_radio, flow_tab_smartracklist, track_mix_contextual, siri_flow, dynamic_page_radio, artist_smartradio, widget_smartracklist, smartracklist_page, siri_shortcuts_user_radio, flow_tab_flow, radio_page, feed_playlist, dynamic_page_livestream, feed_track, track_mix_30s, smartradio_page
E	dynamic_page_playlist, playlist_page, playlist_radio

Table 4. Classification scheme used to derive affordance categories on Deezer.

Affordance nomenclature

For full transparency the classification scheme used to classify affordances is depicted in Table 4. Listening events whose affordance is tagged as “external” (i.e. access was achieved off-platform for instance, via an API) are filtered since we are interested in studying on-platform rather than off-platform confinement.

Filtering strategies

To account for the variation of monadic session types (in terms of affordances) across users we introduce two user filtering strategies filtering strategy which provide an affordance- and user-centric perspective respectively. The former constructs one set of users per affordance $F \in (A, E, P, Q)$, first filtering sessions which do not emerge from the affordance associated with the set and then next, filtering users that have < 3 sessions within this set. This results in 4 sets of users who actively engage with each of the affordances in question. The sets have the following respective user populations $n_A = 12,411$, $n_E = 10,923$, $n_P = 28,896$ and $n_Q = 36,004$. The latter filtering strategy which provides a user-centric perspective imposes the more strict constraint that users must have ≥ 3 session in each affordance and thus, constructs a set of so-called multi-affordance users. It is meant to construct a reference user set to sense-check the conclusions derived at the session level thus, reducing the risk of ecological fallacy. Applying this filtering yields a user set of size $n = 1672$.

Data availability

To elicit reproducibility we publicly release, alongside this work, the dataset used to derive all user-centric diversity conclusions. More precisely, this dataset is restricted firstly to streams which were enacted within monadic sessions (i.e. those that only use one affordance) and subsequently to users who engage with ≥ 3 sessions in each of the Deezer platform’s 4 affordances thus, focusing on “multi-affordance” users. We hope that the release of such a dataset will allow more researchers to perform organic appraisals of algorithmic confinement using multi-affordance usage logs which, to date, remain scarce due to these type of datasets generally being exclusively reserved for those working in industry partnerships. We also release the code needed to reproduce all experiments with the user-centric dataset. Access to the code and dataset can be found at: <https://github.com/deshiini/SciReports-FilterBubble/tree/main>.

Received: 29 April 2024; Accepted: 9 October 2024

Published online: 03 February 2025

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Acknowledgements

This paper has been partially realized in the framework of the “RECORDS” grant (ANR-2019-CE38-0013) funded by the ANR (French National Agency of Research)

Author contributions

D.S., V.C. and C.R. designed the study. D.S. performed the calculations. D.S. designed the computational framework and analysed the data. D.S. took the lead in writing the manuscript. D.S. and C.R. wrote the paper. C.R. supervised the project.

Declarations

Competing Interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-024-75967-0>.

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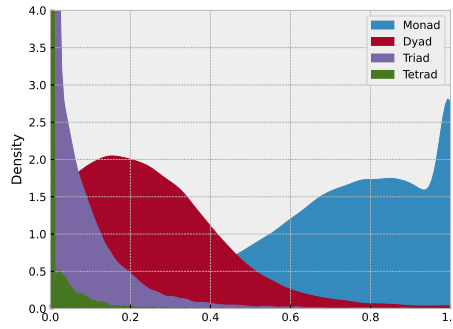
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A Session Scale Exploration

As a groundwork to leveraging the session scale to study diversity we characterise the structure of typical music streaming platform sessions which, to the best of our knowledge, remains surprisingly rarely explored in literature.

A.1 Affordance variety

We first explore the affordance variety of sessions (i.e. in absolute numeric terms how many unique affordances are there). Do users remain *monadically* confined within one affordance or do they rather switch between a plethora of affordances during a session? For each user u , we compute the proportion of their sessions which are monads, dyads, triads or tetrads i.e., drawing on 1, 2, 3, or 4 distinct affordances (in any order), defined as p_1, p_2, p_3, p_4 and shown in table 1. Figure 1 depicts the distribution of each p_n over all users. Quickly, the answer to our former question becomes clear: the vast majority of users rely mainly on monadic sessions, as $\langle p_1 \rangle = 73.9\%$, ($\sigma_{p_1} = 21.3\%$). By contrast, dyadic sessions only make up on average $\langle p_2 \rangle = 22.0\%$, ($\sigma_{p_2} = 18.4\%$) of sessions, and higher orders are even more strikingly rare: triadic sessions compose an average of $\langle p_3 \rangle = 3.77\%$, ($\sigma_{p_3} = 8.70\%$) and tetrad sessions $\langle p_4 \rangle = 0.31\%$, ($\sigma_{p_4} = 2.53\%$). When users do decide to draw upon multiple affordances in a platform engagement session this is typically of the dyadic nature. It is extremely uncommon for users to engage in triad and to an even lesser degree, tetrad sessions. The duration of the engagement exhibits likewise, a marked level of variability between these session types. Formally, let a session S be defined as a multiset of artist IDs which correspond to each stream within the session. Similarly, S_n denotes a multiset for an n -adic session. As shown in table 1, the average length of sessions $\langle |S_n| \rangle$ increases super-linearly with n , the number of affordances adopted, thus making monadic sessions markedly shorter and more bursty than other session types.



Supplementary Figure 1. Kernel density estimates of frequency distributions of monads, dyads, triads and tetrads among user sessions.

session type	$n = \#F$	$\langle p_n \rangle (\%)$	$\langle S_n \rangle$
monad	1	73.9	9.16
dyad	2	22.0	17.87
triad	3	3.77	29.07
tetrad	4	0.31	40.02

Supplementary Table 1. Attributes of each of the session types.

A.2 Affordance preference

We next study the degree to which users tend to show preference for one particular affordance over another at the session scale. In prior works⁷ Deezer users have been found to exhibit a general preference for streaming with the organic affordances (without differentiating P and Q). This is no different in the case of our dataset: for a given user we compute the proportion of streams p_F accessed through affordance $F \in (A, E, P, Q)$, the list of all four affordances. Over all users, we observe averages of $\langle p_P \rangle = 34.88\%$, $\langle p_Q \rangle = 40.39\%$, $\langle p_A \rangle = 13.20\%$ and $\langle p_E \rangle = 11.52\%$, whereby around three quarters of songs are on average accessed organically. Restricted to monadic sessions noted as $\dot{p}$, these proportions reflect a slight weakening in users' preference for recommendation affordances: $\langle \dot{p}_P \rangle = 36.27\%$, $\langle \dot{p}_Q \rangle = 40.83\%$, $\langle \dot{p}_A \rangle = 12.97\%$, and $\langle \dot{p}_E \rangle = 9.93\%$. In

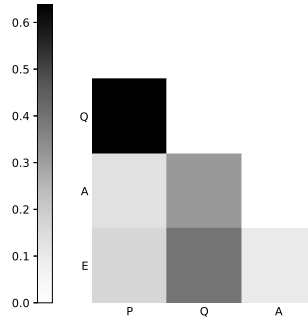
addition, recommendation affordances are associated with longer monadic session lengths: $\langle |S_A| \rangle = 10.62$, $\langle |S_A| \rangle = 12.69$ vs. $\langle |S_Q| \rangle = 8.72$, $\langle |S_P| \rangle = 8.65$. Simply put, at the session scale affordances which rely on external curations be that algorithmic (A) or the human kind (E) are adopted less often by users to reach content - even less so in isolation - yet at the same time, they tend to encourage more engagement than organic sessions. Indeed, it appears they play a specific role of assistance within users' bursty on-platform consumptions.

A.3 Affordance combinations

To explore differing attraction and repulsion dynamics between affordances, we now focus on dyadic sessions, the most popular mixed session type. When engaging in platform access are there certain affordance pairs which users will typically exploit to reach content? Attraction and repulsion refers to preferring one pair of affordances over another: a dyadic session relies by definition, on two affordances used in any order (e.g., 12 playlist streams followed by 3 algorithmic and ending with 5 playlist streams). For a given user, let D represent the symmetric affordance co-occurrence counts such that $D_{i,j}$ reflects the total number of dyadic sessions utilising both affordances F_i and F_j where $F \in (P, Q, A, E)$. Since users exhibit a skewed adoption of affordances without adequate normalisation we are likely to observe high co-occurrence counts for organic affordance pairs even absent of any attraction dynamics. As is common in computational linguistics², we control for this popularity effect by casting D to a Positive Point Wise Mutual Information (PPMI) transition matrix, T . More formally:

$$T_{i,j} = \max \left(\log_2 \frac{P_{i,j}}{P_i P_j}, 0 \right) \quad (1)$$

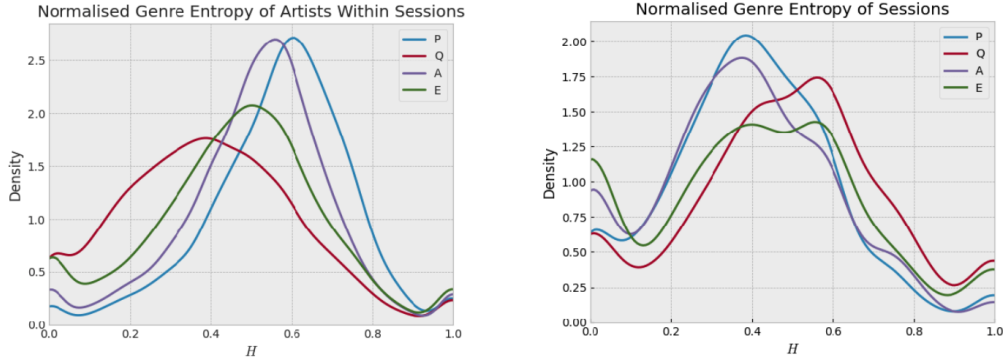
where $P_i = \sum_k D_{i,k} / \sum D$ and $P_{i,j} = D_{i,j} / \sum D$ and $\sum D$ reflects the element-wise summation of D . We then report $\langle T \rangle$ over all users in our dataset as depicted visually in figure 2. Even after accounting for users overarching preference for organic affordances we observe that the strongest attraction dynamics exists nevertheless between P and Q . By contrast, there exists very weak attraction between both the A and E affordances and furthermore, between $P \leftrightarrow A$ and $P \leftrightarrow E$. These findings suggest that Deezer users adopt organic affordances with an interesting duality. On the one hand, users appear to over-utilise organic affordances in isolation as reflected in both the over-representation of P and Q within monadic sessions and the strong $P \leftrightarrow Q$ attraction dynamics within dyadic sessions. On the other hand, users prefer to complement recommendation-based sessions with Q content over all other affordances.



Supplementary Figure 2. Upper triangle of $\langle T \rangle$, the average PPMI matrix for dyadic sessions.

B Diversity measurements

To check the replicability of the diversity differences we draw between affordances we repeat our experiment pipeline on an additional anonymised user dataset supplied from Deezer. This dataset contains the streams of approximately 50K randomly sampled *active* users, this time over the period of March 2022 (i.e. one year prior to our previous analysis). As previously described, we conduct both an affordance- and user-centric analysis and repeat each of our experiments, sequentially transitioning from a low scale intra-session diversity analysis to an inter-session analysis and then finally, to an inter-affordance similarity analysis. Within these experiments we find that the relative differences we observe between affordances are for the most part comparable to the results we attain in our main body of work. See for comparison the summary presented in Table 2.



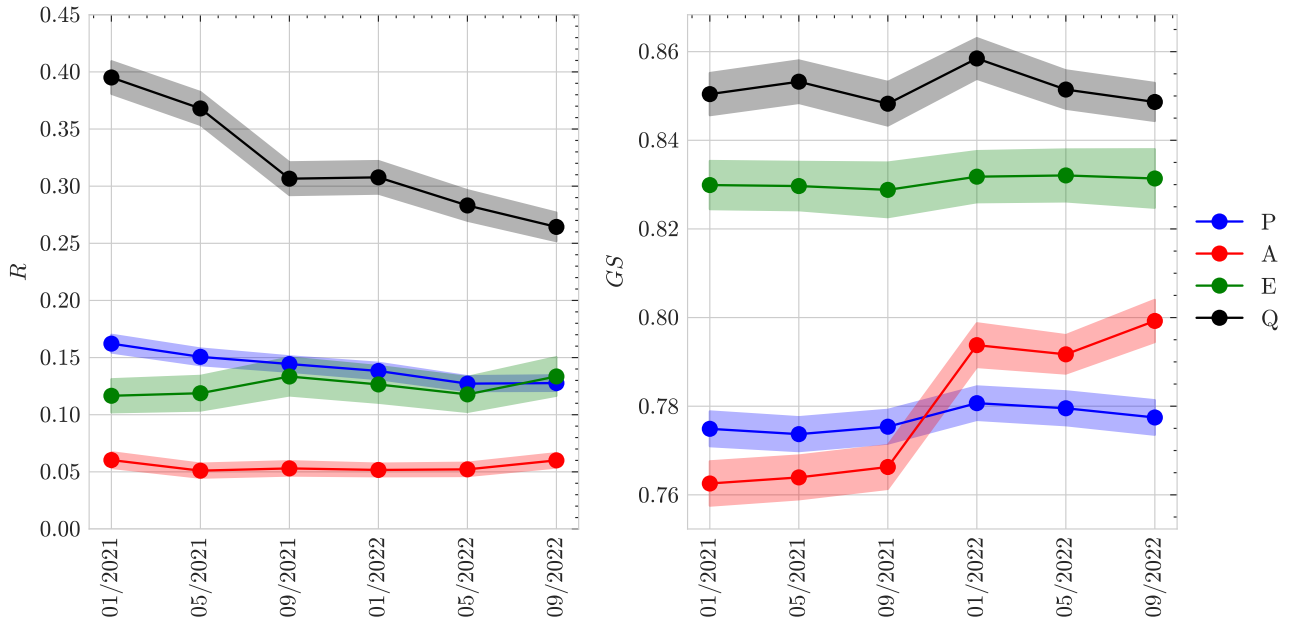
Supplementary Figure 3. Diversity comparison between the GS score and normalised genre entropy for the multi-affordance users in the 2022 50K user dataset. Left depicts intra-session diversity. Right depicts inter-session diversity whereby sessions are assigned to genres based on the most occurring within the session.

dataset			spatial disparity				novelty			
			A	E	P	Q	A	E	P	Q
2023	affordance-centric	Intra-session	0.829	0.874	0.809	0.919	0.189	0.286	0.335	0.475
		Inter-session	0.870	0.793	0.860	0.793	0.098	0.130	0.244	0.134
	user-centric (z-normed)	Intra-session	-0.337	0.285	-0.829	0.881	-0.542	-0.134	-0.031	0.707
		Inter-session	0.402	0.123	0.313	-0.839	-0.200	-0.208	0.394	0.284
2022	affordance-centric	Intra-session	0.807	0.860	0.798	0.923	0.116	0.251	0.336	0.543
		Inter-session	0.868	0.859	0.859	0.793	0.116	0.132	0.273	0.076
	user-centric (z-normed)	Intra-session	-0.461	0.237	-0.813	1.036	-0.618	-0.284	-0.019	0.920
		Inter-session	0.256	0.261	0.282	-0.799	-0.037	-0.160	0.527	-0.063

Supplementary Table 2. Summary of the spatial disparity and novelty findings for the 50K March 2023 dataset (top) and for the March 2022 dataset containing active users (bottom). Results for both sub-data set types (affordance- or user-centric) are detailed as well as for both intra- and inter-session perspectives and all affordances.

C GS score & Genre Coherence

We next analyse the extent to which the dense diversity differences we observed across affordances would be consistent if we had used a more coarse genre-based measure[?]. To this end we compute for our March 2022 50K active users dataset, normalised genre entropy scores at both the intra- and inter-session scale. Since genre information is only provided for the song level by Deezer we assign each artist’s genre to be the most occurring over all songs which they are accredited in our dataset. In the case of ties a random selection is made. We then compute intra-session disparity by computing the normalised genre entropy of artists featured within sessions (i.e. by normalising entropy respective to the length of the session i.e. $\log_2(|S|)$). At the inter-session scale we first assign a session’s genre to be the one which is the most frequent (ties are resolved with a random selection) and compute normalised genre entropy scores from the distribution of genres attained over all of their sessions. Figure 3 describes these normalised genre entropy distributions per. affordance (i.e from an affordance centric perspective). Analysing these distributions it appears that the relative diversity differences we observe between affordances would yield the same conclusions independent of if we applied genre entropy or the GS score.



Supplementary Figure 4. Repeated intra-session analysis using an affordance-centric filtering on a novel active user dataset. The x axis depicts monthly collections of monadic sessions for the months of January, May and September in 2021 and 2022. Solid lines depict mean concentration scores from both a discrete (left) and dense (right) perspective whilst the enclosing shaded regions depict 95% confidence intervals.

D Monthly Generalisability Experiments

In the main body of our work we solely analyse listening histories from the month of March. If the concentration differences we derive between platform affordances were subject to seasonal effects for instance, annually following a period of 12 months it may hold that the differences which we ultimately end up describing may not be generalisable to other months of the year. To analyse the extent to which this may or rather, may not be the case we repeat our intra-session concentration computations on a novel dataset supplied by Deezer spanning over 2 years (2021-2022) of listening histories by 50K randomly sampled users who were active and the beginning and start of each year. Due to computational constraints we work with a subset of this dataset by randomly sampling 20K users and retrieve solely their monadic sessions from 3 evenly spread months, January, May and September across the years of 2021 and 2022. For each unique month-year pair we randomly sample $n = 1500$ users who streamed in this period and once again, conduct an affordance-centric concentration analysis from both a dense and discrete perspective as previously detailed. As depicted in Figure 4 above we observe that the relative affordance differences are largely consistent across months particularly, when comparing between those in the same year. Furthermore, we observe that these differences are also largely consistent with those previously derived as shown in Table 2 in the main text. Nevertheless, we remark that there appears to be some variance regarding dense A concentration scores derived in the year of 2021 and 2022. We theorise this may stem from internal changes in functioning of Deezer’s algorithmic devices. In the year of 2022, we observe little empirical evidence that the month of March may be an outlier. The relative affordance differences look relatively stable within years suggesting that there are minimal seasonal effects which may create marked levels of variance from one month to the next.

Tracing Desired, Exposed and Adopted Novelty on Music Streaming Platforms

Foreword

We now turn to examining how Deezer users' selections of algorithmic recommendations influence their consumption diversity. While such dynamics have been more extensively examined in fields like political science (Bakshy et al., 2015; Robertson et al., 2023), this kind of research is far more scarce within literature focusing on capturing algorithmic confinement on music streaming platforms. This research is planned to be submitted to the World Wide Web conference (WWW26) and is therefore currently presented in a pre-print format.

Leveraging a rich combination of declared survey data, algorithmic exposure metrics, and detailed user consumption logs, this study investigates the relationship between three key dimensions: the level of algorithmic novelty users desire, the novelty they are exposed to, and the novelty they ultimately choose to consume. Our findings reveal a notable tension: although users often report preferring a balanced degree of novelty, in practice they tend to consume less novelty than is offered. Moreover, when presented with more novel recommendations, users frequently respond by gravitating toward familiar content, favouring exploitation over exploration.

Taken in combination with the previously outlined research from this thesis, these results suggest that algorithmic recommendations may, in fact, be even more novel than previously contended (relative to organic activity). This once again, challenges existing techno-determinist narratives that position algorithmic recommendation devices as fostering algorithmic concentration by design. Rather quite the contrary - from a discrete perspective they offer many opportunities for diversity which users actively go against.

Tracing Desired, Exposed and Adopted Algorithmic Novelty on Music Streaming Platforms

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Abstract

Almost a decade after the Filter Bubble's instantiation, concerns that algorithmic guidance may limit diversity remain as widespread as ever. Nonetheless, supporting empirical evidence has, over the years, weakened. Across some of the most widely adopted search engines and social media platforms, several works have found that users' own selections of algorithmic recommendations may principally be responsible for constraining the diversity of media which they end up consuming. Phrased differently, users may be in what literature has coined, "*self-imposed filter bubbles*". On music streaming platforms, similar concerns have frequently been raised over the role which algorithmic guidance plays in influencing users' consumption behaviours particularly in terms of novelty and diversity. Nonetheless, existing empirical literature geared at appraising these dynamics generally leverage solely consumption logs and thus fail to disentangle the impact of user choices from algorithmic decisions. Drawing on both behavioural and survey data from users of the French music streaming platform Deezer, we examine the relationship between the levels of algorithmic novelty users desire, the levels exposed to them and the levels ultimately selected to be consumed. In turn, our work demonstrates that despite commonly expressing a desire for balanced novelty, Deezer users oppose algorithmic novelty, selecting less novelty to consume than what is exposed to them algorithmically. At the same time, our work underscores the complexities of designing platform strategies geared at nudging users towards diversity.

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; • **Social and professional topics** → User characteristics; • **Information systems** → Multimedia streaming.

Keywords

musical recommendation systems, exposure and selection, filter bubble, consumption diversity, user behaviour

ACM Reference Format:

Dougal Shakespeare and Camille Roth. X. Tracing Desired, Exposed and Adopted Algorithmic Novelty on Music Streaming Platforms. In . ACM, New York, NY, USA, 11 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

To disentangle the role of algorithmic guidance in shaping users' online media consumption, it is essential to study not only what users *select*, but also what they are *exposed* to and thus, choose not to consume. In Political Science, research across multiple platforms has shown that while algorithms often propose diverse, cross-cutting media, users tend to select in ways that reduce that diversity [Bakshy et al. 2015; Robertson et al. 2023]. This has led to a more nuanced view of filter bubble dynamics [Borgesius et al. 2016; Pariser 2011] - one that emphasises the agency of users within the human-machine feedback loops ubiquitous across modern day platforms [Brinkmann et al. 2023]. In recent years, some scholars have even begun to refer to such dynamics as *self-imposed filter bubble dynamics* [Ekström et al. 2022]. Nevertheless, this strand of literature commonly speaks of average effects and thus, it remains largely unclear how this behaviour can vary for different types of users or even under different algorithmic behaviours (e.g. when recommending different levels or types of diversity).

In the case of music streaming platforms, similar concerns have also arisen around how platforms and the algorithmic recommendation devices they contain influence music listening, particularly in terms of diversity, novelty, and the volume of music consumed [Hesmondhalgh et al. 2023]. Yet, quantitative studies in this space typically rely on selection logs alone. For example, algorithmic concentration is often inferred by comparing diversity in consumption data from algorithmic affordances (like Spotify's Discover Weekly or Deezer's Flow) with that from less algorithmic ones (such as personal playlists or editorial curations) [Anderson et al. 2020; Mok et al. 2022; Shakespeare et al. 2025; Villermet et al. 2021]. While such research reveals how algorithmic devices are used in practice, it says little about the extent to which the observed diversity levels stem from algorithmic or, rather, user decisions. Users rarely consume recommendations blindly; they actively appreciate and choose what to consume based on a range of factors such as their openness [Anderson et al. 2021], goals [Garcia-Gathright et al. 2018], and algorithmic literacy [Gran et al. 2021]. Without access to exposure or non-selection logs, such user-driven patterns remain obscured, limiting our understanding of how algorithmic guidance functions and influences user behaviour in-practice.

The contribution of this work addresses this limitation by examining empirically the degree to which users of the French music

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Conference'17, July 2017, Washington, DC, USA

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ACM ISBN 978-x-xxxx-xxxx-x/YY/MM

<https://doi.org/XXXXXXX.XXXXXXX>

streaming platform Deezer select different levels of novelty to that which algorithmic devices expose. Developing upon the methodology of existing literature from Political Science, our work provides a nuanced account of how this dynamic varies depending on both the type of users considered and the behaviour of algorithmic recommendation devices. Focusing on one of the most widely adopted algorithmic devices on-platform, *Flow*, our work traces the relationship between the different levels of algorithmically recommended novelty desired by users, exposed to them and then finally, selected to be consumed by them. In turn, our work reveals a striking paradox: users select less novelty than what is recommended to them algorithmically even when they express desiring more.

2 Related Work

In the following section, we review the literature most relevant to this work. We begin by discussing empirical studies on self-imposed filter bubbles, which have inspired the present research. We then situate our work within the broader literature on filter bubbles and music streaming. Finally, we review research on user rejection behaviours (i.e., skipping) on music streaming platforms - one of the main signals we use to disentangle differences between the kinds of information exposed by algorithms and selected by users.

2.1 Self-imposed Filter Bubbles

Empirical works which provide evidence for self-imposed filter bubbles tend to stem from the field of Political Science and are generally conducted by either platform employees or researchers working closely with platforms. Aside from some recent efforts more geared towards capturing users' attention (e.g. via eye tracking) rather than selection behaviours [Michiels et al. 2022], the majority of these works rely on empirically comparing the recommendations which algorithmic guidance exposes to users with that which they select to uncover differences in diversity [Bakshy et al. 2015; Robertson et al. 2023]. This is also the approach taken in this work. Existing empirical research suggests that perhaps surprisingly, on both search engines [Robertson et al. 2023], social media platforms [Aiello and Barbieri 2017; Bakshy et al. 2015] and digital news outlets [Haim et al. 2018] confinement in users' consumptions may emerge mainly from users' self-imposed so-called *organic* [Roth 2019] rather than *algorithmic* personalisation processes. These results are commonly interpreted as providing empirical support for the selective exposure hypothesis, a core concept in communication and media studies which describes the propensity for individuals to consume media that reinforces their prior beliefs [Katz et al. 2017].

2.2 Filter Bubbles & Music Streaming

Driven by growing concerns about artist fairness [Ferraro et al. 2021] and how new technologies may reinforce or disrupt social structures and positions [Airola 2021], an expanding body of interdisciplinary research has sought to understand the role which streaming platforms and algorithmic devices play in shaping and supporting diverse music listening practices [Hesmondhalgh et al. 2023]. In recent years, a novel strand of this literature has emerged, which aims to understand the outcome of algorithmic adoption on listening diversity by benchmarking it against listening diversity under so-called *organic* affordances which rely less on *programmed*

guidance (e.g. personal playlists, the search bar, etc.) [Anderson et al. 2020; Mok et al. 2022; Shakespeare et al. 2025; Villermet et al. 2021]. Findings in this area suggest that, across several of the most widely adopted music streaming platforms, users' adoptions of algorithmic devices may bring about less semantic diversity (understood as spread in a dense, high dimensional geometric item embedding space) [Anderson et al. 2020; Shakespeare et al. 2025] but more novelty [Shakespeare et al. 2025; Villermet et al. 2021] than what they attain organically. These results complicate and nuance the conventional *filter bubble* narrative [Borgesius et al. 2016; Pariser 2011], indicating that in practice, the adoption of algorithmic guidance both broadens and narrows a user's listening experiences depending on the item representation method chosen [Mok et al. 2022; Shakespeare et al. 2025]. Nonetheless, a key limitation of this branch of literature is that the sets of recommendations algorithmically exposed to users are not usually analysed in parallel. As such, it remains unclear if these differences in diversity stem principally from algorithmic or user decisions.

2.3 Skipping Behaviours in Music Streaming

Whilst algorithmic exposure signals have rarely been leveraged to assess diversity-constraining dynamics on music streaming platforms, one particular subset has received significant attention within the Music Information Retrieval community, skips [Lamere 2014]. Research on skipping can be broadly conceptualised under two fractions depending on whether it tries to understand skipping behaviours in terms of variations of song features (e.g. audio fluctuations, changes in loudness etc.) [Donier 2020; Lamere 2014; Montecchio et al. 2020] or user features (e.g. platform behaviours, demographics, etc.) [Anderson et al. 2021; Deshmane et al. 2022; Meggetto et al. 2021, 2023].

Regarding the former, this type of research attempts to shed light on the typical locations within a song where skips occur and to identify any internal features of the song that may be related to skips. As early as 2014, a preliminary analysis of skipping behaviours on the Spotify platform [Lamere 2014] demonstrated that the majority of skips occur within the first 20 seconds of listening to a song and furthermore, that song skips follow a so-called U-shaped distribution which describes the propensity for songs to be skipped at the beginning and to a lesser extent, end of a track. Similarly, Sguerra et al. [2022] show (although do not explicitly mention) that on the Deezer platform, this U-shape distribution relates not just to the stream time of the song but also to the number of times in which the song is repeatedly exposed to the user. Montecchio et al. [2020] show how the skipping distribution of a song closely relates to its musical structure which leads the authors to suggest that it may also be seen as an intrinsic property of the song itself. In a later work, Donier [2020] explore this concept more formally with a kernel-based approach to model skip intensity. In turn, they demonstrate that so-called "salient musical events" mostly corresponding to within-song transitions are good indicators of skipping spikes. Furthermore, their model tends to be generally stable across songs with the exception of those tagged as classical or jazz which are noted to attract markedly lower skip rates in comparison to songs belonging to other genres. Similarly, Ng and Mehrotra [2020] show

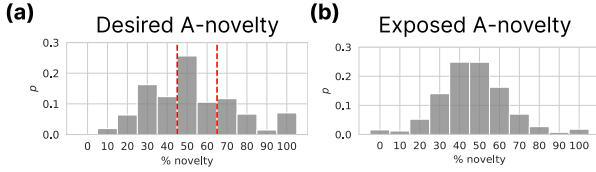


Figure 1: Distribution of desired (a) vs actual exposed (b) A-novelty in Flow's recommendations. Red lines on Figure denote tertile bounds which separate users into r , m and v classes from left to right.

how audio state fluctuations at the session level are also associated with an increased probability of users skipping.

Research on skipping which places the user rather than item at the forefront - the approach also undertaken in our work - is, by contrast to the previously detailed research strand, less common. Meggetto et al. [2021] focus on characterising the sequential skip patterns of first sessions and then later, users. Their work shows how a user's skip typology varies alongside their session length, time of day of platform usage and platform affordance (e.g., algorithmic recommendation, personal playlists etc.) adopted to reach content. In a more recent follow-up work Meggetto et al. [2023] focus on further understanding *why* users reject songs on streaming platforms. By applying a deep reinforcement learning model to predict users' song skips, their work finds that the selection/rejection status of previously consumed item is the feature which is most strongly correlated with users' skipping behaviours. Literature has also indicated that skipping may also be strongly associated with users' personality types. Anderson et al. [2021] find that on the Spotify platform, emotionally stable users tend to have lower skip rates.

3 Dataset

We work with the rich combination of survey data and granular platform interaction logs from a cohort of users from the French music streaming platform, Deezer. On the platform interaction side, we focus primarily on analysing interactions with one of Deezer's most popular algorithmic devices, *Flow*. This device follows what has been coined a '*lean back formulation*' [Meggetto et al. 2023] and is designed to act as an endless radio in which users can neither view nor select in advance future recommendations. As such, each next track is a surprise for users and the main way of progressing from one recommendation to the next is either by letting the current track finish or alternatively, to 'skip'. Flow offers a unique opportunity for studying differences between the media selected by and exposed to users since by design, it ensures that the union of streams and skips constitutes all algorithmic exposures. As such, this simplifies our method and ensures we do not have to approximate exposures from recommended item lists using more complex methods like eye tracking or position bias correction. We study 3 months of platform interaction logs (02/22, 03/22, 04/22) and only consider users who are active in the first and last week of our observation, have minimal drop-off (max 1 week) and have at least one monadic (i.e. containing only Flow) session in 03/22 or 04/22. This

yields a total of $n = 5250$ users. We process listening histories by splitting them into sessions of continuous engagement whereby a new session is defined when ≥ 20 minutes passes between engagements. Interactions with tracks are labelled as skips if the stream duration is < 30 seconds and otherwise as streams. Each listening history is assigned an affordance, indicating how the track was accessed: algorithmically via one of the platform's recommendation device such as Flow (A), editorially through a Deezer curator's playlist (E), or organically by the user for example, through the search bar or their own personal playlists (O). Alongside our survey respondents' platform interaction logs, we also work with a random control dataset which is processed using the same data processing pipeline ($n = 4802$). We use this dataset to assert the generalisability of the conclusions we derive from processing respondents listening histories. On average, survey respondents are slightly older (39.0 vs. 37.5 years) and heavier users of algorithmic devices (38.5% vs. 33.9% of streams) but otherwise are broadly representative of the Deezer population. For further details of the dataset's representativeness we refer readers to the Supplementary Materials.

On the survey side, we make use of a subset of the survey responses collected through the RECORDS research project¹ between the period of February and June 2023 and previously utilised by Marey et al. [2024]. Whilst the full dataset has not yet been publicly released at the time of publication it is anticipated that this will be made open access as the project reaches a conclusion in 2025. We focus primarily on the questions concerning users' preferences for novelty in Flows recommendations, what we coin *A-novelty* from hereon out.

4 Desired A-novelty

We first focus on measuring how much novelty Deezer users would like Flow to expose to them algorithmically. To this end, we analyse users' responses to the following survey question: *Q. 'What percentage of discoveries would you like to see in Flow?'* In the survey, users were presented with a slider movable in intervals of 10 from [0..100] to indicate what percentage of novelty they desire Flow to recommend. As depicted in Figure 1, users tend to be relatively consistent in the amount of A-novelty they prefer. Letting p denote the proportion rather than percentage of desired novelty, we observe that on-average users prefer a balanced mix of novel and redundant recommendations ($\langle p \rangle = 0.523$, $\sigma = 0.219$). Nonetheless, whilst less common, there also exist a marked proportion of users who want Flow to recommend almost exclusively novelty or redundancy. Segmenting users into tertiles based on their responses yields 3 types of users which we label based on their desire for redundant (r , $n = 1943$), mixed (m , $n = 1897$) or novel (v , $n = 1410$) recommendations. Examining what, aside from their expressed desires for A-novelty, makes each of these groups distinct we observe a significant difference in terms of how they use the platform.² In particular, m users rely less on A adoption ($\langle p_A \rangle = 0.363$) and lean more towards O affordances ($\langle p_O \rangle = 0.554$). In contrast, r users rely more heavily on A guidance ($\langle p_A \rangle = 0.404$) and less on O ($\langle p_O \rangle = 0.515$) relative to other user groups. Interestingly, we

¹<https://records.huma-num.fr/>

²Significance computed via two-way ANOVA with $\alpha = 0.05$. See Supplementary Material.

observe no significant differences between the groups in terms of E adoption.

5 Exposed A-Novelty

We next examine the extent to which Flow offers users opportunities for novel consumptions in its recommendations and how this aligns with users' expressed desire of A-novelty. At this stage of our analysis, we focus exclusively on sessions that only make use of the Flow affordance in the period of 03/22 and 04/22. Whilst this might seem like a strict requirement, these kinds of sessions are actually surprisingly common. In our dataset, 66.17% of Flow streams and skips are enacted within sessions that only made use of the Flow affordance.

We define a recommendation in a session, S as novel if it has not previously been streamed one month prior to the start of S . For each Flow session we compute the proportion of recommendations which were novel and then per user, average over all of their monadic Flow sessions to form one data point per user which we define as p_{nov} . As shown in Figure 1, where we plot the distribution of these scores over all users, at an aggregate level, the amount of novelty users desire roughly aligns with what they receive ($\langle p_{nov} \rangle = 0.516$). Nevertheless, users who desire the most or least A-novelty are not the ones who receive this in practice. Running an ANOVA we find no significant differences between r , m and v users in terms of the levels of A-novelty exposed to them ($p = 0.923$, see Supplementary Materials). As such, typically, v users tend to receive less novelty than they desire whilst r users tend to receive more.

Whilst the amount of exposed A-novelty does not vary alongside a user's desired A-novelty, it does however, vary alongside algorithmic engagement. As shown in Figure 2, exposed A-novelty increases alongside the length of sessions ($r = 0.106^{***}$),⁴ and exposed A-novelty decreases alongside the number of monadic Flow sessions a user has ($r = -0.308^{***}$). This creates a polar opposite dynamic whereby increased Flow engagement within sessions tends to be associated with greater levels of exposed novelty whereas more frequent, perhaps more regular, enactments of Flow sessions are associated with reduced A-exposed novelty. These findings are inline with the work of Lathia et al. [2010] who show that activity and inter-event time are negatively related to temporal next-session A-novelty.

6 Adopted A-novelty

Having established that Deezer users are typically exposed to a balanced mix of novelty and redundancy when using Flow (even when it is not desired) we now turn our attention towards understanding the tendency for users to adopt A-novelty. To this end, we measure the extent to which users bias their streams towards *exploiting* familiar recommendations over *exploring* novel ones, how this preference differs across r , m , and v user groups, and how it is influenced by both user and algorithmic behaviour. We understand the behaviour of biasing streams towards exploration as being the

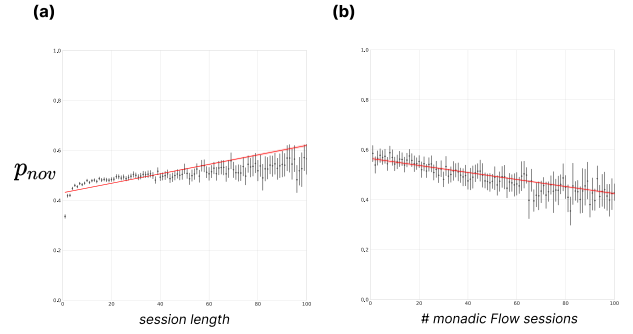


Figure 2: Relation between Flow adoption and exposed Novelty. Figure (a) - shows within session adoption and (b) shows the relation to between-session adoption.

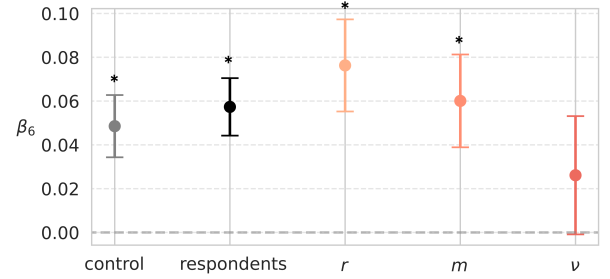


Figure 3: Coefficient values associated with increasing % A-novelty exposed by Flow on users' exploitation behaviours. β_6 is extracted from fixed effect model, * marks significance at $\alpha = 0.05$ and bars show 95% CI.

scenario in which, by skipping recommendations, users end up consuming more novelty than they otherwise would have. Likewise, the inverse holds regarding biasing streams towards exploitation.

Method: We segment each Flow session into 2 sub-lists, redundant and novel, and compute on each sub-list, the proportion of recommendations which are played (i.e. P) rather than skipped. Thus, each session has two data points, $p_{P|nov}$ and $p_{P|red}$. These are combined over all users to form our dependent variable Y . For each data point we assign a binary identifier R which is set to 1 when the redundant part of the session is considered and 0 otherwise. We assign L to be the session length and p_{nov} to be the proportion of novel recommendations exposed in the session. In addition, let p_γ define the proportion of recommendations (either relating to those that are redundant or novel) that are semantically similar to a user's recent consumptions. As detailed in the Supplementary Materials, this is computed as the proportion of recommendations which fall within the hypersphere constructed by a user's recent consumptions (i.e. in the past month) positioned in high-dimensional geometric item embedding space. The motivation for including this variable as a control is to understand if Deezer users' tendency to bias their streams towards exploration or exploitation can be explained more generally, by their preference

³*, ** and *** used throughout to denote significance at $p < 0.05$, 0.01 and 0.001 respectively.

⁴Interestingly, it appears that perhaps by design, first recommendations are more redundant than others.

for engaging within semantic similarity or dis-similarity. Using each of these variables, we then apply the following fixed effect model.

$$Y = \beta_0 + \beta_1 R + \beta_2 L + \beta_3 p_{nov} + \beta_4 p_Y + \beta_5 LR + \beta_6 p_{nov}R + \beta_7 p_YR + \alpha + \epsilon \quad (1)$$

β_0 represents the intercept. β_1 captures a user's tendency to bias their selections (i.e. streams) more towards exploration ($\beta_1 < 0$) or exploitation ($\beta_1 > 0$) when using Flow which more precisely, is characterised by the change in streaming probability when redundant as opposed to novel tracks are recommended. β_2 and β_3 capture the effects of user and algorithmic behaviour in particular, the change in streaming probability associated with a 1 unit increases in session length and exposed novelty respectively. The former variable and coefficient (β_2) is incorporated into our model based on an understanding that music streaming users streams lean more into exploitation with increased algorithmic engagement [Villermet et al. 2021] whilst the latter (β_3) is included based on literature that has shown both online and offline, consumers' affect for [Chmiel and Schubert 2017] and likelihood of streaming [Sguerra et al. 2022] music commonly varies as a function of the number of exposures (following an inverted U-shaped curve). β_4 is associated with p_Y and captures the effect of a 1 unit increase in the proportion of recommendations which are within the semantic hypersphere (which defines a user's recent consumptions) on stream rate. β_5 , β_6 and β_7 are associated with the interaction terms which capture the effect which changes in L , p_{nov} and p_Y can have on users biasing their streams towards exploration or exploitation. α captures the user level fixed effects. Standard errors are clustered. Full results tables are detailed in the Supplementary Materials. We apply this model on each of user classes (r, m, v) to study variations in each of these dynamics which occur alongside users' declared preference for exposed A-novelty.

Results: Even after controlling for the semantic similarity of recommendations we observe that, much like on social media platforms where users have been shown to select less diversity than they are exposed to [Bakshy et al. 2015], likewise music streaming users select less novelty than is exposed to them algorithmically. More precisely, we find that users tend to prefer biasing their streams towards exploiting rather than exploring when using Flow. Put simply, redundant recommendations are significantly more likely to be streamed than novel ones ($\beta_1 = 0.0970^{***}$). Examining variations between r , m and v users we find that surprisingly, this behaviour is largely independent of a user's desires for exposed A-novelty. Even when users declare wanting to see more A-novelty than redundancy, they still bias their streams towards recommendations that are redundant.

Regarding how this behaviour varies alongside the levels of and types of diversity recommended to users, we observe some highly non-trivial dynamics. When Deezer users are recommended a greater proportion of novelty in Flow sessions, they tend to lean more into exploitation ($\beta_6 = 0.0573^{***}$) and furthermore, this effect varies strongly alongside a user's openness to exposed novelty. As depicted in Figure 3, when r users are exposed to higher levels of novelty within sessions, they tend to lean further into exploitation ($\beta_{6_r} = 0.0763^{***}$), increasingly favouring familiar tracks over

novel ones whereas this dynamic is markedly less for m users ($\beta_{6_m} = 0.0601^{***}$) and even less for v users ($\beta_{6_v} = 0.0261$). Furthermore, whilst Deezer users preference for exploitation can not solely be explained by them favouring recommendations that are semantically similar over those that are not, the semantic similarity of algorithmic recommendations can affect how the degree to which users lean towards exploration or exploitation. Deezer users tend to be more open to using Flow for exploration when recommendations are less diverse and thus, align more closely with a user's past consumptions ($\beta_7 = -0.0351^{***}$). Regarding variations across user groups, we observe similar relative difference to before (when p_{nov} was considered). The less users desire exposed A-novelty the more susceptible they are to this dynamic ($\beta_{7_v} = -0.0235^{**}$, $\beta_{7_m} = -0.0332^{***}$, $\beta_{7_d} = -0.0441^{***}$). Put differently, as both the dense (semantic) and discrete (novel) diversity of recommendations increases, users lean more into biasing their streams towards exploitation.

Regarding algorithmic engagement, we observe that longer sessions tend to attract significantly lower stream rates than shorter ones ($\beta_2 = -0.0018^{***}$) even after controlling for exposed p_{nov} which as we previously demonstrated, also increases alongside session length. Several short, bursty sessions likely create more novelty for users than prolonged sessions which amount to the same overall length. Nevertheless, that does not mean that users become more exploitative with longer sessions - we observe no significant effect associated with the LR interaction term ($p = 0.1191$).

Overall, these findings clearly highlight Deezer users' tendency to favour familiarity over novelty when using algorithmic recommendation devices which allow for both. Users generally choose to consume less novelty than they are exposed to irrespective of how much novelty they express wanting. At the same time, these findings also underscore the complexity of designing platform-side interventions geared at fostering more diverse consumptions practices. As we show, increasing the levels of both dense (geometric, semantic) and discrete (novelty-based) diversity in algorithmic recommendations can cause users to rather surprisingly, bias their streams even further towards exploitation. These results align with existing literature which has demonstrated that increasing the diversity of algorithmic music recommendations does not necessarily cause users to consume more diversely [Chen et al. 2013] and literature that suggest that in context of online music consumption behaviours, preference for dense and discrete consumption diversity may be orthogonal concepts [Mok et al. 2022; Shakespeare et al. 2025].

7 A-novelty $\rightarrow$ O-novelty

When engaging with algorithmic guidance that permits varied usage styles, Deezer users tend to lean towards exploitation. But what happens when users *do* adopt A-novelty? What knock-on effects can this have on their subsequent organic platform interactions? For example, a user might discover a new artist or genre through a Flow recommendation and then organically explore the artist's catalogue within the same session, leading to a sudden increase in novelty within their organic streams. In the following section, we focus on understanding what are the short term within-session effects of streaming A-novelty upon the novelty of users' future

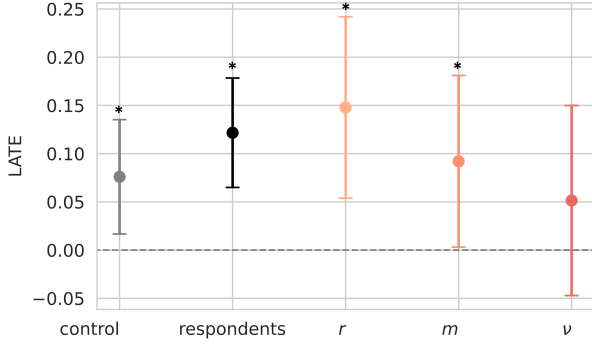


Figure 4: Local Average Treatment Effects (LATE) computed from DID applied to each user group. * marks significance ($\alpha < 0.05$).

organic interactions? To this end, we employ a fuzzy difference-in-differences (DID) model with propensity score matching.

Data Processing: To generate a dataset which is suitable for this form of analysis we filter skips and collect all sessions in 03-04/22 that contained a sequence of consecutive organic streams, followed by Flow streams followed again by organic streams. This results in 3776 sessions which were enacted by a total of 960 users. We split each session into sub-sessions of consistent affordance usage, O_1, A, O_2 . A acts as the treatment and the period of O_1 and O_2 captures pre and post treatment time blocks respectively. For each sub-session we compute the proportion of streams that were novel, stack these per user and then take column-wise means to produce three data point per user, $(p_{nov|O_1}, p_{nov|A}, p_{nov|O_2})$. We consider a session as treated when $p_{nov|A} > \langle p_{nov|A} \rangle$. Since we deal with thresholds rather than a control which never received any treatment, our DID is fuzzy and we deal with local average treatment effect (LATE), identifying the impact of treatment for users whose adoption behaviour changes around the threshold (i.e. the average $p_{nov|A}$).

DID estimation relies on the parallel trends assumption, which assumes that in the absence of treatment, the treatment and control groups would have exhibited similar trends over time. While our data suggest that this assumption holds in the pre-treatment period (see Supplementary Materials), it remains fundamentally untestable, as it concerns an unobservable counterfactual scenario. To enhance the credibility of our analysis, we additionally implement propensity score matching (PSM) prior to applying the DID estimator. By matching treated and control observations on relevant covariates, PSM increases the balance between groups, thereby improving the plausibility of the parallel trends assumption and reducing potential bias in the estimated treatment effects.

Propensity Score Matching: We match treated and control units on both *user-level* and *session-level* features to improve covariate balance prior to applying DID. At the user level, we include algorithmic adoption rates (p_A) and also their registration year which captures how users make use of the platform and how long they have been on the platform respectively. At the session level, we

match on the *average session length* (i.e. per user) and the *average organic novelty at session start* ($p_{nov|O_1}$). These variables collectively form our covariate matrix X , which we use to estimate propensity scores via the following logistic regression:

$$G \sim \text{logit}(X) \quad (2)$$

where G is a binary indicator equal to 1 if the treatment is applied, and 0 otherwise. We then perform *nearest neighbour matching* with a caliper of 0.1 to pair treated units with similar control units.

Post-matching diagnostics (see Supplementary Materials) indicate that covariate balance between the groups is substantially improved and all standardized mean differences are < 0.25 , the threshold commonly applied for two groups to be considered balanced [Stuart 2010].

DID: After matching, we then apply our DID model which can be expressed as:

$$Y = \beta_0 + \beta_1 T + \beta_2 G + \beta_3 GT + \epsilon \quad (3)$$

Y represents the outcome ($p_{nov|O_t}$), T is binary indicator variable which is 0 before intervention and 1 after. β_3 which is associated with the interaction term captures the LATE. We apply PSM and DID at both an aggregate level and also split per r, m, v user groups.

As depicted in Figure 4, we observe that consuming A -diversity has a significant positive effect on users' organic diversity ($\beta = 0.122^{***}$) indicating that organic knock-on effects can and do happen. Nonetheless, the effects are not equal for all user types. Examining differences in effect sizes between r, m and v users we observe that surprisingly, the users who benefit the most from A -novelty are r users who in principal, oppose A -novelty the most ($\beta = 0.148^{**}$). At the other end of the spectrum, for v users, it appears that on the whole, their organic novelty would have been similar even in the absence of consuming algorithmic novelty ($\beta = 0.051$). This appears to indicate that although r users oppose A -novelty, it can, when it aligns with their taste (i.e. since it was streamed), have a greater impact on keeping their organic consumptions novel.

8 Discussion

Familiarity plays a crucial role in moulding our relationship with music, influencing both our preferences and subsequent listening behaviours. Repeated exposure to certain songs or genres can deepen our appreciation [Chmiel and Schubert 2017], as familiarity often brings comfort, predictability, or even nostalgia, which can be pleasant for a listener. However, excessive repetition of favourite songs can limit opportunities for novel music discoveries that, though uncertain, might ultimately be more rewarding. Such forms the classical exploitation, exploration trade-off common across both algorithmic optimisation problems and also ubiquitous in user decision making processes. Striking the right balance between listening to familiar favourites or rather, taking gambles on new discoveries is thus essential for a dynamic and enriching listening experience.

Modern-day so-called lean-back algorithmic devices, such as Deezer's Flow are designed to respect this balance, promoting a mix of novelty and redundancy in their song recommendations. The contribution of this work explores whether this balance is likewise respected by users in their engagement with Flow's recommendations. Comparing declared survey response data with behavioural platform interaction data, our work traces the alignment between

the levels of *A*-novelty users desire, algorithms expose and then finally, users choose to consume. Our work demonstrates that in practice, Deezer users generally favour redundancy over novelty in algorithmic recommendations. In line with recent empirical findings from Political Science [Aiello and Barbieri 2017; Bakshy et al. 2015; Haim et al. 2018; Robertson et al. 2023], we find that users select less novelty than they are recommended, even amongst users who express wanting more *A*-novelty. At the same time, our work underscores the complexity of designing platform intervention strategies geared at *nudging* users towards more diverse consumption practices. Rather conversely, Deezer users react to more novelty by leaning more into exploitation and furthermore, tend to be more explorative when recommendations are less semantically diverse and to use Pariser's metaphor, within their *bubble* [Pariser 2011]. Put differently, users react to increasing dense and discrete diversity in algorithmic recommendations by biasing their streams more towards exploitation. Nevertheless, when *A*-novelty aligns with users' tastes (i.e. when it is streamed) our work finds that it has a positive effect on users' organically driven listening novelty even at the within-session scale and surprisingly, for users who oppose it in principal. Thus, users who oppose *A*-novelty the most may in fact be the ones who also benefit from it the most.

Taken together, these findings not only offer a comprehensive account of how human selections of algorithmic recommendations can limit diversity (specifically in terms of discrete novelty) thereby challenging dominant techno-deterministic narratives, but also underscore the nuanced challenges of increasing novelty in recommendations across different user groups, offering valuable insights for recommender system design moving forward.

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9 Supplementary Material

9.1 Dataset Representativeness

Variable	Respondents	Control
Age	38.96 (12.12)	37.50 (10.84)
p_A	0.38 (0.25)	0.34 (0.26)
p_O	0.53 (0.26)	0.56 (0.27)
p_E	0.08 (0.11)	0.10 (0.14)
registration year	2013.85 (4.29)	2014.39 (4.13)

Table 1: Descriptive statistics for respondents and control group. Mean shown with standard deviation in parentheses.

9.2 Exposed A-novelty

2-way ANOVA to check for statistical differences in exposed novelty between r , m and v users.

	sum_sq	df	F	PR(>F)
C(cluster)	0.004639	2.0	0.079613	0.923475
Residual	152.873058	5247.0	—	—

Table 2: ANOVA results

9.3 Embedding Space Usage

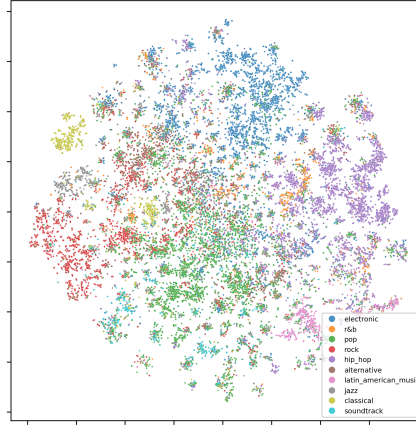


Figure 5: t-SNE projection of the artist embedding space derived from Deezer users' playlists. For clarity, the visualization includes only artists belonging to the top 10 most popular genres in the dataset.

9.3.1 Embedding Construction: As per. existing literature [Anderson et al. 2020; Mok et al. 2022], we generate a 128-dimensional artist embedding space by applying SVD to a PPMI normalised artist-artist co-occurrence matrix constructed from a random sample of Deezer users personal playlists ($n = 100,000$). This forms our embedding matrix, E . As depicted in Figure 5, semantically similar artists that belong to the same genre tend to be positioned closely in vector space. Since even before embedding, some artists may be too *niche* to gather enough co-occurrence information to accurately represent their true structure we choose to filter these artists prior to applying SVD. We define an artist as *niche* if it is featured in less than 10 unique playlists. When computing dense measurements of concentration we map recommended tracks to their associated first assigned artists. This decision was made to both limit the necessary compute power as well as ensure that we minimise the number of cases for which we have to filter tracks due to their nicheness.

9.3.2 Computing the dense concentration of Flow recommendations: We use this embedding to compute the dense concentration of Flow's recommendations. This is then used as a control within our fixed effect model. To this end, let the weighted centroid of a user's recent streams (i.e. in the past month) be defined as μ . We define a Flow recommendation as densely concentrated when it exists within the hypersphere defined by a user's past recent consumption history, H , which are spread from μ in vector space. Put differently, a Flow recommendation, r is deemed to be densely concentrated (γ) if $\cos(E_r, \mu) \leq GS(\mu, H)$ where GS represents the generalist-specialist score as defined by [Anderson et al. 2020; Mok et al. 2022] and $\cos$ represents cosine distance. Within our fixed effect model we compute the proportion of items which are concentrated under this dense formulation which we define as p_γ . This is computed for the sub-list of recommendations that are both redundant and novel within a given Flow session.

9.4 Adopted A-novelty

Results tables from fixed effect models applied to each user group.

(a) User group: Control							(b) User group: Respondents						
	β	Std. Err.	T-stat	P-value	Lower CI	Upper CI		β	Std. Err.	T-stat	P-value	Lower CI	Upper CI
R	0.1037	0.0055	18.965	0.0000	0.0930	0.1144	R	0.0970	0.0052	18.629	0.0000	0.0868	0.1073
L	-0.0019	0.0001	-18.299	0.0000	-0.0021	-0.0017	L	-0.0018	7.854e-05	-22.429	0.0000	-0.0019	-0.0016
p_{nov}	-0.0514	0.0068	-7.6011	0.0000	-0.0646	-0.0381	p_{nov}	-0.0562	0.0058	-9.7102	0.0000	-0.0676	-0.0449
p_γ	0.0528	0.0038	13.843	0.0000	0.0453	0.0603	p_γ	0.0571	0.0034	16.767	0.0000	0.0504	0.0637
$R : L$	7.374e-05	3.943e-05	1.8699	0.0615	-3.553e-06	0.0002	$R : L$	6.124e-05	3.929e-05	1.5586	0.1191	-1.577e-05	0.0001
$R : p_{nov}$	0.0485	0.0073	6.6918	0.0000	0.0343	0.0628	$R : p_{nov}$	0.0573	0.0067	8.5649	0.0000	0.0442	0.0705
$R : p_\gamma$	-0.0258	0.0048	-5.3745	0.0000	-0.0352	-0.0164	$R : p_\gamma$	-0.0351	0.0044	-7.9589	0.0000	-0.0437	-0.0264

(c) User group: r							(d) User group: m						
	β	Std. Err.	T-stat	P-value	Lower CI	Upper CI		β	Std. Err.	T-stat	P-value	Lower CI	Upper CI
R	0.1016	0.0084	12.126	0.0000	0.0852	0.1180	R	0.0966	0.0087	11.050	0.0000	0.0795	0.1138
L	-0.0016	0.0001	-11.637	0.0000	-0.0018	-0.0013	L	-0.0018	0.0001	-15.046	0.0000	-0.0020	-0.0015
p_{nov}	-0.0690	0.0096	-7.1825	0.0000	-0.0879	-0.0502	p_{nov}	-0.0539	0.0093	-5.8192	0.0000	-0.0721	-0.0358
p_γ	0.0584	0.0055	10.702	0.0000	0.0477	0.0691	p_γ	0.0534	0.0055	9.7610	0.0000	0.0427	0.0641
$R : L$	-3.335e-05	5.804e-05	-0.5746	0.5655	-0.0001	8.041e-05	$R : L$	3.658e-05	6.820e-05	0.5364	0.5917	-9.709e-05	0.0002
$R : p_{nov}$	0.0763	0.0107	7.1089	0.0000	0.0552	0.0973	$R : p_{nov}$	0.0601	0.0108	5.5560	0.0000	0.0389	0.0813
$R : p_\gamma$	-0.0441	0.0070	-6.2745	0.0000	-0.0579	-0.0304	$R : p_\gamma$	-0.0332	0.0074	-4.4692	0.0000	-0.0477	-0.0186

(e) User group: ν						
	β	Std. Err.	T-stat	P-value	Lower CI	Upper CI
R	0.0899	0.0100	8.9733	0.0000	0.0702	0.1095
L	-0.0021	0.0001	-15.047	0.0000	-0.0023	-0.0018
p_{nov}	-0.0405	0.0114	-3.5473	0.0004	-0.0629	-0.0181
p_γ	0.0595	0.0071	8.4087	0.0000	0.0457	0.0734
$R : L$	0.0003	8.087e-05	3.1306	0.0017	9.467e-05	0.0004
$R : p_{nov}$	0.0261	0.0138	1.8960	0.0580	-0.0009	0.0531
$R : p_\gamma$	-0.0235	0.0085	-2.7461	0.0060	-0.0402	-0.0067

Table 3: Results from fixed effects models across user groups. 95% confidence intervals shown.

9.5 DID Application

9.5.1 Propensity Score Matching. To assert the alignment of our matching we compute standardised mean differences for each user groups and variable we consider. As depicted in Table 4, these are < 0.25 which is usually the threshold applied for two groups to be considered balanced [Stuart 2010].

Group	$p_{nov O_1}$	p_A	registration year	$\langle L \rangle$
all	-0.0163	-0.1306	-0.0451	-0.1134
r	0.0127	-0.2112	-0.0827	-0.0242
m	0.0803	-0.0349	0.0051	0.1136
ν	0.1618	0.1395	0.0559	0.0413

Table 4: Standardized mean differences between control and treatment groups after matching.

9.5.2 Parallel Trends. To check for parallel trends we examine trends between treatment and control groups' organic novelty pre-treatment. To this end we fit the following model.

$$Y = \beta_0 + \beta_1 Gt + \beta_2 G + \beta_4 t + \epsilon \quad (4)$$

Where Y represents the pre-treatment novelty, G the indicator variable which is 1 when we consider the treatment and 0 otherwise and t represents the time step. Examining the significance of the coefficient associated with the interaction term (i.e. β_2) enables us to assert if we have reason to reject the parallel trends assumption pre-treatment. Since before aggregating, sessions have a variable number of O streams which lead up to Flow ones, we run our model several times on sessions of different lengths. As depicted in Figure 6, we filter to only consider sessions which have a pre-treatment number of streams then aggregate as before at the user scale before subsequently running our model. As depicted in Figure 6 where p -values are shown, we observe no significant effect associated with β_2 and thus, no reason to reject the parallel trends assumption pre-treatment.

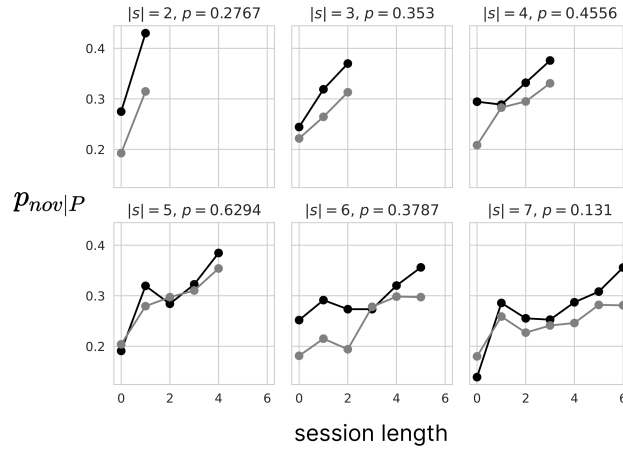


Figure 6: Pre-treatment stream rates on different session lengths. Black lines display treatment group and grey lines the control. p -values on each subplot are extracted from the interaction term run on the linear model to check for parallel trends pre-treatment.

Part III

Reflections

Discussion

Summary

Through several in-depth empirical analyses of the items Deezer users are exposed to, those they choose to consume, and the contexts of consumption, this thesis contributes towards an enriched understanding of how multi-affordance music streaming platforms are adopted and the degree to which algorithmic, and more broadly programmed, guidance shapes diverse and/or confined consumption behaviours in practice rather than in principle. In turn, this thesis challenge many of the dichotomies (e.g., passive vs. active, exploration vs. exploitation) and paradigms (e.g., filter bubbles, algorithmic confinement) that have driven research investigating the relation between algorithms, platforms and cultural consumption practices. Taken together, these results contribute to challenging and refuting pervasive techno-determinist narratives within these domains by affirming both the organic agency of users and the multipolar nature of algorithmic confinement.

In our 2021 ISMIR publication, we addressed RQ1: *“How do users adopt music streaming platforms, and more precisely, how does this vary under the different affordances they embody?”* This research contributes towards a more thorough understanding of the extent to which music streaming platforms, by design, may facilitate or even push users towards relying on passive information retrieval, as several more qualitatively minded scholars have discussed in recent years (Prey, 2019; Hesmondhalgh, 2022). While earlier literature demonstrated that music streaming users primarily rely on organic affordances for consumption (Beuscart et al., 2019; Villermet et al., 2021), the usual origins of organically streamed tracks (i.e., which affordance they were first discovered through) remained unclear. Programmed affordances (algorithmic and editorial) by design facilitate more passive information retrieval. Thus, if organic streams mostly originate from programmed affordances, then showing that most activity is “organic” does not prove that users rely on streaming platforms for active rather than passive information retrieval. Our research

advances the debate by analysing the interdependence between organic and programmed affordances. In turn, we were able to demonstrate that Deezer users’ organic consumption rarely originates from programmed guidance. Taken in combination with prior literature, this research demonstrates that music streaming platforms are not adopted principally for passive information retrieval. This offers a significant contribution to both Media Studies and consumption oriented Sociology, which have until now largely addressed these questions from a theoretical rather than empirical standpoint. Put simply, our research demonstrates quite clearly that music streaming users may be more active participants in shaping their own listening experiences than they have typically been portrayed.

With the motivation of constructing and applying an appropriate item embedding space to capture the diversity of Deezer users’ consumption geometrically (i.e., via spread in vector space), the research published and presented at CompleNet in 2024 represents a substantial advancement in the growing field of interpretable embeddings (Şenel et al., 2018; Gastaldi, 2021). This work introduces not only a novel framework for assessing interpretability but also a new account of the performance of several classic node embedding models under our evaluation framework, which were in turn assessed as candidate models for our use case. This research contributes significantly not only to this thesis, by grounding the subsequent methodological designs of later studies, but also to broader scholarship concerned with understanding distance in embedded space, such as research on generating explainable algorithmic recommendations (Zhang et al., 2020). Although not directly linked to the core research questions of this thesis, this study plays an important role in identifying which models are most suitable for constructing high-dimensional item embedding spaces to measure the diversity of users’ consumption, especially across different affordances. Our findings show that Singular Value Decomposition (SVD) is, perhaps unexpectedly, the most interpretable model tested, even when examining how higher-order neighbourhoods are projected in latent vector space. This outcome informed our decision to apply SVD to generate the item embedding spaces in the two more impact-oriented studies that follow.

The subsequent research, published in Scientific Reports, addressed RQ2: “*To what extent can multi-affordance user-platform interaction data provide novel insights into the role of music streaming platforms’ personalized algorithmic recommendation devices in fostering either confined or diverse consumption practices?*” and RQ3: “*How do choices in scale and information representation fundamentally influence the conclusions drawn from such analyses?*” The contribution of this research advances the literature on aggregate algorithmic confinement appraisals by highlighting the importance of scale and

information representation. This is achieved by reconciling existing studies which suggest, despite using different temporal scales, that algorithmic confinement does and does not occur under dense (Anderson et al., 2020) and discrete (Villermet et al., 2021) formulations of concentration, respectively. Holding scale constant, our research finds the same diversity differences emerge between affordances, supporting the view that these differences in the literature do not necessarily arise from variations in scale. Crucially, these appear to stem from dense and discrete consumption diversity being *orthogonal* concepts, not only at the user level as prior literature has suggested (i.e., the most novel users are not the most diverse Mok et al. (2022)) but also, as we demonstrate, within individual users’ consumption. This leads us to propose an updated version of the filter bubble narrative, which provides a significant development to existing unidimensional appraisals conducted to date: algorithmic devices can function both as diversity-constraining forces, fostering semantically homogeneous listening experiences, and as novelty-expanding forces for users. Put simply, algorithms appear to push not listened to before but yet similar music. Given that music consumers tend to prefer redundancy even without algorithms or platforms (Chmiel and Schubert, 2017; Sguerra et al., 2022), we can theorise that the high level of novelty introduced by algorithms may help explain why Deezer users overwhelmingly favour organic over algorithmic affordances.

At the same time, this research also sheds light on the impact of research design when studying algorithmic confinement on music streaming platforms. Specifically, it exemplifies the marked effects of transitioning from micro to macro scales. Within sessions, and under a dense, geometric framing of concentration, algorithmic devices may create more diverse listening experiences than users achieve organically. Yet, over time, as we demonstrate, they fail to expand into dissimilar, diverse recommendations and instead repeatedly exploit the same regions of embedded space. This finding has several implications for existing literature. On one hand, it highlights the shortcomings of extrapolating and comparing conclusions derived from literature geared at conducting temporal appraisals of algorithmic confinement. Given that within this strand, a range of temporal scales are usually applied, it should come as no surprise that the results from this literature are also likewise heterogenous. On the other hand, from a practitioner’s standpoint, it underscores the limitations of conducting beyond-accuracy evaluations of recommender systems solely at the session level or within recommendation "lists", a relatively common practice in recommender system evaluation (Kaminskas and Bridge, 2016). Since user evaluations of recommendations are inherently temporal and we hypothesise that users’ appreciation of algorithms are built over time, such evaluations risk creating a semantic

mismatch between what users perceive and what metrics measure.

Moreover, the findings and methodology from this research also provide a substantial development of existing multi-affordance literature which in turn, allows us to derive richer measurements of algorithmic confinement which may be more closely aligned with how users experience and perceive diversity on the platform (Porcaro et al., 2022). By splitting the organic affordance into personal playlists (P) and query-based retrievals (Q), we demonstrate that Q adoption tends to spread user attention most widely across semantic vector space, whereas P adoption results in far more homogeneous consumption. Similarly, by modelling similarity between affordances, a step not yet taken in existing multi-affordance research, we show that the most distant, diverse regions of a platform’s informational landscape tend to be reached via human editorial curation. From a sociological perspective, this has significant repercussions: eclectic listening experiences may not be primarily algorithmically constructed but rather shaped by human curation, underscoring the potential power of these actors as new forms of cultural intermediaries and distillers of *eclecticism*.

In the final research article, we focused on understanding the impact of users’ selections on the discrete diversity of their consumption (i.e., novelty). This addressed RQ4: “*To what degree are users’ selections of algorithmic recommendations also responsible for inducing confined consumption practices?*”. Leveraging the combination of desired, exposed, and selected algorithmic diversity, rarely considered in unison within Music Information Retrieval (MIR) literature, this research makes a significant and novel contribution both to MIR and, more broadly, to aggregate algorithmic confinement literature by underscoring the role user agency plays within human–machine feedback loops and, more specifically, filter bubbles. In particular, one of the main novelties of this work lies in its articulation of declarative data with behavioural data. Within existing algorithmic confinement studies this rich combination of data has to date, rarely been exploited especially in terms of comparing declared preference data to consumption data stemming from actual on-platform interaction logs. Much like in studies of search engines and social media platforms in the context of politically charged information (Robertson et al., 2023; Bakshy et al., 2015), we show that Deezer users tend to select less diversity than they are algorithmically exposed to. Combining behavioural and declarative data reveals a striking behaviour–claim contradiction: when given the choice, users consistently lean toward exploitation when interacting with algorithmic systems, even when they express a desire for more diversity. Moreover, this pattern cannot be explained simply by general user tendencies to engage with either semantically similar or dissimilar recommendations. Taken together with the

earlier findings of this thesis, we see that the tendency for algorithmic affordances to foster high levels of novelty stems not from how users leverage these affordances (i.e., since their selections reduce novelty) but from how algorithms function by design and the types of recommendations they produce. This supports the hypothesis that music streaming users favour organic affordances over algorithmic ones because they seek higher levels of redundancy and exploitation in their consumption, a phenomenon that predates music streaming platforms and their integration into the everyday (Fuentes et al., 2019). At the same time, this work also offered valuable insights for practitioners developing platform intervention strategies. As we show, and in line with existing literature (Chen et al., 2024), increasing both dense and discrete diversity in algorithmic recommendations may not be the optimal strategy to nudge users toward more diverse consumption. Instead, users tend to respond by biasing their listening further toward exploitation. Put differently, and to borrow Pariser’s metaphor (Pariser, 2011), users may be more explorative when recommendations remain within their “bubble,” whereas pushing them outside may encourage retreat into familiar territory. Once again, these findings underscore the orthogonality of dense versus discrete diversity in online music consumption. This tension not only exists between users (Mok et al., 2022) and within users’ consumption across different affordance channels, but also in how users interact with algorithmic devices themselves. Given that algorithmic confinement measurements can vary greatly across scales, it is notable that this relationship emerges consistently across the contributions of this thesis.

Taken together, the scientific contributions of this thesis cascade to construct a significant challenge to existing techno-determinist narratives that have long shaped discourse in fields such as Human Computer Interaction and Media Studies. As we show, contemporary music streaming users are far from passive recipients of platforms and algorithms. At least in the case of Deezer, music streaming platforms are not commonly leveraged for passive information retrieval and moreover, programmed affordances which facilitate more passive information retrieval do not necessarily lead to more confined consumption practices. As we have demonstrated, they can in several contexts, yield greater levels of dense and discrete diversity which may we hypothesise, even have the potential to afford more *active* listening practices.

At the same time, this thesis clearly underscores the importance of adopting a multipolar perspective when examining algorithmic confinement dynamics on music streaming platforms. Dense and discrete diversity are orthogonal concepts not only *between* users but also *within* users’ own consumptions. This suggests that simple uni-dimensional “filter bubble” appraisals are insufficient to capture these dynamics and arguably, how users

experience diversity in practice. Finally, and perhaps more crucially, this research sheds light on the organic nature of discrete concentration in music streaming consumption: users consistently bias their selection of algorithmic recommendations towards redundancy, even when controlling for semantic similarity and even when expressing a desire for novelty. All things considered, by affirming both the organic agency of users and the multipolar nature of algorithmic confinement, this thesis provides a significant challenge to pervasive techno-determinist narratives and repositions the user as a central actor in human-machine feedback loops and algorithmic confinement processes.

Future Developments

While this thesis has addressed many empirically unexplored research questions at scale, several others remain largely unexamined, yet represent promising directions for future research. Whilst such is always the case for research under time and funding constraints, we nonetheless detail several of these most promising directions here.

First and foremost, the causal assessment of algorithmic confinement through counterfactual methods remains one of the most promising yet under explored directions for future research in multi-affordance contexts. As outlined in the literature review, isolating a counterfactual signal that captures user behaviour in the absence of algorithmic guidance (i.e., organic minus algorithmic, or $O \setminus A$) is essential for moving from descriptive accounts to claims of causal effect. However, achieving this is methodologically challenging. Randomly assigning users to treatment or control groups is largely infeasible in practice, as it would require platform-level interventions that risk user churn and are therefore unlikely to be implemented. In the absence of randomised experiments, researchers must rely on quasi-experimental methods that aim to approximate causal effects in naturalistic settings. While this thesis takes a step in that direction by applying a difference-in-differences (DID) approach to estimate the impact of novel algorithmic listening on subsequent organic novelty, such methods still assume the presence of the algorithm and thus do not fully capture a true $O \setminus A$ counterfactual. Constructing such a counterfactual remains significantly more complex. Nevertheless, a range of established causal inference techniques, such as instrumental variables, matching methods, and others (Liu et al., 2021; Bailey et al., 2024), offer promising avenues for multi-affordance research. While these tools may require methodological innovation to suit the specific challenges of streaming data and platform-mediated behaviour, they represent a critical next step for advancing causal understanding in this domain.

Similarly, an additional direction for future research involves exploring the temporal dynamics of diverse listening experiences and how these vary across different affordances and types of guidance. Understanding the longer-term effects of both engaging with and missing out on diversity could provide important insights into user behaviour and platform impact. While this thesis captures effects primarily at the within-session scale, future work could investigate how such effects evolve over time, potentially using methods such as lagged difference-in-differences or other longitudinal modelling approaches. This would help determine whether the influence of diverse listening sustains, diminishes, or compounds throughout a user's platform lifecycle.

Another promising direction for future research would be to integrate affordance adoption histories with user interviews to better understand what drives users to favour organic affordances in the majority of their platform interactions. While data-augmented interviews have previously been employed within the RECORDS project (Cura et al., 2022), they have not explored these specific themes. Pursuing this direction could provide a richer understanding of whether users themselves explicitly cite a desire for control as a core reason for predominantly engaging with organic affordances. Such insight would offer valuable context to the behavioural patterns observed in this thesis and help bridge empirical findings with user-reported motivations.

In a similar qualitative direction, another prominent path for future sociological works may lie in further understanding the practices of editorial curators. As we demonstrated within this thesis, editorial affordances may, we contend, act as new forms of cultural intermediaries which expose eclecticism. Nonetheless, we cannot yet claim that the music exposed by editorial affordances are devoid of algorithmic intervention. In practice, editorials can rely on algorithmic tools to monitor trends and user behaviour. Furthermore, the playlists themselves which are exposed via the platform's interface are inherently algorithmically tailored to be personalised. For instance, an avid hip-hop fan may see a 90s Hip Hop editorial playlist on their home screen but not a Rock playlist. A prominent direction for future works may thus lie in qualitatively understanding the practices of editorial curators and how much they rely on these kinds of devices in their practice.

Part IV

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Part V

Appendix

Résumé de la thèse en français

Introduction

« *Chaque format [musical] successif a moins de présence physique tout en permettant davantage de stockage et de plus grandes possibilités de programmation pour l'utilisateur* »

(McCourt, 2005, p. 249)

Contexte

Avec les plateformes de streaming musical représentant désormais plus des deux tiers des revenus mondiaux de la musique (International Federation of the Phonographic Industry (IFPI), 2025), leur influence sur la recherche et la consommation musicales ne doit pas être sous-estimée. Les plateformes de streaming musical offrent un accès sans précédent à des catalogues de musique en constante expansion : Spotify héberge à lui seul plus de 100 millions de titres (Spotify Newsroom, 2025), tandis que le service français Deezer en propose plus de 120 millions (Deezer, 2025). Plus de musique qu'il n'est possible d'écouter en une vie entière. Ainsi, le consommateur de musique d'aujourd'hui n'est plus contraint par la *rareté*, comme autrefois lorsque le marché était dominé par les ventes physiques. À l'inverse, en l'absence d'un design de plateforme approprié, il peut même être frappé par la *surcharge de choix* (Romero Meza and D'Urso, 2024).

Pour aider les utilisateurs à naviguer dans ces vastes entrepôts culturels en ligne, il est devenu courant que les plateformes intègrent diverses *affordances* programmées (Davis, 2020). Comme pour l'écoute de la radio traditionnelle, la musique mise en avant est généralement pré-programmée, organisée en listes thématiquement cohérentes et présentée sous une forme prête à être consommée. En même temps, les affordances programmées diffèrent de la radio traditionnelle sur un point crucial : la musique proposée est souvent

hautement personnalisée, façonnée par des systèmes algorithmiques exploitant les habitudes de consommation des utilisateurs. Des exemples typiques de ce type d’affordances incluent les dispositifs de recommandation algorithmique tels que le « Discover Weekly » de Spotify ou le « Flow » de Deezer, ainsi que des playlists éditoriales humaines ou hybrides diverses étiquetées « Top 50 Global », « RapCaviar » ou « Rock Essentials ». ²¹ Ces types d’affordances programmées sont en un sens de nouvelles formes d’intermédiaires culturels qui se sont développées et façonnées parallèlement à l’adoption massive des plateformes de streaming musical. Malgré un nombre croissant de travaux penchant vers l’idée que ces affordances, en particulier celles de nature algorithmique, pourraient favoriser une consommation passive (Hesmondhalgh, 2022; Prey, 2019) et confinée (Pariser, 2011), leur impact réel sur la consommation musicale *en pratique plutôt qu’en principe* demeure à ce jour peu documenté.

En parallèle, la littérature existante qui explore la manière dont les utilisateurs se servent en pratique des plateformes de streaming musical contemporaines indique que, pour la plupart des utilisateurs, les affordances programmées ne jouent pas un rôle marqué dans la recherche et la consommation musicales. En pratique, les plateformes de streaming musical n’exigent pas des utilisateurs qu’ils adoptent les affordances programmées. Ces plateformes sont relativement uniques en ce qu’elles offrent également plusieurs possibilités d’engagements musicaux *organiques* (c.-à-d. sans guidage programmé) et, de plus, il s’agit de l’un des modes d’accès les plus largement adoptés sur ces plateformes (Villermet et al., 2021; Beuscart et al., 2019). Par exemple, les playlists personnelles organiques permettent aux utilisateurs d’enregistrer et de revisiter leurs musiques préférées tandis que des fonctionnalités classiques comme la barre de recherche permettent d’effectuer des recherches organiques de morceaux, similaires à une recherche Google. En principe, un utilisateur peut passer tout son temps sur la plateforme sans jamais recourir aux affordances programmées.

En bref, les plateformes de streaming musical sont devenues des environnements *multi-affordances*, une sorte de « couteau suisse » offrant diverses manières de découvrir et d’écouter de la musique. Cette diversité dans le design, cruciale car *offerte* mais non imposée, ouvre, comme nous le démontrerons à travers les contributions de cette thèse, plusieurs opportunités méthodologiques nouvelles pour examiner et réévaluer les inquiétudes existantes autour du guidage algorithmique et, plus généralement, des plateformes de streaming musical. Cette thèse mobilise la combinaison unique des historiques d’écoute

²¹Le travail éditorial est d’ailleurs souvent appuyé par des outils algorithmiques (Eriksson, 2020), leur permettant de rester à jour avec les nouvelles sorties et tendances émergentes.

des utilisateurs de la plateforme Deezer, et notamment leurs adoptions et rejets de divers contenus issus d’interactions avec les affordances organiques, algorithmiques et éditoriales de cette plateforme. Sous cet angle, elle souligne à la fois l’agentivité organique des utilisateurs dans l’émergence d’expériences d’écoute confinées, ainsi que le rôle potentiellement diversifiant que peuvent, dans plusieurs cas, jouer les affordances programmées. Pris ensemble, les apports scientifiques de cette thèse représentent un défi majeur aux discours techno-déterministes existants en vigueur dans des domaines tels que l’Interaction Homme-Machine et les Études des Médias.

Objectifs de Recherche

En faisant progresser le champ émergent des études sur les plateformes de streaming musical multi-affordances (Anderson et al., 2020; Villermet et al., 2021; Epps-Darling et al., 2020; Mok et al., 2022) à travers l’analyse d’historiques d’usage multi-affordances rétrospectives et anonymisés issus de la plateforme française Deezer, la contribution de cette thèse est double. Premièrement, elle cherche à comprendre comment, en pratique, les plateformes de streaming musical multi-affordances sont adoptées par les utilisateurs finaux et quels types de pratiques de consommation et de recherche d’information elles facilitent. Cela constitue une contribution précieuse à la Sociologie de la Consommation et aux Études des Médias. Deuxièmement, cette thèse vise à illustrer comment le développement des plateformes multi-affordances et l’analyse de données granulaires multi-canaux dérivées des interactions des utilisateurs sur la plateforme peuvent fournir de nouvelles perspectives sur le débat récurrent autour des bulles de filtres (Michiels et al., 2022; Borgesijs et al., 2016; Pariser, 2011), en mettant en lumière la manière dont les utilisateurs se comportent à la fois lorsqu’ils adoptent et en l’absence relative de dispositifs algorithmiques.

À cette fin, nous centrons cette thèse autour des questions de recherche suivantes.

- **(RQ1)** Comment les utilisateurs adoptent-ils les plateformes de streaming musical et, plus précisément, comment cela varie-t-il selon les différentes affordances qu’elles incarnent ?
- **(RQ2)** Dans quelle mesure les données d’interaction multi-affordances utilisateur-plateforme peuvent-elles fournir de nouvelles perspectives sur le rôle des dispositifs de recommandation algorithmique personnalisée dans l’émergence de pratiques de consommation confinées ou diversifiées ?

- **(RQ3)** Dans quelle mesure les choix concernant l'échelle et le niveau de représentation de l'information influencent-ils fondamentalement les conclusions tirées de telles analyses ?
- **(RQ4)** Dans quelle mesure la manière dont les utilisateurs adoptent, ou rejettent, les recommandations algorithmiques, conduit à des pratiques de consommation confinées ou non ?

Aborder chacune de ces questions de recherche permet de développer une compréhension nuancée de l'interaction entre le design algorithmique et la consommation culturelle, constituant le fondement de cette thèse.

Contributions Scientifiques

Les recherches présentées dans cette thèse ont été réalisées dans le cadre du projet financé par l'ANR, « RECORDS » ²² qui, aux côtés de l'un de ses principaux partenaires Deezer, a facilité l'accès confidentiel aux historiques d'écoute de Deezer qui constituent la base d'une grande partie des analyses menées ici. Dans le cadre d'une affiliation conjointe avec le Centre d'analyse et de mathématique sociales (CAMS), le Centre national de la recherche scientifique (CNRS) et le Centre Marc Bloch (CMB), les recherches ont été effectuées en grande partie au sein de l'équipe de Sciences Sociales Computationnelles du CMB ²³ et encadrées par Camille Roth. Le financement de la thèse a été assuré par l'équipe via un mélange de fonds ANR provenant de la subvention "RECORDS" (ANR-2019-CE38-0013) et d'autres financements obtenus par l'équipe.

La contribution de cette thèse repose sur quatre articles académiques. La thèse étant fondée sur des articles, les chapitres sont principalement construits à partir de ces publications.

Le premier article répond à la RQ1 et a été publié lors de la Conférence de l'International Society of Music Information Retrieval en 2021. Il propose une exploration préliminaire de la manière dont différentes affordances attirent des dynamiques d'engagement variées et, de plus, de leur interdépendance.

Shakespeare, Dougal, et Camille Roth. « *Tracing Affordance and Item Adoption on Music Streaming Platforms.* » Proceedings of the 22nd International Society

²²<https://records.huma-num.fr/>

²³<https://cmb.huma-num.fr/>

for Music Information Retrieval Conference. 2021. (Shakespeare and Roth, 2021)

Le deuxième article, publié à CompleNet en 2024, présente ensuite une analyse plus théorique de l’interprétabilité des espaces de plongement d’objets en haute dimension qui seront déployés par la suite pour construire des mesures de concentration dense. Ce chapitre, bien que ne se rapportant pas directement à nos principales questions de recherche, constitue un prérequis nécessaire pour mener nombre des analyses suivantes en établissant que les principaux outils utilisés pour capturer la diversité sont solides et interprétables.

Shakespeare, Dougal, et Camille Roth. « *Interpreting Node Embedding Distances Through n -Order Proximity Neighbourhoods*. » Complex Networks XV. CompleNet-Live 2024. Springer Proceedings in Complexity. Springer, 2024. (Shakespeare and Roth, 2024) ²⁴

Le troisième article, publié dans *Scientific Reports* de Nature en 2025, répond aux RQ2 et RQ3 et se concentre sur une analyse approfondie de la manière dont les utilisateurs adoptent les affordances des plateformes de streaming musical pour atteindre des niveaux variés de concentration et, de plus, comment cela varie en fonction du type de représentation de l’information et de l’échelle mobilisée.

Shakespeare, Dougal, Victor Chareyron et Camille Roth. « *Reframing the Filter Bubble Through Diverse Scale Effects in Online Music Consumption*. » Scientific Reports, 2025. (Shakespeare et al., 2025)

Le dernier article de cette thèse adopte une approche mixte et répond principalement à la RQ4. Cette recherche s’appuie sur des données d’enquête riches et à grande échelle issues du projet RECORDS (Renisio et al., 2024) afin d’éclairer la manière dont les utilisateurs varient typiquement dans leur ouverture – capturée à travers des comportements sur la plateforme et des attitudes hors plateforme – à la nouveauté dans les recommandations musicales générées algorithmiquement. Cette recherche présentée dans la thèse est actuellement disponible sous forme de pré-print. Elle est prévue pour soumission à la Conférence internationale World Wide Web 2026.

Shakespeare, Dougal, Camille Roth. « *Tracing Desired, Exposed and Adopted Algorithmic Novelty on Music Streaming Platforms* », Pré-print, 2025.

²⁴Reproduit avec l’autorisation de Springer Nature.

Revue de littérature

La question des affordances sur les plateformes de streaming musical?

« *Des quantités stupéfiantes de musique sont créées et produites chaque année, et la grande majorité n'est pas destinée à une écoute attentive.* »

(Kassabian, 2013, p. 1)

Comme première étape vers la contextualisation des contributions de cette thèse, nous centrons notre attention sur la compréhension des types d'interactions que les plateformes de streaming musical *affordent* et, de plus, sur celles qui sont plus ou moins courantes. Nous commençons par une revue de littérature plus théorique sur les *affordances*, une terminologie et un cadre que nous utiliserons tout au long de cette thèse pour désigner les différentes opportunités d'écoute que les plateformes de streaming musical offrent mais n'imposent pas à l'utilisateur. À cette fin, nous retraçons la trajectoire de la théorie des affordances à travers différents champs académiques, jusqu'à ses applications à la technologie et à l'étude des plateformes. En nous appuyant sur ces fondations, nous décrivons ensuite la littérature sur deux formes distinctes d'interactions que l'on attribue aux plateformes de streaming musical : différentes formes de recherche d'information et différentes formes d'attention. Pour cela, nous nous concentrons sur la compréhension de ces types d'interactions à travers une dichotomie *passif-actif*, à la fois bien établie dans la littérature existante en sciences sociales et en études des médias sur le comportement des consommateurs de musique, et suffisamment large pour englober des études plus quantitatives qui mettent en lumière la manière dont les utilisateurs se servent des plateformes en pratique plutôt qu'en principe. Ensemble, ces deux perspectives offrent un cadrage distinct mais complémentaire qui met en évidence la manière dont les plateformes de streaming musical sont de plus en plus ancrées dans le *quotidien* des utilisateurs et le degré auquel ceux-ci sont disposés à renoncer au contrôle *organique* au profit d'un guidage externe, qu'il soit humain ou algorithmique.

Le cadre conceptuel des affordances

Le cadre associé aux *affordances*, bien qu'aujourd'hui solidement établi dans les Études des Médias et l'Interaction Homme-Machine, a été introduit pour la première fois par le psychologue cognitif et écologique J.J. Gibson dans son ouvrage « *The Senses Considered as Perceptual Systems* » (Gibson, 1966) et développé plus en profondeur dans ses travaux

ultérieurs (Gibson, 2014b). Si le verbe *to afford* existait avant la formulation de Gibson, le terme « affordance » fut créé pour combler ce que Gibson considérait comme un vide sémantique. Plus précisément, et selon ses propres mots :

« Les affordances de l'environnement sont ce qu'il offre à l'animal, ce qu'il lui procure ou lui fournit, pour le meilleur ou pour le pire. Le verbe to afford se trouve dans le dictionnaire, mais le nom affordance n'y figure pas. Je l'ai inventé. J'entends par là quelque chose qui renvoie à la fois à l'environnement et à l'animal d'une manière qu'aucun terme existant ne permet. Cela implique la complémentarité de l'animal et de l'environnement. » (Gibson, 2014a, p. 119)

En termes simples, Gibson décrit une affordance comme un construit relationnel qui capture les possibilités d'action qu'un environnement (par ex. une plateforme) offre à un agent (par ex. un utilisateur). Selon Gibson, celles-ci existent indépendamment des perceptions des agents (c.-à-d., le fait qu'aucun agent ne les perçoive n'en nie pas l'existence). Par un exemple classique plus concret, une porte peut *afforder* l'action de tirer ou pousser, ce qui peut être médié par son design (par ex. la conception de la poignée). De la même manière, les plateformes de streaming musical *affordent* différents types de guidage et d'écoute (par ex. issus de dispositifs algorithmiques, de la recherche, de playlists éditoriales, etc.), qui peuvent eux aussi être influencés par la manière dont la plateforme est conçue. La théorie des affordances capture le fait que « *technologies produce fields of action (including unexpected actions), but that not all actions are possible* » (Jordan, 2012, p. 302). Ce cadrage situe la théorie des affordances de Gibson entre le déterminisme technologique et le constructivisme social – une position intermédiaire équilibrée souvent citée comme l'une des forces de cette théorie (Hutchby, 2001; Scarlett and Zeilinger, 2019).

Suite au développement des affordances par Gibson, la théorie a connu une certaine dérive sémantique lorsqu'elle a été étudiée plus largement par les chercheurs, d'abord dans le champ du Design (Norman, 1988, 1999; Davis, 2020), puis plus tard dans l'Interaction Homme-Machine (Gaver, 1991) et les Études des Médias (Scarlett and Zeilinger, 2019). Concernant le premier champ, peut-être le développement le plus influent de la théorie des affordances provient du travail de Norman (Norman, 1988, 1999). Contrairement au cadrage théorique de Gibson, parfois considéré comme plus flou, Norman adopte une perspective plus pratique, centrée sur le design, définissant une affordance comme "*the perceived and actual properties of the thing, primarily those fundamental properties that determine just how the thing could possibly be used*" (Norman, 1988, p. 9). Plus précisé-

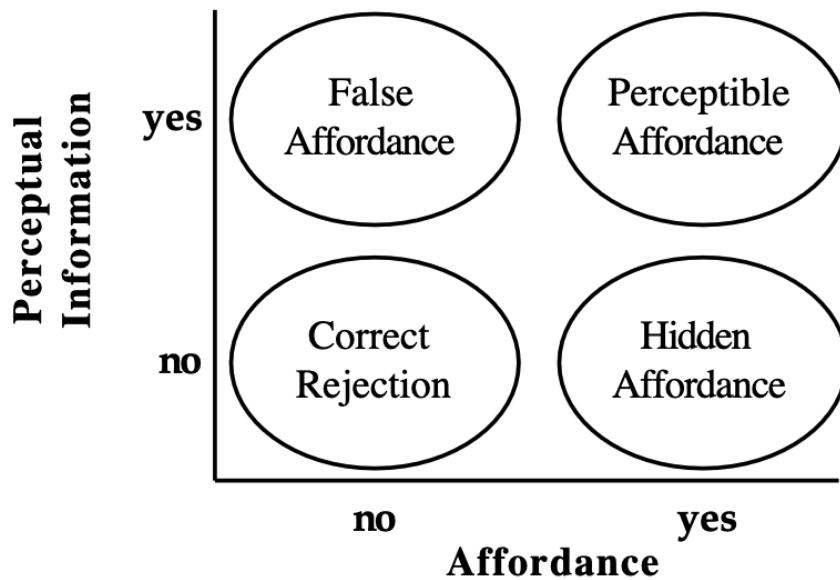


Figure 4: Distinction entre affordances perçues et actionnables telle que définie dans le travail de Gaver (1991)

ment :

« Les affordances donnent de fortes indications sur la manière dont les choses fonctionnent. Les plaques sont faites pour être poussées. Les boutons sont faits pour être tournés. ... Lorsque les affordances sont exploitées, l'utilisateur sait quoi faire simplement en regardant : aucune image, étiquette ou instruction n'est nécessaire. » (Norman, 1988, p. 9)

Norman soutient en effet que les affordances peuvent être développées *par design*, ce qui met davantage l'accent sur les affordances perçues. Cela marque un départ significatif de la formulation originale de Gibson qui positionnait les affordances comme indépendantes des perceptions des agents.

Depuis Norman, de nombreux chercheurs ont entrepris de reformuler le cadre des affordances, en particulier dans le but d'étudier les technologies et, plus récemment, les plateformes en ligne. À noter, Gaver (1991) développent l'idée de « *Technology Affordances* » en mettant l'accent sur les affordances dans le contexte des interfaces utilisateurs. Comme illustré dans la Figure 4, Gaver distingue non seulement les affordances qui sont ou non perçues – la distinction clé entre Norman et Gibson – mais aussi celles qui sont correctement ou incorrectement perçues, ajoutant ainsi plus de nuance au cadre des affordances. De manière similaire, Hutchby (2001) développe en profondeur la notion d'affordances en lien avec les technologies, sous un angle plus sociologique. Plus récemment, et directement lié à l'étude des effets du guidage algorithmique, Panigutti et al.

(2025) montrent que les études d’audit algorithmique visant à comprendre les effets du design d’une interface sont en fait souvent structurées autour de l’analyse des résultats des affordances technologiques.

Dans le contexte des *platform studies*, utiliser le cadre des affordances permet non seulement de reconnaître les propriétés matérielles des plateformes mais aussi l’agence des utilisateurs – une dualité qui échappait souvent aux tentatives contemporaines de décrire les interactions homme-machine sur plateformes (Davies, 2018; Michiels et al., 2022; Pariser, 2011). Dans cette thèse, nous adoptons le terme *affordance* justement parce qu’il capture les diverses manières dont les utilisateurs interagissent avec les plateformes tout en mettant en avant le rôle de l’agence utilisateur dans la construction de ces interactions et comportements en aval. Plus précisément, nous utilisons le terme *affordance* pour décrire les différents moyens par lesquels les plateformes permettent aux utilisateurs d’écouter de la musique. Comme illustré dans la Figure 5, il existe trois principales affordances qui peuvent être adoptées pour faciliter l’écoute et la découverte musicale sur les plateformes comme Deezer : *organique* (*O*), *algorithmique* (*A*) et *éditoriale* (*E*) (Villermet et al., 2021; Beuscart et al., 2019; Louail and Barthelemy, 2017). Ces affordances sont généralement positionnées de manière proéminente sur les écrans d’accueil des plateformes. De ce fait, nous soutenons que les utilisateurs *perçoivent* généralement ces mécanismes comme disponibles, même s’ils ne les adoptent pas activement. En ce sens, notre usage du terme *affordance* s’aligne à la fois sur les formulations de Gibson et de Norman puisque l’ensemble des affordances perçues approche l’ensemble des affordances disponibles, dissolvant de fait la distinction ontologique entre affordances perçues et réelles dans le contexte de cette thèse.

Streaming musical et Passive Information Retrieval

Les plateformes modernes de streaming musical hébergent une abondance sans précédent de musique, avec des catalogues si vastes qu’aucun auditeur ne pourrait en consommer plus qu’une fraction au cours d’une vie. Face à cette profusion, ces plateformes sont conçues pour guider les utilisateurs vers la musique qu’ils sont susceptibles d’apprécier, réduisant ainsi la *surcharge de choix* (Romero Meza and D’Urso, 2024). Pour ce faire, les plateformes proposent couramment diverses formes d’*affordances programmées* (Anderson et al., 2020), telles que les systèmes de recommandation algorithmique et les playlists éditoriales humaines, qui aident les utilisateurs à découvrir de nouvelles musiques en accord avec leurs goûts. En même temps, les plateformes n’imposent pas aux utilisateurs

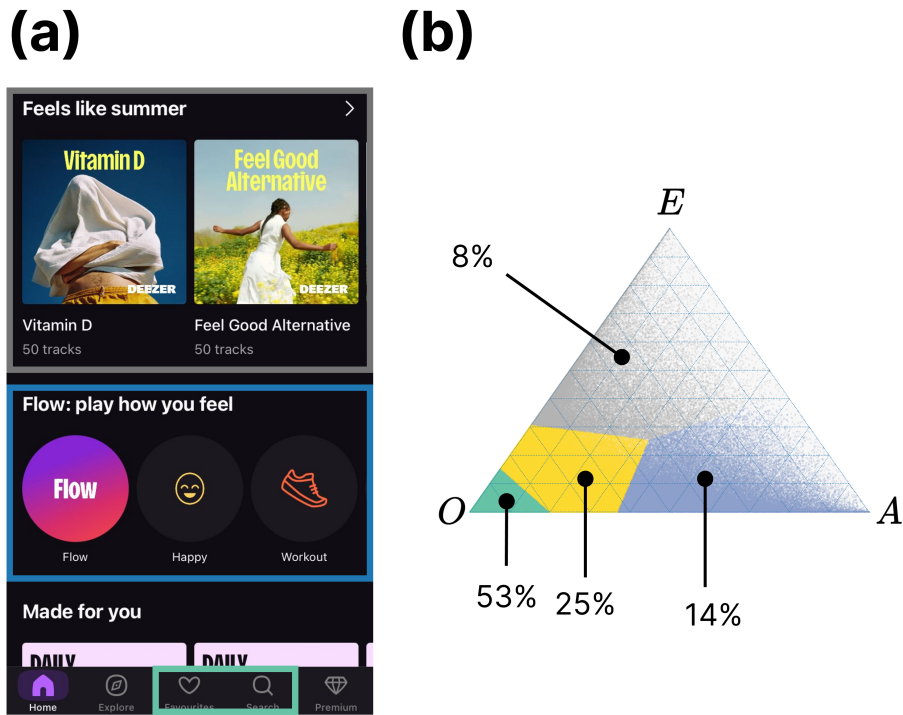


Figure 5: **Affordances sur la plateforme Deezer.** La figure (a) montre l'écran d'accueil actuel de Deezer, qui propose différentes affordances *organiques* (O , vert), *algorithmiques* (A , bleu) et *éditoriales* (E , gris) pour écouter de la musique. La figure (b) montre la distribution d'environ 500,000 utilisateurs aléatoires de Deezer dans un espace ternaire, en fonction de la proportion moyenne de leurs écoutes provenant des affordances O , A et E (voir les Matériaux Supplémentaires pour plus de détails). Les frontières sont tracées à l'aide d'un algorithme k-means ($k = 4$).

teurs d'adopter ces guidages programmés ; ils peuvent également écouter de la musique de manière *organique* (c.-à-d., en relative absence de guidage programmé) à travers des affordances comme la barre de recherche, leurs propres playlists personnelles ou encore en parcourant manuellement les discographies d'artistes. En termes simples, les plateformes de streaming musical se sont développées non seulement verticalement (c.-à-d., en profondeur, en se spécialisant sur une fonction), mais aussi horizontalement (c.-à-d., en largeur, en se diversifiant à travers de nombreuses fonctions), de sorte qu'il n'existe plus une seule façon d'accéder au contenu sur plateforme.

Bien que les affordances programmées simplifient la navigation dans les vastes catalogues des plateformes, elles rendent aussi, par construction, les utilisateurs moins maîtres de leurs choix et donc plus passifs dans les processus de recherche d'information. Par exemple, lorsqu'ils utilisent un dispositif algorithmique comme le Flow de Deezer, les utilisateurs ne contrôlent pas directement la recherche ; les morceaux sont sélectionnés automatiquement et il n'est pas possible de choisir directement les recommandations suivantes dans une liste préétablie. À l'inverse, lorsqu'ils consomment de la musique via leurs playlists personnelles organiques, les utilisateurs contrôlent la sélection, à la fois à un

niveau sémantique plus général (par ex. choisir parmi leurs playlists par genre ou par humeur) et au niveau des morceaux, mais aussi dans la construction même de la playlist. De même, les playlists éditoriales se situent entre les deux – les utilisateurs ne contrôlent pas les morceaux inclus dans la playlist mais ils choisissent à un niveau sémantique (par ex. consommer « 90's Boom-Bap », « 80s Throwback ») et au niveau des morceaux. En somme, les plateformes modernes de streaming musical offrent une grande diversité de moyens pour rechercher et écouter de la musique, chacun de ces moyens externalisant le processus de recherche d'information à des degrés divers. Néanmoins, il n'est pas nécessairement *exigé* que les utilisateurs se servent de la plateforme d'une manière particulière. Étudier l'adoption relative d'affordances plus programmées ou plus organiques offre donc un moyen original et fructueux d'analyser dans quelle mesure les consommateurs modernes de musique souhaitent garder le contrôle lorsqu'ils écoutent de la musique en ligne.

Alors que plusieurs chercheurs ont exprimé des inquiétudes sur le fait que les plateformes de streaming musical encouragent les consommateurs à externaliser passivement « *the creation, maintenance, and storage of their music collections* » (Morris, 2015, p. 168), les recherches empiriques existantes sur le comportement réel des utilisateurs sur plateforme suggèrent que ces craintes peuvent être surestimées (Beuscart et al., 2019; Villermet et al., 2021). Par exemple, sur Deezer, les flux qualifiés d'organiques – résultant des interactions des utilisateurs avec des affordances comme leurs playlists personnelles ou la barre de recherche – constituent la majorité de l'activité, aussi bien à l'échelle de la plateforme (Beuscart et al., 2019) qu'à celle de l'utilisateur moyen (Villermet et al., 2021). Des effets similaires ont également été observés sur Spotify. Dans un rapport interne produit par Spotify en 2018, seulement 31% de tous les flux étaient attribués aux affordances programmées telles que les dispositifs de recommandation algorithmique et les playlists éditoriales (United States Securities and Exchange Commission, 2018) – le reste étant présumément organique. Pris ensemble, la littérature existante suggère que, sur plusieurs des plateformes de streaming musical les plus répandues, les utilisateurs montrent souvent une préférence pour une recherche d'information active et contrôlée.

Pourquoi les utilisateurs de streaming musical préféreraient-ils garder le contrôle de la recherche d'information et, ainsi, adopter fréquemment les affordances organiques ? À quels types de comportements ce contrôle permet-il d'accéder ? Une explication possible, appuyée par la littérature existante, pourrait se situer dans le niveau de consommation exploitative, que l'on peut désigner par « concentration discrète » puisqu'elle consiste en une consommation répétée des mêmes items, et que les affordances organiques facilitent. Bien que nous n'entrons pas encore en profondeur dans cette littérature – une revue plus

détaillée sera présentée plus loin – il reste pertinent d’en donner déjà une vue d’ensemble pour commencer à aborder ces questions. Beuscart et al. (2019) montrent qu’une fraction significative des flux organiques d’un utilisateur Deezer provient généralement de ses propres playlists personnelles et que, par construction, ces flux reflètent donc probablement des interactions avec de la musique déjà connue. De même, Hosey et al. (2019) et Li et al. (2019) montrent que les utilisateurs initient souvent des recherches sur les plateformes de streaming musical avec un morceau spécifique en tête, ce qui suggère que le contenu ainsi accédé a souvent déjà été consommé, peut-être même sur la plateforme. Villermet et al. (2021) montrent que les flux organiques tendent à être plus redondants que ceux issus des affordances algorithmiques ou éditoriales, renforçant encore cette hypothèse. De même, Beuscart et al. (2019) trouvent que les morceaux découverts via des affordances organiques sont plus susceptibles d’être ré-exploités que ceux découverts via des affordances algorithmiques sur Deezer. Pris ensemble, cet ensemble de travaux suggère que la recherche organique, contrôlée par l’utilisateur, pourrait jouer un rôle distinct dans l’encouragement à des formes d’engagement redondantes et exploitatives. Ce recours à de telles affordances pourrait signaler une orientation plus large vers l’exploitation dans les pratiques de streaming musical, rappelant des usages associés aux médias précédents comme les iPods et les lecteurs MP3.

En résumé, bien que les recherches publiques sur les préférences des utilisateurs pour le maintien d’un contrôle actif de la recherche d’information sur les plateformes de streaming musical restent limitées, la littérature existante pointe de manière consistante dans la même direction. Lorsque le choix leur est donné, les utilisateurs tendent à s’engager principalement avec des affordances organiques, suggérant qu’ils souhaitent rester activement impliqués dans leurs pratiques d’écoute (Villermet et al., 2021; Beuscart et al., 2019; United States Securities and Exchange Commission, 2018). Cela remet en question les récits techno-déterministes existants qui dépeignent les utilisateurs comme des récepteurs passifs du guidage algorithmique en ligne (Davies, 2018) et souligne que, du moins sur les plateformes de streaming musical, les utilisateurs peuvent et exercent une véritable agence sur l’adoption de ces dispositifs. Cela renforce une fois de plus notre choix d’utiliser la « théorie des affordances » pour décrire les facettes de la recherche et de l’écoute offertes par une plateforme.

Bien que cette littérature constitue une base solide, elle bénéficierait d’une compréhension plus approfondie des interdépendances entre affordances. Par exemple, si la playlist personnelle d’un utilisateur se compose en grande partie de morceaux découverts grâce à des recommandations algorithmiques, il devient plus complexe d’évaluer si l’usage de

telles affordances traduit une préférence pour une recherche d'information active ou passive. Cette thèse vise à combler cette lacune en examinant comment les items circulent entre affordances, ce qui permet de mieux clarifier ce que signifie réellement l'engagement organique. Ce faisant, elle cherche à contribuer à une compréhension plus nuancée de l'agence des utilisateurs dans la consommation musicale médiée par plateforme.

Streaming musical et écoute passive

Une autre question étroitement liée à la manière dont les utilisateurs s'approprient et s'engagent avec les plateformes de streaming musical concerne la perception plutôt que le comportement. Lorsque les utilisateurs adoptent des plateformes comme Spotify ou Deezer pour écouter de la musique, dans quelle mesure les utilisent-ils pour écouter activement ? La musique est-elle couramment placée au centre de l'attention, à l'avant-plan, ou au contraire, dans quelle mesure a-t-elle tendance à se fondre dans l'arrière-plan et à servir de stimulus complémentaire à une autre activité ? Dans quelle mesure les plateformes permettent-elles et même favorisent-elles ces pratiques ? Pour répondre à ces questions, nous commençons par revisiter les différentes manières dont les pratiques d'écoute musicale ont été conceptualisées et catégorisées, avant même le passage de la musique enregistrée du format physique au numérique puis aux plateformes. Cela permet d'adopter une perspective plus holistique sur les inquiétudes contemporaines selon lesquelles les plateformes de streaming musical favoriseraient une écoute passive (Prey, 2019).

Dans les sciences sociales, les pratiques d'écoute des consommateurs de musique ont depuis longtemps été caractérisées de manière minutieuse à travers une dichotomie *passive* vs *active* (Hesmondhalgh, 2022). Une grande partie de cette littérature remonte au philosophe et musicologue allemand Theodor Adorno, qui a décrit en détail les consommateurs selon le degré d'attention accordé à l'écoute et tenté de comprendre les forces à l'origine de pratiques d'écoute différentes. Cela est illustré de manière exemplaire dans sa publication de 1945 « *A Social Critique of Radio Music* » (Adorno, 1945), dans laquelle il classe les consommateurs sous la dichotomie actif-passif pour théoriser sur les effets de la standardisation croissante de la production musicale sur l'écoute. Selon Adorno, à mesure que les sociétés passaient de systèmes féodaux à des systèmes industriels, la musique, elle aussi, passait d'une fonction sociale à un bien de consommation produit en masse. En termes simples, la production culturelle serait soumise aux mêmes logiques de standardisation et d'efficacité que la production matérielle, ce qui, au fil du temps, conduirait à une augmentation de l'écoute passive, appelée *commodity listening*, conçue « *supprimer autant que*

possible tout effort de la part du récepteur » (Adorno, 1945, p. 231). Les développements technologiques facilitant la diffusion de la musique à grande échelle, comme la radio à l'époque, renforceraient ce processus. Bien que ces dispositifs donnent aux consommateurs l'illusion du choix, en pratique, selon Adorno, les curation proposées sont souvent prévisibles, formulées et standardisées, favorisant ainsi une écoute passive, distraite, en arrière-plan. Ce raisonnement rappelle fortement les critiques contemporaines qui avancent que les dispositifs de recommandation algorithmique et les plateformes fonctionnent de manière similaire (Seaver, 2019; Prey, 2019).

Comme le notent Hesmondhalgh (2022), une lecture de « *A Social Critique of Radio Music* » offre plusieurs enseignements encore précieux aujourd'hui, utiles pour contextualiser la littérature contemporaine sur les liens entre plateformes de streaming musical, leurs affordances et l'attention des auditeurs. D'abord, ce texte montre que les inquiétudes concernant les avancées technologiques destinées à diffuser la musique en masse, et plus précisément la crainte qu'elles encouragent une écoute passive, ne sont pas propres aux plateformes de streaming. Ensuite, il suggère aussi que si les inquiétudes contemporaines sont largement influencées par une pensée adornienne, elles peuvent être infondées. Bien que le texte fournisse des descriptions intéressantes sur la marchandisation croissante de la musique, les affirmations plus causales concernant l'effet sur l'attention des auditeurs demeurent peu solides (Sterne, 2003; Hesmondhalgh, 2022). Par exemple, les genres considérés comme exigeant une écoute attentive se sont révélés être fortement liés à la compétence, à la modernité et à la classe sociale (Bourdieu, 1984), soulevant des questions sur la mesure dans laquelle la lecture causale d'Adorno était fortement biaisée par sa propre position sociale. Ainsi, en tant que chercheurs, nous devons être particulièrement prudents face aux travaux affirmant que les plateformes de streaming musical *causent* une écoute passive. De telles affirmations peuvent simplement reproduire des inquiétudes anciennes qui n'étaient pas fondées à l'origine (Hesmondhalgh, 2022).

En tenant compte de cette mise en garde, les recherches contemporaines sur l'écoute passive dans le contexte du streaming musical évitent généralement les affirmations causales larges sur les effets de la standardisation de la production sur l'attention. À la place, elles se concentrent de plus en plus sur la manière dont la musique s'intègre dans le *quotidien* (DeNora, 2000), et sur la façon dont les plateformes de streaming permettent et soutiennent ce mode d'engagement parallèle, parfois en arrière-plan. Cette littérature, couvrant la sociologie (DeNora, 2000), les études des médias (Pedersen, 2020; Prey, 2019) et le marketing (Fuentes et al., 2019), se connecte directement à l'étude de l'écoute active/passive puisque, par extension, elle analyse les scénarios où la musique n'est pas forcément le

seul centre d'attention. Fuentes et al. (2019) étudient les pratiques d'écoute fondées sur les activités de 15 consommateurs suédois, qu'ils nomment « *Soundtracking* ». Par le biais d'entretiens, les auteurs identifient l'écoute fondée sur l'activité comme la fonction principale des plateformes de streaming musical.²⁵ Des résultats similaires ont aussi été observés aussi bien aux débuts des plateformes (Schedl et al., 2015) qu'avant leur adoption massive (Kamalzadeh et al., 2012; North et al., 2004).

L'analyse du design des dispositifs de recommandation algorithmique des plateformes fournit également des éléments allant dans le sens de l'idée que les plateformes de streaming sont souvent utilisées pour l'écoute liée aux activités. Emblématique du « *contextual turn* » dans la conception des systèmes de recommandation au cours de la dernière décennie (Pagano et al., 2016), l'équipe R&D de Deezer a consacré des efforts importants au développement d'embeddings utilisateurs destinés à capturer les contextes et activités d'écoute, afin de les déployer dans de futurs systèmes de recommandation (Marey et al., 2024). De même, depuis 2021, le dispositif de recommandation « Flow » de Deezer est capable d'utiliser des embeddings de morceaux liés à l'humeur (Deezer, 2023), permettant ainsi aux utilisateurs d'adapter les recommandations non seulement à des activités et contextes spécifiques (par ex. morceaux dynamiques pour courir) mais aussi à leurs états psychologiques propres. Ces développements font écho aux conclusions de Pedersen (2020), qui soutiennent que les plateformes de streaming recommandent la musique de façon à « *amplifier et encourager les pratiques d'écoute ubiquitaire des utilisateurs* ». Dans l'ensemble, cette littérature souligne la prévalence de l'écoute fondée sur l'activité et le rôle que jouent les plateformes de streaming pour la faciliter.

Ces dernières années, plusieurs chercheurs ont également commencé à établir plus concrètement des liens entre la capacité des plateformes de streaming à favoriser l'écoute fondée sur l'activité et l'écoute inattentive (Kamalzadeh et al., 2012; Pedersen, 2020; Prey, 2019), aussi appelée « *lean-back listening* » (Pelly, 2018). En particulier, Prey (2019) discute du lien entre l'externalisation de la recherche musicale vers les plateformes et l'écoute passive. Prey décrit comment le design des plateformes a évolué pour devenir de plus en plus fondé sur la curation, reposant largement sur des affordances basées sur les playlists, mises en avant dans l'image de marque et l'interface des plateformes. Ce changement est dit orienter l'exposition musicale (et parfois la production) de plus en plus autour des humeurs et des activités. Comme la musique est adaptée au consommateur, plutôt que l'inverse, l'écoute serait impliquée comme devenant moins attentive et

²⁵Cette pratique peut être favorisée par la nature *in-pocket*, portable, des appareils couramment utilisés pour accéder aux plateformes de streaming (aujourd'hui les smartphones) – sur Deezer, deux tiers des flux proviennent de l'écoute mobile (Beuscart et al., 2019).

donc plus passive. Bien que de tels arguments puissent sembler intuitifs – notamment au vu des théories cognitives sur les goulots d’étranglement attentionnels (Tombu et al., 2011) – les preuves empiriques les soutenant demeurent limitées. À ce jour, la grande majorité des travaux existants reposent soit sur des entretiens à petits échantillons avec des consommateurs, soit sur des revues de littérature citant souvent d’autres travaux non empiriques. Si l’écoute fondée sur l’activité peut transformer la musique en activité de soutien induisant une écoute plus inattentive, une activité peut tout autant soutenir une expérience d’écoute. En même temps, l’idée selon laquelle l’attention des utilisateurs de streaming est plus souvent passive est aussi en tension avec la littérature empirique existante sur les comportements de saut de morceaux. Dans un rapport fondé sur les données publiques publié par Spotify en 2014, environ 50% de tous les morceaux joués étaient passés (Whitman, 2014), ce qui implique que les utilisateurs gardent bel et bien *le contrôle* des plateformes et effectuent des évaluations actives.

En résumé, l’examen de la littérature qualitative et du design des plateformes de streaming musical fournit de solides indices selon lesquels ces plateformes sont couramment utilisées en parallèle avec d’autres activités. Les appareils couramment utilisés pour héberger les plateformes (les smartphones) facilitent inévitablement cette pratique. Néanmoins, la mesure dans laquelle cela se traduit par une écoute passive reste discutable, malgré certains chercheurs qui théorisaient en ce sens (Prey, 2019). Il est possible, comme le soutiennent Hesmondhalgh (2022), que cela ne soit que la reproduction d’*anciennes inquiétudes*. Tester empiriquement de telles affirmations, bien que hors de portée de cette thèse, reste un défi majeur pour la communauté Music Information Retrieval dans les années à venir.

L’hypothèse de la bulle de filtrage

« *Les métaphores peuvent être véritablement utiles pour aborder un ensemble complexe de problématiques, mais seulement si elles sont bien choisies, bien expliquées, et qu’elles n’entravent pas une analyse plus approfondie par leur simplicité intuitive apparente.* »

(Bruns, 2019, p. 8)

Dans la section suivante, nous nous concentrons sur l’exposé de l’hypothèse de la bulle de filtres telle que formulée par Pariser (2011). Nous abordons ensuite des lectures critiques plus récentes de cette théorie avant de proposer notre propre définition du *confinement algorithmique*, utilisée dans les sections suivantes de cette revue de littérature.

Présentation de l'hypothèse

Les recherches critiques sur les « nouveaux médias » – tels que la radio, le cinéma, la télévision et aujourd’hui le streaming musical – sont profondément influencées par le climat sociopolitique dans lequel ils sont produits. En période de relative stabilité, par exemple l’après-guerre où la « *société de consommation* » s’est développée (Baudrillard, 1998), les recherches tendent à être moins critiques, tandis qu’en période d’instabilité elles adoptent un regard plus critique, cherchant souvent à mettre en évidence de forts effets des médias dès le départ (Silverstone, 2005). Cette perspective offre un contexte utile pour comprendre le flux apparemment incessant d’universitaires (Brinkmann et al., 2023), de décideurs politiques (The New York Times, 2017), et de médias généralistes (Levy, 2023) qui, ces dernières années, ont affirmé, souvent avec des intentions très différentes, que les réseaux sociaux et plus récemment les dispositifs algorithmiques des plateformes de streaming contraignent plutôt qu’ils ne favorisent la diversité des pratiques de communication et de consommation.

À première vue, cette inquiétude peut sembler récente, mais ses origines remontent aux débuts d’Internet. En 1995, Negroponte (2015) prévenait que le web produirait inévitablement « *une sorte de journal imprimé dans une édition unique* » et, de manière similaire, Sunstein (2006) arguait que le web isole les utilisateurs dans des « *cocons informationnels* ». Ces concepts ont contribué à façonner le ton et le contenu de l’ouvrage très cité du journaliste et militant Internet Eli Pariser, « *The Filter Bubble: What the Internet Is Hiding from You* » (Pariser, 2011), publié en 2011, qui reste à ce jour l’une des théories les plus citées concernant le rôle du guidage algorithmique en ligne dans le renforcement de pratiques de consommation confinées. Paru peu après que Google a pour la première fois intégré la personnalisation algorithmique dans son moteur de recherche (Areeb et al., 2023), la thèse de Pariser affirme que les dispositifs algorithmiques personnalisés en ligne exposent les utilisateurs à une information pauvre en diversité, limitant ainsi la mesure dans laquelle ils accèdent à des contenus variés comme, par exemple, des articles de presse issus d’un bord politique opposé ou des genres musicaux extérieurs à leurs préférences. Cette notion est ce que Pariser appelle désormais célèbrement la *filter bubble*. Plus précisément, selon ses principes, le flux d’information entre utilisateurs et algorithmes fonctionne comme suit dans un environnement en ligne. Les dispositifs algorithmiques filtrent l’information pour générer des recommandations personnalisées. Les utilisateurs appliquent ensuite un deuxième filtre en choisissant quelles recommandations sélectionner, consommer ou parfois rejeter. Cette interaction est consignée par la plateforme et utilisée comme retour dit *implicite* pour l’algorithme (Oard et al., 1998).

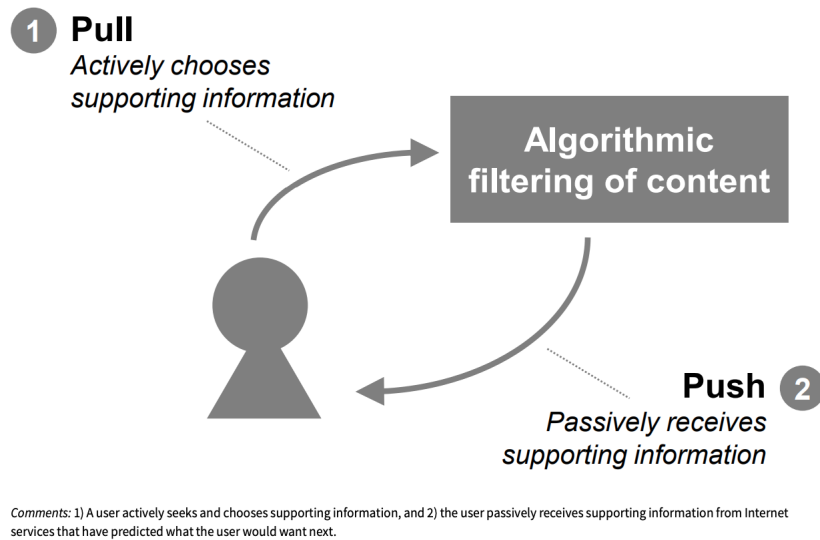


Figure 6: Boucle de rétroaction théorisée dans l'hypothèse de la bulle de filtres. Image issue des travaux de Dahlgren (2021).

L'algorithme est alors ré-entraîné avec ce retour et, à une itération ultérieure, délivre de nouvelles recommandations, plus personnalisées, et ainsi de suite. Comme illustré dans la Figure 6, cela établit une boucle de rétroaction continue entre utilisateurs et algorithmes qui, selon Pariser, peut conduire les dispositifs algorithmiques à exposer des recommandations de moins en moins diversifiées au fil du temps.²⁶ Ces types de boucles de rétroaction se retrouvent aujourd'hui presque universellement sur les plateformes (Brinkmann et al., 2023) et, en tant que tel, l'étude des dynamiques de type bulle de filtres n'est pas réservée à une plateforme particulière mais constitue plutôt un cadre général permettant de comprendre les interactions homme-machine et leurs effets sur la diversité de consommation.

Dans le cas des plateformes de streaming musical, les effets de type bulle de filtres peuvent avoir des répercussions majeures pour un large éventail d'acteurs, y compris les plateformes, les artistes et, peut-être surtout, les utilisateurs. Du point de vue des plateformes, la littérature a montré que des pratiques de consommation plus diverses sont généralement associées à un engagement accru (Anderson et al., 2020) et que la diversité des recommandations algorithmiques joue un rôle essentiel pour garantir la satisfaction des utilisateurs (Nguyen et al., 2014). Ainsi, trouver un équilibre approprié dans la diversité des recommandations peut, du point de vue des plateformes, sembler une stratégie efficace pour stimuler l'engagement et réduire le churn. Cela fournit un exemple clair de pourquoi les dynamiques de bulle de filtres peuvent susciter des inquiétudes pour ces acteurs. De même, du point de vue des artistes, les effets de type bulle de filtres peuvent également être indésirables. Des recommandations personnalisées et plus confinées peuvent réduire

²⁶Cela est dit avoir un effet déstabilisant sur le processus démocratique, bien que la manière précise reste largement spéculative.

les chances pour un artiste d’atteindre des publics diversifiés par le biais de l’algorithme, les contraignant potentiellement à se reposer sur un seul segment de marché et augmentant le risque d’échec si les préférences évoluent dans ce groupe. Enfin, du point de vue des utilisateurs, des recommandations moins diversifiées peuvent non seulement réduire la satisfaction (Nguyen et al., 2014), mais aussi perturber les structures de classe existantes, dont plusieurs chercheurs affirment qu’elles sont étroitement liées à des goûts omnivores et diversifiés (Peterson, 1997). En somme, l’effet des bulles de filtres sur les plateformes de streaming musical ne doit pas être minimisé et, comme cela a été démontré dans l’étude du confinement politique en ligne (Robertson et al., 2023), peut avoir des répercussions prévues comme imprévues pour un large éventail d’acteurs.

Lectures critiques

Bien que les bulles de filtres puissent sembler inquiétantes pour les acteurs de l’écosystème du streaming musical, dans quelle mesure constituent-elles, en tant que cadre, un concept scientifiquement solide ? Si la thèse de Pariser a été rapidement adoptée et relayée par les médias généralistes (Levy, 2023; The New York Times, 2017), cela ne signifie pas qu’elle ait reçu le même accueil positif dans le milieu académique. Certains chercheurs sont même allés jusqu’à qualifier les bulles de filtres de « métaphores qui nous ont trahis » (Bruns, 2019). La recherche s’est rapidement employée à examiner de manière critique les fondements de la thèse de Pariser, d’abord sur le plan théorique, avant même sur le plan empirique (Bruns, 2019; Michiels et al., 2022; Dahlgren, 2021; Davies, 2018). Parmi les plus notables, Dahlgren (2021), puis plus tard Michiels et al. (2022), proposent chacun une lecture critique du texte original de Pariser.

Du côté le plus critique, Dahlgren (2021) appellent à l’abandon complet de la terminologie, estimant qu’elle véhicule un excès de connotations négatives infondées. De plus, selon Dahlgren, il n’est pas clair que les dynamiques de confinement imaginées par Pariser émergent même en l’absence de dispositifs algorithmiques. Il avance donc que le concept même de bulle de filtres relève du « *déterminisme technologique* » (Davies, 2018) – si un algorithme est impliqué dans une dynamique, doit-on nécessairement conclure que l’algorithme en est la cause ? Dahlgren critique aussi la formulation de Pariser quant aux mécanismes qui produiraient une bulle de filtres ainsi que les effets supposés sur la société. Par exemple, Pariser soutient que les bulles de filtres résultent d’une personnalisation importante, à la fois du côté utilisateur et du côté plateforme, créant une boucle de rétroaction où le contenu devient de plus en plus homogène à mesure que l’engagement algorithmique augmente. Néanmoins, du côté utilisateur, la littérature montre que les

processus de personnalisation ne sont pas si simples. Bien que plusieurs études empiriques appuient le rôle de la personnalisation dans la consommation des médias, en accord avec la théorie de l'*exposition sélective* – un cadre psychologique postulant que les individus gravitent vers des médias renforçant leurs croyances préexistantes (Katz et al., 2017)²⁷, cela n'implique pas que les consommateurs évitent totalement de s'exposer à du contenu diversifié (Garrett, 2009). Ainsi, la formulation de Pariser concernant la personnalisation des utilisateurs dans cette boucle est probablement trop stricte et, en fin de compte, néglige le rôle de l'agence humaine dans les processus de sélection dits *organiques* (Roth, 2019).

Si Dahlgren plaide pour l'abolition de la bulle de filtres en tant que *signifiant* (Barthes, 1957), d'autres chercheurs adoptent une position plus réformiste. En particulier, Michiels et al. (2022) reconnaissent que la terminologie conserve une certaine utilité, susceptible d'être reformulée. S'appuyant sur l'idée que la bulle de filtres est un processus dynamique et non statique, les auteurs proposent une définition affinée de la bulle de filtres, formulée comme suit :

« Une bulle de filtres technologique est une diminution, au fil du temps, de la diversité des recommandations d'un utilisateur, dans n'importe quelle dimension de la diversité, résultant des choix effectués par différents acteurs de la recommandation. » (Michiels et al., 2022, p. 2)

Cette définition repose sur quatre dimensions censées aider à son opérationnalisation. La première dimension est la diversité. Une bulle de filtres correspond à une diminution de la diversité « dans n'importe quelle dimension de la diversité » (Michiels et al., 2022), qu'il s'agisse de la diversité des sources (par ex. sources d'actualité), de la diversité des sujets (par ex. sémantique des articles), de la diversité des points de vue (par ex. sentiments exprimés) ou, plus généralement, de la diversité fondée sur différents modes de représentation de l'information (Mok et al., 2022). La deuxième dimension, la *recommandation*, précise qu'une bulle de filtres concerne une diminution de la diversité des recommandations reçues par l'utilisateur, et non des consommations. La troisième dimension, le *temps*, souligne que le résultat d'une bulle de filtres est le produit d'un processus longitudinal. Enfin, la quatrième dimension, les *acteurs de la recommandation*, encourage une

²⁷Historiquement, la plupart des travaux empiriques explorant cette théorie sont motivés par la vérification de la théorie psychologique de la dissonance cognitive (Morvan and O'Connor, 2017), qui affirme que lorsqu'un individu est confronté à des idées contrastantes, certains mécanismes de défense mentaux sont activés afin de produire une cohérence entre nouvelles idées et croyances préexistantes, conduisant à un équilibre cognitif. Plus récemment, des travaux empiriques ont aussi été motivés par la vérification de la théorie de l'identité sociale (Knobloch-Westerwick et al., 2020), qui avance que les consommateurs privilégient les médias qui présentent leur groupe d'appartenance sous un jour favorable (Tajfel et al., 1979).

perspective multi-acteurs (par ex. utilisateurs et fournisseurs) afin de mieux comprendre comment une bulle de filtres se constitue.

Bien que la conception de Michiels et al. soit plus fondée que celle de Pariser, elle présente aussi ses propres limites. Première-ment, affirmer que les bulles de filtres doivent être évaluées uniquement à partir des expositions algorithmiques limite fortement la capacité de la plupart des chercheurs à travailler empiriquement sur ce sujet. En pratique, l'accès à ce type de données – signaux d'exposition algorithmique – est, sauf partenariats privés avec les plateformes, difficile à obtenir, bien que mobilisé dans cette thèse. Deuxième-ment, se focaliser uniquement sur les expositions algorithmiques peut omettre le rôle de la personnalisation initiée par l'utilisateur dans la création du confinement. Prenons l'exemple d'un utilisateur de longue date qui, en vieillissant, suit le schéma apparemment courant de préférences d'écoute de plus en plus restreintes (Harrison and Ryan, 2010; Glevarec et al., 2020), schéma qui se serait produit même dans un monde contrefactuel où les plateformes de streaming n'auraient jamais existé. Même si un système de recommandation fonctionnait de manière non contraignante (par ex. en reflétant simplement l'activité organique de l'utilisateur), l'étude des seuls signaux d'exposition n'indiquerait pas forcément cette réalité.

À la lumière de ces limites – concernant à la fois la formulation originale de Pariser (Pariser, 2011; Dahlgren, 2021) et ses extrapolations ultérieures (Michiels et al., 2022) – nous choisissons de nous abstenir en grande partie d'utiliser le terme « bulle de filtres » dans cette revue de littérature et, dans une certaine mesure, dans cette thèse. Nous privilégions à la place le terme « *confinement algorithmique* », qui compare l'impact algorithmique à une référence (éventuellement contrefactuelle) où les algorithmes sont partiellement ou totalement absents. Plus précisément, nous définissons le confinement algorithmique comme suit :

Confinement algorithmique : Le confinement algorithmique correspond à une situation où les médias exposés ou sélectionnés par des utilisateurs via des algorithmes sont plus concentrés que les sélections/expositions impliquant une absence partielle ou totale d'algorithmes.

Nous soutenons et démontrons que cette formulation est suffisamment générale pour capturer la littérature étudiée, qu'elle s'intéresse aux effets des dispositifs algorithmiques ou aux plateformes. En outre, cette définition permet d'intégrer des travaux reposant aussi bien sur des méthodes temporelles qu'agrégées, ainsi que les études multi-affordances de

pointe.

Tester le confinement algorithmique

« *Comprendre le fonctionnement et les effets des algorithmes doit alors être attentif à leur déploiement contextuel et contingent selon la situation, le temps et l'espace. En pratique, cela signifie que des interactions uniques ou limitées avec les algorithmes ne peuvent pas être simplement extrapolées à l'ensemble des cas.* »

(Kitchin, 2019, p. 16)

Nous nous tournons à présent vers une littérature à vocation plus quantitative, visant à tester empiriquement les dynamiques de confinement algorithmique. Dans quelle mesure la littérature suggère-t-elle que le *confinement algorithmique* est un phénomène répandu à travers les plateformes et, en particulier, sur les plateformes de streaming musical comme Spotify et Deezer, qui constituent le cœur de ce travail ? Peut-être de manière surprenante, la réponse à cette question reste ambiguë. Bien que certains travaux mettent en évidence ce type de dynamique (Anderson et al., 2020; Bandy and Diakopoulos, 2020), d'autres suggèrent qu'elle est peu courante – plusieurs recherches montrent qu'au contraire, le recours au guidage algorithmique favorise des pratiques de consommation plus diversifiées (Robertson et al., 2023; Bakshy et al., 2015; Cardenal et al., 2019; Möller et al., 2018; Epps-Darling et al., 2020; Villermet et al., 2021; Zhou et al., 2010; Fortunato et al., 2006; Aiello and Barbieri, 2017; Nguyen et al., 2014; Datta et al., 2018; Haim et al., 2018; Beuscart et al., 2019). Ces résultats apparemment contradictoires peuvent s'expliquer, au moins en partie, par les différences d'approche concernant la manière dont les scores de concentration sont calculés et comparés entre contextes plus ou moins algorithmiques. Dans la littérature, cela a principalement été fait via des comparaisons *agrégatives* ou *temporelles*. Pour respecter cette distinction, nous segmentons cette partie de notre revue en deux sections basées sur la manière dont les scores de concentration sont construits et comparés. Pour une synthèse concise de cette littérature, nous renvoyons le lecteur au tableau 2.

Comparaisons temporelles

Une méthode couramment mobilisée dans la littérature pour évaluer le confinement algorithmique consiste à suivre l'évolution de la diversité des expositions algorithmiques ou des sélections des utilisateurs dans le temps ou au fil de leur *cycle de vie sur plateforme*

Auteur	Plateforme / Type RS	Analyse	Type de données	Algorithmes / plateformes limitent-ils la diversité ?
Lathia et al. (2010)	RS vidéo	Temporelle	Simulation	y + n (selon activité et échelle)
Nguyen et al. (2014)	RS vidéo	Temporelle	Empirique	n
Su et al. (2016)	Réseaux sociaux	Agrégative	Expérimentale	y (avec + personnalisation)
Datta et al. (2018)	Streaming musical	Agrégative	Empirique	n
Geschke et al. (2019)	Réseaux sociaux	Temporelle	Simulation	y
Aridor et al. (2020)	n/a	Temporelle	Simulation	n
Roth et al. (2020)	Streaming vidéo	Temporelle	Simulation	y
Holtz et al. (2020)	Streaming musical	Agrégative	Expérimentale	y (avec + personnalisation)
Brown et al. (2022)	Streaming vidéo	Temporelle	Expérimentale	y
Huszár et al. (2022)	Réseaux sociaux	Agrégative	Empirique	y (avec + personnalisation)
Godinot and Tarissan (2023)	Streaming musical	Temporelle	Simulation	n
Chen et al. (2024)	Streaming musical	Agrégative	Expérimentale	aucun effet (avec RS diversifiées)
Sutkennik et al. (2024)	Streaming vidéo	Temporelle	Empirique	y + n (selon échelle)
Coavoux and Aussant (2024)	Streaming (musique, vidéo)	Agrégative	Empirique	n
Littérature multi-affordances				
Sharma et al. (2015)	E-commerce	Agrégative	Empirique	n
Aiello and Barbieri (2017)	Blogging	Agrégative	Empirique	n
Beuscart et al. (2019)	Streaming musical	Agrégative	Empirique	n (moins de concentration top-head)
Anderson et al. (2020)	Streaming musical	Agrégative / Temporelle	Empirique	y
Epps-Darling et al. (2020)	Streaming musical	Agrégative	Empirique	n
Villemet et al. (2021)	Streaming musical	Agrégative / Temporelle	Empirique	n
Mok et al. (2022)	Streaming musical	Agrégative / Temporelle	Empirique	n/a

Table 2: Synthèse des travaux les plus pertinents et influents de la dernière décennie sur l'effet des dispositifs algorithmiques des plateformes sur la diversité des pratiques d'engagement des utilisateurs. La littérature est présentée dans l'ordre chronologique et segmentée selon que l'étude portait ou non sur l'aspect multi-affordances d'une plateforme.

(Mok et al., 2022). Ces travaux abordent le confinement algorithmique puisqu'ils comparent la diversité des sélections/expositions algorithmiques à celle d'étapes antérieures où, par construction, les utilisateurs avaient reçu moins de guidage algorithmique. Brown et al. (2022) étudient comment le système de recommandation de YouTube dirige les utilisateurs vers des espaces idéologiques de plus en plus confinés à mesure que l'adoption progresse. Pour ce faire, leur étude recrute des participants déjà utilisateurs actifs de YouTube avant le début de l'expérience et les assigne aléatoirement à une vidéo initiale ainsi qu'à un ensemble de règles dictant comment adopter les recommandations (par ex. suivre la deuxième recommandation affichée), contrôlant ainsi l'activité courante des utilisateurs pouvant influencer la diversité des recommandations reçues. En suivant les expositions aux recommandations au fil d'une chaîne d'interactions, les auteurs constatent que, quel que soit le point de départ ou le comportement initial, le système de recommandation de YouTube oriente progressivement les utilisateurs vers des contenus de plus en plus confinés (modérément conservateurs). Toujours sur le thème du streaming vidéo, Sukiennik et al. (2024) examinent comment les recommandations personnalisées d'une grande plateforme chinoise de streaming évoluent en termes de concentration au fil du temps. Leurs résultats montrent que les recommandations adoptées tendent en moyenne à devenir plus confinées thématiquement, mais que ce phénomène dépend fortement du niveau de granularité de classification des sujets – à un niveau plus fin de classification, l'augmentation de la concentration apparaît moins évidente.

En l'absence de données empiriques détaillées au niveau utilisateur, il est courant que ce type de recherche repose sur des simulations du comportement de sélection des items par les utilisateurs, voire des algorithmes eux-mêmes (Lathia et al., 2010; Roth et al., 2020; Godinot and Tarissan, 2023). Bien que souvent fondées sur des hypothèses simplifiées concernant le comportement des utilisateurs, ces recherches sont précieuses car elles permettent de distinguer l'effet du filtrage par les utilisateurs et celui des algorithmes dans le confinement algorithmique et, de plus, d'identifier les comportements qui conduisent à des niveaux différents de diversité. Lathia et al. (2010) étudient la relation entre le confinement des systèmes de recommandation basés sur le filtrage collaboratif et les comportements des utilisateurs à travers une approche par simulation. Mesurant la concentration de manière discrète, via la répétition, leurs résultats montrent que le confinement algorithmique dépend fortement de la manière dont les utilisateurs interagissent avec les plateformes. Une faible activité intra-session (capturée par le nombre d'items notés) est associée à des recommandations plus concentrées et répétitives tandis qu'à une échelle plus globale, une activité réduite (capturée par le nombre total d'items notés) est

associée à davantage de diversité. De manière similaire, sur YouTube, Roth et al. (2020) étudient comment, à court terme et à l'échelle de la recommandation suivante, le suivi des recommandations entraîne un confinement thématique. Interrogeant les recommandations vidéo de YouTube pour générer un réseau de recommandations, puis simulant les consommations des utilisateurs dans ce réseau, leurs résultats indiquent que, dans ce contexte, suivre les recommandations de YouTube renforce le confinement thématique, en particulier pour les contenus étiquetés comme musicaux et pour les contenus plus niches. Néanmoins, la recherche suggère que les dynamiques de confinement à plus long terme peuvent être moins triviales. Générant des recommandations basées sur le filtrage collaboratif à partir de journaux d'écoute musicaux rétrospectives et mesurant la diversité de manière réseau-théorique fondée sur la relation entre utilisateurs, items et catégories d'items, Godinot and Tarissan (2023) montre que, théoriquement, la diversité de consommation musicale des utilisateurs augmente de manière sous-linéaire avec le nombre de recommandations adoptées. Toutefois, la dynamique varie à nouveau selon la manière dont les utilisateurs interagissent avec la plateforme dans la simulation.

Alors que plusieurs des travaux présentés soulignent que les interactions des utilisateurs avec les recommandations algorithmiques tendent à devenir plus concentrées dans le temps (Roth et al., 2020; Lathia et al., 2010; Brown et al., 2022; Sukiennik et al., 2024), dans quelle mesure les consommations des utilisateurs auraient-elles été encore plus confinées s'ils n'avaient pas suivi l'algorithme ? Dans leur étude de référence de 2014, Nguyen et al. (2014) mènent la première recherche portant à la fois sur la diversité des contenus recommandés, l'évolution du comportement des utilisateurs dans le temps et l'effet de l'adoption de recommandations sur la diversité des contenus consommés. Bien que quelque peu datée en termes de dispositifs algorithmiques et de plateformes étudiées²⁸, leur travail fournit néanmoins une intuition solide sur la manière dont les premiers systèmes de recommandation pouvaient influencer la diversité de consommation des médias dans le temps. En analysant conjointement la diversité de consommation des utilisateurs de la plateforme MovieLens et leur historique d'adoption de recommandations sur la période 2008-2010, leurs résultats montrent que, indépendamment de l'adoption ou non des recommandations, celles-ci devenaient de plus en plus confinées à mesure que le système s'ajustait aux préférences des utilisateurs. Dans le même temps, les consommations des utilisateurs devenaient elles aussi plus confinées, mais de manière significativement atténuée chez ceux qui *adoptaient* les recommandations algorithmiques. Les auteurs interprètent ces résultats en suggérant que les dispositifs de recommandation algorithmique

²⁸Systèmes de recommandation rudimentaires basés sur le filtrage collaboratif et plateforme MovieLens où les consommations se faisaient hors-plateforme.

peuvent, contrairement aux récits traditionnels de la bulle de filtres (Pariser, 2011), avoir un effet d’amortissement sur le confinement des pratiques de consommation qui pourraient, de toute façon, émerger *organiquement* même en l’absence d’algorithmes.

Plus récemment, l’idée que les consommations en ligne tendent à se refermer dans le temps même sans dispositifs algorithmiques a été appuyée par plusieurs travaux empiriques (Mok et al., 2022) et par simulation (Aridor et al., 2020; Geschke et al., 2019). D’un point de vue plus théorique, Aridor et al. (2020) explorent les mécanismes qui entraînent le resserrement de la diversité dans les consommations et les recommandations algorithmiques. En développant un modèle mathématique du comportement utilisateur qui intègre l’aversion au risque ainsi que des composantes idiosyncratiques et de valeur commune dans l’évaluation des items, leurs travaux confirment les conclusions de Nguyen et al. (2014), suggérant que les recommandations algorithmiques limitent le confinement qui se serait autrement produit *organiquement*. Lorsqu’ils simulent un scénario où les utilisateurs suivent aveuglément les recommandations²⁹, l’adoption des recommandations conduit à une dynamique claire de confinement. Toutefois, lorsque les processus de sélection des utilisateurs sont modélisés de manière plus réaliste, les résultats convergent avec ceux de Nguyen et al. (2014). De même, Geschke et al. (2019) recourent à un modèle basé sur des agents pour simuler les processus décisionnels qui influencent la consommation des contenus recommandés sur les réseaux sociaux. Fondé sur des mécanismes cognitifs comme le biais de confirmation, leur modèle montre que des dynamiques de confinement émergent même en l’absence de filtres sociaux ou technologiques supplémentaires, bien que leur présence amplifie la concentration des engagements. Enfin, Mok et al. (2022) montrent que, sur la plateforme Spotify, les flux organiques des utilisateurs deviennent de plus en plus exploitants dès l’inscription.

En résumé, la littérature existante suggère que, dans plusieurs cas, les interactions des utilisateurs avec le guidage algorithmique peuvent devenir plus confinées à mesure que l’adoption progresse (Roth et al., 2020; Lathia et al., 2010; Brown et al., 2022; Sukiennik et al., 2024; Mok et al., 2022). Néanmoins, ce phénomène n’est pas toujours causé par le guidage algorithmique. Plusieurs travaux montrent non seulement que les consommations des utilisateurs tendent à se refermer dans le temps même en l’absence de dispositifs algorithmiques (Geschke et al., 2019), mais aussi que, dans plusieurs cas, l’adoption du guidage algorithmique peut ralentir ce processus et ainsi maintenir une diversité plus longue dans les consommations (Nguyen et al., 2014; Aridor et al., 2020).

²⁹Fait surprenant, cette hypothèse est fréquemment adoptée dans de nombreuses études d’audit algorithmique (Aridor et al., 2020).

Comparaisons agrégées

Un autre courant de littérature étroitement lié s'attache à évaluer le confinement algorithmique à travers l'étude de dynamiques *agrégatives* plutôt que temporelles. Ce type de travaux vise généralement à analyser l'effet de l'adoption de dispositifs algorithmiques plus personnalisés en remplacement de dispositifs moins personnalisés (Holtz et al., 2020; Huszár et al., 2022; Su et al., 2016; Chen et al., 2024) ou, plus largement, l'effet de l'adoption de plateformes par rapport à un contexte sans plateforme (Coavoux and Ausant, 2024; Datta et al., 2018).

Concernant le premier cas, sur plusieurs plateformes, la littérature suggère qu'une personnalisation accrue des dispositifs de recommandation algorithmique peut entraîner une concentration plus forte des consommations des utilisateurs. Les méthodes utilisées pour parvenir à ces conclusions reposent soit sur des expériences de terrain randomisées (Holtz et al., 2020; Huszár et al., 2022; Chen et al., 2024), soit sur des expériences naturelles permettant d'établir une assignation aléatoire ou quasi-aléatoire à des conditions traitement et contrôle (Su et al., 2016). Huszár et al. (2022) étudient comment l'introduction d'un fil d'actualité plus personnalisé sur Twitter a pu rendre la consommation d'informations des utilisateurs plus homogène du point de vue partisan. En se concentrant sur 5% de l'ensemble des utilisateurs de Twitter et en assignant aléatoirement un groupe de contrôle à un fil chronologique inverse et les autres à une version algorithmique personnalisée, leurs résultats montrent que cette dernière a conduit à une consommation plus biaisée vers des sources partisans. De manière similaire, Su et al. (2016) montrent que des algorithmes de recommandation plus personnalisés sur les réseaux sociaux peuvent induire un confinement algorithmique, non seulement au niveau individuel mais aussi collectif. En exploitant une expérience naturelle semi-randomisée et des données Twitter pour une analyse causale, ils étudient l'effet de l'introduction de la fonctionnalité « Who to Follow » en 2010. Leurs résultats indiquent que l'introduction d'un dispositif plus personnalisé a probablement conduit à des comportements de suivi plus biaisés vers les utilisateurs populaires, renforçant une dynamique de type « the rich get richer ».

Dans le domaine du streaming musical, Holtz et al. (2020) fournissent également des preuves convaincantes que l'accroissement de la personnalisation peut rendre les consommations plus confinées. Étudiant le cas de la recommandation de podcasts sur Spotify, les auteurs conduisent une expérience de terrain à grande échelle impliquant 900,000 utilisateurs. Leurs résultats montrent que l'augmentation de la personnalisation du système de recommandation réduit la diversité de consommation des podcasts et les rend en outre plus

distinctes d'un utilisateur à l'autre – le paradigme typique de la bulle de filtres décrit par Pariser (2011). Néanmoins, ces effets ne semblent pas durables. Lorsque les recommandations des groupes en traitement reviennent à l'algorithme par défaut moins personnalisé, la diversité des consommations se rétablit rapidement, ce qui suggère une certaine plasticité et adaptabilité des pratiques de consommation aux évolutions des plateformes. Dans la même veine, Chen et al. (2024) étudient comment des modifications apportées aux dispositifs de recommandation sur la plateforme chinoise NetEase Cloud Music affectent la diversité des consommations. Conduisant une expérience de terrain à grande échelle sur 10 millions d'utilisateurs, ils examinent si la mise en place d'un système de recommandation davantage orienté vers la diversité que vers la seule *précision* – voir (Kaminskas and Bridge, 2016) pour une revue de ces approches dites « beyond accuracy » – peut amener les consommations musicales à se diversifier. Leurs résultats montrent qu'il n'existe pas de preuves claires que des recommandations plus diverses entraînent une diversité accrue des consommations. Toutefois, elles pourraient, contrairement à ce qu'avancent certaines recherches antérieures (Anderson et al., 2020; Holtz et al., 2020), réduire l'engagement global.

Si la littérature tend à montrer que, tant sur les réseaux sociaux (Huszár et al., 2022; Su et al., 2016) que sur les plateformes musicales (Holtz et al., 2020; Chen et al., 2024), l'accroissement de la personnalisation des dispositifs algorithmiques peut avoir un effet négatif sur la diversité des consommations³⁰, qu'en est-il si l'on s'intéresse plus largement à l'effet de l'adoption des plateformes elles-mêmes ? Par exemple, dans quelle mesure la littérature suggère-t-elle qu'un consommateur aurait des pratiques d'écoute plus ou moins diversifiées dans un monde contrefactuel où il n'aurait jamais adopté Spotify ou Deezer ? Dans les travaux récents de Coavoux and Aussant (2024), ce scénario est exploré en profondeur. Analysant les consommateurs de musique en France et utilisant l'enquête « Pratiques Culturelles » de 2018, leur étude cherche à identifier l'effet de l'adoption des plateformes sur la diversité tout en contrôlant les effets de causalité inverse (par ex. l'effet des goûts diversifiés sur la probabilité d'utiliser une plateforme). Dans trois domaines – musique, films et séries TV – l'adoption des plateformes de streaming est associée à une augmentation de la diversité des consommations, bien que dans une moindre mesure pour la musique. De même, Datta et al. (2018) montrent, à partir d'un panel d'historiques d'écoute sur diverses plateformes numériques, que l'adoption des plateformes de streaming est associée à une augmentation significative de la quantité et de la diversité des musiques consommées durant les premiers mois suivant l'adoption. Bien que ces effets

³⁰Peut-être parce que les recommandations, en devenant plus précises, perdent en sérendipité.

diminuent progressivement, même après six mois, les adopteurs consommaient toujours plus de musique avec une diversité supérieure à celle observée avant l'adoption.

En résumé, la littérature suggère que si l'introduction de dispositifs plus personnalisés sur les plateformes peut entraîner un resserrement des interactions médiatiques et sociales des utilisateurs (Holtz et al., 2020; Su et al., 2016; Huszár et al., 2022), l'adoption des plateformes contenant ces dispositifs peut, pour l'instant, favoriser des pratiques médiatiques plus riches et diversifiées (Datta et al., 2018; Coavoux and Aussant, 2024).

Tester le confinement algorithmique à l'aide de comparaisons organiques

« [Les recherches sur les bulles de filtres] sont passées de la considération de la curation algorithmique comme force décisive façonnant l'engagement en ligne à la reconnaissance de l'influence des choix des utilisateurs eux-mêmes sur les algorithmes de classement des réseaux sociaux et moteurs de recherche. »

(Brinkmann et al., 2023, p.16)

Dans le chapitre suivant, nous présentons la branche récente et à la pointe de la littérature sur le confinement algorithmique qui consiste à appliquer des points de référence *organiques* afin d'évaluer les effets relatifs du guidage algorithmique sur la diversité des consommations. Cette littérature est particulièrement pertinente pour cette thèse pour plusieurs raisons. Premièrement, les plateformes étudiées sont le plus souvent des plateformes de streaming musical, ce qui rend les enseignements tirés particulièrement applicables à l'étude de Deezer. Deuxièmement, la majorité des contributions scientifiques de cette thèse suivent un cadre méthodologique proche de celui mobilisé dans ces travaux, en comparant les comportements algorithmiques aux comportements organiques en termes de flux, de sélections ou parfois de scénarios organiques contrefactuels. Ainsi, les méthodes et conclusions de cette littérature sont hautement transposables à cette thèse.

Établir des comparaisons organiques à partir de signaux multi-affordance

Depuis sa création en tant que « Blogmusik » en 2007 (Deezer Jobs, 2025), Deezer a considérablement évolué en termes d'affordances offertes aux utilisateurs pour explorer, exploiter et organiser la musique. Aujourd'hui, à l'instar de Spotify, les utilisateurs peuvent accéder à la musique via un large éventail d'affordances organiques, algorithmiques et éditoriales, ce qui ouvre de nouvelles opportunités de recherche sur le *confinement algorithmique* à travers les riches signaux multi-affordances générés par les interactions des

utilisateurs avec ces dispositifs (Anderson et al., 2020; Epps-Darling et al., 2020; Villermet et al., 2021; Mok et al., 2022; Beuscart et al., 2019; Aiello and Barbieri, 2017; Sharma et al., 2015; Chang and Ugander, 2022). Globalement, ces travaux (principalement centrés sur les plateformes de streaming musical) proposent plusieurs approches pour étudier le confinement algorithmique sur les plateformes multi-affordances.

Par exemple, pour réaliser des *évaluations agrégatives*, les comportements des utilisateurs sous affordances organiques (O) peuvent être comparés à ceux observés sous d'autres affordances plus ou moins algorithmiques de deux manières : (1) de façon *descriptive*, en reconnaissant que les comportements O des utilisateurs sont conditionnés par leur adoption d'autres affordances (Anderson et al., 2020; Villermet et al., 2021; Epps-Darling et al., 2020), (2) de façon *contrefactuelle*, en appliquant des méthodes de « matching » pour approximer à partir de O le contrefactuel O sans l'existence de l'algorithme (i.e. $O \setminus A$) (Aiello and Barbieri, 2017; Sharma et al., 2015; Chang and Ugander, 2022). Concernant la première approche descriptive, ce type de travaux se concentre souvent sur la mesure de la concentration entre affordances, soit au sein des consommations d'un utilisateur (comparer la diversité de O et de A chez le même utilisateur) (Anderson et al., 2020), soit entre utilisateurs (comment les utilisateurs adoptant O se comparent-ils en moyenne à ceux qui adoptent A) (Villermet et al., 2021). Plus rarement, certains travaux comparent l'activité d'utilisateurs à travers des plateformes offrant des niveaux variés de filtrage social et/ou algorithmique (Nikolov et al., 2015), mais nous n'approfondissons pas ce volet puisqu'il ne concerne pas à proprement parler les plateformes multi-affordances.³¹ De manière similaire, des *évaluations temporelles* peuvent être menées en étudiant comment, après contrôle des variables confondantes, la diversité des consommations d'un utilisateur varie en fonction de ses proportions relatives d'adoption des affordances (par ex. p_A , p_O , etc.) (Villermet et al., 2021; Anderson et al., 2020).

La littérature multi-affordances existante présente un consensus mitigé quant à la mesure dans laquelle les plateformes de streaming et leurs dispositifs algorithmiques favorisent le *confinement algorithmique*. En grande partie, cela dépend de l'échelle à laquelle les mesures de diversité sont réalisées et de la représentation de l'information choisie.³²

Anderson et al. (2020) étudient comment la diversité d'écoute de 100 millions d'utilisateurs de Spotify varie entre affordances « programmées » (algorithmiques ou éditoriales) et affordances organiques. À l'aide d'un espace d'embedding de haute dimension capturant la

³¹Nikolov et al. (2015) montre comment, dans les premières années du web (2006-2010), les sites privilégiant le filtrage social (réseaux sociaux avec fils chronologiques) plutôt que la recherche organique (moteurs de recherche) différaient en termes de diversité encouragée. En exploitant des journaux de clics anonymisés de plus de 100 000 étudiants américains, les auteurs trouvent que l'information extraite via les réseaux sociaux présentait une concentration plus forte que celle issue des moteurs de recherche, révélant ainsi un effet de répétition et de focalisation lié à l'introduction de *filtres sociaux humains* avant même celle de dispositifs algorithmiques personnalisés.

³²Par exemple, représentations denses en espace vectoriel ou représentations discrètes basées sur les items ou catégories.

similarité entre morceaux – une approximation de la similarité sémantique – leur travail réalise à la fois des évaluations agrégatives et temporelles du confinement algorithmique. Pour les premières, les auteurs montrent que la musique consommée via les affordances « programmées » est en moyenne moins diverse (moins étendue dans l’espace d’embedding) que celle écoutée via les affordances organiques. Pour les secondes, ils montrent que réduire l’adoption algorithmique est associé à une diversité accrue. En bref, les utilisateurs qui recourent moins aux algorithmes dans leur usage de la plateforme tendent à être plus diversifiés. Bien que les auteurs parlent davantage en termes de corrélation que de causalité, ces résultats offrent une première intuition de la mesure dans laquelle le guidage programmé favorise le confinement par rapport à d’autres pratiques d’écoute.

Inversement, certains travaux montrent que lorsque la concentration est mesurée de façon discrète (capturant la redondance), le confinement algorithmique est moins marqué (Villermet et al., 2021; Mok et al., 2022). Epps-Darling et al. (2020) démontrent que les streams sur Spotify sont davantage biaisés vers des artistes masculins lorsque les utilisateurs adoptent des affordances organiques plutôt qu’algorithmiques. De même, à une échelle plus agrégative, Villermet et al. (2021) explorent comment différents niveaux d’usage des affordances sur Deezer produisent des degrés variés de concentration discrète. En catégorisant les affordances en trois types – organiques, algorithmiques et éditoriales – leur étude révèle que les affordances organiques sont en réalité les moins diverses lorsqu’on mesure la diversité de manière discrète et basée sur la redondance. Un examen temporel de la concentration ajoute encore de la nuance à ce résultat. Contrairement au travail de Godinot and Tarissan (2023), qui montrent dans un cadre de simulation que la diversité globale augmente de manière sous-linéaire avec l’adoption des recommandations, Villermet et al. observent qu’en pratique, le confinement algorithmique croît avec l’adoption.³³ Fait intéressant, Villermet et al. montrent aussi que cette dynamique n’est pas exclusive aux affordances algorithmiques : un utilisateur recourant massivement aux affordances organiques présente également une concentration discrète plus forte, parfois même davantage que via les affordances algorithmiques. Ces résultats suggèrent que, sous une formulation discrète de la concentration, l’usage accru des affordances organiques est plus fortement associé à la concentration que celui des affordances algorithmiques. En résumé, ces travaux indiquent, à l’instar de Nguyen et al. (2014), des origines organiques des dynamiques de confinement en ligne ainsi qu’un possible décalage entre mesures de

³³Une explication possible à cette contradiction est que Villermet et al. considèrent la concentration à l’intérieur d’une affordance plutôt qu’au niveau global. Il se peut ainsi que l’écoute algorithmique devienne plus saturée à mesure de l’adoption, tout en ayant un effet global d’amortissement si le chevauchement entre affordances organiques et algorithmiques est faible. Ce sujet reste encore peu exploré et met en évidence le manque actuel de compréhension des interdépendances entre affordances.

diversité fondées sur différents modes de représentation (dense vs discret).

Dans la même ligne, Mok et al. (2022) concluent qu’il existe « peu de lien entre les deux perspectives [c’est-à-dire, la diversité construite sur des représentations de l’information denses versus discrètes] ». Leur étude montre que les utilisateurs de Spotify les plus exploreurs et atteignant ainsi une diversité discrète ne sont pas ceux qui sont les plus diversifiés dans un sens géométrique dense. De plus, leurs résultats rejoignent ceux de Villermet et al. (2021) et Datta et al. (2018) : le *narrowing* des consommations n’est pas propre aux affordances algorithmiques, mais dès l’inscription sur la plateforme, l’écoute – algorithmique et, dans une plus grande mesure encore, organique – tend à se concentrer au fil du temps. Si cet effet peut s’expliquer mécaniquement au début du cycle de vie des utilisateurs (tout contenu étant alors nouveau), comprendre pourquoi il se maintient reste une question ouverte.

Concernant les approches plus causales d’évaluations agrégatives exploitant les signaux O pour approximer $O \setminus A$, l’existence et l’ampleur des effets de confinement algorithmique demeurent tout aussi incertaines et dépendantes de l’échelle et de la représentation de l’information. Aiello and Barbieri (2017) étudient l’impact des recommandations algorithmiques sur la diversité des réseaux de suivi sur Flickr et Tumblr. Pour approximer des effets causaux, ils emploient une méthode de matching associant des utilisateurs aux connectivités locales quasi-identiques jusqu’à leurs k premiers voisins. Ces paires sont ensuite divisées en deux sous-groupes : l’un où la connexion au voisin ($k+1$) provient d’une recommandation algorithmique, l’autre où elle provient organiquement. En mesurant la diversité à l’aide de l’entropie thématique, les auteurs trouvent que les recommandations algorithmiques peuvent accroître la diversité à court terme. En revanche, Sharma et al. (2015) concluent qu’à plus long terme, les recommandations algorithmiques ont peu d’effet sur le comportement des utilisateurs sur Amazon. En estimant $O \setminus A$ à partir de O (requête auto-initiée) et A (« articles similaires »), ils estiment que jusqu’à 75% des articles accédés par l’algorithme auraient de toute façon été consultés organiquement en l’absence de guidage algorithmique.

Dans l’ensemble, cette littérature montre que les plateformes de streaming musical affordent de manière différenciée selon la façon dont les utilisateurs les mobilisent (i.e. adoption des affordances), le type de diversité mesurée et l’échelle considérée. Elle remet aussi en question la vision trop simpliste des dispositifs algorithmiques comme forces uniformément contraignantes en termes de diversité. En cherchant à comprendre ce que les utilisateurs font et, de plus, ce qu’ils auraient pu faire en l’absence relative ou to-

tale d’algorithmes, cette littérature propose une évaluation plus nuancée du confinement algorithmique. Comme nous le montrerons, elle permet aussi de réinterpréter les récits existants autour des bulles de filtres.

Tenir compte des sélections organiques

Jusqu’ici, nous avons vu que si l’adoption des plateformes de streaming peut accroître la diversité des consommations des utilisateurs (Coavoux and Aussant, 2024; Datta et al., 2018), cet effet semble être conditionné par le degré auquel les dispositifs algorithmiques sont personnalisés (Holtz et al., 2020; Huszár et al., 2022; Su et al., 2016), la manière dont les utilisateurs exploitent la plateforme (Villermet et al., 2021; Anderson et al., 2020; Nguyen et al., 2014; Aridor et al., 2020) ainsi que le type de mesure de la diversité mobilisé (Mok et al., 2022) (par ex. discrète (Villermet et al., 2021) ou dense (Anderson et al., 2020)). Néanmoins, la grande majorité de cette littérature repose sur l’analyse de signaux de consommation. Le rôle des sélections des utilisateurs dans ces dynamiques – en particulier dans le confinement algorithmique – demeure donc largement obscur. Bien que ces questions restent relativement peu explorées dans la littérature sur le streaming musical, les sciences politiques s’y sont intéressées de près en comparant la concentration des médias auxquels les utilisateurs sont *exposés* par les algorithmes à celle des médias qu’ils *sélectionnent* eux-mêmes (Robertson et al., 2023; Bakshy et al., 2015; Mummolo, 2016). Bien que ces travaux se concentrent principalement sur les réseaux sociaux et l’information politique, ils offrent une source d’inspiration précieuse et un fondement solide pour cette thèse.

En général, ces recherches s’appuient sur la théorie de l’*exposition sélective*, concept central des études en communication et médias qui décrit la tendance des individus à consommer des contenus renforçant leurs croyances préexistantes (Katz et al., 2017), limitant ainsi leur exposition à des items contradictoires et augmentant potentiellement le risque de se retrouver dans des chambres d’écho (principalement partisans) (Iyengar and Hahn, 2009; Sunstein, 2001). Ces études examinent la mesure dans laquelle les dispositifs algorithmiques et les sélections organiques s’alignent plus ou moins sur les préférences préalables des utilisateurs (par ex. orientations politiques) (Bakshy et al., 2015; Robertson et al., 2023). Typiquement, ces préférences sont mesurées autrement que par la consommation, par exemple via l’analyse de la distribution des idéologies politiques sur les pages suivies d’un réseau social. L’alignement est ensuite opérationnalisé par le degré auquel les journaux d’exposition et de consommation dans l’affordance algorithmique restent

confinés aux préférences préexistantes (Bakshy et al., 2015).

Les résultats empiriques existants dans ce champ suggèrent que, de manière peut-être surprenante, sur les moteurs de recherche (Robertson et al., 2023), les réseaux sociaux (Bakshy et al., 2015; Aiello and Barbieri, 2017) et les médias numériques d’actualité (Haim et al., 2018), la concentration émergerait principalement de la personnalisation *organique* plutôt qu’*algorithmique*. Dans les travaux de Bakshy et al. (2015), les sélections d’articles d’actualité par les utilisateurs de Facebook sont analysées comme une chaîne séquentielle de filtrage : d’abord par le réseau social (amis dont les publications apparaissent), ensuite par l’algorithme (classement du fil d’actualité), et enfin par la sélection de l’utilisateur. Les auteurs montrent que les utilisateurs sont exposés à 15% de contenus contradictoires en moins en raison du classement algorithmique (par rapport à ce que partage leur réseau social). Mais surtout, ce chiffre est sans commune mesure avec la réduction causée par les propres choix des utilisateurs : ceux-ci consomment 70% de contenus contradictoires en moins du fait de leurs sélections.

L’idée que les sélections des utilisateurs soient encore plus confinées que les expositions algorithmiques est confirmée par Robertson et al. (2023). Ils montrent qu’en utilisant Google, les utilisateurs choisissent des sources d’information plus conformes à leur identité et moins fiables que celles auxquelles ils étaient exposés initialement dans les résultats de recherche. Cela soutient l’hypothèse de l’exposition sélective et remet en question les travaux qui reposent uniquement sur des journaux de sélection pour évaluer le confinement algorithmique – comme c’est souvent le cas dans la littérature sur le streaming musical. Par ailleurs, des travaux récents montrent aussi que les thématiques des contenus jouent un rôle essentiel dans ce que les utilisateurs consomment ou non, suggérant une multitude de processus de sélection interdépendants qui peuvent se renforcer ou s’affaiblir mutuellement (Mummolo, 2016).

Dans l’ensemble, ces résultats fournissent des preuves solides que, dans le contexte de la consommation en ligne de médias recommandés algorithmiquement et politiquement chargés, les utilisateurs limitent *organiquement* la diversité des informations qu’ils consomment par leurs comportements de sélection et de rejet. Toutefois, la mesure dans laquelle ces enseignements s’appliquent au domaine du streaming musical reste, à notre connaissance, largement inexplorée – un champ de recherche auquel cette thèse entend contribuer.

Contextualiser le confinement organique

« *La musique est prophétie. Ses styles et son organisation économique sont en avance sur le reste de la société, car elle explore, beaucoup plus rapidement que la réalité matérielle ne le peut, toute la gamme des possibilités dans un code donné.* »

(Attali, 1985, p.11)

Sur le plan théorique (Dahlgren, 2021; Bruns, 2019) comme empirique (Nguyen et al., 2014; Aridor et al., 2020; Coavoux and Aussant, 2024; Datta et al., 2018; Villermet et al., 2021), la littérature fournit des éléments solides en faveur de l'idée que des pratiques de consommation musicale confinées peuvent émerger *organiquement*, même en l'absence de dispositifs algorithmiques. De plus, il a été démontré que les interactions des utilisateurs avec ces dispositifs (i.e. leurs sélections et rejets) peuvent également limiter la diversité de leurs consommations médiatiques en ligne (Bakshy et al., 2015; Robertson et al., 2023). Pour commencer à comprendre ce qui alimente ces processus, nous nous détachons des approches purement computationnelles pour nous tourner vers le riche corpus de littérature qualitative issu en grande partie de la psychologie et de la sociologie de la consommation. Dans la section suivante, nous cherchons à comprendre pourquoi la concentration des consommations musicales des individus pourrait émerger organiquement selon deux perspectives – dense et discrète. Cette distinction reflète la littérature empirique centrée sur les plateformes qui montre que la mesure de la diversité des consommations selon ces deux perspectives peut conduire à des conclusions remarquablement différentes (Mok et al., 2022; Anderson et al., 2020; Villermet et al., 2021).

La perspective discrète

La variété peut bien être le sel de la vie (Sheldon et al., 2013), mais selon un riche corpus de littérature en psychologie, la familiarité semble également constituer un moteur substantiel de l'appréciation et de la consommation musicales, même en l'absence d'influence algorithmique. Pour commencer à comprendre pourquoi la consommation musicale des utilisateurs pourrait s'enfermer organiquement dans un sens discret et ainsi manquer de nouveauté, nous centrons notre attention sur la manière dont l'affect et la consommation d'une chanson varient en fonction du nombre d'expositions. Conformément à la littérature dans ce domaine, nous distinguons deux types d'expositions : celles qui sont déterminées de manière externe et celles qui sont autodéterminées.

En considérant d'abord la manière dont l'affect des consommateurs pour la musique

varie avec des expositions répétées qui sont déterminées de manière externe, un corpus bien établi de recherches en psychologie a théorisé que la préférence pour une chanson augmente *organiquement* avec les expositions en raison d’une familiarité accrue – ce que l’on appelle l’« *effet de simple exposition* » (Zajonc, 1968; Chmiel and Schubert, 2017). En termes simples, l’effet de simple exposition est une théorie générale de la formation des préférences par la familiarité, qui s’applique à une variété de stimuli couvrant les domaines visuels, sociaux et auditifs. Même lorsqu’une chanson n’est pas choisie pour être consommée, la recherche a démontré que l’affect des consommateurs pour cette chanson augmente initialement avec le nombre d’expositions (Chmiel and Schubert, 2017; Peretz et al., 1998).³⁴ Néanmoins, cet effet a ses limites ; les recherches suggèrent également que la musique devient « rassasiée » pour les consommateurs. Après un certain seuil d’expositions, on observe une diminution de l’affect, probablement en raison de la saturation/l’ennui l’emportant sur les moteurs d’habituation (Berlyne, 1970). En d’autres termes, le gain d’affect d’une exposition à l’autre diminue avec le nombre d’expositions jusqu’au point où, après un certain seuil où l’affect atteint un sommet, une exposition supplémentaire ne conduit qu’à une réduction de l’affect (Chmiel and Schubert, 2017)³⁵ Cela aboutit à la courbe en U inversé bien documentée, selon laquelle l’affect varie en fonction de l’exposition au stimulus.

La mesure dans laquelle ces dynamiques peuvent être reproductibles dans le contexte d’une exposition autodéterminée, plus fréquente dans les environnements de plateformes de streaming, reste moins claire. Comme le soulignent pertinemment O’Brien (2021), « *peu (voire aucune) études dans la littérature sur la simple exposition examinent les choix libres des individus de revivre à plusieurs reprises le stimulus en question* ». Les recherches axées sur les expositions autodéterminées tendent, dans l’ensemble, à être plus quantitatives et à s’appuyer sur des journaux d’utilisation rétrospectifs issus des plateformes. Ainsi, dans cette littérature, il est plus courant de se concentrer sur la manière dont les comportements de consommation des consommateurs (plutôt que l’affect) varient en fonction des expositions à une chanson. Sguerra et al. (2022) vérifient que, sur la plateforme Deezer, des dynamiques en U inversé apparaissent même si l’on considère les taux de streaming et les expositions autodéterminées à la place de l’affect et des expositions externes respectivement.³⁶ Les auteurs étudient comment la probabilité de streaming d’un titre par les « fans » (utilisateurs ayant > 40 expositions à un titre donné)

³⁴Chmiel and Schubert (2017) ont mené une méta-analyse complète des études examinant la relation entre la courbe en U inversé et la préférence musicale, englobant des recherches couvrant un siècle entier. Leurs résultats révèlent que 87.72% de la littérature soutient la validité du modèle en U inversé.

³⁵Ce processus est étroitement lié aux processus d’« *adaptation hédonique* » (Frederick and Loewenstein, 1999).

³⁶Étant donné que sur la plateforme Deezer, la majorité de l’activité est organique (Villermet et al., 2021), il est probable qu’une proportion marquée de ces écoutes proviennent de sélections faites par les utilisateurs, que ce soit en termes de curation ou de recherche de chansons.

varie de la 1^{re} à la 40^e exposition. Les auteurs constatent que les courbes en U inversé apparaissent également couramment dans ce contexte. Une manière d’interpréter ces résultats est en lien avec les observations de laboratoire précédemment détaillées (Chmiel and Schubert, 2017) – les utilisateurs sont peut-être plus enclins à réécouter (jusqu’au sommet) en raison d’une augmentation de l’affect survenue lors de l’exposition précédente. De plus, à un niveau plus granulaire, Whitman (2014) démontrent que, sur Spotify, la probabilité de streaming d’un titre en fonction du temps d’écoute suit également une forme en U inversé. Néanmoins, la mesure dans laquelle cette association peut résulter de processus psychologiques similaires (c’est-à-dire, ennui, désir de familiarité) ou plutôt de publics distincts reste incertaine. De même, ces dernières années, plusieurs efforts notables ont été réalisés pour modéliser statistiquement les dynamiques de consommation musicale en ligne des utilisateurs (Anderson et al., 2014; Benson et al., 2016). Contrairement aux travaux précédemment détaillés, ces efforts cherchent généralement à comprendre et donc à modéliser les consommations répétées en fonction du temps écoulé entre les consommations d’un même élément (Anderson et al., 2014; Benson et al., 2016) ou à travers les propriétés intrinsèques d’un élément lui-même (par exemple, la qualité captée via la popularité d’un élément (Anderson et al., 2014)).

Le concept de consommations médiatiques répétées autodéterminées a également été largement exploré dans la littérature en psychologie du consommateur. Cette littérature est dans l’ensemble plus théorique qu’empirique et se concentre sur la théorisation des processus psychologiques qui motivent la consommation répétée. En termes simples, ces travaux suggèrent que la consommation répétée peut être un processus d’apprentissage qui expose à la nouveauté, que ce soit du côté de l’*objet consommé* ou du côté du *consommateur* (O’Brien, 2021). Concernant le premier cas, les pratiques de consommation répétée sont censées fonctionner par la nouveauté du stimulus, par exemple, via une découverte nouvelle (par exemple, apprécier la ligne de basse d’une chanson pour la première fois) ou une redécouverte dans la chanson. En même temps, la consommation répétée est également dite avoir le potentiel de produire de nouvelles compréhensions de soi du côté du consommateur. Par exemple, il a été démontré que la consommation répétée de médias est associée à une augmentation des sentiments de maîtrise, de *fandom* et d’affirmation identitaire (O’Brien, 2019), ce qui peut être plaisant et donc inciter les consommateurs à répéter leurs consommations.

En résumé, la littérature en psychologie suggère que la tendance des utilisateurs de streaming musical à répéter leurs consommations et à rester discrètement confinés ne constitue pas nécessairement un phénomène spécifique à la plateforme ni motivé par les

algorithmes. Au contraire, il existe de solides preuves suggérant que ce comportement peut provenir des processus cognitifs et psychologiques sous-jacents des consommateurs (O’Brien, 2019, 2021; Zajonc, 1968; Chmiel and Schubert, 2017), qui à leur tour, peuvent accroître l’affect et inciter les utilisateurs à répéter leurs engagements futurs. Contrairement à d’autres médias comme la télévision et le cinéma, il s’agit d’une propriété particulièrement distincte de ce que la musique permet (Sguerra et al., 2022). Cette littérature, bien que peut-être triviale pour les consommateurs de musique assidus, fournit une base théorique solide pour contextualiser pourquoi les dispositifs algorithmiques ne sont pas toujours la cause directe de la concentration discrète observée dans la consommation des utilisateurs sur les plateformes.

La perspective dense

Alors que le domaine de la psychologie de la musique offre des perspectives précieuses sur les raisons pour lesquelles les consommateurs de musique ont tendance à s’engager de manière répétée avec les mêmes chansons, les perspectives sociologiques sont particulièrement utiles pour comprendre comment différents groupes sociaux peuvent répartir leur attention sur des musiques présentant divers degrés de similarité sémantique (par exemple, couvrant plusieurs genres). Étant donné que la similarité sémantique est souvent opérationnalisée par des distances dans des espaces d’*embedding* d’items, une telle littérature sociologique fournit un prisme utile pour interpréter pourquoi les utilisateurs peuvent *organiquement* présenter différentes dispersions dans un espace vectoriel dense (Anderson et al., 2020).

Historiquement, l’étude de la consommation dans les sciences sociales était souvent secondaire par rapport à l’analyse de la production, ou lorsqu’elle était abordée, encadrée principalement par des critiques de la « société de consommation » (Slater, 2005). Cependant, ces dernières années, la consommation est devenue un axe central, notamment parce que la consommation médiatique est de plus en plus reconnue comme un site clé de l’articulation de la distinction élitaine, c’est-à-dire l’utilisation des préférences culturelles, y compris le goût musical, comme moyen de signaler et de maintenir sa position sociale. Les premières théories de la distinction basées sur l’imitation, telles que celles proposées par Veblen (2017) et plus tard Simmel (1957), suggéraient que les élites maintiennent leur position sociale en faisant évoluer continuellement leurs goûts en réponse à l’imitation des groupes sociaux inférieurs, préservant ainsi l’exclusivité et le pouvoir symbolique de leurs préférences. Plus récemment, les théories basées sur la méconnaissance,

développées notamment par Pierre Bourdieu (Bourdieu, 1984), ont soutenu que la distinction repose moins sur des changements réactifs de goût et davantage sur la capacité des groupes sociaux dominants à institutionnaliser leurs préférences comme légitimes grâce à leur influence sur la production culturelle, l'éducation, les médias et d'autres structures de pouvoir symbolique (Friedman and Reeves, 2020).

Parmi d'autres choses, les résultats de ces processus sont des différences de goûts culturels et de pratiques de consommation entre ceux qui occupent différentes positions sociales (par exemple, différentes classes). En termes plus computationnels, il existe encore aujourd'hui un débat actif parmi les chercheurs en méconnaissance quant à savoir si les mécanismes de distinction entraînent toutes les classes à présenter des goûts de genre concentrés et à faible entropie ou bien si les élites présentent des goûts à haute entropie et équilibrés tandis que ceux disposant de moins de capital économique conservent leurs goûts à faible entropie et biaisés. Concernant la première école de pensée, dans l'encadrement bourdieusien, chaque classe est censée avoir des goûts exclusifs étroits. Par exemple, au moment de la rédaction de la *Distinction* de Bourdieu (années 1970), la classe supérieure française avait plus de chances d'exprimer un goût pour les genres dits « nobles » comme la musique classique, tandis que les classes inférieures étaient plus fortement associées à l'expression d'un goût pour les genres dits « populaires » tels que la pop, le rock and roll et la chanson française.

Depuis Bourdieu, un grand nombre de sociologues aussi bien en France que dans le monde anglophone, en particulier depuis les années 1990, ont cherché à confirmer la reproductibilité de ces résultats et à développer la thèse de l'homologie. Peut-être le défi le plus notable et en outre le plus radical ³⁷ à la thèse de l'homologie de Bourdieu provient de la thèse de l'« *omnivore* », notée pour la première fois par DiMaggio (1987) et développée ensuite dans les travaux de Peterson (Peterson, 1997, 2005). Plus de 30 ans plus tard et en étudiant la distinction de classe en Amérique plutôt qu'en France, la thèse de Peterson proclame qu'un nouvel aboutissement de la distinction de classe a émergé, reposant non pas sur l'expression d'un goût *univore* noble mais plutôt sur un goût large et *éclectique* couvrant à la fois les cultures populaires et nobles. Encore une fois, en traduisant ces résultats dans une terminologie plus centrée sur la concentration et plus adaptée à cette thèse, la thèse de Peterson proclame que les classes supérieures présentent des goûts plus « omnivores », dispersés ³⁸ tandis que les classes inférieures sont associées,

³⁷Dans le monde académique anglophone, il existe des débats sur la mesure dans laquelle la thèse de Peterson remet en cause la théorie de l'homologie de Bourdieu (Erickson, 1996; Chan and Goldthorpe, 2017), ou bien s'aligne sur son cadre général malgré des spécificités différentes (Holt, 1997; Lizardo and Skiles, 2015).

³⁸Peut-être en raison de frontières des préférences culturelles devenues plus fluides, peut-être en raison d'une moindre dépendance au dégoût pour délimiter l'identité sociale.

de manière similaire à ce que suggère Bourdieu, à des goûts « univores » à faible entropie.

Ces dernières années, plusieurs tentatives ont été faites pour appliquer un prisme bourdieusien aux plateformes de streaming musical (Webster, 2020, 2023; Airoidi, 2021; Coavoux and Aussant, 2024). Bien que rares, et de fait davantage de recherches soient nécessaires, la littérature existante suggère que, lorsqu'elles sont adoptées, l'effet relatif que les plateformes de streaming musical peuvent avoir sur la division de classe omnivore–univore est soit désamplificateur (Webster, 2023), soit neutre (Coavoux and Aussant, 2024). Webster (2023) affirment que l'adoption des plateformes de streaming musical peut affaiblir les processus de distinction en prenant en charge le travail de curation musicale et en réduisant le temps nécessaire pour apprécier la musique pour elle-même. De même, Coavoux and Aussant (2024) montrent que l'effet de l'adoption des plateformes sur la diversité de la consommation musicale est indépendant de la position sociale – l'adoption augmente légèrement la diversité dans les deux cas et donc, si tous adoptaient les plateformes de streaming, cela aurait un effet négligeable sur la division omnivore–univore préexistante. Néanmoins, si en théorie l'effet des plateformes de streaming sur la diversité de consommation est indépendant du capital économique, leur adoption ne l'est clairement pas. Tout comme au début des années 2000, un statut social plus élevé était associé à une adoption plus grande des dispositifs d'écoute offrant plus de flexibilité aux utilisateurs (López-Sintas et al., 2008), il en va de même pour l'adoption des plateformes de streaming musical (Coavoux and Aussant, 2024). Pour une expérience sans publicité et utilisable, les utilisateurs doivent payer un abonnement mensuel, créant ainsi une barrière substantielle pour ceux qui disposent de moins de capital économique. Ainsi, les *paywalls* peuvent en réalité amplifier les distinctions de classe puisque ces opportunités de diversification ne sont disponibles que pour quelques-uns (DiMaggio, 2004; Coavoux and Aussant, 2024). Dans une direction plus théorique, Airoidi (2021) font un effort notable pour traduire une pensée bourdieusienne vers une perspective socio-technique, traitant ainsi les dispositifs algorithmiques comme des agents sociaux qui participent activement à la vie sociale. Nous renvoyons le lecteur à ce texte s'il souhaite comprendre plus en profondeur comment la sociologie de la consommation de pointe aborde actuellement l'étude des plateformes de streaming musical et des dispositifs de recommandation algorithmique.

En résumé, ce qui peut initialement apparaître comme un simple choix mathématique entre mesurer la diversité de la consommation musicale à l'aide de représentations discrètes ou continues d'items géométriques est, comme le montre cette section, profondément lié à des perspectives disciplinaires plus larges. L'approche discrète tend à s'aligner sur les perspectives psychologiques, offrant un aperçu des significations et des origines de

la diversité au niveau individuel. En revanche, l’approche géométrique continue et dense s’aligne plus étroitement sur la littérature sociologique visant à comprendre comment la diversité varie entre de plus grands groupes sociaux. Ensemble, ces perspectives fournissent un contexte essentiel pour la décision d’employer les deux approches dans cette thèse et aident à clarifier pourquoi une concentration organique peut encore émerger dans chaque cadre.

Résumé des contributions scientifiques

Tracing affordance and item adoptions on music streaming platforms

Cette recherche a été publiée lors de la conférence de l’International Society of Music Information Retrieval en 2021, au début de mon doctorat. S’appuyant sur les travaux antérieurs du projet RECORDS (Villermet et al., 2021; Beuscart et al., 2019), elle explore le phénomène relativement peu étudié des transferts d’items entre affordances – en particulier ceux qui proviennent de découvertes programmées (par ex. via l’affordance A ou E) et qui réapparaissent ensuite dans des consommations organiques – et développe davantage les taxonomies d’utilisateurs proposées par Villermet et al. (2021).

La motivation de cette étude des transferts d’items entre affordances venait d’un intérêt pour la compréhension du caractère réellement organique des interactions sur Deezer. Au moment de la publication, les comportements organiques étaient de plus en plus utilisés comme référence pour évaluer la concentration algorithmique (Anderson et al., 2020). L’objectif plus large était donc de mieux comprendre si une telle approche était pertinente ou, à tout le moins, devait être reformulée. Cela faisait de ce projet un point de départ naturel pour ma thèse et une occasion de tester des hypothèses fondamentales avant de les approfondir.

En définitive, cette recherche montre que, pour la grande majorité des utilisateurs de Deezer, les morceaux écoutés de manière organique proviennent rarement de découvertes réalisées via des affordances programmées telles que les dispositifs de recommandation A ou les playlists E . Utiliser soit les morceaux écoutés organiquement, soit l’ensemble des écoutes organiques comme référence pour évaluer l’influence algorithmique apparaît donc comme une approche raisonnable et défendable. Ces résultats soutiennent également l’idée que les utilisateurs de plateformes de streaming musical préfèrent généralement rester *aux commandes* de leurs pratiques d’écoute (Beuscart et al., 2019). De manière cruciale,

l'activité organique continue de représenter la majorité de l'activité sur la plateforme, même en tenant compte de l'affordance initiale adoptée pour la découverte.

Interpreting node embedding distances through n -order proximity neighbourhoods

Les travaux antérieurs issus ou liés au projet RECORDS se sont principalement appuyés sur des mesures discrètes de concentration pour évaluer la diversité d'écoute des utilisateurs selon différentes affordances de plateforme (Louail and Barthelemy, 2017; Beuscart et al., 2019; Villermet et al., 2021). Bien que fructueuses en soi, ces approches ne tiennent pas compte des degrés variés de similarité entre les items et laissent donc subsister un écart sémantique entre la manière dont la diversité d'écoute est perçue et la manière dont elle est évaluée quantitativement (Porcaro et al., 2022). Pour y remédier, la contribution de cette thèse consiste à appliquer un espace d'*embedding* dense et de haute dimension pour capturer la similarité entre artistes / morceaux puis, en aval, la concentration des écoutes des utilisateurs sous différentes affordances. Cette approche permet d'obtenir une mesure plus granulaire et géométrique de la similarité entre items, offrant – comme nous le montrerons – une perspective distincte sur le rôle des dispositifs algorithmiques dans le renforcement de la concentration des écoutes.

Dans ce chapitre, présenté à la conférence Complex Networks en 2024 et publié dans les actes via Springer ³⁹, nous nous concentrons sur l'identification du modèle le plus adapté pour générer un espace d'*embedding* d'items destiné à ces fins. La construction d'espaces d'*embedding* n'est pas propre à l'étude de l'écoute musicale : un large éventail de techniques a émergé de domaines tels que les réseaux complexes et le traitement automatique du langage naturel (NLP). Beaucoup de ces méthodes peuvent être facilement transposées à des données musicales, moyennant un certain niveau d'abstraction entre les objets musicaux (par ex. artistes ou morceaux) et leurs équivalents dans d'autres domaines. Par exemple, considérer une playlist comme un document et les morceaux comme des termes permet d'appliquer directement des modèles d'*embedding* issus du NLP aux données d'écoute. De même, l'application de modèles d'*embedding* de nœuds est triviale dès lors que les matrices de co-occurrence artistes/morceaux basées sur l'usage sont traitées comme la matrice d'adjacence d'un réseau. Cette dernière perspective a guidé notre exploration de diverses techniques d'*embedding* de nœuds pour construire un espace d'*embedding* centré sur les artistes. Dans cette étude, nous avons évalué la pertinence des

³⁹Reproduit avec l'autorisation de Springer Nature.

embeddings en examinant leur interprétabilité au sein du réseau de co-occurrence original. Ce choix était intentionnel : le réseau encode des informations structurelles riches, allant des connexions de premier ordre (directes) aux relations d'ordre supérieur (multi-sauts), que nous avons cherché à préserver et à refléter de manière fidèle dans l'espace appris.

Nos résultats montrent que la décomposition en valeurs singulières (SVD) classique surpasse des méthodes plus sophistiquées telles que node2vec et DeepWalk en matière de préservation de la structure locale et globale du réseau. Bien que moins spécialement conçue pour les tâches spécifiques aux réseaux, la SVD s'est révélée plus performante pour conserver les voisinages de premier, second et supérieur ordre. Pour nos objectifs, la SVD offre donc le meilleur compromis : une méthode à la fois interprétable et efficace pour capturer des relations structurelles complexes dans le réseau.

Reframing the filter bubble through diverse scale effects in online music consumption

Publié en libre accès dans *Nature Scientific Reports* en 2025, ce travail présente une analyse originale de l'influence de l'échelle temporelle et du mode de représentation de l'information (dense ou discret) sur les conclusions relatives au confinement algorithmique. Nous montrons ainsi qu'au fil du temps, l'adoption des dispositifs algorithmiques tend à produire moins de concentration discrète (i.e. davantage de nouveauté), tout en générant simultanément plus de concentration dense (i.e. une moindre dispersion dans l'espace vectoriel). Cette double dynamique apporte de la nuance et enrichit les récits autour de la bulle de filtres.

Les deux chapitres précédents de cette thèse ont fortement orienté la direction méthodologique de ce travail. Premièrement, en s'appuyant sur le constat que les streams organiques sont en grande partie indépendants des streams algorithmiques (Shakespeare and Roth, 2021), nous avons choisi d'utiliser des benchmarks *organiques* pour évaluer le confinement algorithmique. Deuxièmement, en nous fondant sur les enseignements tirés des expériences menées dans Shakespeare and Roth (2024), nous avons choisi d'appliquer la SVD pour générer les représentations denses des items. Cela garantit que les mesures denses de confinement algorithmique dérivées restent interprétables.

À la suite d'une série de discussions avec le responsable R&D de Deezer, nous avons également pu publier à la fois le code nécessaire à la reproduction des expériences de cet article et un sous-ensemble anonymisé du jeu de données utilisé. Ce dernier est

mis à disposition en libre accès afin de faciliter la reproductibilité et d’encourager de futures recherches dans ce domaine. Ces types de jeux de données sont particulièrement précieux pour le champ du Music Information Retrieval, car les traces multi-affordances sont généralement rarement accessibles en dehors de partenariats entre l’industrie et le monde académique.

Tracing desired, exposed and adopted novelty on music streaming platforms

Nous nous tournons à présent vers l’examen de l’influence des sélections des recommandations algorithmiques par les utilisateurs de Deezer sur leur diversité de consommation. Alors que de telles dynamiques ont été davantage explorées dans des domaines comme la science politique (Bakshy et al., 2015; Robertson et al., 2023), ce type de recherche reste beaucoup plus rare dans la littérature consacrée au confinement algorithmique sur les plateformes de streaming musical. Ce travail est destiné à être soumis à la conférence World Wide Web (WWW26) et est donc présenté ici sous forme de pré-print.

En mobilisant une combinaison riche de données d’enquête déclarées, de mesures d’exposition algorithmique et de journaux de consommation détaillés, cette étude analyse la relation entre trois dimensions clés : le niveau de nouveauté que les utilisateurs déclarent souhaiter, la nouveauté à laquelle ils sont exposés et la nouveauté qu’ils choisissent effectivement de consommer. Nos résultats révèlent une tension notable : bien que les utilisateurs déclarent souvent préférer un degré équilibré de nouveauté, dans la pratique ils consomment moins de nouveauté que ce qui leur est proposé. De plus, lorsque les recommandations présentent davantage de nouveauté, les utilisateurs tendent fréquemment à se tourner vers des contenus familiers, privilégiant l’exploitation à l’exploration.

Pris en combinaison avec les travaux précédemment présentés dans cette thèse, ces résultats suggèrent que les recommandations algorithmiques pourraient en réalité être encore plus novatrices que ce qui a été soutenu jusque-là (relativement à l’activité organique). Cela remet une nouvelle fois en question les récits techno-déterministes qui dépeignent les dispositifs de recommandation comme générant par conception de la concentration algorithmique. Bien au contraire – d’un point de vue discret, ils offrent de nombreuses opportunités de diversité que les utilisateurs contournent activement.

Discussion

Principaux résultats

À travers plusieurs analyses empiriques approfondies des items auxquels les utilisateurs de Deezer sont exposés, de ceux qu'ils choisissent de consommer et des contextes de consommation, cette thèse contribue à une compréhension enrichie de la manière dont les plateformes de streaming musical multi-affordances sont adoptées et du degré auquel l'orientation algorithmique, et plus largement programmée, façonne des comportements de consommation diversifiés et/ou confinés en pratique plutôt qu'en principe. Cette thèse remet ainsi en cause de nombreuses dichotomies (par ex., passif vs. actif, exploration vs. exploitation) et paradigmes (par ex., bulles de filtres, confinement algorithmique) qui ont orienté la recherche sur la relation entre algorithmes, plateformes et pratiques de consommation culturelle. Pris ensemble, ces résultats contribuent à contester et à réfuter les récits techno-déterministes omniprésents dans ces domaines en affirmant à la fois l'agence organique des utilisateurs et la nature multipolaire du confinement algorithmique.

Dans notre publication ISMIR 2021, nous avons abordé la RQ1 : « *Comment les utilisateurs adoptent-ils les plateformes de streaming musical, et plus précisément, comment cela varie-t-il selon les différentes affordances qu'elles incarnent ?* » Cette recherche contribue à une compréhension plus approfondie de la mesure dans laquelle les plateformes de streaming musical, par conception, peuvent faciliter ou même inciter les utilisateurs à recourir à une récupération passive de l'information, comme plusieurs chercheurs plus qualitatifs l'ont souligné ces dernières années (Prey, 2019; Hesmondhalgh, 2022). Alors que la littérature antérieure a montré que les utilisateurs de streaming musical s'appuient principalement sur les affordances organiques pour leur consommation (Beuscart et al., 2019; Villermet et al., 2021), l'origine habituelle des morceaux diffusés organiquement (c.-à-d., l'affordance par laquelle ils ont été découverts) restait incertaine. Les affordances programmées (algorithmiques et éditoriales) facilitent par conception une récupération plus passive de l'information. Ainsi, si les écoutes organiques proviennent majoritairement des affordances programmées, alors montrer que la plupart de l'activité est « organique » ne prouve pas que les utilisateurs s'appuient sur les plateformes de streaming pour une récupération active de l'information. Notre recherche fait progresser le débat en analysant l'interdépendance entre affordances organiques et programmées. Nous avons ainsi pu démontrer que la consommation organique des utilisateurs de Deezer provient rarement d'une orientation programmée. Combinée à la littérature antérieure, cette recherche dé-

montre que les plateformes de streaming musical ne sont pas adoptées principalement pour une récupération passive de l'information. Cela apporte une contribution significative aux Études des Médias et à la Sociologie de la consommation, qui jusqu'à présent avaient largement abordé ces questions sous un angle théorique plutôt qu'empirique. En termes simples, nos recherches montrent clairement que les utilisateurs de streaming musical peuvent être des participants plus actifs dans la construction de leurs expériences d'écoute que ce qui a souvent été dépeint.

Avec la motivation de construire et d'appliquer un espace d'*embedding* d'items approprié pour capturer géométriquement la diversité de la consommation des utilisateurs de Deezer (c.-à-d., via la dispersion dans l'espace vectoriel), la recherche publiée et présentée à Complemet en 2024 représente une avancée substantielle dans le domaine croissant des *embeddings* interprétables (Şenel et al., 2018; Gastaldi, 2021). Ce travail introduit non seulement un cadre novateur d'évaluation de l'interprétabilité mais aussi une nouvelle évaluation des performances de plusieurs modèles classiques d'*embedding* de nœuds dans ce cadre, modèles qui ont été examinés comme candidats pour notre cas d'usage. Cette recherche contribue de manière significative non seulement à cette thèse, en ancrant les conceptions méthodologiques des études ultérieures, mais aussi à un champ plus large de travaux portant sur la compréhension de la distance en espace d'*embedding*, comme les recherches sur la génération de recommandations algorithmiques explicables (Zhang et al., 2020). Bien que non directement liée aux questions centrales de cette thèse, cette étude joue un rôle important en identifiant quels modèles sont les plus adaptés à la construction d'espaces d'*embedding* d'items en haute dimension pour mesurer la diversité de consommation des utilisateurs, en particulier à travers différentes affordances. Nos résultats montrent que la Décomposition en Valeurs Singulières (SVD) est, peut-être de manière inattendue, le modèle le plus interprétable testé, même lorsqu'il s'agit d'examiner comment les voisinages d'ordre supérieur sont projetés dans l'espace vectoriel latent. Ce résultat a guidé notre décision d'appliquer SVD pour générer les espaces d'*embedding* utilisés dans les deux études plus orientées vers l'impact qui suivent.

La recherche suivante, publiée dans *Scientific Reports*, a abordé la RQ2 : « *Dans quelle mesure les données d'interaction multi-affordances utilisateur-plateforme peuvent-elles fournir des perspectives nouvelles sur le rôle des dispositifs de recommandation algorithmique personnalisée des plateformes de streaming musical dans l'encouragement de pratiques de consommation confinées ou diversifiées ?* » et la RQ3 : « *Comment les choix d'échelle et de représentation de l'information influencent-ils fondamentalement les conclusions tirées de telles analyses ?* » La contribution de cette recherche fait progresser

la littérature sur les évaluations agrégées du confinement algorithmique en soulignant l'importance de l'échelle et de la représentation de l'information. Ceci est accompli en réconciliant des études existantes qui suggèrent, malgré l'utilisation d'échelles temporelles différentes, que le confinement algorithmique a et n'a pas lieu respectivement sous des formulations denses (Anderson et al., 2020) et discrètes (Villermet et al., 2021). À échelle constante, nos recherches montrent que les mêmes différences de diversité apparaissent entre affordances, soutenant l'idée que ces différences dans la littérature ne proviennent pas nécessairement de variations d'échelle. De manière cruciale, elles semblent découler du fait que la diversité dense et la diversité discrète sont des concepts *orthogonaux*, non seulement au niveau des utilisateurs comme la littérature antérieure l'a suggéré (c.-à-d., les utilisateurs les plus novateurs ne sont pas les plus divers Mok et al. (2022)), mais aussi, comme nous le démontrons, au sein même de la consommation individuelle des utilisateurs. Cela nous conduit à proposer une version actualisée du récit de la bulle de filtres, qui apporte un développement significatif aux évaluations unidimensionnelles menées jusqu'à présent : les dispositifs algorithmiques peuvent fonctionner à la fois comme des forces de confinement de la diversité, favorisant des expériences d'écoute sémantiquement homogènes, et comme des forces d'expansion de la nouveauté pour les utilisateurs. En termes simples, les algorithmes semblent pousser vers de la musique encore jamais écoutée mais néanmoins similaire. Étant donné que les consommateurs de musique tendent à préférer la redondance même sans algorithmes ni plateformes (Chmiel and Schubert, 2017; Sguerra et al., 2022), nous pouvons théoriser que le haut niveau de nouveauté introduit par les algorithmes peut aider à expliquer pourquoi les utilisateurs de Deezer privilégient massivement les affordances organiques par rapport aux affordances algorithmiques.

En même temps, cette recherche met également en lumière l'impact du design de recherche lorsqu'il s'agit d'étudier le confinement algorithmique sur les plateformes de streaming musical. Plus précisément, elle illustre les effets marqués du passage de micro à macro-échelles. À l'intérieur des sessions, et sous un cadrage dense et géométrique de la concentration, les dispositifs algorithmiques peuvent créer des expériences d'écoute plus diversifiées que celles obtenues organiquement par les utilisateurs. Pourtant, au fil du temps, comme nous le montrons, ils n'élargissent pas vers des recommandations dissimilaires et diversifiées et exploitent à la place de manière répétée les mêmes régions de l'espace d'*embedding*. Cette constatation a plusieurs implications pour la littérature existante. D'une part, elle met en évidence les limites des extrapolations et comparaisons des conclusions issues d'une littérature orientée vers des évaluations temporelles du confinement algorithmique. Étant donné que dans ce courant, une variété d'échelles temporelles

sont généralement appliquées, il n’est guère surprenant que les résultats soient également hétérogènes. D’autre part, du point de vue des praticiens, elle souligne les limites des évaluations « au-delà de la précision » des systèmes de recommandation uniquement au niveau de la session ou des « listes » de recommandations, une pratique relativement courante dans l’évaluation des systèmes de recommandation (Kaminskas and Bridge, 2016). Puisque les évaluations par les utilisateurs des recommandations sont intrinsèquement temporelles et que nous faisons l’hypothèse que leur appréciation des algorithmes se construit dans le temps, de telles évaluations risquent de créer un décalage sémantique entre ce que perçoivent les utilisateurs et ce que mesurent les métriques.

De plus, les résultats et la méthodologie de cette recherche apportent également un développement substantiel à la littérature existante sur les multi-affordances, permettant ainsi d’obtenir des mesures plus riches du confinement algorithmique, plus proches de la manière dont les utilisateurs expérimentent et perçoivent la diversité sur la plateforme (Porcaro et al., 2022). En divisant l’affordance organique en playlists personnelles (P) et recherches par requêtes (Q), nous démontrons que l’adoption de Q tend à disperser l’attention des utilisateurs le plus largement dans l’espace vectoriel sémantique, tandis que l’adoption de P entraîne une consommation beaucoup plus homogène. De même, en modélisant la similarité entre affordances, une étape encore non entreprise dans la littérature existante sur les multi-affordances, nous montrons que les régions les plus éloignées et diversifiées du paysage informationnel d’une plateforme tendent à être atteintes via la curation éditoriale humaine. D’un point de vue sociologique, cela a des répercussions significatives : les expériences d’écoute éclectiques ne sont peut-être pas principalement construites algorithmiquement mais plutôt façonnées par la curation humaine, soulignant le pouvoir potentiel de ces acteurs en tant que nouvelles formes d’intermédiaires culturels et de distillateurs d’*éclectisme*.

Dans le dernier article de recherche, nous nous sommes concentrés sur la compréhension de l’impact des sélections des utilisateurs sur la diversité discrète de leur consommation (c.-à-d., la nouveauté). Cela a abordé la RQ4 : « *Dans quelle mesure les sélections des recommandations algorithmiques par les utilisateurs sont-elles également responsables de l’induction de pratiques de consommation confinées ?* ». En exploitant la combinaison de la diversité souhaitée, exposée et sélectionnée, rarement considérée conjointement dans la littérature MIR (Music Information Retrieval), cette recherche apporte une contribution significative et novatrice tant à la MIR qu’à la littérature plus large sur le confinement algorithmique agrégé en soulignant le rôle de l’agence des utilisateurs dans les boucles de rétroaction homme-machine et, plus spécifiquement, dans les bulles de filtres. En partic-

ulier, une des principales nouveautés de ce travail réside dans son articulation des données déclaratives avec les données comportementales. Dans les études existantes sur le confinement algorithmique, cette combinaison riche de données a été jusqu'à présent rarement exploitée, notamment pour comparer des données de préférence déclarée avec des données de consommation issues de journaux d'interactions effectives sur la plateforme. Tout comme dans les études sur les moteurs de recherche et les plateformes de réseaux sociaux dans le contexte d'informations politiquement sensibles (Robertson et al., 2023; Bakshy et al., 2015), nous montrons que les utilisateurs de Deezer ont tendance à sélectionner moins de diversité que celle à laquelle ils sont exposés algorithmiquement. La combinaison des données comportementales et déclaratives révèle une contradiction frappante entre comportement et déclarations : lorsqu'ils ont le choix, les utilisateurs se tournent systématiquement vers l'exploitation dans leurs interactions avec les systèmes algorithmiques, même lorsqu'ils expriment un désir de plus de diversité. De plus, ce schéma ne peut pas être expliqué simplement par des tendances générales des utilisateurs à interagir avec des recommandations sémantiquement similaires ou dissemblables. Pris ensemble avec les résultats antérieurs de cette thèse, nous voyons que la tendance des affordances algorithmiques à favoriser de hauts niveaux de nouveauté ne découle pas de la manière dont les utilisateurs exploitent ces affordances (puisque leurs sélections réduisent la nouveauté), mais plutôt de la manière dont les algorithmes fonctionnent par conception et des types de recommandations qu'ils produisent. Cela soutient l'hypothèse selon laquelle les utilisateurs de streaming musical privilégient les affordances organiques par rapport aux algorithmiques parce qu'ils recherchent des niveaux plus élevés de redondance et d'exploitation dans leur consommation, un phénomène qui précède les plateformes de streaming musical et leur intégration dans le quotidien (Fuentes et al., 2019). En même temps, ce travail a également offert des perspectives précieuses pour les praticiens développant des stratégies d'intervention sur les plateformes. Comme nous le montrons, et conformément à la littérature existante (Chen et al., 2024), accroître à la fois la diversité dense et discrète dans les recommandations algorithmiques n'est peut-être pas la stratégie optimale pour inciter les utilisateurs à une consommation plus diversifiée. Au contraire, les utilisateurs tendent à réagir en biaisant encore davantage leur écoute vers l'exploitation. En d'autres termes, pour reprendre la métaphore de Pariser (Pariser, 2011), les utilisateurs peuvent être plus exploratoires lorsque les recommandations restent à l'intérieur de leur « bulle », tandis que les pousser à l'extérieur peut encourager un retour vers un territoire familier. Une fois de plus, ces résultats soulignent l'orthogonalité entre diversité dense et diversité discrète dans la consommation musicale en ligne. Cette tension existe non seulement entre utilisateurs (Mok et al., 2022) et au sein des consommations individuelles à travers différents

canaux d'affordance, mais aussi dans la manière dont les utilisateurs interagissent avec les dispositifs algorithmiques eux-mêmes. Étant donné que les mesures de confinement algorithmique peuvent varier considérablement selon les échelles, il est notable que cette relation émerge de manière constante à travers les contributions de cette thèse.

Pris ensemble, les apports scientifiques de cette thèse se combinent pour constituer un défi majeur aux récits techno-déterministes existants qui ont longtemps façonné le discours dans des domaines tels que l'Interaction Homme-Machine et les Études des Médias. Comme nous le montrons, les utilisateurs contemporains de streaming musical sont loin d'être des récepteurs passifs des plateformes et des algorithmes. Du moins dans le cas de Deezer, les plateformes de streaming musical ne sont pas couramment utilisées pour une récupération passive de l'information et, de plus, les affordances programmées qui facilitent une telle récupération ne mènent pas nécessairement à des pratiques de consommation plus confinées. Comme nous l'avons démontré, elles peuvent dans plusieurs contextes générer des niveaux plus élevés de diversité dense et discrète qui peuvent, nous l'hypothétisons, avoir le potentiel de favoriser des pratiques d'écoute plus *actives*.

En même temps, cette thèse souligne clairement l'importance d'adopter une perspective multipolaire pour examiner les dynamiques de confinement algorithmique sur les plateformes de streaming musical. La diversité dense et discrète sont des concepts orthogonaux non seulement *entre* utilisateurs mais aussi *au sein* des consommations des utilisateurs eux-mêmes. Cela suggère que les évaluations unidimensionnelles simples de type « bulle de filtres » sont insuffisantes pour capturer ces dynamiques et, sans doute, la manière dont les utilisateurs expérimentent la diversité en pratique. Enfin, et peut-être plus crucialement, cette recherche met en lumière la nature organique de la concentration discrète dans la consommation musicale en streaming : les utilisateurs biaisent systématiquement leurs sélections de recommandations algorithmiques vers la redondance, même en contrôlant la similarité sémantique et même lorsqu'ils expriment un désir de nouveauté. Tout bien considéré, en affirmant à la fois l'agence organique des utilisateurs et la nature multipolaire du confinement algorithmique, cette thèse constitue un défi majeur aux récits techno-déterministes omniprésents et repositionne l'utilisateur comme acteur central dans les boucles de rétroaction homme-machine et les processus de confinement algorithmique.

Recherches futures

Bien que cette thèse ait abordé à grande échelle de nombreuses questions de recherche encore peu explorées empiriquement, plusieurs autres demeurent largement inexaminées et

représentent pourtant des pistes prometteuses pour de futurs travaux. Si cela est presque toujours le cas pour des recherches menées sous contraintes de temps et de financement, nous détaillons néanmoins ici certaines de ces directions parmi les plus prometteuses.

Avant tout, l'évaluation causale du confinement algorithmique par des méthodes contrefactuelles demeure l'une des directions les plus prometteuses mais encore sous-explorées pour la recherche future dans des contextes multi-affordances. Comme exposé dans la revue de littérature, isoler un signal contrefactuel qui capture le comportement des utilisateurs en l'absence de guidage algorithmique (c.-à-d., organique moins algorithmique, ou $O \setminus A$) est essentiel pour passer d'exposés descriptifs à des affirmations d'effet causal. Toutefois, y parvenir est méthodologiquement difficile. L'assignation aléatoire d'utilisateurs en groupes traitement ou contrôle est largement infaisable en pratique, car elle requerrait des interventions au niveau de la plateforme qui risquent de provoquer du *churn* et sont donc peu susceptibles d'être mises en œuvre. En l'absence d'expériences randomisées, les chercheurs doivent recourir à des méthodes quasi-expérimentales visant à approximer des effets causaux dans des environnements naturalistes. Bien que cette thèse fasse un pas en ce sens en appliquant une approche *différence-en-différences* (DID) pour estimer l'impact de l'écoute algorithmique novatrice sur la nouveauté organique subséquente, de telles méthodes supposent encore la présence de l'algorithme et ne capturent donc pas pleinement un véritable contrefactuel $O \setminus A$. La construction d'un tel contrefactuel demeure significativement plus complexe. Néanmoins, un ensemble de techniques établies d'inférence causale, telles que les variables instrumentales, les méthodes d'appariement, et d'autres (Liu et al., 2021; Bailey et al., 2024), offrent des voies prometteuses pour la recherche multi-affordances. Bien que ces outils puissent nécessiter des innovations méthodologiques pour être adaptés aux défis spécifiques des données de streaming et des comportements médiés par plateforme, ils représentent une étape cruciale pour faire progresser la compréhension causale dans ce domaine.

De même, une direction supplémentaire pour les recherches futures consiste à explorer les dynamiques temporelles des expériences d'écoute diverses et la façon dont elles varient selon les affordances et les types de guidage. Comprendre les effets à plus long terme de l'exposition à la diversité, comme de son absence, pourrait apporter des éléments importants sur le comportement des utilisateurs et l'impact des plateformes. Alors que cette thèse capture principalement des effets à l'échelle intra-session, de futurs travaux pourraient étudier comment ces effets évoluent dans le temps, en mobilisant par exemple des méthodes telles que des différences-en-différences à retards (*lagged DID*) ou d'autres approches longitudinales. Cela aiderait à déterminer si l'influence de l'écoute diverse

se maintient, s'amenuise ou se renforce au cours du cycle de vie d'un utilisateur sur la plateforme.

Une autre piste prometteuse pour de futurs travaux serait d'intégrer les historiques d'adoption des affordances avec des entretiens utilisateurs afin de mieux comprendre ce qui pousse les utilisateurs à privilégier les affordances organiques dans la majorité de leurs interactions sur la plateforme. Bien que des entretiens augmentés par les données aient déjà été employés dans le cadre du projet RECORDS (Cura et al., 2022), ils n'ont pas exploré ces thèmes spécifiques. Suivre cette direction pourrait permettre de mieux comprendre si les utilisateurs citent explicitement un désir de contrôle comme raison centrale de leur engagement prédominant avec les affordances organiques. Un tel éclairage offrirait un contexte précieux aux motifs comportementaux observés dans cette thèse et aiderait à relier les constats empiriques aux motivations déclarées par les utilisateurs.

Dans une direction qualitative similaire, une autre voie importante pour de futurs travaux sociologiques pourrait consister à approfondir la compréhension des pratiques des curateurs éditoriaux. Comme nous l'avons montré dans cette thèse, les affordances éditoriales peuvent, soutenons-nous, agir comme de nouvelles formes d'intermédiaires culturels qui exposent l'éclectisme. Néanmoins, nous ne pouvons pas encore affirmer que la musique exposée par les affordances éditoriales est dépourvue d'intervention algorithmique. En pratique, les éditoriaux peuvent s'appuyer sur des outils algorithmiques pour suivre les tendances et le comportement des utilisateurs. En outre, les playlists elles-mêmes, exposées via l'interface de la plateforme, sont par nature personnalisées de manière algorithmique. Par exemple, un fan assidu de hip-hop peut voir une playlist éditoriale « Hip Hop 90s » sur sa page d'accueil mais pas une playlist Rock. Une direction majeure pour de futurs travaux pourrait donc consister à comprendre qualitativement les pratiques des curateurs éditoriaux et dans quelle mesure ils s'appuient sur ces types de dispositifs dans leur pratique.